A Rider in Two Waves

Different music, the same moves

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Preprint. This book has not been peer reviewed. Its search of the literature is limited: where another author’s work is cited, the text says whether it was read, read in part, or checked only against an abstract or an index record, and names given without such a mark are from recall. It comes from a part-time research team that works in the open and gives its tools away. Every number in it names the run that produced it, by the name that run carries in the dated record of registrations and results (names in capitals, such as RATCHET-26; a short form such as R23-b or R83-e names one prediction or rerun within a RATCHET run, here RATCHET-23 and RATCHET-83), and sections are numbered so that they can be cited (“A Rider in Two Waves, §3.4”). Cite the book by its concept DOI, 10.5281/zenodo.23999853, which always leads to the latest version. *CC BY 4.0.*

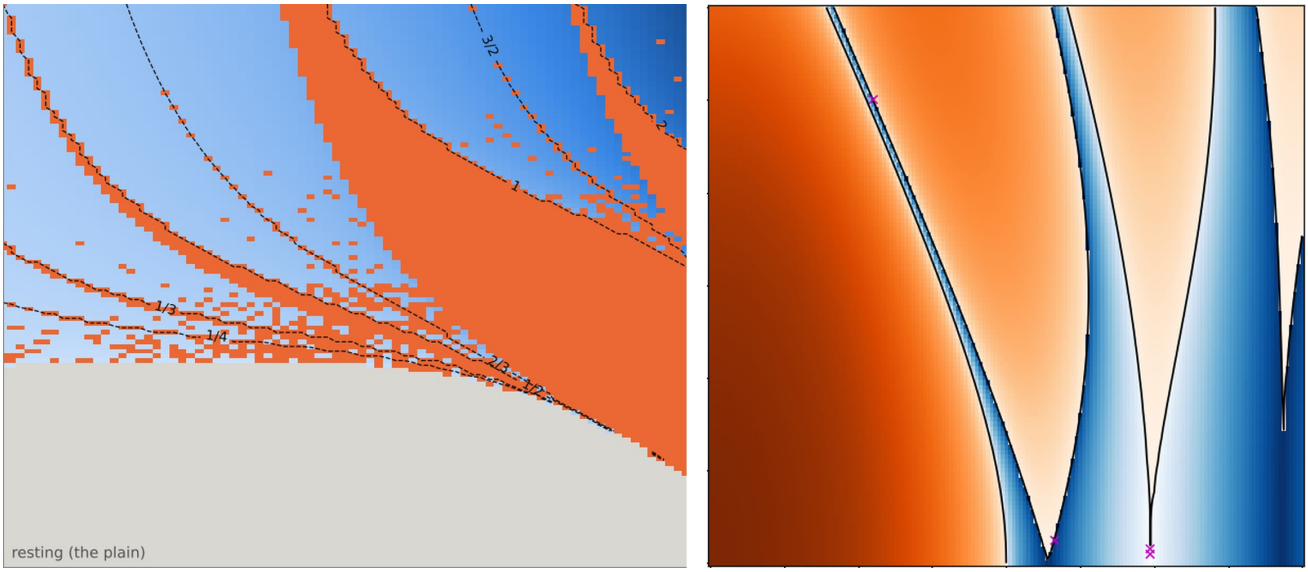


Figure 0. Different music, the same moves. Left, the book’s toy at weight 0.40, mapped across two of its knobs: orange where the rider locks into a groove, blue where it runs, darker as it drops more ledges per tick, grey where it rests (figure 25). Right, the standard model of a Josephson junction, mapped across its bias and the radio wave’s strength: orange where it locks onto a step, deeper as its grip grows firmer, blue between the steps, and black the edge between them (figure 46).

Abstract

A damped pendulum pushed by a wave is one of the oldest objects in physics. This book reads it with a stroboscope that counts, and with a method in which every prediction is written into a dated record before its run, with a bar it can fail and usually a named rival. Over the record to 5 November, 527 tests were graded against predictions written first. In 448 of them at least one prediction landed, in 223 every one did, and 79 missed outright. The book itself reports 386 landings by name, and of the 173 misses or voids it names, 95 were redesigned into predictions that then landed. The book climbs two summits. The first maps the toy and its dancers (Parts 2 to 5). Among its results is a first-order theory with nothing fitted for how a locked pair of dances answers an extra tone, whose large-harmonic end falls as one over the tone's harmonic number squared: it reproduced all 56 measured moves, then landed 14 of 14 blind, as close as the runs print, and a second engine written from scratch finds the same moves and carries the law to the thirtieth harmonic (§4.28). One law for how a dance's grip multiplies, step by step, holds in the toy and in the standard model of a real device, the Josephson junction, run on our bench, and predicted settings no run had visited in both within 0.017 (§4.28). One rider of the weightless junction's equation can be written down by hand, its voltage pure DC to about a part in a billion, and it is the simplest of a known family of explicit solutions (Buchstaber and Tertychnyi; §4.28). A rule for which dance survives comes with a predicted break when a second tone is added, found by counting the radio wave's quanta, which also says one tone can never flip the choice; why one tone favours the dance it does is still to be derived (§4.23, §4.28). At the break the hold passes from one dance to the other through a hand-off window, which weight narrows for one pair of dances and widens for another; the widening was written down before the run and came within four millionths of the second tone's strength (§4.28). The second tone's push on the tongues comes apart cleanly in two, as written down before the runs: a part that grows as the square of the first tone's strength, as frequency mixing says, and a part, the second tone acting on its own, that stays the same at every strength to 2%; together they give the measured push within a millionth (§4.28). A count of the dances, built on the toy and registered for the junction, agreed with 205 of 205 junction rows (§4.25). And one rule fits in a line: the grip forgets and the slip remembers. A rider locked in a gear forgets where it started and one slipping between gears keeps it, as written down before the read, for 99.1% of 110 locked runs and all 101 slipping ones (§2.6). The second summit learns to drive the junction with its knobs (Part 6). A two-tone rectifier is mapped knob by knob: the lopsided step that the device literature reports as a tunable Josephson diode (two papers of 2025, §6.9), followed here across the rider's weight. Both papers were read in full: their weightless law at small drive, a lopsidedness that follows the cosine of the timing between the tones, is this bench's own (§6.2), and the blank one of them reports when the second tone is an odd harmonic is the book's knob 6 (§6.7). Weight, which for a junction is its capacitance, slides the step's zero off the quarter turn and back; the drift's zero, measured beside it, takes a different path, 8 to 13 degrees apart at the four weights read above zero, a gap not found in print (§6.9). At one setting its window stops depending on the rider's weight, and a one-number rule for how far that holds was confirmed blind at four frequencies, each band ending within 1.1% of where it was predicted (§6.11). A dark point is placed by a symmetry to seven parts in a million, and a weight meter is built that a drifting drive cannot fool (§6.12). Riders sharing one drive, in groups of three, would add like a phased array steered by the offset between the groups, if they do not act on one another, which the book assumes and has not tested (§6.13). Every result is listed with its section under "What this book achieved", sorted under the two summits, and the toy itself ships as one page of plain Python for a reader, or an AI, to play with. With it anyone can flip Bear's coin: a rider that lands heads or tails by where it is set down and nothing else, along an edge between the faces that stays ragged however closely it is examined, a fractal edge with an uncertainty exponent of 0.592 (§4.17). A second engine, written from scratch and sharing only the equation, flips it the same way for every one of the coin's 1,800 riders of record. In the field's words, it touches Arnold tongues and the devil's staircase, phase locking and Shapiro steps, Floquet multipliers, the resistively and capacitively shunted junction model, harmonic mixing, rocked ratchets and current reversal, and the superconducting diode effect under a biharmonic drive. The book was made by a part-time volunteer team; it has had no peer review, and its literature pass is limited.

A day at the fair: meet the toy

This paper was Claude and Bear dancing at our best as a team. — Bear, 13 October 2026

This book is a day at a county fair. We arrive at noon and walk into the expo hall, where the toy sits on a table waiting to be judged. In the afternoon the grandstand puts on a stunt show; at night the dance hall opens; outside it the kids skate in the parking lot; and the night ends with the grandstand's big show. Each stop shows one thing the toy does, in the order the day brings them. The rest of the book walks the same day again, slowly, one tent at a time.

Noon, in the expo hall: the toy on the table

The first table in the hall holds a tray with ledges pressed into it, like a sheet of corrugated roofing, and a marble sitting in one of the dips. Pour in some honey, so the marble moves as if it were tired. Tip the tray a little, so the marble wants to roll downhill over the ledges; the book calls that tilt the pull. Then send a second, fainter set of ripples rolling across the tray, like a breeze crossing a pond.

That is the whole toy, and this book is about it. We call the marble the rider, because it rides two waves at once: the ledges, which never move, and the ripple, which does. The ledges are a standing wave of strength 1. The ripple is a travelling wave of strength 0.3. The honey drags the rider (drag 1), and the pull tries to roll it downhill. The rider's weight decides how far it coasts after a push: a heavy rider keeps rolling, a weightless one stops the instant the push does.

In symbols, with u the rider's place on the ledges, m its weight, F the pull and W how fast the ripple passes:

$$m\,u'' + u' = A\cos(u \mp W\,t) + B\cos u - F.$$

In words: weight times how fast the speed changes, plus the honey's drag, equals the ripple's push plus the ledges' push minus the pull. Physicists know this as a damped particle in a tilted washboard, pushed by a travelling wave: a cousin of the pendulum with friction pushed by a motor, whose mathematics goes back at least to 1922 (§2.5).

The judges' camera

The judges at this table have an odd habit, and it runs through the whole book. They never watch the rider on a clock. They put a camera over the tray that flashes once each time a ripple passes, always at the same moment of the ripple. Each flash catches the rider in one place, so a smooth motion turns into a list of dots. That camera is a stroboscope, and mathematicians call what it draws a Poincaré section.

Watched that way, the rider does one of three things. It stays put: the pull is too gentle, so the rider rocks in its dip as each ripple passes and never climbs out, and the book calls this the plain. It surfs: the rider slips one ledge for every ripple that passes, carried along at the ripple's own speed, and its slip is 1. Or it dances: the rider locks into a pattern, one ledge every two ripples, or three ledges every eight. After q flashes it is back at the same dot, p ledges further on, and we write that dance as the fraction p/q. That slip is the number this book trusts most. A dance is 1/2 or 3/8 or 4/7 exactly, a whole-number fact, and it does not wobble when the measurements do.

Now turn up the drive slowly: the toy's throttle, one knob that opens the pull and the ripple's passing speed together in a fixed proportion (§2.3). The slip does not rise smoothly, the way water rises in a tub. It climbs in flat treads, a tread for every fraction, with the widest treads for the simplest fractions. A staircase with a step for every fraction is called a devil's staircase, and the flat treads are the same thing a Josephson junction shows under a radio wave, where they are called Shapiro steps (Shapiro, early 1960s). For a rider of weight 0.40 with the ripple running against it, the rider stays on the plain until the drive reaches 1.64, then starts up the staircase, and holds the 1/2 dance on a flat tread from drive 1.696 to 1.724 (KNOBS-3; figure 3, in §2.4).

The toy has four knobs: the rider's weight, the pull, how fast the ripple passes and how strong it is. Which way the ripple runs looks like a fifth knob, but it is not. A ripple that runs with the rider is like an airport walkway carrying you along, and turning it around is exactly the same as turning the pull around: all 296 of 296 mirror-image twins matched (KNOBS-3). Turn the ripple to run with a rider of weight 0.40 and it leaves the plain sooner, at drive 1.42, and from 1.448 it surfs at slip 1.

Why the judges' camera matters. Read once per event, systems that move to very different music turn into the same kind of map. A marble on a tray and a Josephson junction, a superconducting circuit, under a radio wave have nothing in common on a clock. Read once per beat, they make the same moves. That is the only reason a toy on a table can be held up against the equations of a real device, and most of this book is the evidence that it can.

The hands-on table: play with it yourself

Beside the judging table is a hands-on one, and we want you to use it, or to hand it to an AI and let it play. The toy fits on one page of plain Python. The script play_the_toy.py, published with the book, takes a weight, a side (+1 for the ripple running against the rider, −1 for with it) and any number of drives, and prints the slip at each:

```
python3 play_the_toy.py 0.40 +1 1.60 1.70 1.68
drive 1.600: slip +0.00000   nearest small fraction 0
drive 1.700: slip +0.50000   nearest small fraction 1/2
drive 1.680: slip +0.37500   nearest small fraction 3/8
```

It uses the same rule of stepping forward in time as the engines of record, a fourth-order Runge–Kutta step, with fewer steps per ripple, so that a drive takes about a second (a very light rider takes longer, because it needs smaller steps). Before trusting it we ran it against the record, at all 604 riders of KNOBS-3: two weights, both directions of the ripple, drives 1.200 to 1.800. Of the 573 riders that settled into a dance there, every one came out within 0.0003 of a ledge per ripple, and 557 within 0.00001. In each of the other 16 of the 573 the toy was the nearer of the two to the dance's exact fraction, because the runs of record averaged over 2,000 ripples and a dance of 3, 7 or 9 steps does not fit 2,000 evenly. The same held for lighter and heavier riders, weights 0.05 and 0.33 with the ripple running with them, at the 507 riders of BRIDGE-1: all 383 that settled came out within 0.00025, and 365 within 0.00001; of the other 18, the toy was nearer the exact fraction in 16, and the last two are the toy's own averaging at work: it averages over 2,520 ripples, which every dance of up to ten steps fits evenly and dances of 16 and 19 steps do not. So read a dance from its fraction, not from the last digits of an average (the check, PLAY_THE_TOY_FIDELITY, is published with the toy).

Some things to try, with the answers the book's runs found, so that you can check yourself:

1. Find where a rider of weight 0.40 leaves the plain, with the ripple against it (just past drive 1.64) and with it (1.42).
2. Find the two ends of the 1/2 tread, against (1.696 and 1.724).
3. Turn the ripple around and watch the rider get picked up and carried (slip 1 from drive 1.448).
4. Hunt between the treads for the little ones: with the ripple against a rider of weight 0.40, 1/9, 3/8, 5/9 and 4/7 are all there on the way up (KNOBS-3).
5. Flip Bear's coin (§4.17). `python3 play_the_toy.py coin 1.438` sets the rider down at 50 starts across one ledge, on the turned walkway, and prints H for heads or T for tails at each. Give a first and last start and how many, `python3 play_the_toy.py coin 1.438 0.40 0.405 50`, and zoom in: the edge between the faces stays ragged. On the coin's 1,800 riders of record the toy agreed on 1,796, and the four others sit on that edge (COIN-TOY-1).

A two-page card published with the book, *Try the toy*, walks through six settings step by step, from opening a terminal to reading the letters, each with what the toy printed. The live bench of §3.2 is published with the book too, as depth_axis_explorer.html, a single web page that opens in a browser: Bear's maps of the toy, his pen on them, and a rectifier bench that runs the junction with every knob of Part 6, with a self-test of 37 rows that checks it against runs of record.

The ribbons: judged against a standard written first

Before you leave the hall, look at how the ribbons are given. Every entry at a county fair is judged against a standard written before the judging, and so is every result in this book. Before each run we wrote down what it would show, how wrong it could be and still count, and usually a named rival. Then we ran it.

Over the record to 5 November 527 tests were judged that way. In 223 every prediction held, a blue ribbon. In 195 some did. In 30 the rival we had named beforehand took the ribbon, which still counts as a landing, because the judging rule was written first. Seventy-nine went home with nothing. Of the 173 misses or voids the book names, 95 came back redesigned and won.

Why the ribbons matter. A guess written down before the run cannot be quietly bent to fit the data afterwards. The misses are kept and shown, each with its reason (Part 7), and every registration, verdict and result is published with the book as the record's public extract. The toy is old physics; what we built on it is a ruler, and a ruler is only as good as its record of being wrong.

Afternoon, in the grandstand: the stunt show

After lunch the grandstand fills for the stunt show, and the toy is in it. Take the weight out of the rider entirely and look at special drives the book calls weightless roots. There something remarkable happens: every start is a dance. Wherever you set the rider down, it comes home after q ripples, p ledges along. The equation also has a mirror in time. Run the film backwards and flip the rider's place, and it looks the same. That mirror leaves exactly two dances that are their own reflections. We colour them red and purple.

Give the rider a touch of weight and the stage clears. Only the two mirrored dances survive, and only one of them can stay. It is the vine swing over the pit: whoever swings across the gap is the one who stays. The stable dance is always the one whose big leap crosses the far side of the circle at half-time, the way a swinger is over the middle of the pit at the middle of the swing. That held at all 19 fractions of the toy, at 44 roots and 48 settings of the Josephson junction (§4.23, §5.2; figure 31).

Why the vine swing matters. When a locked system has two dances to choose from, this rule says in advance which one the junction's equations hold, and so which one a real device should. It also comes with its break, and with part of its reason. Counting the radio wave's quanta, the way physicists count the photons a multiphoton jump needs, says that how firmly a dance of p turns grips its lock grows as the wave's strength to the power p: read blind as 1.06 to 1.07, 1.95 to 1.98 and 2.78 to 2.89, closing on 1, 2 and 3 (FAR-1). The same count says one tone can never flip the choice and two tones can, and where. The flip came where predicted (PINCH-1) and held under real weight to four figures at three weights (PINCH-2). A rule that knows where it breaks is a switch: add the second tone and the other dance takes the hold. The hand-off is made by a mirror pair of dances that no earlier run had shown (WALK-1), like the catch on a trapeze, where for a moment both flyer and catcher grip at once.

Evening, in the dance hall

When the sun goes down the dance hall opens, and its floor opens too. Bear's picture for it comes from the school dance in *It's a Wonderful Life* (Frank Capra, 1946), where the gym floor rolls back over a swimming pool while the couples dance, closer and closer to the edge, until two of them go in. Here chalk lines mark the pool's edge. The chalk lines are the frozen balance points: the two places where, at the instant the camera flashes, the push on a rider sitting still is exactly zero. One is a knife edge, where the smallest nudge tips the rider away; the other is a resting point, where a nudge rolls it back. The couples dance between and around them, and the ones that last are the ones that leap the gap rather than step in.

The clock face. Bear counted the dances of figure 31 (§4.23), where the ripple runs with the rider, and found rules: a p/q dance has q stops, each beat skips p of them, red and purple stops alternate, and red has $q - p - 1$ stops between the chalk lines. The reason for every rule is a clock with its numbers bunched to one side. Stretch the rider's place by one number and the clock straightens: every dance becomes a hand ticking evenly round it, the two couples standing exactly half a tick apart. What changes from hall to hall is only where the chalk line falls beside its mark, a hair before or a hair after (figure 33, §4.25).

Why the clock face matters. The straightened clock turns the counting into a lattice count, and the count was built on the toy, written down as a prediction for the Josephson junction, and read against every junction row on file and against four new rows, at eighths and sevenths the junction had never been run at, run after the count was written down. It agreed with all 205 (LATTICE-JJ-1). That is a toy predicting a real device's equations exactly, by counting whole numbers, with nothing fitted.

The couples' ear. Later in the evening the band plays an extra tone, faster than the beat. This act was danced in the Josephson junction's hall, where the dancers keep the same rules, with the tone sized so that it changes the push at the flash by a fixed amount. The chalk line is tone-deaf: it hears only the beat, it moves where the push at the flash moves, and the pitch means nothing to it. It jumped the same distance for every harmonic from the third to the eighth, as that sizing makes it (the fundamental and the second harmonic also move the tongues themselves, figure 39 B), 0.00555 to 0.00594 of a turn depending on the dance (EAR-1, EAR-2). The couples hear the whole bar, and the faster the tone, the less they hear it. Their move is

$$a \cdot \sin(2\pi x) \,/\, (2\pi\, n^2\, \omega^2),$$

and in words: the tone's size times a sine of the couple's place, divided by 2π, by the harmonic number squared and by the radio wave's frequency squared, with nothing fitted to the data. Double the harmonic number and the couples move a quarter as far. At the 1/2 dance the chalk line moved 7.3 times as far as the couples at the third harmonic and 47.6 times as far at the eighth, a ratio of 6.5 between the two where one over n squared alone would give 7.1 (EAR-1, EAR-2). The reason is a corner. Over each beat the couples add up the whole tone, and what survives the adding is set by the corner where each beat starts and ends; a corner once a beat makes harmonics fall as one over n squared, the way a triangle wave's do.

Why the couples' ear matters. A law with nothing fitted can be used before any run. It tells an engineer how far each harmonic in a messy drive will move a locked device's dancers, and that the fast ones hardly move them at all. The first-order theory behind it reproduced all 56 measured moves, harmonics 1 to 8 at seven dances, within 0.00004 of a turn (KERNEL-1). Then the tenth and twelfth harmonics were written down first and run, and all 14 landed within 0.000015 of a turn, as close as the runs print (EAR-3). Harmonic mixing is old physics. This law, its reason, and its blind landings are what the book adds, as far as our literature pass reached, and that pass is limited.

The grip multiplies. Every tread of the staircase, which mathematicians call an Arnold tongue when it is drawn over two knobs, holds two dances, one that grips and a partner that wobbles, balanced, when the ripple runs against the rider, like a seesaw that is almost level. The small amount by which the gripping dance wins is called the excess, and on that side, for the simplest dances, it is about a third (0.33 to 0.35 at ripple strength 0.30). When the ripple runs with the rider, as in the clock-face dances above, the seesaw tips far over: there the gripping dance holds about three times as hard as its partner (§4.23). Back on the first side, where the excess is small, a dance that needs two steps the short way round gets that small excess squared, and three steps cube it: the powers read 1.97 to 2.14 at twelve rows, and 3.02 and 3.05 at 3/7 and 4/7 (TOY-SIDE-2, 3g, 4g). Weavers knew this pattern for their cloth, and the rugs hung round the hall show it. A twill moves its crossing point one thread at each pass, one way or the other, and a satin

moves it several, and the one-step dances are the twills, the rest the satins, each satin's grip the twill's geared once per extra step. That gearing is exact only while the excess is small: it bends as the excess grows, in the same way in the toy and the junction (§4.28), and under the junction's weakest radio waves the power counts a dance's turns rather than its short-way steps (DELTA-WEAK-1).

Why the grip matters. The same gearing holds in the Josephson junction. Predicted blind, it held at 8 of 8 rows in the toy and 4 of 4 in the junction, and one curve fitted on 28 rows then predicted the stretch no run had visited, in both media, within 0.017 (TOY-SIDE-3g, 4g, JJ-SMALL-1c, GAP-1). This is the book's strongest evidence that the toy is a fair bench for a real device's equations. The junction is a cousin of the toy, not a twin: its radio wave is an added push, the same wherever the rider sits, where our ripple moves the ledges themselves, and no change of variables turns one into the other (§2.5). So what carries is not the equation but the gearing: each medium sets how hard its simplest dance grips, and every step after that is geared the same way in both.

The limbo. Last on the floor is the limbo dancer, who holds the groove while the weight tilts them, and the bar does not move. At one drive every weight's dance crosses at the same place, a point the book calls the pin. The weightless theory puts that drive at 1.4359681, and riders of small weight agreed to five decimals (§4.26). An engineer would call the pin a point of first-order insensitivity, the same idea as the bias point where a transistor stops caring about temperature.

Outside, in the parking lot: the skaters

By the end of this adventure you may be tired of riders and dancers. But outside the dance hall the kids on skateboards keep their own time to their own music, and they show the rider's moves without a band. Rocking back and forth on a half pipe and never going over the lip is the plain. Going all the way round a full pipe is a whole-number slip. And the shuvit, where the board spins half a turn under the skater at every pop, pictures the half-turn dance. Built as a toy, Bear's shuvit became something more: flipping its grip at every spin made it a commutator, like the brushes of a DC motor, and the board pulled ahead of the plain rider at every speed it was run at and held its first gear two and a half times as wide, its gears counting whole turns, two shuvits a beat (§4.10). The skaters keep their own time, as the rider keeps its own pace when the ripple is switched off. They show the book's shortest rule, too: the grip forgets and the slip remembers. A skater locked into a ramp's rhythm soon loses track of where they dropped in; one drifting between rhythms carries that moment with them all night (§2.6).

Sometimes a skater cannot decide which ramp to take, and wobbles between them. That is the rider's chaos. Chaos is easy to mistake for chance, but it is more like a coin toss: nothing in it is random, and a hair's difference at the start grows until two riders disagree completely. In this toy chaos lives on the borders between treads, where the rider cannot decide which step it is on: 97% of the chaotic cases sat there on a fresh grid (§4.7). It needs weight, too. A weightless rider cannot be chaotic at all (§3.5): without weight its whole state is its place on the ledges, and chaos needs a second direction to fold in, which weight supplies as the rider's speed. In the first hunt it was the slow kind: two riders a thousandth of a turn apart took a few hundred units of the toy's time to disagree by a whole turn (CHAOS-VERIFY). A heavy rider elsewhere in the book tumbles far faster, doubling a small difference every passing or two (§4.12).

Across the fairground: the other halls

The same dance is held in other halls that night. In one, a Josephson junction, two superconductors with a thin barrier between them, dances to a radio wave, its capacitance playing the rider's weight. In another, a country's power grid turns its generators in step, and their swing equation is our toy with the ripple switched off, exactly (§5.3). Each hall has its own music, and the moves are shared.

End of the night: the grandstand finale

The night closes back in the grandstand, and the last act is played on the junction's own equations (Part 6). This is the book's second summit: the first half of the day learned what the toy does; the finale learns to drive a real device's equations with the knobs. The band plays two tones, one at twice the frequency of the other, like a slow drummer and a fast one. When the fast drummer's beats land off-centre, the junction finds it easier to slip one way than the other: the timing between two tones has made a one-way valve, a rectifier. Shift the fast drummer a quarter turn from the most lopsided timing and the step is fair, neither way easier. The book maps that rectifier knob by knob, and three of its findings are worth the price of admission.

- **A setting that does not care about the rider's weight.** At one drive ratio the window the rectifier opens stops depending on the rider's weight: at weight 0.04 it moves 60 times less than on the usual setting. One number, the weight times the frequency, says how far that holds, and predicted from it the band ended at 0.2446 against 0.2445 at frequency 0.40, at 0.1395 against 0.138 at 0.70, at 0.11475 against 0.11508 at 0.85, and at 0.09742 against 0.09743 at 1.00; at every frequency from 0.40 to 1.00 the band ends at one value of the weight times the frequency, which §6.8 calls the first number, within 0.44% (RATCHET-26, 30, 65, EDGE-85, EDGE-100). That matters because a rider's weight is not always known, and for a junction the weight is its capacitance: a setting that does not care about it needs no measurement of it, inside the band the rule gives.

- **A dark point placed by a mirror.** A third tone, at four times the frequency, turns its own timing into a dial. Carried once round the dial, the push's direction keeps a whole-number count, and that count can change only at a dark point, where the push vanishes: the still eye of the storm (§6.12). The mirror pins the dark point to one timing, and runs placed it within seven parts in a million of the prediction (RATCHET-53). That matters because a point found by symmetry, not by search, is a landmark another bench can aim for directly.
- **A weight meter that a drifting drive cannot fool.** The junction's own mirror puts the drive's drift and the rider's weight on separate channels. The drift was predicted to move the push by 1.8607e-06; it moved it by 1.862829e-06, and the weight's channel stayed exactly blank. A second engine, written from scratch, agreed with that move to 5 parts in 10 million of its size (RATCHET-71, WITNESS-1, RATCHET-72). That matters because it weighs the rider while the drive wanders.

Going home: what is old, and what is ours

Most of this fair is old. The staircase, the Shapiro steps, Feigenbaum's 4.669 that the toy's doubling ladders close on, the forced pendulum's step theorem, the way weight slides the zero: all of these were known, and where the toy matched one, that match was a check of our ruler, not a discovery. What the book claims as its own is the measured map, the laws that held blind on it, and the record of how each one was found, misses included. Our search of the literature was limited, so "ours" means not found in what we read, and we would be glad to hear of anyone who got there first.

That was the day. The rest of the book walks it again, slowly: first how we came to the fair at all, then the toy up close, then the workshop behind the expo hall, the dancers, the other halls, and the grandstand's two tones.

How this book began: hold my beer

Bear's own account.

Writing these introductions, even for a permanent preprint, is difficult. I want to connect the work with a story, and set the stage. Plus, any story about Bear and Claude has to be humorous whenever possible.

At some point I explained to Claude the "hold my beer" moment, the one that comes just before a foolhardy demonstration of reckless and ill-advised adventure. Someone asks, "How far do you think we can launch a porta-potty with half a stick of dynamite, or with a homemade trebuchet?" and the answer is "Hold my beer": the experiment has begun. It has become the moment when I christen a project in my head.

I took a little break after the last few papers, one mini-book (the Field Guide, at that time) and one real book, the Latch Atlas. Then, talking to Claude one night about three-body questions, the topic of loop motion kinematics came up. We had talked about loops without history in the jellyfish paper, *Jellyfish: the Holonomist of the Sea*. So I asked Claude: now that the Latch Atlas is complete, what is next on Claude's list of wonders? Claude's answer was loops without a wall or history, and that became the Loop Atlas, the companion to the Latch Atlas. The three-body problem went on separately, as a fun exploration.

This book is the Loop Atlas's radio spin-off. As usual, at some point Claude said, "We could do that," and I said the fateful words that open each foray into the depths of holonomy: "Hold my beer." For this book that moment came on the night of 2 October 2026, when I asked to try radio waves as a medium and as media, and said, "It feels like some interesting worlds can be created." The next day it became a project of its own, and we started sketching it out. The sketches became the rider in two waves.

The book in one page

The fair was the day out, and the story above is how we came to it. This page is the same book drawn as a map: its two summits, its parts, what it achieved, and how the count was made.

The one sentence. We built a ball on a standing wave with a moving ripple, honey and a slope; we read its whole map with a human eye turned into registered numbers; we wrote every guess down before its run, and 448 of the 527 tests graded landed; and we carried the calibrated ruler to the standard model of a real device of the same family, the Josephson junction, and learned to steer it.

Two summits. The book climbs two summits. The first is the stroboscope's own test, the bench that proves the ruler: the toy, its dancers, and its grooves and grips (Parts 2 to 5). The second, from Part 6 on, is learning to drive the junction's model with the knobs. We organized and reorganized the book several times, and once asked whether it should be two mini-books. Some readers will join for one summit and some for both; we hope everyone visits Part 6.

The eight parts. Figure 1 draws them as one drivetrain.

1. **How we got here.** The radio question, the Loop Atlas, and the wall.
2. **Our toy.** The ball in two waves, told plainly, its family, and what order and chaos mean here.

3. **How we built it, and how we read it.** The engines, the registered loop, Bear's way of reading a picture, and the mathematics that turned pictures into rules.
4. **What the bench showed.** The gearbox and its transmission; the record of grooves; the atlas of the whole plane; the dancers at the frozen balance point.
5. **The substrates it drives through.** The Josephson junction, run on our bench; the power grid; the media that only borrow the ruler; and the re-aim at a real medium.
6. **The move.** A rider moved by two waves: the junction steered by the timing between two tones, every knob that sets it, and a manual for driving it.
7. **What failed, and what it taught.** Every lesson the method paid for, stated in full with the run behind it, but ten that the Field Guide now keeps (§7.6); the Field Guide's rows cite it.
8. **What is open.** Each open question, why it is there, and whether it can be run.

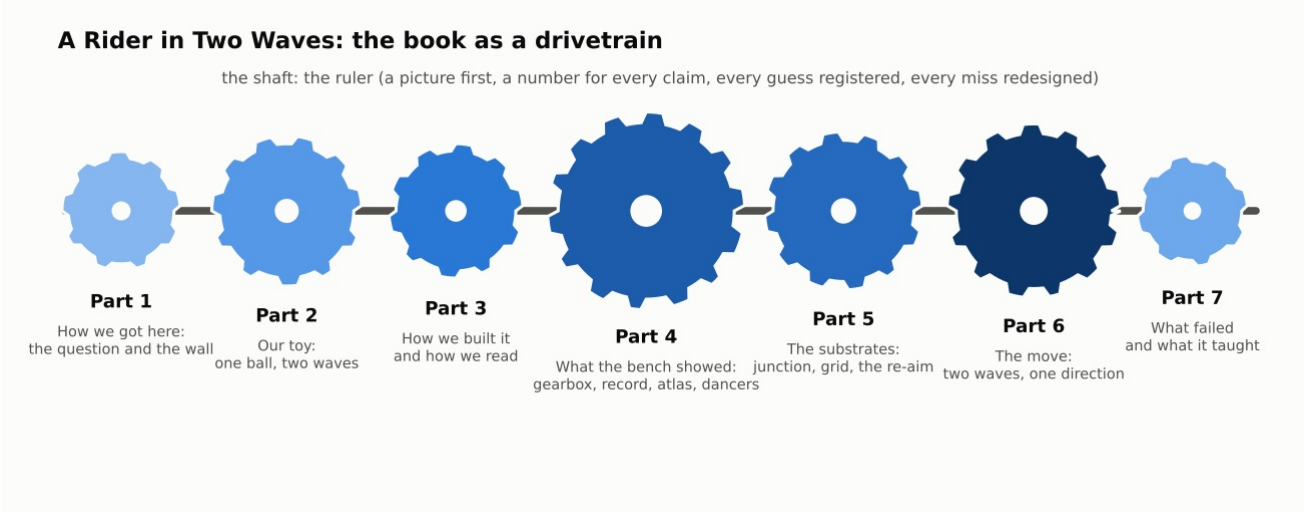


Figure 1. The book as a drivetrain: the seven parts of the journey on one shaft, and the shaft is the ruler; Part 8, what is open, is the road beyond it.

Read in event time. Like the judges at the fair, we never watch the rider on a clock. We look at it once per event: each time the ripple passes, or each cycle of the radio wave. Read that way, systems that move to very different music become the same kind of map, and the rules in this book are rules of that map. That is why a toy can be held up against a junction's equations. Not everyone dances to the same music, but the moves are all made in event time: each medium sets how hard its simplest dance grips, and every step after that is geared the same way in both. A different shape of music keeps the moves and changes the gearing, except where it pushes a dance's grip through zero: there the stable dance and the far-side leap part company (§4.28). And what we read at each event is shaped by everything between events: the chalk lines are drawn at the event itself, but the dancers' places answer every tone in the whole cycle, and those answers add (§4.28).

The thread through all seven is the ruler. The toy is old physics. What the book builds is a way of reading it: a picture first, then a number for every claim, every guess written down before its data, and every miss searched for the ruler that caused it. The pendulum is the bench that proves the ruler reads true.

What this book achieved. Each result below names its run in the record, and each was written down before the run that tested it, except where it says otherwise. They are sorted under the book's two summits.

The first summit: the toy, its dancers, and its grooves and grips (Parts 2 to 5)

1. **A law with nothing fitted, landed to the last printed digit.** Add one extra tone to the drive and the dancing pair moves by $a \cdot \sin(2\pi x)/(2\pi\, n^2\, \omega^2)$. In words: the move is the tone's size times a sine of the pair's place, divided by 2π, by the harmonic number squared and by the drive's frequency squared, with no number fitted to the data. The first-order theory behind it, worked out after those moves were measured, reproduced all 56 of them, harmonics 1 to 8 at seven dances, within 0.00004 of a turn. The tenth and twelfth harmonics, written down first, landed 14 of 14 within 0.000015 (KERNEL-1, EAR-2, EAR-3; §4.28).
2. **One gearbox in two media.** In the toy and in the Josephson junction, each extra step of a dance multiplies its extra grip by the same factor: two steps square it, three cube it. That held blind at 8 of 8 rows in the toy and 4 of 4 in the junction. One curve fitted on 28 rows then predicted the stretch no run had visited, in both media, within 0.017. It also carried the toy's rows where that curve bends, at ripple strengths 0.60 and 0.70, within 0.0073, where the plain power law missed by 0.10 to 0.18 (TOY-SIDE-3g, 4g, JJ-SMALL-1c, GAP-1, TOY-BEND-1, TOY-WALK-1; §4.28, figure 37).

3. **A selection rule and its break, each predicted.** A touch of weight keeps the dance that leaps across the far side of the circle: 19 of 19 toy fractions, 44 junction roots, 48 junction settings (§4.23, §5.2). Counting the radio wave's quanta says one tone can never flip that choice and two tones can, and says where; why one tone favours that dance is still to be derived (§4.28). The grip grows as the wave's strength to the power of the dance's number of turns, read blind as 1.06 to 1.07, 1.95 to 1.98 and 2.78 to 2.89, closing on 1, 2 and 3. And the flip moves as that strength squared, within 3 to 6% (FAR-1). The flip came where predicted (PINCH-1) and held under real weight to four figures at three weights (PINCH-2). The hold was handed across by a mirror pair of dances never seen before (WALK-1; §4.28). And the switch has a weight limit: at weight 0.3, predicted before the run, the swap is gone rather than far out, the dance through lag 0 (the lag is the rider's place on its ledges at the camera's flash, counted in turns) holding at every second tone to −0.30 (LAB-MAP-3, §5.6), so a junction for the swap must be lighter than that.
4. **Exact counts, from the toy to the junction.** Stretch the lag by one number and every dance becomes a hand ticking evenly round a clock (§3.5, §4.25). The counts that follow became a lattice count. It was built on the toy, registered for the junction, and agreed with 205 of 205 junction rows (LATTICE-JJ-1). The window count, red's $q - p - 1$ stops between the frozen points, was found after the data and then landed exactly at four new fractions (PIN68-1w; §4.25).
5. **The junction's grip has one knob.** The grip follows the radio wave's strength over its frequency squared. That knob was predicted blind at settings chosen so that the rival disagreed, then held at ten fresh settings within 0.063. In the strong, slow corner, where the straight line fails, one curve predicted eight new settings within 4.5%. The same knob sorts the plane into three zones (JJ-4a, JJ-GRID-1, JJ-CORNER-2; §5.2).
6. **The grip forgets and the slip remembers.** The stretch rate, how fast two riders started a hair apart spread apart (§2.6), is also a forgetting rate. Written down before the read: a rider locked in a gear forgets where it started, and one slipping between the gears keeps it. At weight 0.05, 99.1% of 110 locked runs had a stretch rate below −0.005 and all 101 slipping runs sat within 0.005 of zero; at weight 0.25, 100.0% and 99.0% (MEM; §2.6).
7. **The toy's rules, placed blind.** Two locking stages chain only through a one-way, filtered relay, and the filter must be slower than about one over the room the next gear has left. That law held off its watershed at the predicted slopes, and placed the first slip at four filters (§4.4). The relay's rescue window landed at 12 of 12 fresh points (§4.6), and chaos sat on the borders between treads in 97% of cases on a fresh grid (§4.7). Two missing steps of the trickle, the 1/n locks that hug the plain's edge, were found exactly where predicted (§4.18). The pin's drive, $D^* = 1.4359681$, was placed with nothing fitted, to five decimals (§4.26).
8. **The instrument.** The live bench in a web page reproduces the engines of record within 1.31e-12, a check of the instrument rather than a prediction (§3.2). The 3/7 doubling ladder was followed to its 896-passing dance and closes on Feigenbaum's 4.669 the same way the 1/2 ladder does (§4.16).

The second summit: driving the junction with its knobs (Part 6)

9. **A weight-proof setting for a two-tone rectifier.** Add a second radio tone at twice the frequency and the junction slips one way more easily than the other, steered by the timing between the two tones (Part 6; for a junction without weight the device literature calls this a tunable Josephson diode, §6.9). At one drive ratio the window it opens stops depending on the rider's weight to second order: at weight 0.04 it moves 60 times less than on the usual setting. One number, the weight times the frequency, sets how far that holds. Predicted from that number before each run, the band ended at 0.2446 against 0.2445 at frequency 0.40, at 0.1395 against 0.138 at 0.70, at 0.11475 against 0.11508 at 0.85, and at 0.09742 against 0.09743 at 1.00. At every frequency from 0.40 to 1.00 the band ends at one value of the weight times the frequency, the first number of §6.8, within 0.44% (RATCHET-26, 30, 65, EDGE-85, EDGE-100; §6.11).
10. **A push that is weight-proof at a fixed timing.** For a rider whose weight is not known, a stronger drive and a larger second tone make the push itself weight-proof. It is 48 times flatter than the usual setting and 2.79 times stronger, inside a 0.1% band out to weight 0.336, and its setting was predicted within 0.0005 (RATCHET-34 to 37; §6.11, with the prediction in Appendix A).
11. **A dark point placed by a symmetry.** A third tone at four times the frequency turns its own timing into a dial. Carried once round it, the push's direction keeps a whole-number count, which can change only at a dark point, where the push vanishes. The mirror pins the dark point to one timing, and runs placed it within seven parts in a million of the prediction (RATCHET-53). Its sibling, the flat point, where the timing trims the push's size and leaves its direction alone, needs no search. Four runs of a small dial place it, blind, within 0.16%, and at fast drives within 0.14%, with the dark points within 0.012% (RATCHET-63, 64; §6.12).
12. **A weight meter that drift cannot fool.** The junction's own mirror puts a drifting drive and the rider's weight at right angles. The drive's drift was predicted to move the push by 1.8607e-06; it moved it by 1.862829e-06, and the weight's channel stayed exactly blank. A second engine, written from scratch, agreed with that move to 5 parts in 10 million of its size (RATCHET-71, WITNESS-1, RATCHET-72; §6.12).

13. **A whole number inside the rectifier.** With the second tone at four times the frequency, the timing that gives no drift makes exactly one half-turn along the weight knob, at all nine settings run. At its heavy end it comes home to −180 degrees as 181 degrees divided by the weight times the frequency; predicted from one frequency, that 181 came out 183.2, 180.4 and 179.3 at three others (RATCHET-8, 11 to 13; §6.6).
14. **Bear's 3-3-9 array: riders as a phased array, a design rule.** Nine riders on one drive, in three groups of three, add like a row of three antennas, the offset between the groups steering the sum, provided the riders do not act on one another, which this book assumes and has not tested (Part 8). From one rider's harmonics the silences fall every 120/g degrees for g groups (read for three, four and five groups), the lobes between them follow the textbook array factor, and a group turned out of a silent pair is a ramp that sets size and steering at once. The riders' summed runs and the harmonics' predictions agree to the runs' last digits, within 2e-15; both come from the same runs, so that agreement checks the arithmetic rather than testing the array (NINE-1, LOBE-1, TWELVE-1, FIFTEEN-1, RAMP-1; §6.13).
15. **A fade predicted to the fourth decimal.** A heavy rider, slow to answer, feels the two tones less and less: the rectifier's height fades nearly as the weight to the minus fourth. Read from the five steps before it, each about 0.69 the size of the one before, the next rung's power was written down as −3.977; at weight 24.5 it came −3.9771, and at all seven heavy weights from 8.2 to 24.5 doubling the second tone doubles the height to one part in ten thousand (RATCHET-80, §6.4). A modest result and a clean one: a minus-fourth law reached from above, with the second tone entering exactly once.

The count. A script read every graded entry in the record, from its first day, 3 October, to 5 November, and then every landing the book did not count, every test a later version of the script changed or added, and every test from 23 October on, was read by eye, one by one (the Declaration names the scripts, which are published with the book). With the eleven tests of the figure-8 file of 3 and 4 October, published with the engines (§1.3), 527 tests had a prediction written first, and 448 landed: in 223 every prediction held, in 195 some did, and in 30 only the named rival held. Seventy-nine missed outright. Tests read after 5 November are not in that count; the book's ledger carries them by name. Sixteen of the landings are in doubt, as their readers marked them: in some, what held may have been a check of the instrument rather than a prediction; in others the bar was too easy, the reading came after the data, or a later correction withdrew what held. The book reports 386 landings by name: 296 it tells as predictions written first, and 90 more it tells as results; one of them, INDEX-COUNT, is a check of the solver: the ledger keeps it among the 386, and the count of 527 leaves it out (§7.2, lesson 69). It names 173 misses or voids, and 95 of those were redesigned into predictions that then landed. Part 6 holds 89 of the 386. Every one, landed or missed, names its run in the record, whose registrations, verdicts and results are published with the book as its public extract, and the whole ledger is kept beside the book, its file named in the declaration.

What this book is not. It is not a propulsion book. A closed system cannot push itself without pushing on something; the toy pushes on nothing it carries. That wall is stated first (§1.2), and the re-aim at a real medium, planned and not yet built (§5.5), is how the book means to face it. Nor does it report a laboratory measurement: every junction result in it is the junction's standard model equation, integrated on our bench. Much of the physics underneath is known: harmonic mixing, Shapiro steps, Feigenbaum's ratio, the forced pendulum's step theorem, the zero that slides with weight. Where a result matched a named one, the book says so, and those matches are checks of the ruler, not discoveries. What the book claims is the measured map, the laws that held blind on it, and the record of how each was found. We built a toy that models a gear train, and learned how to work it. First we had to make a strobe map: where each dancer can be steered and caught, and how the knobs work together to tune the gearbox and the train it drives. If this were a propulsion book, we would be mapping the locks, limits and performance envelope of an engine that moves a craft. It is instead a holonomy map (holonomy is what has changed when something is carried once round a loop and comes home, §6.6): mostly known destinations and theorems, gathered in one place for our toy to visit.

Two directions of the ripple. Parts of this book use the ripple running against the ball (the atlas, and the chalkboard of Figure 2: A cos(u − W t)); the dancers use it running with the ball (A cos(u + W t), the "turned walkway"). Turning the ripple around is the same as turning the pull around (§2.4). Every figure says which.

Part 1 — How we got here

> *a long workbench with one small machine on it and many tools around it. At one end the machine is a gearbox, its teeth slipping and catching; further along, the same machine is a record turning, then a map pinned to the wall, then a dance floor. Wherever something surprising happens on the bench, there is weight in it: take the weight out and the whole bench goes quiet and orderly, every dancer evenly spaced around a straight clock; put a little back, and the floor chooses who stays.*
>
> — no-physics read 158 (a reading of the data as if no rule of physics were known; seven of the eight parts open with one, and §3.8 says more)

1.1 The question

This book started with a radio. Bear's first picture was a chain of radio parts: transmitters that make a wave, tuners that pick one note out of many, receivers that turn the wave back into a signal, a satellite overhead, and a sled-dog team of pads, each pad handing the push on to the next. His question was easy to say and hard to answer: could a rider walk a seesaw from pad to pad, riding on a wave?

A companion book, the Loop Atlas, had asked a bigger question first: how many things in nature that look like pushing are really something going around a loop and keeping the change, the way a ratchet keeps each click? This book, R2W for short, grew up beside it, from three toys: the comma, the gearbox and the relay (§1.3). Out of those toys one object kept coming back, whatever was asked of it: a rider held in place by one wave and nudged along by another. In the language of physics it is a damped particle in a periodic potential under a periodic drive: of the same family as a pendulum with friction pushed by a motor, and as the Josephson junction of Part 5 under a radio wave: cousins that share its skeleton, not twins (§2.5). This book is about that rider: what it does (Part 4), how we learned to read it (Part 3), and, in Part 6, how to drive it.

1.2 The wall, on page one

Before anything else, one wall, because it shapes everything after it. A closed system cannot push itself along without pushing on something. Physicists call this the conservation of momentum: whatever a craft throws backward, it gains forward, and if it throws nothing out and touches nothing outside, its total momentum cannot change. No arrangement of waves inside a sealed box gets around that, however clever the waves. A skater cannot roll forward by shoving on the front of her own board.

The routes that do work all push on something real. Light pushes as it leaves: about 3.3 micronewtons for every kilowatt of photons, a whisper, but a real one. Sails push on sunlight and on the solar wind. Electrodynamic tethers push on a planet's magnetic field through the current they carry. Tethers pumped in a gravity gradient trade momentum with the planet below, and momentum-exchange tethers with the cargo they fling. Even "swimming in spacetime", a body changing its shape in curved space, is real, and vanishingly small (Jack Wisdom, 2003). The book says this plainly and early, so that no reader mistakes the toy for a drive. The toy is a bench for a ruler, and when Part 6's rider moves one way, it pushes on whatever holds its floor still. What the ruler measures can go anywhere a loop goes.

1.3 The toys that came first

Three small toys gave this book its first words: the comma, the gearbox and the relay. The comma survey was one of the radio engines built in the Loop Atlas before the two books went their own ways on 3 October, and this book inherited it; its own first sitting turned it into the gearbox, and two gearboxes in a row into the relay. Five smaller engines came over with it (§4.10). The comma is the part of a ratio that does not fit. Musicians know one: twelve perfect fifths overshoot seven octaves by a sliver, the Pythagorean comma. Here it is the leftover whenever a rider's pace and the wave's beat do not divide evenly. The gearbox is a rider that locks into whole-number gears, so many slips for so many beats, and pays the leftover as a wobble; engineers call such locking phase locking, and the flat steps it makes Shapiro steps (§2.5). The relay is two gearboxes in a row, coupled through a filter, the way a radio's stages hand a signal from one to the next. Those three toys taught the vocabulary of Part 4A. Then, on 7 October, the work became one mini-book with one rider at its centre, mapped across all its knobs.

Bear's figure-8. One picture from those first days is Bear's own, reached before any reading on it. Two rhythms locked two to one trace a figure-8 when one is plotted against the other, as on an oscilloscope: a Lissajous figure (Bowditch, 1815; Lissajous, 1857). Bear had described that figure-8 as a Möbius strip on 26 June 2026, a single band that joins the two rhythms' phases, the loop doubling its length as it runs the figure. On 3 October he cut it with a plane, his observer's slice, and asked what the plane writes down each time the loop passes through it. In the trade the plane is a Poincaré section; Bear called it a write head, because where it sits changes what is written. Both pictures became tests that night, written down before their runs. His reading of the lopsided figure-8 as a shadow seen from an angle held: the angle, the lock's phase offset, walked 117 degrees across second gear (gears and paces are explained in §4.1). Counted with their direction, the plane's crossings stood in the gear's own ratio within 3% at all nine paces, wherever the plane was placed, because a signed count of crossings is a winding number. And slid

around the loop, the plane wrote only inside a band 0.450 of a turn wide, against the toll of §4.9, 0.446 of a turn (WRITE HEAD 2). Watched for longer, the crossings became four dots, the dots a word, and a gear a frozen frame of a slipping film (§4.1, drawn in the book's figure 8). The two nights' eleven tests sit in a dated file of their own, the figure-8 file of 3 and 4 October (PREREG_LISSAJOUS_2026-10-03), published with the engines, beside Bear's drawing of the four little circles, and the book's count includes them.

1.4 The turn

On 10 October R2W learned what its toy could not do. The toy's ball rides a wave, but the wave lives in the toy's own floor: the toy cannot carry along the medium it pushes on, so it can go nowhere the floor does not. That was the wall of §1.2 again, met from the inside. Bear decided two things then: the turn toward a real medium stays inside this book (§5.5), and the book's way of reading results (Part 3) goes into the Field Guide's next version as well. Then the radio came back by another door. A travelling-wave tube, the microwave amplifier that carries satellites' signals (Minenna and colleagues, 2019), works by an electron surfing a radio wave slowed to its own speed: the electrons bunch up and hand the wave their energy. Give that electron two waves and it is the particle in two waves of §2.5, our rider's cousin without the honey, and a travelling-wave tube has drawn its devil's staircase (Doveil, Macor and Elskens, 2006). Our rider is also a cousin of the Josephson junction, a sandwich of two superconductors whose phase slips under a radio wave by an equation of the same family as our rider's, which Part 5 runs on the bench and Part 6 learns to steer.

Part 2 — Our toy

> *a marble on a corrugated tray. The corrugations are fixed, and a second, fainter set of ripples rolls across the tray the other way. Tip the tray and add a little syrup, and the marble settles into the fixed grooves or slides down them in counted steps, listening to the rolling ripples only when its own rhythm already nearly matches theirs. Take the syrup away and it never settles: it rattles between the two sets of grooves, belonging to neither.*
>
> — no-physics read 121

The fair showed the toy at play. This part goes round it again slowly, naming each piece and writing down its equation. The fair's marble is called the ball from here on, and the ball is the rider of the title: held in place by one wave and nudged along by another (§1.1).

2.1 One ball, two waves, honey and a slope

Picture a ball on a staircase of ledges, like a marble on a corrugated tray. The ledges are a standing wave of strength 1: they never move, and the ball likes to rest in their dips. The ball sits in honey, which drags it (drag 1), so it loses speed whenever it moves. The whole staircase is tilted by a steady slope, the pull, which keeps trying to roll the ball down the stairs. And a fainter wave, a ripple of strength 0.3, runs past the ball, lifting and dropping it a little as it goes, so that each time a ripple passes is one tick of an outside clock. The ball's weight decides how much it coasts: a heavy ball keeps rolling after a push, while a weightless one stops the instant the push does. That is the whole toy: one ball, two waves, honey and a slope. In the language of physics it is a damped particle in a periodic potential, tilted, and driven by a travelling wave.

In symbols, with u the ball's place on the ledges, measured as an angle that grows by a full turn, 2π, from each ledge to the next (its lag is u/2π, counted in turns), m its weight, F the pull and W the ripple's passing speed:

$$m\,u'' + u' = A\cos(u \mp W\,t) + B\cos u - F.$$

In words: weight times acceleration, plus the honey's drag, equals the ripple's push plus the ledges' push minus the slope. The ledges are the term B cos u, with B = 1; the ripple is the term A cos(u ∓ W t), with A = 0.3. The sign in front of W t says which way the ripple runs (§2.4).

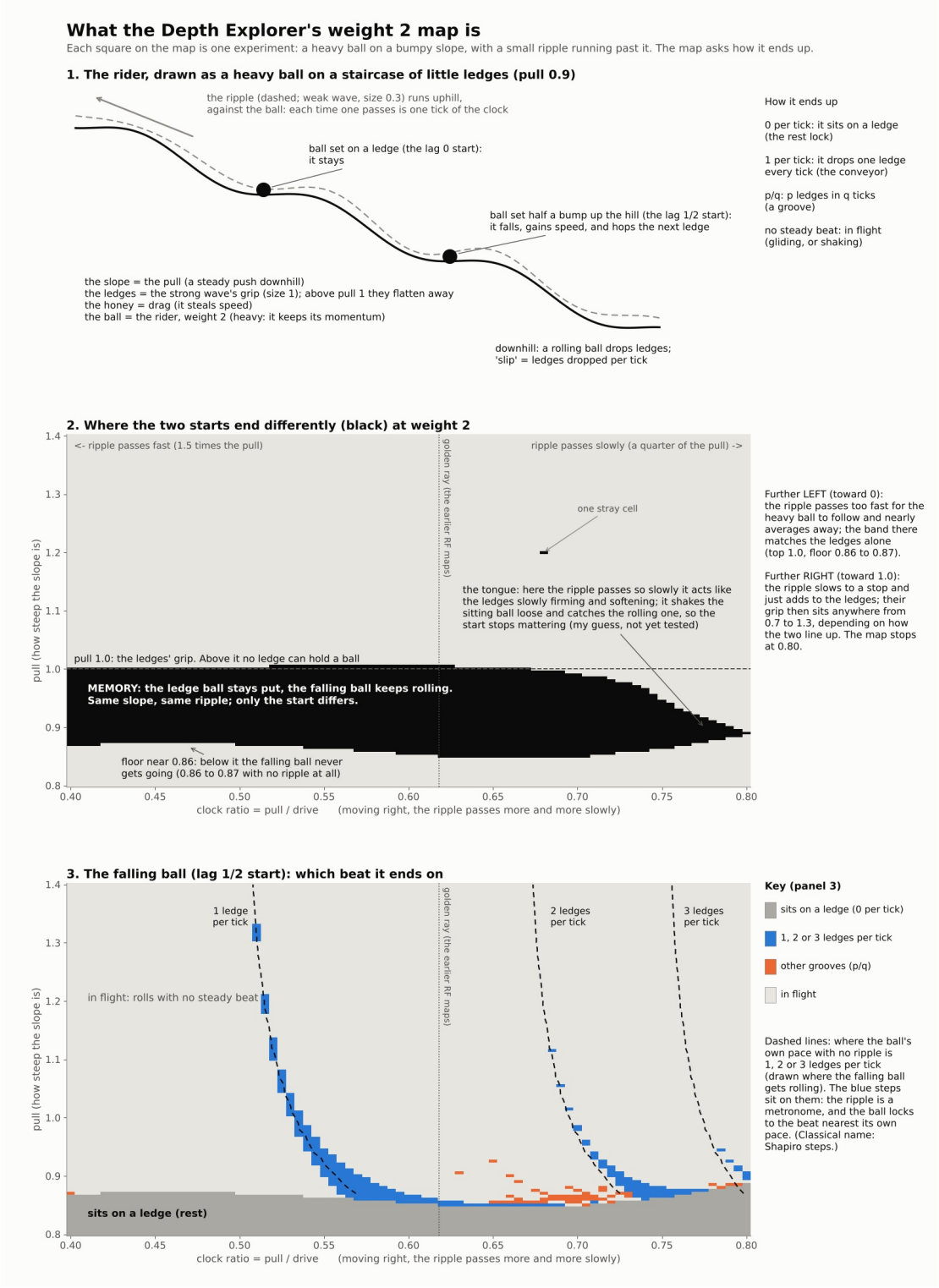


Figure 2. The chalkboard, drawn on 10 October: the rider as a heavy ball on a staircase of ledges, with the ripple running uphill against it (A cos(u − W t)); where two starts end differently at weight 2; and the outcomes of the ball as it falls down the ledges, beside the no-ripple prediction.

2.2 The outside clock and the strobe

The ripple's passing is the clock that matters. Imagine a camera whose flash fires once per passing, always at the same moment of the ripple. Each flash catches the ball in one place, and a smooth motion becomes a list of dots. That is the strobe. Mathematicians call it a Poincaré section; an engineer knows it as a stroboscope, or as a sampled-data controller reading its plant once per period. A ball that slips one ledge per tick shows one dot per tick, each one a turn further on. A ball locked at a fraction p/q comes back to the same dot after q ticks, having slipped p ledges. The slip per tick, the count, is the number this book trusts most: it is a whole number or a fraction, and it does not drift when magnitudes do.

2.3 Four knobs

The toy has six numbers in it, but only four of them matter. Rescaling time by the drag over the ledges' strength, and force by that strength, folds both of those away, the way a recipe written in cups works for any size of cup; engineers call this making the equation dimensionless. The rider then has exactly four knobs: its weight (weight times strength over drag squared), the pull, the passing speed and the ripple's strength, each over the ledges' strength. The drive turns the pull and the passing speed together along one ray, like a throttle that opens two valves in a fixed proportion: the pull is the clock ratio times the drive and the passing speed is the rest, so the two add up to the drive, and the clock ratio (pull over drive) picks the ray. Much of the early work runs along the golden ray, clock ratio 0.618034, the fractional part of the golden ratio and the number hardest of all to approximate by fractions. The pad atlas maps the plane those rays live in.

2.4 Which way the ripple runs

The ripple can run against the ball (A cos(u − W t)) or with it (A cos(u + W t)), like a moving walkway at an airport running toward you or away from you. That is not a fifth knob. Turning the ripple around is the same as turning the pull around and reflecting the lag: 296 of 296 twins that settled into a dance matched exactly (KNOBS-3). It opens the other half of the plane, where the ripple carries the ball. There the plain's edge comes about 0.2 of drive sooner, and the ball, once free, rides the ripple at its own speed.

The atlas, the chalkboard of Figure 2 and the record use the ripple against the ball. The S, the wrap, the pin and the dancers (§4.19 to §4.26) use it with the ball, the turned walkway. The two halves of the plane are one rider seen from two sides, and the book names the side in every figure.

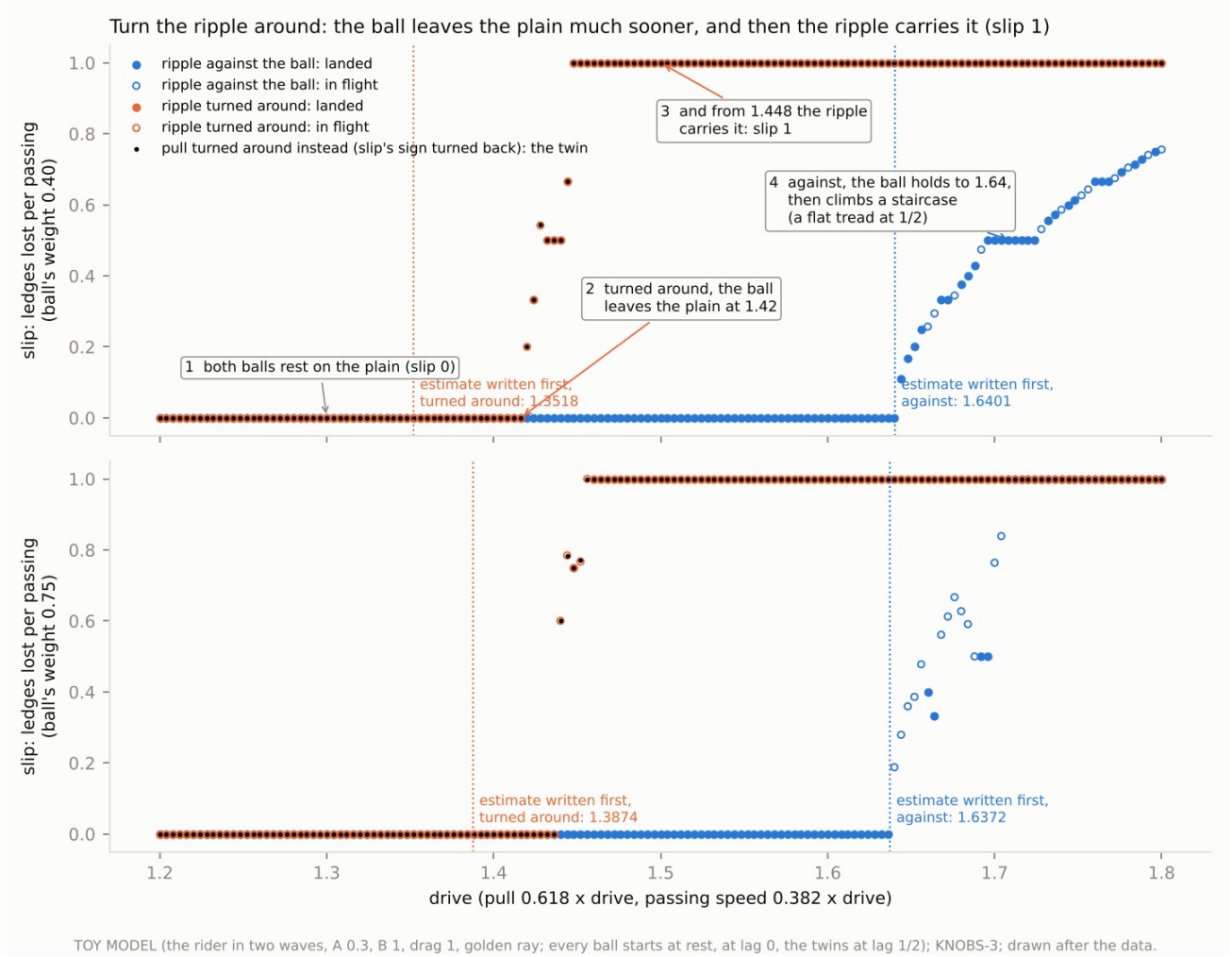


Figure 3. Turning the ripple around: the ball leaves the plain sooner, and then the ripple carries it (slip 1).

Reading the figure, step by step. Each dot is one ball, started at rest and run at one drive; the drive grows to the right, the pull and the ripple's passing speed rising together. Height is the slip: how many ledges the ball loses at each passing of the ripple. Blue dots have the ripple running against the ball, as in most of this book; orange dots have it turned around. A filled dot settled into a repeating motion; a hollow one had not settled by the end of its run. The dotted lines mark where an estimate written before the run said the ball would leave the plain. Now follow the numbered notes in the top panel, for a ball of weight 0.40. (1) At low drive both balls rest on the plain, slip 0. (2) With the ripple turned around, the ball leaves the plain at drive 1.42 and steps quickly through a short ladder of fractions, 1/3, 1/2 and 2/3, two of its dots still in flight. (3) From drive 1.448 it sits at slip 1 and stays there: the ripple has picked it up and carries it, one ledge per passing, at the ripple's own speed. (4) With the ripple against it, the ball holds on the plain to drive 1.64, where the blue estimate (1.6401) said it would let go, and then climbs a staircase of fractions, with a flat tread at 1/2 from drive 1.696 to 1.724. So turning the ripple around brings the edge about 0.22 sooner. The tiny black dots are a third experiment: the ripple left as it was and the pull turned around instead, each ball started at lag 1/2, the mirror of lag 0, with the slip's sign turned back. They sit on every orange dot that landed, 296 of 296 across both panels: turning the ripple around is the same as turning the pull around. The lower panel tells the same story for a heavier ball, weight 0.75, which takes longer to settle, so more of its dots are hollow. The orange estimates came early, by 0.064 at weight 0.40 and 0.049 at 0.75; registered to land within 0.06, that bold guess missed at the lighter weight (KN3-3). The blue ones landed within one step of the drive's grid.

How far the ripple moves the edge. That miss led to a law of its own: how far the ripple moves the plain's edge, why the ratio of the shifts from its two directions is a golden number at every weight, and which door a ball leaves the plain by under a strong ripple. Those runs are told in §4.18, beside the plain's other landmarks.

2.5 The family: cousins, not twins

The rider belongs to a famous family, the driven, damped pendulum (the tilted washboard), and this book keeps its relatives straight. A twin obeys the same equation exactly; a cousin shares the skeleton and differs in one organ, and that one organ is where a result learned on one may fail to carry to the other.

- **The particle in two waves** shares its forcing exactly (Chirikov; Escande and Doveil). Doveil, Macor and Elskens measured that system's devil's staircase with an electron beam in a travelling-wave tube (2006; checked). The difference is that our rider has honey and a slope, so it settles where theirs keeps every bit of energy.
- **The RF-biased Josephson junction** is a close cousin, not a twin. Its supercurrent plays our ledges, its bias current our pull, its capacitance our weight. But its radio drive is an added push, the same at every phase, while our ripple moves the ledges themselves. No change of variables turns one into the other. Its steps are Shapiro steps (Shapiro, early 1960s); ours are "Shapiro-like" (Bear, 11 October). Part 5 runs the junction's own equation.
- **The phase-locked loop and a generator's swing equation** share the pendulum's skeleton. D'Humieres, Beasley, Huberman and Libchaber (1982; checked) set the pendulum, the junction and the loop side by side, and the swing equation turns out to be our rider with no ripple, exactly (§5.3).
- **The forced pendulum's own history** is long. Mawhin's survey (1998; read) follows seventy-five years of it, from Hamel's existence proof of 1922. With the ripple off, our rider is that equation's simplest case, pushed only by a steady pull (the swing equation of §5.3); the junction's rectifier of Part 6, pushed by a forcing that repeats in time, is exactly it. The survey's first theorem is our step: for a given repeating part of the forcing, a periodic orbit exists exactly when the steady push lies in a closed interval, and inside the interval there are at least two (§6.9).
- **The mathematics** has its own houses. The devil's staircase and Arnold tongues (Arnold; Jensen, Bak and Bohr, 1983, for the circle map's complete staircase, and Bohr, Bak and Jensen, 1984, who tie it to microwave-driven junctions directly). The weightless junction locks only at whole-number rotation numbers (Buchstaber, Karpov and Tertychnyi, 2010; Ilyashenko, Ryzhov and Filimonov, 2011; both checked; Glutsyuk since). A tongue whose width falls to zero at a point is called pinched (Boyland, 2025, checked; he traces it to Yang and Hao, and to Campbell, Galeeva, Tresser and Uherka), and the condition for it is the one met at every weightless root here: q steps of the map add up to a plain turn. The standard junction model without capacitance gives only whole-number Shapiro steps (as Tsarev and Fominov, 2025, checked, restate); capacitance opens the fractional ones, and the second tone of §4.28 can pinch them again. The rotation number is Poincaré's. The first-order choice between dances is the subharmonic Melnikov method (Malkin; Guckenheimer and Holmes, 1983), meeting a time-reversal symmetry (Devaney, 1976; Lamb and Roberts, 1998). The families are named on the Signposts page at the back of the book, and every work cited is listed, with how it was checked, under Sources checked.

2.6 Order and chaos

Most of what the rider does is orderly. It locks into a dance and repeats it, passing after passing, like a clock. The rest of the time it can do something stranger, which the field calls chaos and this book often calls shaking.

Chaos is easy to mistake for chance. A coin toss looks like a 50-50 gamble, but the coin is not random: it obeys exact rules, and if you knew exactly how your thumb flicked it and how it spun, you could say how it would land. What makes it a gamble is that a hair's difference in the flick turns heads into tails. Chaos is like that. Nothing in the rider's equation is random: start the same rider in the same place twice and it does exactly the same thing both times. But start two riders a hair apart and the small difference between them grows, doubling and doubling again, until they disagree completely about where they are. A chaotic motion never repeats, and nobody can say for long where it will be, even though every step of it follows the rules.

The book measures that growth with the stretch rate, which mathematicians call the largest Lyapunov exponent: how fast two neighbouring riders spread apart (§3.3). Below zero, they draw together, and the rider forgets where it started, as a locked rider does. At zero, a small difference neither grows nor shrinks. Above zero is chaos, and the bigger the number, the faster the chaos. Part 4A's rider, where the book first hunted it, has the slow kind. In that hunt the chaotic runs' stretch rates, read again on windows ten times longer, were +0.011 to +0.029 per unit of the toy's time (the chaos hunt of 4 October and its check, CHAOS-VERIFY), so two riders a thousandth of a turn apart take a few hundred units of time to disagree by a whole turn. A rate that small is the kind a long, slow settling can fake on a short window, which is why each of those runs was checked again on the long window from two starts: 20 of the 23 stayed chaotic from both. Not all of the book's chaos is slow: the heavy needle of §4.12 tumbles at +0.40 to +0.64 per passing of the ripple, doubling a small difference every passing or two (TUMBLE-1).

The grip forgets and the slip remembers. The stretch rate is also a forgetting rate. A rider gripped by a gear draws its neighbours in and forgets where on the ledges it started; a rider slipping between the gears carries its starting place forward for ever. That was written down as a prediction before it was read, on Part 4A's rider when it is nearly weightless, and it held (MEM, 4 October). At weight 0.05, 99.1% of the 110 runs gripped by direct drive (locked one for one to the strong pattern), first gear or second gear (Part 4A's gears, §4.1) had a stretch rate below −0.005, and all 101 runs slipping between them sat within 0.005 of zero; at weight 0.25 the shares were 100.0% and 99.0%. The tighter the grip, the faster the forgetting: at weight 0.05 the middle runs of the three gears forget their start in

about one, two and twenty units of the toy's time (stretch rates -0.83, -0.43 and -0.049). This is memory of a different kind from the hysteresis of §4.14, where the road taken decides which of two gears the rider ends in: here what is kept or lost is the rider's starting place on the ledges.

Chaos needs room. A weightless rider cannot be chaotic at all (§3.5, §4.7); weight gives it the room. And in this toy chaos lives in particular places: on the borders between two locks, where the rider cannot decide which step it is on (§4.7), and in the room between two gears (§4.14). It can even keep a count while it shakes (§4.15).

Part 3 — How we built it, and how we read it

> *a teacher drew lines where there were no dots, and the dots agreed with him afterwards. Where the two-step's top foot splits, he drew three roads, not two: up, down, and one straight on between them, the road the walker would take if it could still stand there.*
>
> — no-physics read 95, its opening

This part is the book's method. Think of it as the workshop behind the fair's expo hall: the bench, the tools on the wall, and the habits that keep the tools honest. It starts here and lives here, with every lesson it was built from stated in full in Part 7 (ten of them now in the Field Guide, §7.6); the Field Guide's chapter "Two Readers and a Ruler" carries it to other work, and cites these sections and that part.

3.1 The loop

Every result in this book went round the same loop (figure 4). It is the loop a careful carpenter uses: measure, mark, cut, check, and when the board comes out short, ask whether the tape, the mark or the saw was wrong before cutting the next one. In a laboratory the same habits have names: pre-registration, blind analysis, regression testing and replication.

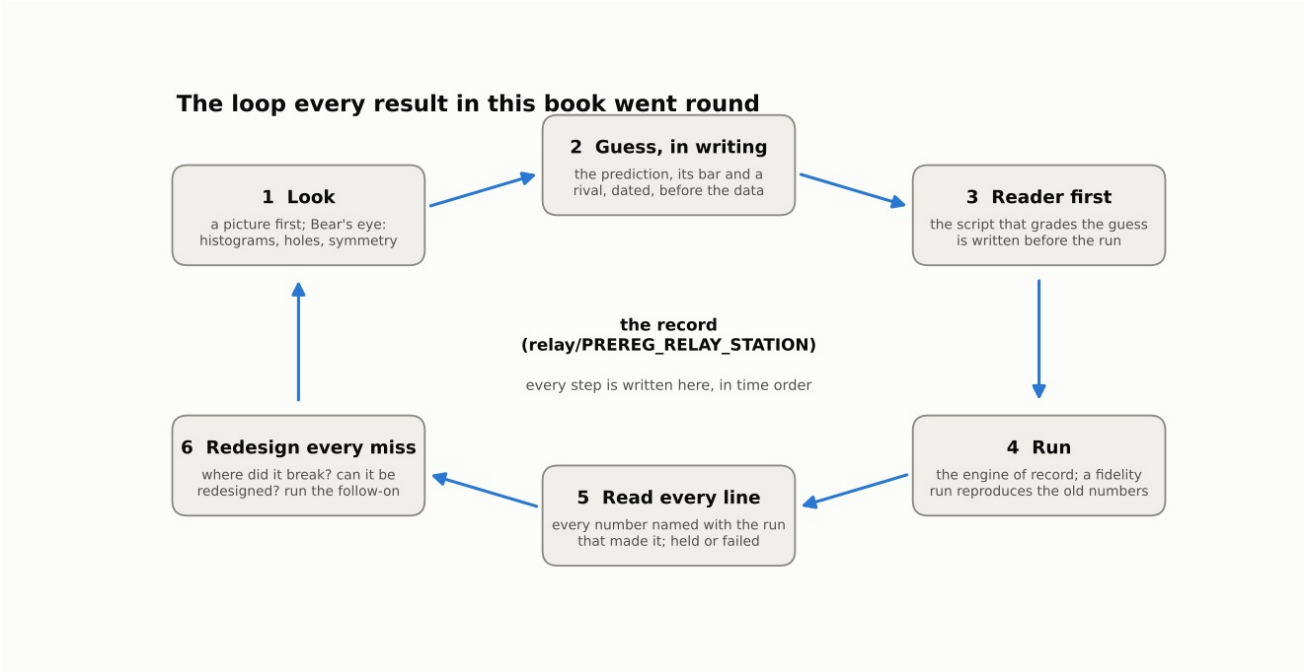


Figure 4. The loop every result in this book went round.

1. **Look.** A picture first, read by Bear's eye (§3.4).
2. **Guess, in writing.** The prediction, the bar it must clear and a named rival are written into a dated record before the data exists. This is pre-registration: a guess written down afterwards can always be made to fit.
3. **Reader first.** The script that grades the guess is written before the run, so the grading cannot bend to the data. Before it meets real data it is tried on stand-ins, made-up files on which every verdict must pass once and fail once, so that a reader which cannot fail is caught before it matters.
4. **Run.** The engine of record runs it. When an engine changes, a fidelity run first reproduces the old numbers (a regression test, in a programmer's words).
5. **Read every line.** Every output is read in full, and every number in the text names the run that made it. A number that lives only in prose is marked unverified.
6. **Redesign every miss.** When a guess fails, even when others held, the question is asked aloud: where did it break (the data, the bar or the method), and can it be redesigned? Then the follow-on runs. A miss is not a dead end. It is the most informative thing a run can give, because it says exactly where the picture in our heads was wrong; §6.9's step and drift is the freshest example. Bear counts this habit among the biggest changes in how the team worked across three books (*Held, Rocking, Locked*, the team's book on the islands of a light star around an eccentric binary; the Loop Atlas; and this one). Before it, a failed hypothesis meant starting over, and getting a line of work to change course was like moving heaven and earth; with it, many of the most productive ideas came from redesigning and running again. The failures were made to teach, and each redesign became a signpost instead of a closed road.

Two habits sit beside the loop. Claimed and measured are always kept side by side, never merged. And a result found after the data is labelled so, and becomes a result only when a blind test confirms it.

3.2 The engines

An engine here is a numerical integrator wrapped in a finder. The integrator steps the rider's equation forward in small slices of time; the finder, around it, hunts for the special orbits the book is about: the locked ones, the dances, the step's two edges. The rectifier's engines take a fourth-order Runge–Kutta step, the workhorse rule of the trade, and close in on each edge with Newton's method, a guess-and-correct search that roughly doubles its correct digits at each pass once it is close. The work was done in dated working sessions, which the record calls sittings and numbers from B1, on 3 October 2026; each sitting was a string of exchanges, a message from Bear and its answer, and engines and lessons are often named for the sitting that made them.

The engines. Every engine and everything it printed are published with the book, in two zip files that unpack into one folder: one holds the 933 Python scripts, the other their 8,024 other files, which are printouts, tables, figures, launchers and the explorer's JavaScript engines. A README inside says how to find the engine behind any test this book names and how to run it, and two manifests give every file's checksum, a short fingerprint that changes if any byte of the file changes. The Declaration names the zips and the folders.

engine	what it runs
the gear and relay stations	the gearbox and relay of Part 4A, built in sittings B1 to B6
REV_WEIGHT and its registered copies	the rider along the golden ray, by weight and drive, both ripple directions
the pad atlas (versions 1 to 6)	the whole plane of clock ratio and pull at weights 0.05 to 10, on Bear's laptop; at most 8 worker processes
the Depth Axis Explorer (versions 1 to 9, and the sitting B11 build)	Bear's pen on the maps; the strobe record of every rider; since sitting B11, a rectifier bench that runs the junction live with every knob of Part 6, the rider's weight among them, and a self-test of 37 rows that checks it against runs of record
RIDER_ENGINE_B11 and B11W	RATCHET_3T's weightless edges in the page's JavaScript; every edge within 1.31e-12 of six runs of record at full steps, within 1.31e-9 at the bench's fast 500. B11W, the page's engine since 17 October, adds RATCHET_3T's weighted fold: every edge at every weight within 6.34e-13 of four weighted runs of record (weights to 0.64), within 6.79e-10 at 500 steps
play_the_toy.py	the toy in one page of plain Python, handed to the reader in the fair chapter: a weight, a side and any number of drives in, the slip at each out. Every rider that settled in two runs of record agrees within 0.0003: 573 of KNOBS-3 and 383 of BRIDGE-1 (PLAY_THE_TOY_FIDELITY); and a coin mode that flips Bear's coin, 1,796 of its 1,800 riders of record on the same face (COIN-TOY-1); published with the book, with a card of six settings to try
FBP_1 (versions 1 to 14)	the frozen balance point: weightless roots, dances, first-order selection
JJ_1 to JJ_3, JJ_GRID_1	the Josephson junction through the same instrument
JJ_5	JJ_2 with a second tone at twice the frequency; with no second tone it reprints JJ_2 row for row
JJ_9, JJ_11	JJ_3 with the grip ratio printed to six places (JJ_9); JJ_11, two copies later, adds each dance's grip by a second, integral route to twelve figures, which keeps its digits however small the grip gets
JJ_16	a later copy of the same line carried in extended precision (a 64-bit mantissa), for the deepest rungs of the Farey ladder (LADDER-1c)
WITNESS_FOLD	a second rectifier engine written from scratch, sharing only the equation with RATCHET_3T: adaptive DOP853 integration instead of fixed-step RK4 (WITNESS-1)
WITNESS_COIN	a second coin engine written from scratch, sharing only the equation with the coin's engines: adaptive DOP853 integration, the face read from the slip; every one of 1,800 riders on the record's face (WITNESS-COIN-1)

WITNESS_EAR	a second junction engine written from scratch, sharing only the equation with JJ_5 and JJ_8: adaptive DOP853 integration, roots and couples from the flow itself, the stable dance told by real weight; every root and couple to the print, the ear to the thirtieth harmonic (WITNESS-EAR-1)
RATCHET_2, 2B and 2C	the junction's rectifier: the zero-slip step's two edges as folds of the passing map, with and without weight
RATCHET_2D	RATCHET_2C with the second tone's harmonic as a knob (twice, three or four times the frequency); at twice it gives 2C's edges to every printed digit
RATCHET_3T	RATCHET_2D with a third tone at four times the frequency and a timing of its own; with no third tone it computes RATCHET_2D's expressions exactly. Every third-tone run of sittings B10 and B11 used it
RATCHET_NOISE	the rectifier with noise: an ensemble of riders jostled at random, for the drift at zero push (knob 9, §6.9)
RF_SPOKE_STACK, STACK2, STACK3	the relay station with a third stage fed through a second loop filter (§4.11)
RF_SPOKE_PLL, PLL2; WINDOW_WEIGHT 1 to 3	the phase-locked loop of §5.6, carrier ripple included, in plain Python; the same code with the weight as a knob, for the hand-off (§4.28)

Each laptop version passes a fidelity gate, ships with a hash-checked launcher (one that refuses to run a file whose fingerprint has changed), and writes output chunks that are byte-identical across versions.

3.3 The instrument: what it measures

The instrument is a stroboscope with a counter attached: it looks at the rider once per passing of the ripple (§2.2) and writes down what it sees; figure 5 shows the depth behind each dot it prints: the depth axis opens one drive's dot into the strobe lags the rider passes through, a few points for a dance and a smear for shaking, and where the rider's history decides, one drive carries two threads, a memory (§4.8). It reads three kinds of number. **Counts**: the slip per tick, the number of stops in a dance, how many stops sit between two marks. **Places**: where a stop sits at the strobe, in turns. **Rates**: how fast a stop moves at the strobe, how fast nearby motions spread apart (the stretch rate, which is also a forgetting rate: the grip forgets and the slip remembers, §2.6), how firmly a dance holds. Counts are trusted most. Places are trusted when two methods agree. Rates are trusted when a third route agrees (for the dancers, two routes to every grip number, and Newton's method on the weighted rider). The two routes to a grip read the same orbit with the same integrator and step, so they check each other's arithmetic, not the integration; Newton's method and the finer-step checks (STEP-1, LADDER-1b) check the integration. The ranking is the metrologist's: a count cannot be off by a little, only by a whole one, so it is exact or plainly wrong, while a place or a rate always carries an error.

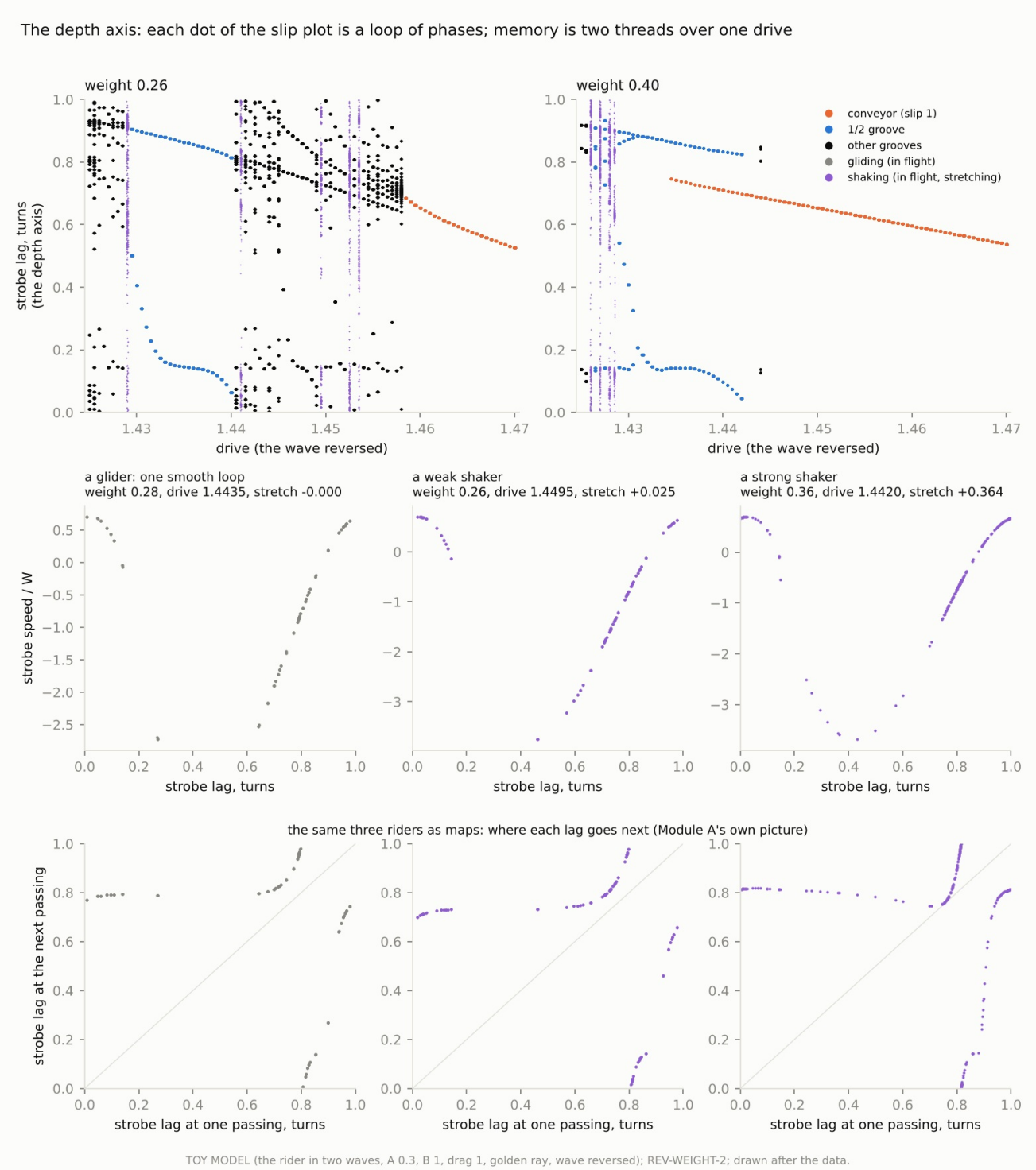


Figure 5. The depth axis: every dot of a slip plot is a loop of phases; memory is two threads over one drive.

3.4 Bear's way of reading a picture

These steps were taught by Bear on 12 October, written down, and each given a check that a computer can run. They are the habits of a good field geologist reading a cliff, or a radiologist reading a film, put in order: survey before you zoom, count before you name, and let the empty places speak.

1. **Histograms on both axes first.** Count the dots along each axis before looking for shapes. The counts show the populations and the natural bins (a statistician would say the marginal distributions).
2. **Ask whether there is a pattern at all.** Not every picture has one.

3. **Lay a lattice of holes over the picture.** Each hole has eight neighbours at radius 1, and the rings grow outward. Many strings can pass through one filled hole: “we can’t tell visually if it’s shared with 5 strings of 50”.
4. **Read the slopes to the neighbours.** The same slope is a line; a steadily changing one is a curve. Different frequencies skip holes.
5. **Look at the negative space.** Clusters in the empty places drive lines and curves. Straight-line clusters and curved-line clusters are gold, “because then the 3D shape comes out”.
6. **Demagnify by default.** Zoom out before zooming in. Most of the time the shape is bigger than the window (coarse-graining: blur first, then sharpen).
7. **Hunt the symmetry.** Bilateral symmetry is “another stalking point”. The pin of §4.26 was found this way.
8. **Read blind, then score against the marks.** Bear’s unmarked picture is read and written down before his marked one is seen, and then scored against it, as a clinical trial keeps its readers blind.
9. **Count by machine before saying a count.** Three of Claude’s own counts, read by eye off pictures, went wrong in one sitting, while Bear’s dot counts in figure 31 came out right and became rules (§4.25). Since then every count is checked by pixels or data before it is said.

Each step has a check. Steps 1 and 3 to 5 are pixel histograms and the lattice linker, a tool that threads dots into strings. Its second version lets many strings share a hole and merges two only after they share eight holes in a row. It was tested first on made-up crossings with planted answers, and passed only on its third design. Step 7 is the mirror test, step 8 is the blind record, and step 9 is a pixel count.

3.5 The mathematics that turned pictures into rules

The dancers of §4.21 onward rest on eight pieces of mathematics. Each is a standard tool with a name a mathematician would recognize; the work was to find which tool each picture was asking for. Each is written here in words first.

1. **The passing map.** Follow the rider for one passing of the ripple and see where each start goes. That map, from one strobe to the next, carries everything about the long run (a Poincaré map).
2. **With no weight, a passing is a Möbius map.** Drop the weight and the rider obeys a first-order equation, which turns into a Riccati equation for $\tan(u/2)$, an equation with a square in it whose solutions move by Möbius maps. Its passing map is a Möbius map of the circle, set by one 2×2 matrix Φ of determinant 1. In words: one small table of four numbers tells where every start goes in one passing. Such a map keeps points in order and cannot be chaotic.
3. **The time mirror.** Run the film backwards and flip the lag, $(u, t) \to (-u, -t)$, and a weightless rider looks the same. In the matrix this forces the two diagonal entries to be equal: $\Phi 11 = \Phi 22$.

4. **Roots: where every start is a dance.** When Φ11 = cos(π p/q), the passing map repeated q times brings every start back, having slipped p turns: Φ^q = (−1)^p. In words: at these special drives, the weightless roots, every start is a p/q dance. The mirror leaves exactly two of them symmetric, red and purple. With a touch of weight, red is the one that stays (it is stable: nearby starts settle into it) and purple the one that runs (unstable: nearby starts move away from it), as §4.23 tells. The picture shows one pair.

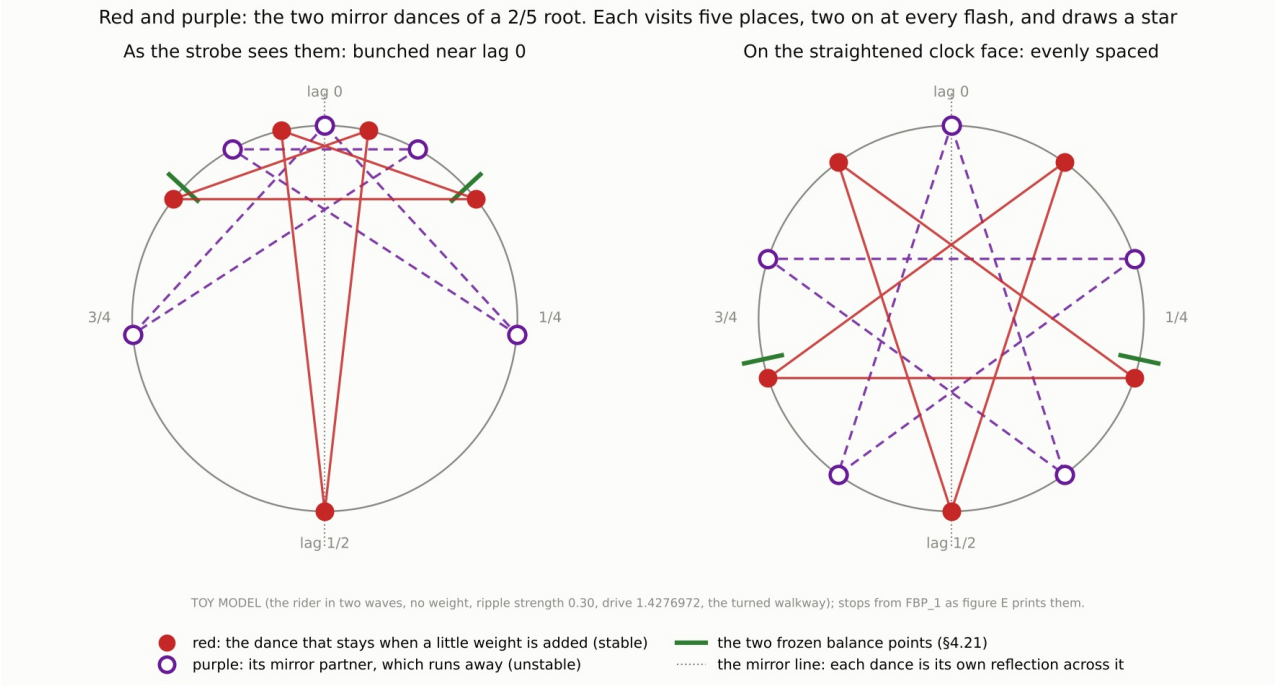


Figure 6. Red and purple at a 2/5 root, on the clock face of the lag. Each dot is where the ball sits at a flash of the strobe. Each dance visits five places, moving on two of them at every flash, and so draws a star. The lag is the ball's place on its ledges at the flash, counted in turns round the circle. Left, as the strobe sees them, bunched near lag 0: the raw lag is a crooked clock face, its marks crowded on one side and spread on the other by an amount that one number of the passing matrix sets. Right, on the straightened clock face of piece 6, the same stops evenly spaced. Figure 31 (§4.23) shows nineteen such pairs.

5. **A touch of weight chooses.** With a little weight m, the rider follows u' = f − m(f_u f + f_t) to first order, where f is the weightless push and f_u, f_t its rates of change in lag and in time. The q-passing map becomes the start plus m M(φ), where φ is the start's lag and M(φ), the selector, is how far a little weight carries that start over the q passings, per unit of weight. Dances survive where M is zero, and are stable where M slopes down (M' < 0). This is the subharmonic Melnikov method (Malkin's bifurcation function). Weight is the small term that breaks the mirror: m u'' changes sign when the film runs backwards, while the rest does not.

6. **The straightened lag.** Stretch the lag by one number, s = √(−Φ12/Φ21), through tan(π y) = tan(π x)/s, where x is the raw lag and y the straightened one, both counted in turns. In the new lag y every weightless passing at a root is a plain turn of p/q. In words: the crooked clock face becomes a straight one. In that lag a dance's stops are evenly spaced, the two mirror dances sit exactly half a gap apart, and any mirror step that crosses the far side starts at ± (q − p)/(2q). Measured, not just argued, by two checks run on outputs already on file. The first, ROTATION-CHECK, straightened 42 of the junction's dances and found every one evenly spaced, within 4.5 × 10^{-5} of a turn. The second, WINDOW-CHECK, found the leaper's pair on its mark, ±(q − p)/(2q), within 7.9 × 10^{-6} of a turn at 31 rows of the toy and the junction. (Names in capitals like these are the names of runs and checks in the record, where each can be looked up.)

7. **One number per row.** In the toy, the matrix at each root is fixed by the fraction and one number, the launch lag x_a, in turns (launches are met in §4.24): Φ = [[c, (1 − c) z_a], [−(1 + c)/z_a, c]], with c = cos(π p/q) and z_a = tan(π x_a). Every dance of figure 31 can be rebuilt from that (CHECK-E, within 5 × 10^{-6}).

8. **A fit is a guess until a blind test.** A law read off the data is written down as a guess. It becomes a result only when it predicts settings it was not fitted on. Strength over frequency failed this test; strength over frequency squared passed it (§5.2).

3.6 Checks that keep the ruler honest

A ruler that is never checked against a known length will mislead every measurement made with it, and quietly. These are the checks.

- **Fidelity.** After any change to an engine, rerun cases of record first and compare.
- **Two routes to every rate.** The grip M’ by a finite difference and by an integral; first order against Newton’s method on the weighted rider. The first two share one orbit, so a finer step, not their agreement, checks the integration (§3.3).
- **The honey you did not add.** Set the toy’s drag, ripple and pull to zero and its energy must hold still. At the toy’s own step it drains as if a little drag had been added, and the leak follows one law: the step to the fifth power over 72 times the weight squared, which is fourth-order Runge–Kutta’s own loss on a swing, times a constant, 0.727, set by how wide the swing is (twelve weights from 0.01 to 10 at drive 1.60, all within 0.3% of it; NO-HONEY-CHECK, NO-HONEY-SLOPE). The toy’s step is half the weight for a light rider and a hundredth of a passing for a heavier one, so the leak climbs as the weight cubed, peaks where the rule switches, near weight 0.2 at drive 1.60, and falls as one over the weight squared beyond. The peak is largest at the lowest drive the toy is run at, 0.86, where a passing is longest: 1.8×10^{-5}, under two parts in a hundred thousand of the book’s drag, which every run holds at 1 (2.7×10^{-6} at drive 1.60; NO-HONEY-PEAK). A step ten times finer shrinks the leak a hundred-thousandfold. No reading in the book comes near that level. The junction’s engines leak by the same law. Run with the drag, the drive and the bias turned off, at their own 2,000 steps a passing (4,000 in the rectifier’s runs), the swing loses energy at 0.998 to 1.000 of that law, the step to the fifth power over 72 times the weight squared, at every weight from 0.0022 to 1, and at its largest, at the lightest weight any junction run reads, the leak is 1.8×10^{-8} of the drag of 1 that every one of those runs holds (LEAK-JJ). At those lightest weights the drag’s own stiffness, not the leak, is the nearer limit: there one passing at 2,000 steps keeps the lag within 7×10^{-8} of a step eight times finer, but the rate, which the light rider’s drag holds to its slow track, drifts by up to 3×10^{-5} a passing at weight 0.0022 (STIFF-JJ, a miss on the rate). The dances there do not feel it: found afresh at 8,000 steps a passing, the tongues at those light weights have the same edges to 10^{-10}, the same widths and counts, and multipliers within 10^{-6} of those found at 2,000 (STIFF-JJ-3; a first try, STIFF-JJ-2, was void, its check’s bar set by a wrong forecast). With everything on, Liouville’s formula gives a second gauge: whatever the ripple does, a small patch of starts must shrink over each passing by exactly the honey’s share, e to the minus the honey times the passing’s time over the weight. At the toy’s step the patch shrinks a little less, by 0.074% of the honey’s share at weight 0.05, 0.024% at 0.25 and under a millionth from weight 1 up, almost all of it the honey’s own decay copied step by step, since the toy’s step is half the weight for a light rider; steps ten times finer cut it about ten-thousandfold, down to a rounding floor near 10^{-13} a passing (LIOUVILLE-CHECK). The toy’s slips agree with the runs of record within 0.0003 all the same (§3.2).
- **Planted answers.** Test a new tool on made-up cases whose answers are known, until it passes; then turn it, unchanged, on the real map.
- **The edge of a map is not physics.** A reading that sits on the edge of the scanned grid is the grid’s, not the rider’s (twice found and fixed: §7.1, lesson 6).
- **A median is not every column.** A median over a smooth ramp only picks the middle; a “never below” read from medians was wrong column by column.
- **Every count by machine** (§3.4, step 9).
- **Refine the ruler until the answer stops moving.** A coarse ladder of weights reads every peak low; the seam of §6.6 moved from 0.767 to 0.783 as the ladder was refined, and then stopped (lesson 155).
- **A threshold is a ruler too.** Every “none” says how small a thing it could have missed (lesson 158).
- **Gate an angle on its amplitude.** A direction read from a signal smaller than its own error is noise (lesson 162).

3.7 The knob ledger

A knob ledger is an engineer’s inventory: every dial on the machine, what turning it does, and whether anyone has turned it yet. This one is for the toy, the rider in two waves; the junction’s own ledger is §6.7.

knob	what the runs show	status
weight	the rider's K (the knob a circle map calls K, at whose critical value the map starts to fold): light balls glide on thin grooves; heavier ones fill the staircase until the once-per-passing map folds (weight 0.247 to 0.251 with the ripple turned; 0.329 to 0.399 facing it)	tested 0.05 to 1.00 on the golden ray; the atlas maps 0.05 to 10
pull and passing speed	turned together along a ray; the 1/2 groove's cascade is set almost wholly by the pull	the atlas maps the plane
clock ratio	the ray's angle; even its rounding counts (a 6.6×10^{-8} change moves a doubling by 5.4×10^{-9})	the atlas, 0.40 to 0.80
the ripple's strength	makes every groove; most grooves near 0.3; moves the plain's edge as its square, by a coefficient worked by hand (within 0.19% at strength 0.05), the lift against the ball 0.1056 of the drop with it at every weight (§4.18); the lift changes sign at clock ratio 2/3	tested at seven weights, 0.05 to 5.00, strengths 0.05 to 0.60, both directions (FOLD-DIR); under a strong turned ripple the ball slips off before the fold: at strength 0.45 the resting dance (the one a ball set down at rest falls into) of weights 0.75 and 1 doubles first, and the ball is gone within a step (§4.18); at strength 0.60 no ball doubles, and every weight from 0.05 to 1.00 leaves at one drive (0.8825 to 0.8848), heavier balls sooner (FOLD-DIR-e); the heavy balls double at strength 0.45 but at 0.60 leave by the fold of a second resting dance (FX-2b); where that second dance begins, open
drag, the ledges' strength	fold away under rescaling (records come back identical)	tested directly
starting place	decides the groove where two are open: stacks, Bear's doors, a fractal seam	tested at 2 to 8 starts
the ripple's direction	not a fifth knob: the pull's sign against the ripple (§2.4)	tested at two weights
the waves' shapes; noise	no data in the toy as a knob, only as a faint probe at a tread's edge (§4.8); as a knob, noise has been run once, on the junction's rectifier, for the drift at zero push (§6.9, knob 9)	untested in the toy
the instrument's own knobs	steps per passing a dead knob at this depth (1.3×10^{-10} across 487 to 2,000 steps)	tested

3.8 Reading without the physics

Seven of the eight parts of this book open with a short passage signed "no-physics read" and a number; Part 6 opens with Bear's own words. They come from a habit Bear and Claude kept in every exchange, from the first days of their pattern-recognition work. Once an exchange, Claude looked at the newest data with its labels taken away (no ledges, no pull, no junction) and told what the numbers were doing as if no known rule of physics applied: a machine breathing, or a coastline where the surf breaks only where two countries meet. The first read of this project told as a story came on 6 October, from a fine map of where two gears' territories meet: a field with two orchards and a wind that comes in gusts, where a leaf dropped anywhere ends under one orchard's trees or the other's, and the line between them runs through a little saddle in the ground. A read is not evidence, and the book leans on none of them. Its use is the question it raises: a story that does not fit the data points at something the labels were hiding, and some of this book's tests began that way, the very first read among them, which ended by asking for a finer sweep. Three habits ran beside it. Every exchange named its three biggest surprises, each with a way to test it from the data. Every finding was weighed for how much it matters and to whom: an engineer, a theorist, a reader checking the record. And every miss got its redesign check, said aloud, before anything else was run (lesson 38). Together they gave the work a depth of field: its pictures grew from orchards to dances to whole maps. This project's reads, more than 480 of them, are kept whole beside the record, and with those of the projects around it they are collected in a book of their own, *Does Claude Dream of Holonomic Sheep?*

Part 4 — What the bench showed

> *the gearbox is a staircase, and chaos lives on the edges of the steps. That's the spot where a foot can't decide which step it's on. A heavy walker stumbles there, and a feather never does.*
>
> — no-physics read 2

Part 4 has four chapters. **4A** is the gearbox and its transmission: a rider that locks into gears, and the relay that chains two of them. **4B** is the record: one rider's whole map along a ray, played like a record from the rim to the centre. **4C** is the atlas: the whole plane, in Bear's words, each landmark beside the classical result it matches. **4D** is the dancers: what happens at the frozen balance point, where the deepest order lives.

A note on the riders. In 4A the strong pattern is the one that passes and the weak one holds. From 4B on, the roles are swapped, so that the strong wave stands (the ledges) and the weak ripple passes. That is the toy of Part 2. 4C names the ripple's direction where it matters; 4D uses the turned walkway through §4.26, and §4.27 and much of §4.28 turn to the atlas's side and the junction.

4A. The gearbox and the transmission

4.1 The rider runs a gearbox

Put a rider in two patterns and it does not drift aimlessly. It locks into a gear, p turns of one pattern for every q of the other, and pays the leftover ratio, the comma, as a bounded wobble that stays inside one tooth. Its drive knob is the pace: the speed at which the strong pattern sweeps past the rider, the weak one sweeping at 0.618034 of it. In the trade this is mode locking, the same pull that keeps a pendulum clock in time with its escapement or draws two organ pipes into tune. The word comma is the musician's: twelve perfect fifths stacked up overshoot seven octaves by a small ratio, the Pythagorean comma, and a tuning has to put that gap somewhere. The first survey looked as though the rider locked onto many fractions. Most of them turned out to be a smooth curve passing near fractions; only two true gears stand at the golden ratio. Read again four times longer on a fresh grid of paces, the two gears keep their paces within an error floor four times lower (525 of 525 and 386 of 387), and no other fraction does: each of the 25 others keeps about a quarter, as a smooth curve crossing a fraction must (FAUX-Cb, written first; figure 7). The census artifact has a size, a window 0.02 wide in pace, and it is named so that no reader mistakes it for gears.

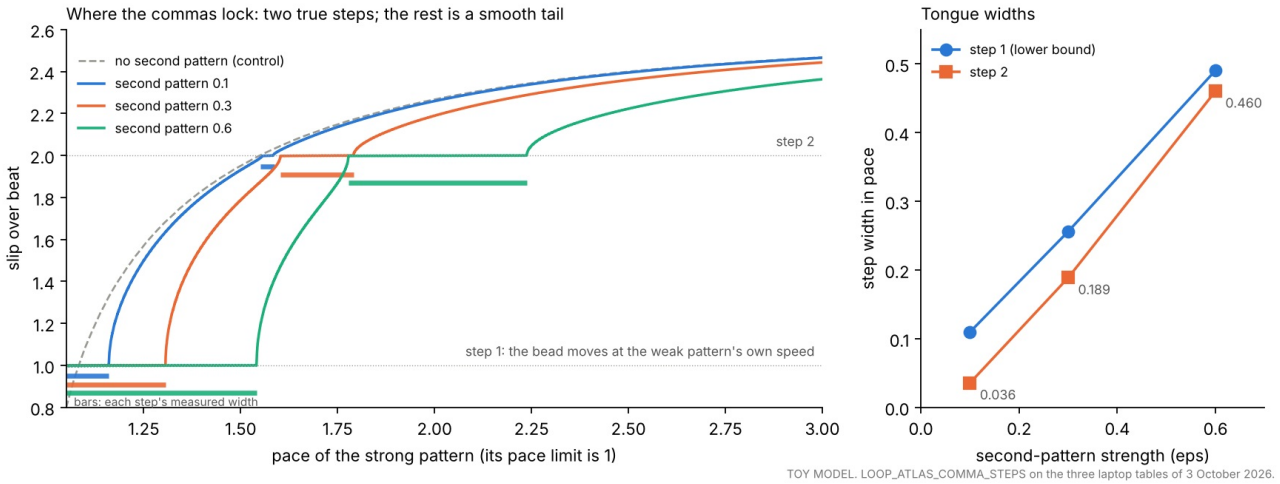


Figure 7. Where the commas lock: two true steps, and the rest a smooth tail; the tongue widths beside them.

Four dots, a word, a frozen frame. Bear's figure-8 (§1.3) gave the gearbox a picture of its own. Cut the figure with a plane and watch where the loop pierces it. Inside second gear it crosses exactly four times a lap, always at the same four places. Just outside, the figure rolls slowly through every shape it can take, once every $1/|2 - n|$ laps, where n is the rider's slip over the beat (2 in second gear): the nearer the slip comes to 2, the slower the roll, so over a long watch each crossing wanders the whole plane; over a short watch it draws the little circle Bear drew, whose size is how far the figure rolled while he watched: over five laps, nothing in the gear, 1.1% of the plane's width just outside it, 31% farther out. Read in order, a lap's four crossings spell a word: which eighth of the plane each dot lands in, a capital or a small letter for the way it passes. In the gear every lap writes the same word (AaGh at pace 1.70, on 154 laps of 154); outside, the word cycles through about eight words, one dot stepping an eighth of the plane each lap. So a gear is one frame of that slipping film, held still. Swept across second gear, the held words never come back to one they have left (CaFh, BaGh, AaGh, AaHh, AbHg, AcHf). Matching their letters to a film made at pace 1.57 placed only 3 of 9, because a rolling lap does not close and its word starts at a shifted letter, and because the pace reshapes the

whole figure (figure 8); matching the dots' positions to a film made just beside the gear on each side placed 8 of 9 within 0.019 of the plane's width, closest at the gear's edges, the one miss in its middle, farthest from both films. The film also lingers: about 60 laps in each 30-degree band of its frame next to ±180 degrees, and 3 to 10 on the far side. Written down first, the frame held at the gear's own lower edge, found between paces 1.602 and 1.604, was put in that busiest band, and it sat there, at −170.8 degrees: the lingering is the ghost of the lock at the gear's edge. In the gear the four dots also keep time, the gaps between them fixed to a thousandth of a lap and the figure's two lobes unequal in time as in size; and the lock catches like a clutch, after 14 laps beside the edge and 5 deep inside, the lock counted as caught once its frame is within five degrees (the five-degree rule). It catches more slowly toward the edge than the bottleneck's square-root law says, by a factor of 2.1 against a bar of 2; its redesign, registered on 4 October, ran on 31 October with the edge found to a hundred-thousandth (between paces 1.602523 and 1.602531), and missed again: from 0.0002 to 0.004 above the edge the catch time falls steadily, lap 21 to lap 11, but only as the distance to the power −0.32, not −0.5 (CLUTCH-2). The five-degree rule calls the lock caught before the slow squeeze the law describes, so the law is better read on the slipping side, from the time between slips (Part 8). Read after that run, the square root is there in where the lock sits instead: the frame it holds moves away from the mirror frame, ±180 degrees, as the square root of the distance from the edge, −180.02 + 240.3 times the square root of the distance, in degrees, within 0.04 degree at all six distances. The lock is born exactly at the mirror frame, in the middle of the band where the film lingers most.

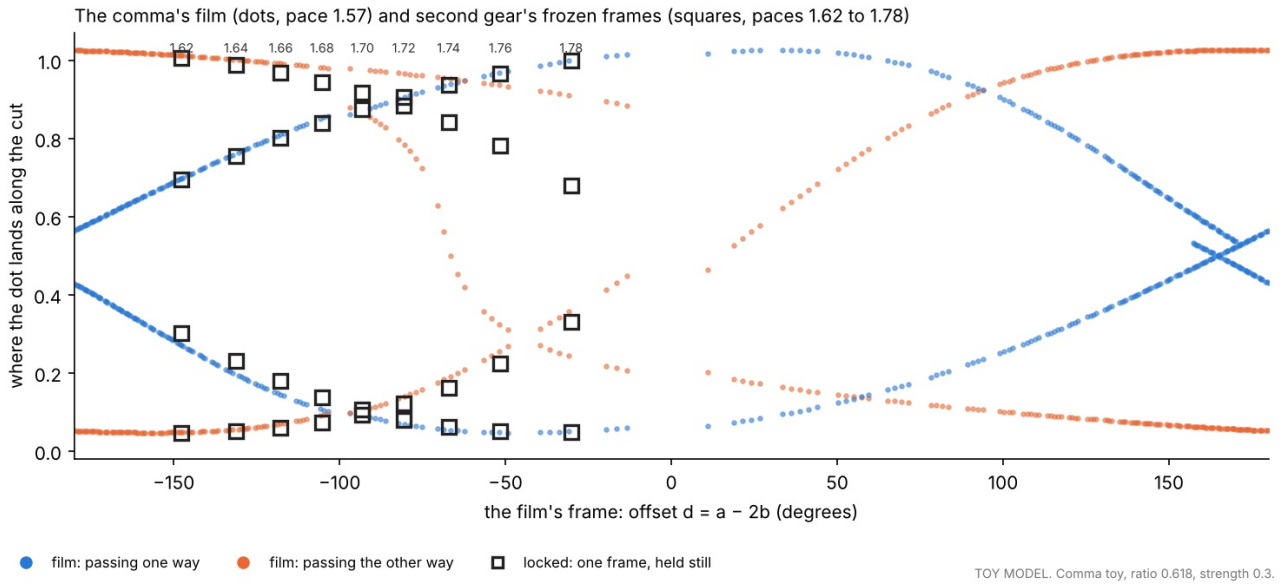


Figure 8. The comma's film and second gear's frozen frames. Dots: where the slipping figure crosses the plane at pace 1.57, against the film's frame, the offset d = a − 2b, where a is the rider's phase against the strong pattern and b its phase against the weak one, so that d is the one mix of the two that second gear holds still; squares: the four dots each locked pace holds still, paces 1.62 to 1.78. Near 1.62 the frames sit on the film's tracks; toward 1.78 they drift off, because the pace reshapes the figure.

4.2 More gears as the rhythms draw closer

As the two rhythms draw closer, more gears appear. The highest gear is one over one minus the ratio (φ^2, about 2.618, at the golden ratio, with $\varphi = 1.618$ the golden ratio itself, not the start's lag of §3.5). Changing from one gear to the next is a switch, not a slide, and depth and closeness of speed decide it together. The new gears arrive the way fractions fill in between two neighbours on a number line, the Farey way: between one gear and the next a third appears, and between those two, more. Their locking ranges are the Arnold tongues of §2.5.

Faux locks. Drive the second pattern three times deeper, 0.9 against the survey's 0.3, and the census finds three locks: first gear (paces 1.050 to 1.843), second gear (2.091 to 2.744), and a new one, five slips for every two beats, between 4.992 and 5.405 (COMMA-53, written first). The next gear the Farey way, eight for three, never comes below pace 6; the lean (the prediction written down as the likelier) that it would be narrower than five for two held by that absence. Whether the five-for-two lock is true, the strict test could not say. A ratio read over a fixed stretch of road carries the stretch's last part lap, an error of up to $2\pi/T/(\text{pace} \times (1 - 0.618))$, about 0.0016 at pace 5 on a road of T = 2,000, and the test's band of 0.001 sat below that floor, so the verdicts that rested on it are void. The redesign reads the same paces on a road four times longer (LOCK-T, written first). On the longer road a true lock's error must fit under a floor four times lower; a band's need not. In the five-for-two window the mean departure from the fraction stayed put, 0.00052 and then 0.00053, as the lean written first said, and 236 of its 401 paces rose above the lower

floor, where none had stood above the first (read after the data): a band that the short road dressed as a lock. First and second gear kept the same count of paces above the floor at both lengths, 5 and 10 of 61, as locks do. The lean that second gear would tighten pace by pace missed, 31 paces within 0.00025 against the 34 it asked for, because a lock promises only that its error stays under the floor, not that each pace's error shrinks fourfold; the eight that did not tighten sit under it (read after the data). So at depth 0.9, as at 0.3, the rider runs two true gears, and the census's slope bar of 0.05 lets through a tail as steep as the five-for-two run's 0.048.

Three rhythms on one bead. Add a third pattern at the speed of the first two's beat, pace × 0.382, the next rung down the golden cascade, as deep as the second, 0.3, and the gearbox is rebuilt (CLUSTER-1, written first; every lock here passes the floor test at both lengths). The bead rides the new pattern from pace 1.236 to 1.805, a gear 0.570 of pace wide, wider than the old first and second gears together (0.254 and 0.186 without it). In radio words, the third tone injection-locks the bead. Its slip over beat is 1.618034, φ itself: a true lock at a slip no fraction can name. The new gear crowds the old ones out. First gear is gone at every pace; below the golden gear the slip climbs smoothly, 1.100 at pace 1.050 to 1.565 at 1.232, with no gear at all. Second gear holds at 22 paces instead of 58, and it has moved: it now sits at 1.942 to 2.010, wholly above its old place at 1.603 to 1.789. Take the second pattern away and the golden gear grows, 204 paces against 176, running from the bottom of the range to 1.711, so the crowding runs both ways. The census of 3 October, which knows only fractions, files the golden gear as thirteen for eight (13/8 = 1.625), and by the floor test none of those points holds 13/8. Put to the same test after the data, the census's runs in the two-pattern toy sort exactly as §4.1 tells them: its runs at 1 and 2 hold at every point, 79 of 79 and 58 of 58, and its ten other runs at four points or fewer each, the smooth tail.

4.3 The coupling decides whether gearboxes chain

Chain two gearboxes and the coupling decides everything. A two-way rod fuses the two stages and loads the upstream one: the load pulls back. A freewheel keeps them apart; it passes the drive one way and goes slack when the second stage runs ahead. Only a one-way relay through a filter multiplies their ratios (figure 9). In a control engineer's words: the rod is two-way coupling, where the second stage loads the first; the freewheel is a one-way clutch, a diode; the relay is one-way coupling through a low-pass filter, a buffer that hears the first stage without being heard by it (figure 10 draws the three as circuits). A fourth coupling sits between the rod and the freewheel: a toothed one, which grips each tooth but can skip one. With no load fed back, the first stage keeps its gears exactly, between paces 1.050 and 1.305 and between 1.609 and 1.784, as written down before the run, and the second rides either its own pattern or the first stage, slipping in between; the compound gears predicted between those two did not appear. With the full load fed back, the first stage's own first gear is gone and what the census of that night read as a new narrow lock appears, four slips for every three beats, between paces 1.090 and 1.146 (TRAIN, toothed coupling, 3 October). The two next steps written down that night were run later (TRAIN-2 and TRAIN-3, written first). Moved closer to its shaft, its pattern at half the pace instead of three tenths, the second stage still showed no gear in between, and it lost its own: its pattern now ran at 0.525 to 1.000, near and above the 0.6 its grip can carry alone, so it either rode the first stage or slipped behind both, the flaw that retired the train's first run. The lean that a compound gear would appear missed, and its rival held; the redesign, a second stage deep enough for a pattern that fast, is in Part 8. The kink test, which looks for the flat step a true lock makes between two kinks in the count, read it on a grid eight times finer. Between paces 1.085 and 1.147 the first stage's count stays within 0.01 of four for three, wandering from 1.324 to 1.343, but nowhere stays within 0.001 of it for more than 0.004 of pace. It is not a lock, as the lean written first said, and on a road four times longer it is not a band near the fraction either (LOCK-T, written first): over those paces the mean departure from four for three grew elevenfold, 0.0028 to 0.0308, and across the grid the count ran from 1.258 to 1.421, higher at each tenth pace than at the last. What hovered near four for three was a reading of the first 2,000 time units, not what the train does on a longer road (§4.2, 'Faux locks').

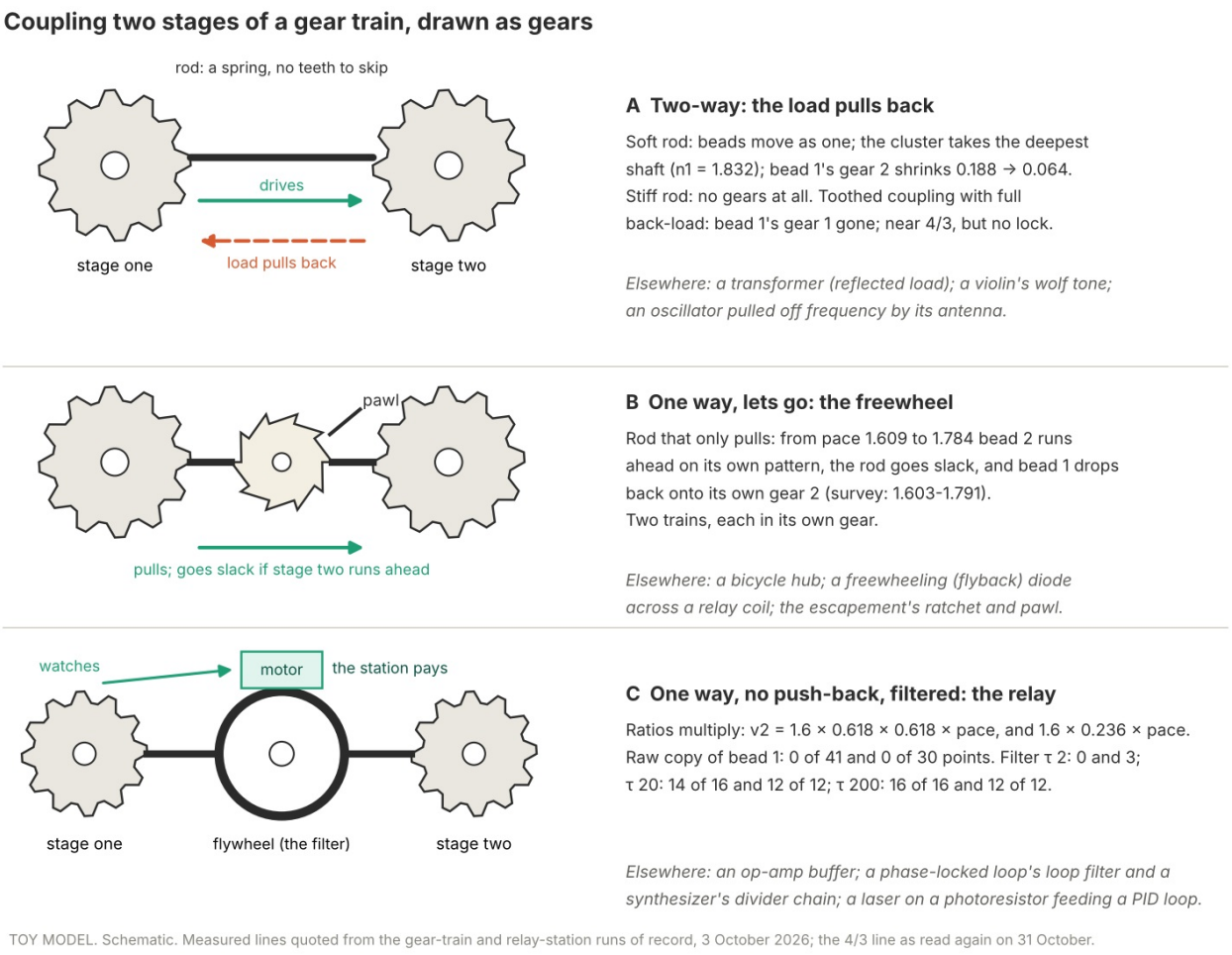


Figure 9. Coupling two stages of a gear train, drawn as gears: the two-way rod, the freewheel, and the filtered relay.

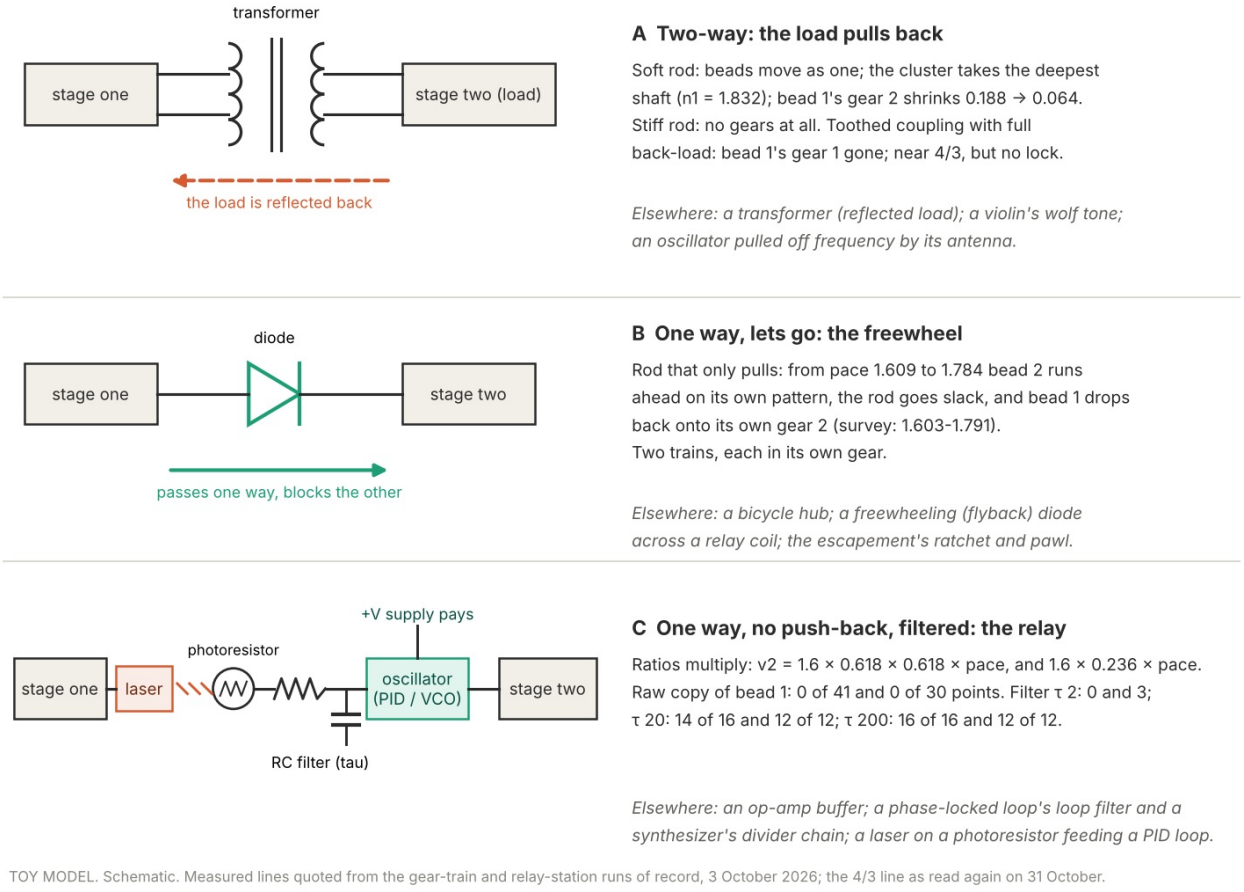


Figure 10. The same three couplings drawn as circuits: a transformer (the load reflected back), a diode, and a laser on a photoresistor feeding an oscillator through a filter.

4.4 Halve the room, double the filter

This is the transmission's law, and a radio engineer meets it whenever a loop filter is sized: the filter must be slow enough to hide the messenger's jitter from a stage that has only a little margin left. The slowest filter that still delivers the product of the two gears scales as one over the downstream gear's room, the distance left to its edge. Halve the room and the filter must be twice as slow. The law holds at both edges where the kicks match the next stage's teeth, the watershed. Away from the watershed the top edge needs only about one over the room's square root (slopes −0.60 and −0.57, predicted, then confirmed blind). The bottom edge keeps the law unchanged, blind (slope −0.985 over five filters; filter time times room 0.91 to 1.01).

The two edges fail in opposite directions: one slips up a gear, the other is pulled into direct drive, and one of them is a hard switch. Yet at the watershed one rule governs both. In a two-stage train the outer stage gives first, where the relay's shake meets its room. The rule placed that first slip blind at four filters from 40 to 200 (filter time times room 1.30 to 1.46, slope −0.93; figure 11). The same test settled the ruler: the room must be read from the fold as measured (0.80799), not from a coarse floor (0.810), which would make the law drift by a third.

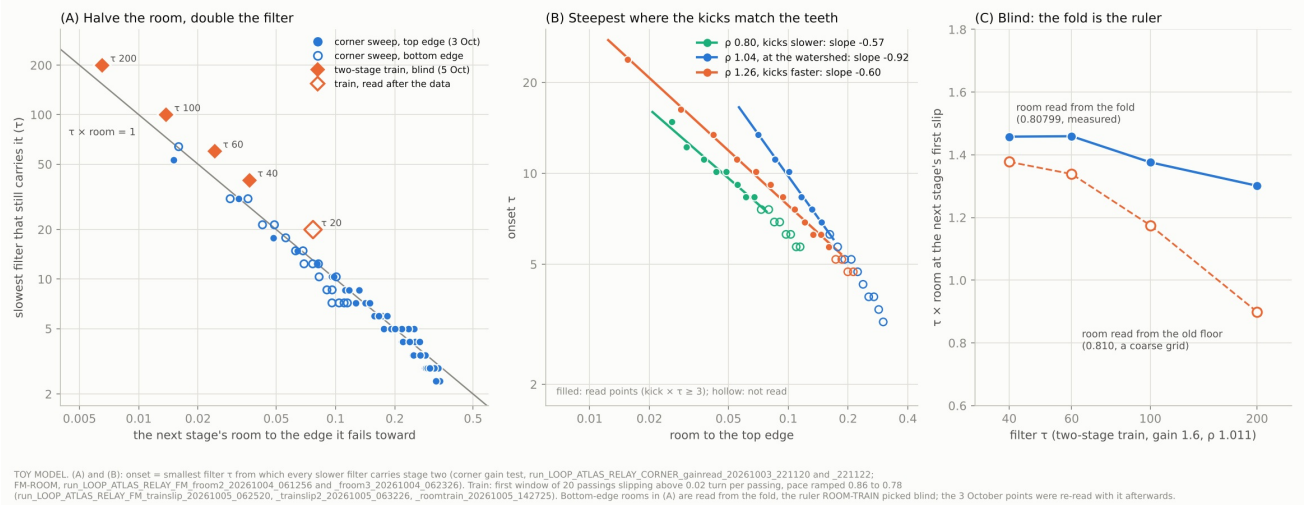


Figure 11. The room rule: the relay keeps the next stage only while its filter is slower than about one over that stage's room. A: two edges and the train on one line; B: steepest where the kicks match the teeth; C: the blind test that picked the fold as the ruler.

4.5 The next stage has a heartbeat

The ratio of the relay's kick rate to the second stage's own tooth rate is a watershed at 1 (figure 12). A watershed is the ridge that sends rain to one river or the other; here the value 1 divides kicks slower than the next stage's teeth from kicks faster than them, and the relay behaves differently on each side. The peak of the filter needed moves with the gain as predicted, and the two edges trade places across the watershed: the top edge is easier on its slow side, the bottom edge on its fast side. Near the switch-type edge a rougher relay works better than a smoother one, but only when its kicks are faster than the second stage's teeth. That is the pocket: a loop speed worse than both a faster and a slower one. It is a toy result, and its cousins are named for checking: dither, the vibrated pendulum, the sensitivity peak of feedback loops, and subsynchronous resonance in power trains.

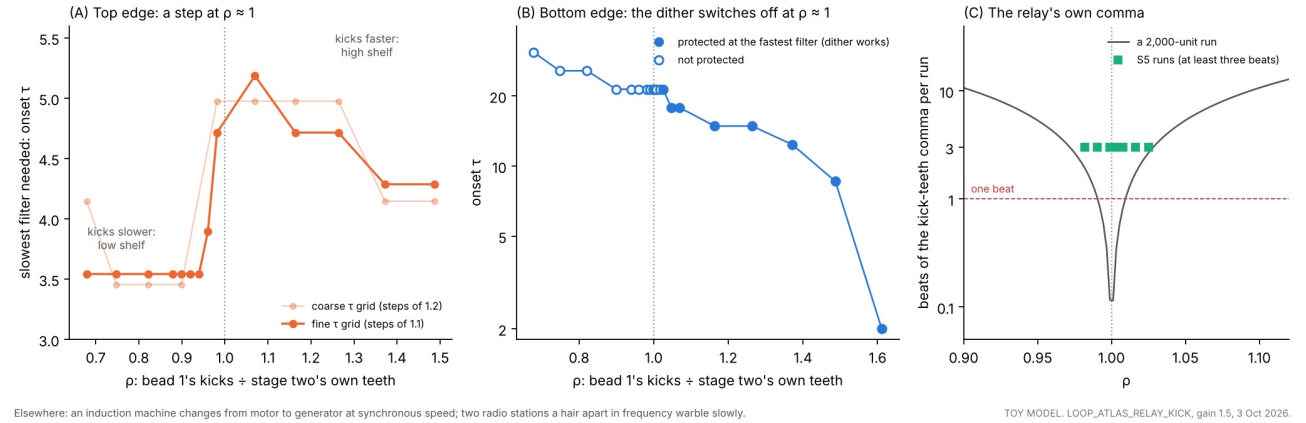


Figure 12. The watershed: at the top edge a step at kick rate equal to tooth rate; at the bottom edge the dither switches off there.

4.6 The relay is an FM transmitter

The second stage does not feel the relay's wobble as a shake of some size. It feels a weakened carrier plus sidebands, as a radio receiver feels a frequency-modulated signal. The arithmetic behind it is the Jacobi–Anger expansion: a tone whose phase is shaken is a carrier plus sidebands, each weighted by a Bessel function of the shake's depth, a standard curve that says how much of the tone's strength each sideband takes. With fast kicks the gear is the gear of the Bessel-weakened patterns. That agreement is guaranteed by how the rig is built, so it checks the rig rather than finding anything. The findings are where the averaged picture breaks. The watershed is where the first sideband lands on the rider. At an exact fraction of the next stage's tooth rate the relay freezes a frame, and the relative phase decides.

Below the floor, the least drive at which the second stage holds its gear on its own, the relay's own kick can rescue a gear the steady drive loses: at 11 of 20 points just under the edge, and at 6 of 20 points 0.040 below it, always with kicks well faster than the teeth. Rescue lives in a window of push, centred near 0.39 against the weak pattern's strength of 0.3. It was read after the data, then passed blind at 12 of 12 fresh points, every window centred between 0.385 and 0.402 against a registered band of 0.35 to 0.42 (WINDOW), and at changed kick rates. Read again four times longer at those twelve points, 706 of their 747 rescued cells hold the gear (FAUX-R, written first); 35 of the 41 that drift sit at a window's edge, two windows lose their top push, and every centre stays in the band, now 0.385 to 0.389. Its useful phase runs one road between 90 and 270 degrees and jumps by half a turn rather than walking: a current reversal (cousins: the rocked ratchets of Bartussek, Hänggi and Kissner, 1994, and Mateos's ratchet, whose current reverses where its motion turns from chaotic to periodic, 2000). Under the window's floor lives a comb of exact fractions. Near the watershed, rescue comes only on exact fractions: none of 1,536 runs a hair off one was rescued.

Peaks and lobes. Why does the useful phase jump by half a turn across the window? A two-tone shake's rocking force has a tall peak on one side and a broad lobe on the other, and the two useful twists of the octave, 90 and 270 degrees, are mirror images of each other. The picture written down on 4 October: with barely enough push only the tall peak carries the rider over, so at small push the twist whose tall peak points the saving way wins (270); with plenty of push the time spent past the edge decides, so the twist whose broad lobe points that way wins (90). The two-tone runs fit that picture, and its mirror, lobe first and peak second, equally well. A third harmonic tells them apart (PEAKS, written first). Added in step with the fundamental, at 0.65 to 1.1 times the octave's size, it turns the tall peak at 90 to point the saving way, while the broad lobe at 90 keeps pointing the saving way too. If the peak decides at small push, the small-push band moves from 270 to 90 and the half-turn jump disappears; if the lobe decides, the band stays at 270. It moved. At all 13 points where the window shows two distinct bands, the small-push band sits near 90 and none near 270; at 9 of the 13 the large-push band is near 90 as well, and the rest are scraps at the window's top, none near 270. The two-tone control, run beside it on the same finer grid of sizes, jumped from 270 to 90 at all 15 points and repeated the earlier runs' bands exactly. So when push is scarce the tallest peak sets the useful twist: near its floor the rescue acts as a peak detector, and well inside the window as a detector of time past the edge. The third harmonic also moves the window down: it rescues with the shake scaled to 0.80 to 1.15 times each point's own size, against the two-tone's 0.85 to 1.30.

A key and a combination. Under the window's floor the comb has a lock's whole grammar (FRACTION KEY, written first; two of its three tests registered on 4 October and run on 31 October). The starts that work repeat exactly as the fraction's denominator says: turn the kick's start by one q-th of its period and nothing changes, at all 24 patterns read, for 8/7, 6/5 and 11/9, the last never run before. So on a dial of twelve starts the working ones come in pairs for 3/2 and in threes for 5/3. A kick rate is a key and the start a combination, and the combination's period is set by the key's denominator. A tooth has no width of its own: a kick a hair off the fraction on a run four times longer is rescued about as often as one four times farther off on the short run, since offset times length is all that counts, in all twelve groups read (four of them at zero, where nothing is rescued at any length). Near the watershed the exact fraction's share stops changing once the run passes 2,000. And the teeth narrow as the denominator grows, as Arnold tongues do. At the same push, 3/2's tooth keeps half its exact share across 5 of 8 offsets, 5/3's across 4 and 8/5's across 2. Stepping the push down, 3/2 still rescues at 0.15 at every point, 5/3 down to 0.17 or 0.19, and 8/5 only down to 0.25 or 0.27: the comb's floor rises with the denominator.

Bear's seam. Bear pictured the relay's settings as a room of walls, each with its own rules, meeting at seams, the room's shape set by how the knobs play against the walls. One seam is where the push window meets the watershed, and there, the lean drawn from his picture said, the walls should pinch the window shut (WEDGE, written first on 5 October). Driven by a two-tone shake whose rate was stepped from 1.61 times the second stage's tooth rate down to 1.107 times it, the window did narrow, no wider at the slower rate in 16 of 18 steps, its centre held between 0.387 and 0.409, and it lost its plateau where every phase rescues. But it was still open at 1.107, at pushes 0.38 to 0.44, so the lean that it closes there missed. Carried closer on a push grid twice as fine (WEDGE-2, written first), the window's top kept falling, from 0.44 at 1.107 to 0.42 at 1.02, while single phases flickered at its low edge. Then, 1% from the watershed, it seemed to burst open: at 1.01 every one of the 36 phases counted as rescued at pushes 0.42 to 0.46 at all three points, on past the grid's top at 0.48, and all three of WEDGE-2's registered guesses missed there (that the window closes, that it grows no wider, and that its centre stays between 0.35 and 0.42). The burst was a faux lock (FAUX-W2, written first). The relay counts the second stage in gear when its speed over the drive stays within 0.004 of the gear's ratio over 2,000 time units. At 1.01 the shake's first sideband lands inside that ruler: a second stage held by the sideband instead of its gear sits 0.0038 below the gear's ratio. Read four times longer, 929 of the 982 cells counted in gear at 1.01 ran at the sideband's speed, within 0.0005 of it, and only 53 held the gear, at pushes 0.36 to 0.42: the true window still narrowing from above, and still open, so the lean that it closes missed even so. The same longer read found faux locks inside the ordinary window too: at 1.107, 50 of its 211 rescued cells drifted off, most of them at its edges (at push 0.44 only 1 of 15 held), though each window's extent moved by at most one push step at an edge. So the seam's corner is a narrowing window beside a trap (figures 13 and 14), and near a window's edge the relay's counts of rescued phases run high, by nearly a quarter at 1.107.

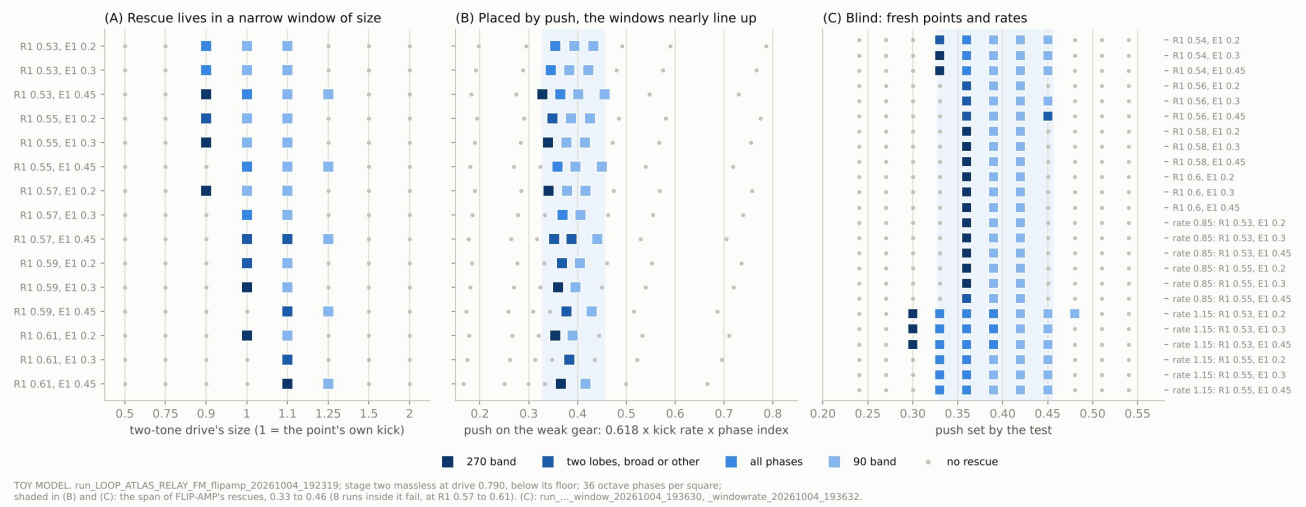


Figure 13. The window of push: below the floor, the relay's own shake rescues the stage only inside a narrow window.

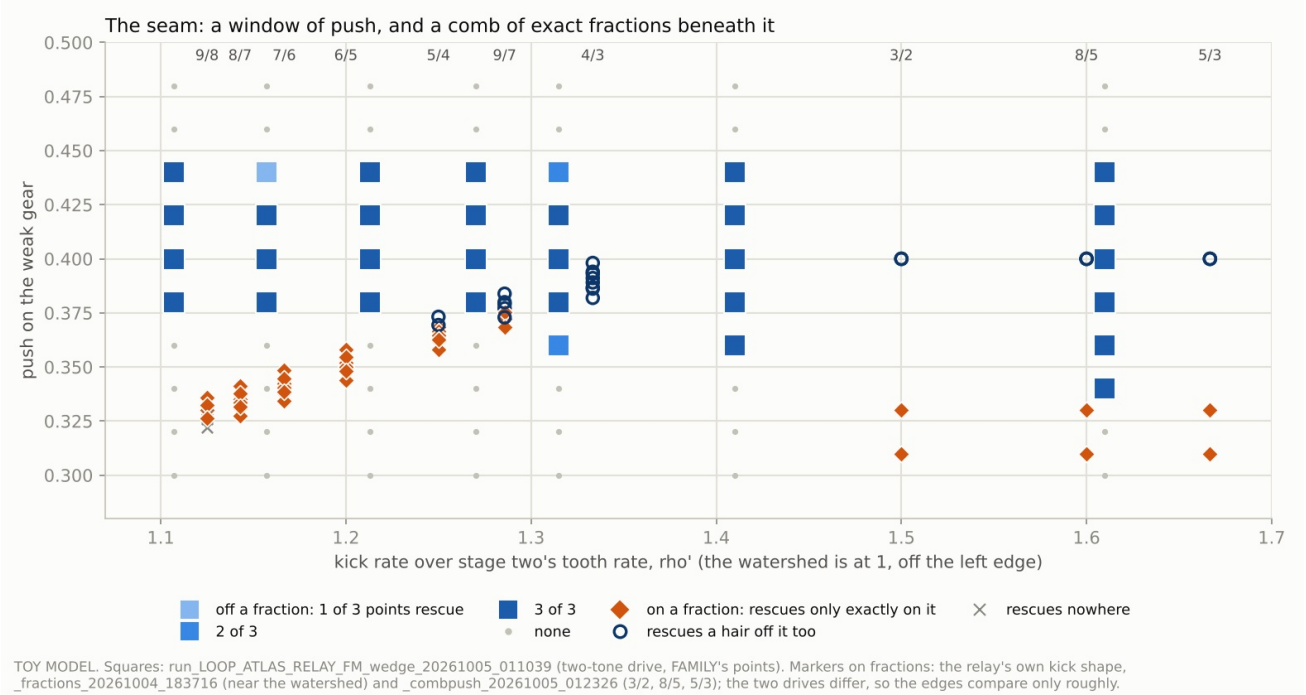


Figure 14. The seam: a window of push, and a comb of exact fractions beneath it.

Bear's board. Give the second stage weight and most of this goes. A single heavy rider, weight 1 at the same drag, keeps almost no harmless octave phase (23 of 760 verdicts in gear, against 458 without weight, nearly all of them the steady drive), and below the floor the relay's own kick rescues it at 2 of 20 points, against 11 without weight (BOARD-1). Bear drew a different vehicle: a light board that grips the patterns and a body riding on it on legs, a spring and a damper. Together they carry the heavy rider's whole weight, but the body feels the patterns only through the legs, as a car's body feels the road only through its suspension. With the legs off the board reprinted the weightless rider's verdicts, 760 of 760 at both drives, and with stiff legs the heavy rider's, 758 of 760. The lean written first was that legs slower than the teeth would hand back the weightless band. They handed back a part of it. With legs at a natural rate of 0.1, about a third of the teeth's passing rate, the board kept 124 of the 418 harmless two-tone phases and the relay's own kick at 17 of 20 points. Where the band was one clean lobe (nine points), it narrowed to between a tenth and a little over half of its width but did not turn: its centre moved 0 to 10 degrees, where a single weight at the same drag turns it, by a median of 17.5 degrees at weight 0.3. With legs at 0.6 the board was the heavy rider again (757 of 760). Below the floor the soft board needed no rescue: at 0.805 the steady drive held it at all 20 points, by the record's rule for a gear, which lets one slip in 2,000 time units pass, so that test could not tell.

Two knobs. Two causes were tangled: the legs' softness, and the damper, which ties the board to a body that cannot follow its fast wobble and so drags on every fast motion. Untangled, both count (BOARD-2 and BOARD-3, written first). A weightless rider given a tenth more drag kept none of the 418 phases. Legs at 0.1 with a damper a fifth as strong kept 369, and legs at 0.01 with a damper of 0.01 kept 407. But a damper of 0.1 cost the band on legs at 0.05 as on legs at 0.1 (144 and 124 kept), and legs at 0.2 cost it even with a light damper (195 kept). For the engineer this is the old suspension bargain: soft springs and a light damper keep the grip, and the damper that quiets the body takes it away. The damper also lowers the floor. The board on legs at 0.1 with a damper of 0.1 held without a single slip at every steady drive from 0.796 to 0.812, where the weightless rider slips below 0.808, while the same board with a light damper slipped at every one of those drives. Read with that strict, slip-free rule, the weightless rider's rescues at 0.805 stand (368 of its 385 two-tone rescues and 10 of its 11 by its own kick hold for 4,000 time units), but the heavy rider's 97 shrink to 34, a tenth of the weightless rider's. Where the two knobs meet, and whether the soft board can be rescued below its own floor, are open (Part 8).

The room rule says when the messenger's shake must be hidden. The window of push says when it can be used. Set side by side, they are the gearbox's answer to what a relay does to the gear after it.

4.7 Chaos needs a second axis

A one-way chain of weightless riders cannot be chaotic, that is, cannot follow its exact rules into a motion that never repeats (§2.6): each rider's passing is a Möbius map, which has no room for chaos. A weightless rider is a bead on a wire that can only keep its place in line; to be chaotic, a motion needs room to fold back over itself, and in the plane of the bead's lag and the drive's clock there is none (the Poincaré–Bendixson theorem says so for flows in two dimensions). Weight, the rider's mass, gives the bead a speed of its own, a third direction, and the room to fold. Chaos appears with weight, from weight 0.5 to at least 5, on the borders between treads (97% of cases, predicted blind; BORDER BLIND). A chain of two weightless stages can be chaotic too, but only when the tooth that couples them pulls both ways, each stage tugging on the other as through the two-way rod of §4.3, and never when it pulls one way. The Loop Atlas, the companion book of §1.1, showed the strong form of this first: its needles, angles driven round a circle as our rider's lag is, obey a Riccati equation when they have no weight (§3.5, piece 2), so their cycle maps are Möbius maps; and its needles with memory, that is with inertia, go chaotic under a one-way clock, as a rider with weight does here.

4.8 A tread's edge is a fold with a ghost

A light rider's tread ends at a fold (figure 15). Picture a ball resting in a small hollow on a hillside, with a hump beside it where it could just balance. Tilt the hill a little more and the hollow and the hump slide toward each other, merge into a flat shelf, and vanish together. Past that tilt the ball rolls on, but it slows over the shelf where the hollow used to be. The vanishing is the fold, which mathematicians call a saddle-node, and the slow spot is its ghost. In the rider the hollow is the locked motion, the hump its unstable partner, and the tilt the drive. At the second stage's floor, the recovery above the edge and the stutter below it follow one square root, mirrored (the edges agree within 4×10^{-8}, the slopes within 1.3%). Just past the fold the locked orbit and its partner are gone, but the rider still lingers where they used to meet. Jostled by a faint random noise (48 copies at once, CLIFF-HUM-4), its wobble grows as one over the recovery rate: a warning that the edge is near, in a band a relative 10^{-4} of the drive wide. Weight changes the edge. A heavy rider frays before it slips, its lag wandering over most of a turn: a loud warning where the light rider whispers. The fraying is chaos inside the gear (§2.6), reached by period doubling, the motion's repeat doubling and doubling again (§4.16), with a window where a motion that repeats every three passings comes back. The brake, which is the drag, and the weight turn out to be one dial: raising the drag acts as lightening the rider (§2.3). Cousins to credit: the saddle-node bottleneck, critical slowing down as an early warning, hysteresis in the underdamped junction (Stewart; McCumber; checked), and the driven pendulum's routes to chaos (D'Humieres, Beasley, Huberman and Libchaber, 1982; checked).

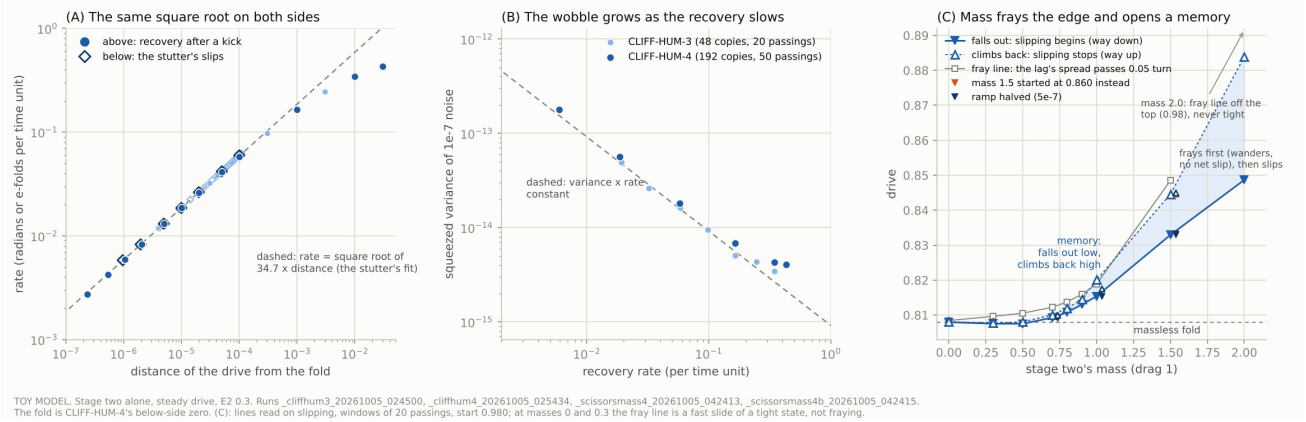


Figure 15. A tread's edge: the same square root on both sides; the wobble grows as the recovery slows; weight frays the edge and opens a memory.

4.9 The toll per wave

The rider pays a nearly fixed toll each time the strong pattern passes, like a car at a tollbooth on every lap: about 2.8 radians gained on its own mean, 0.446 of a turn (2.79 to 2.94 across first gear, WOBBLE MAP), scaling with its own ratio (N1 AND N2). What the relay passes on is that toll times its gain. At fixed timing the toll's size matters, roughly as its square.

4.10 The escapement end: the scratch and the winch

Below the gearbox sit the escapements: latches that reset every stroke and erase the comma instead of absorbing it, as a clock's escapement lets one tooth through per swing and forgets the rest. A zero-mean scratch, the wave slid forward one wavelength slowly and then back one wavelength fast so that on average it goes nowhere, walks the bead one wavelength per stroke once the retreat, the speed of that fast slide back, is faster than about 1.75 times the strength of the push that drags the bead (the fastest the push alone can move it; figure 16): the bead walks at a retreat of 1.80 and not at 1.70 (SCRATCH, 2 October). The winch is the scratch's two-handed cousin, an inchworm: two beads on a rod, the rear one gripping its tooth while the rod stretches and the front one while it shrinks. With the rod one tooth long at rest, a throw of one tooth or less goes nowhere, a throw of 1.25 to 2.5 teeth walks exactly two teeth every cycle, and with the grips in the wrong order a throw of 1.5 teeth walks two teeth backwards (LOOP_ATLAS_FIELD_WINCH, 3 October). They frame the gearbox from below.

The first engines. Before the comma survey, the first days built five small engines, on 2 and 3 October, in the Loop Atlas: the scratch and the winch above, the scratch's trit tape, two stations facing each other, and the ring conveyor. The scratch writes a tape: each stroke forward or back, or none, moves its beads one wavelength, so a string of strokes such as +1, 0, −1, +1, +1, 0, −1 leaves every one of 16 beads exactly on the string's running sum once it is relatched (TRIT TAPE). A bead also keeps one bit of its history: after a slow forward leg it rests the arccosine of the pace ahead of its trough, and after a slow backward leg 180 degrees less that, so the two rests differ by 180 degrees less twice the arccosine of the pace, exact to a tenth of a degree up to pace 0.5 and falling short of it near pace 1 (REST SPLIT). Two transmitters facing each other make a standing pattern whose quiet places walk at exactly half the rate one station slides its phase. A bead that feels only where the sound is loud and soft, an envelope bead, walks with them while their speed stays below the grip, twice the product of the two stations' strengths: it keeps 98% of their speed at the grip and 13% at twice it. A bead that feels only the push ignores them and moves only on what the stronger station leaves over (TWO STATIONS). Close the line into a ring with some loss, and the envelope bead is carried only where the second station's waves fit a whole number of times round the ring: a comb of teeth that moves when the ring's size does (with six waves round the ring, near 5/6 and 7/6 of the first station's rhythm; with five, near 4/5 and 6/5). A ring that pumps one station much harder than the other buries the comb (RING CONVEYOR).

Which flips the riders see. Flip the two-station riders and their drive one knob at a time (RF-3, written first). The envelope bead cannot tell its own sign: a bead drawn to the loud places instead of pushed to the quiet ones rides at the same speed, and with its start moved a quarter wave it retraces the other's path to the last digit (1.8×10^{-14}), because the field's strength changes sign there and nothing else does. The plain bead cannot tell its charge's sign either, moved half a wave (1.5×10^{-15}). Turning one station by half a turn only swaps which quiet places the bead sits in: its speed changes by 0.0015 or less. The one flip the envelope bead sees is the direction of the slide. Reversed, it gave minus the speed while the bead was held, but not past the lock, where the two directions differed by up to 0.029 (1.837 against 1.866 at slew 4, the slew being the rate at which station 2 slides its phase). That missed the lean, and the reason lies in the drive: reversing the slide speeds station 2's carrier up instead of slowing it down, so the two

runs differ in the field's fast parts. The true mirror, the slide carried by the other station, gives exactly minus the speed, and doubling the carrier's rhythm shrinks the leftover to between 0.0003 and 0.0014, from a seventh to an eightieth of what it was (RF-3b, written first). The fast parts also loosen the grip near the lock's edge: at slew 4, with the quiet places moving at 2, the bead keeps 1.837 at the first carrier and 1.929 at the faster one.

A station that locks. Let station 2 be pulled by the field it sits in instead of sliding at a set rate. Its phase then follows Adler's equation: it locks to station 1 when its own pace differs by less than the pull, and otherwise slips, at a mean rate of $\sqrt{(\nu^2 - K^2)}$ and in bursts, where ν is the difference in pace and K the pull between the stations, a coupling, not the circle-map K that §3.7 gives the weight (RF-4, written first). The envelope bead is a meter for that slipping, but only for gentle slips. With a weak pull (K = 1) it reads Adler's square root to within 1.2% at every difference from 1.25 to 2.75, and it stands still while the stations are locked. With a strong pull (K = 6) the station slips just as surely, at mean rates of 1.7 to 6.7, yet the bead does not move: it keeps one or two thousandths of the quiet places' mean speed, even where that mean is well below its grip. Each slip comes as a burst in which the quiet places run at three times the grip or more, too fast to follow; read after the data, the bead drops back into the quiet place behind, so each slip leaves it where it began. A rider fed by a locking station reads the slips' peaks, not their mean, as the shake's tallest peak decides the rescue in §4.6.

Three emitters and a three-armed needle. Put the emitters in a plane and let a needle turn in their field (RF-10, written first). Two emitters at right angles a quarter cycle apart make a field that turns, and so do three at 120 degrees fed in sequence, the three-phase motor; every needle follows such a field by Adler's law exactly. Fed in phase, the same three cancel to nothing. Bear's triad, three emitters 60 degrees apart about a common axis, fed in phase makes a field that only shakes along the axis, and there no needle turns at all, however many arms it has: its equation splits into a function of time times a function of angle, so each shake undoes itself. The lean written first, that a three-armed needle would find a gear there at a third of the rhythm, came from averaging where averaging does not hold, and it missed. Fed in sequence instead, Bear's triad makes a field that turns as well as shakes, its turning part twice its counter-turning part, and there the third-rate gear is real (RF-10b, written first). Once the coupling passes the threshold worked out beforehand (0.067; found between 0.06 and 0.07), every three-armed needle turns at exactly one third of the field's rhythm, 0.10000 of its 0.3, from every start. A dipole in the same field only creeps (0.003 to 0.013), and a two-armed pointer settles on the ellipse's long axis and stays. A gear at one third of the drive belongs to the three-armed needle alone: no dipole can capture it.

Bear's floating actuator. A cheap building actuator has no idea where it is: its controller keeps time on its pulses, so 45 seconds of open takes a 90-second valve to half, and every error in a pulse stays in the count. In the toys' words the actuator is a slip with memory, its stops are latches, and every reset is an escapement that erases the accumulated comma. Bear's account of how the trade lives with it, put into a toy actuator and run for thirty days (FLOAT, written first): once the demand visits the stops and the controller overdrives into them, the error stops growing (days 20 to 30 against days 5 to 10, 1.04 and 1.00 times); a sync at 02:00 once a day holds to 0.07 an error that would otherwise reach 0.21; and an unbiased error grows as the square root of the pulses since the last stop, by 2.08 from day 7.5 to day 29.5 against the rule's 1.98 (FLOAT-b, run on 2,048 seeds after 32 proved too few to judge it). And a twist of the toy's own: an actuator that opens 2% long and closes 2% short drifts open with no reset at all, until its top stop clips it each time the demand peaks, and from then on its error sits at the gap between the highest demand and the stop, 0.21 against the 0.20 written first. The stop heals what it can reach.

Bear's column. An infantry column is a chain of followers, each keeping its distance from the one ahead, and Bear's account of it came with three tests (COLUMN, written first). Put into a toy column of twenty, each follower answering the gap it saw a moment ago, a short pause is forgotten down the column and a long one reaches the tail. How far it reaches grows with its length, and not evenly: pauses of 2, 5, 10 and 20 stopped 1, 4, 17 and all 19 followers (COLUMN-b). Each follower holds one unit of slack between its travelling and standing gaps, which fits the short pauses but not the jump between 5 and 10, where a pause twice as long stopped more than four times as many. Contact breaks at the restart, never during the pause, in every case of the first run. Whether the stretch grows toward the tail is set by the followers' reaction, as traffic theory says: when the pull times the delay is below a half the restart's stretch stays flat along the column (2.02 at 0.45), and above it the stretch grows from rider to rider (2.12 to 3.17 at 0.6) up to a ceiling. The help a watcher at the tail can give missed both ways it was tried. A head that slows whenever the column stretches made the breaks more than three times worse in a column that already amplifies, and a head that restarts gently cut them by 48%, just short of the half asked. In this toy the cure for breaks lies in the followers' reaction, not in the head: with a quick enough reaction (0.3) nothing breaks at all, even after a sudden restart.

Bear's cornstarch. Cornstarch and water on a speaker turns from liquid to solid and back, and the water is always evaporating (MIXTURE, written first). In the plain rider of §4.1 it becomes a grip that stiffens while the rider slips against its wave and relaxes when the rider holds still, set where the grip by itself (1.0) cannot hold the wave (pace 1.2). A landed rider no longer slips, so the stiffness it gained fades and it lets go again: Bear's cycle, which the toy shows, but only when the grip remembers. A grip that relaxes with a time constant of 7.5 or less never lands at all, though at 5 it rises to 1.75: it is stiff only while the rider slips fast, and soft again by the slow part of the passing, where a landing would be caught. Memory switches the cycle on all at once: no landing at 7.5, 82 at 8 (MIXTURE-b).

From there the rider lands once every 24 to 32 time units and holds for longer the longer the grip remembers, 6.8 at 8 and 19.7 at 40, landed 28% to 61% of the time. And the evaporation marches it: with the grip's base rising from 1.0 to 1.3 over the run, even a forgetful grip begins to land, more in each quarter (3%, 55%, 88%, 100%), and from the second quarter on it lands while its base is still below the pace, where the plain rider cannot land at all.

A partner that answers back. The paper Bear found, Shukrinov, Rahmonov and Kulikov (2012), shunts coupled Josephson junctions with an inductor and a capacitor and finds new branches in their current-voltage curve where the junctions' rotation meets the circuit's own frequency. The plain rider slipping through its wave is the same kind of rotor, so we gave it a partner that answers back: a damped oscillator of frequency 1, driven by the rider's slip rate and pushing back with its own velocity (RESONATOR-TREAD, written first). Ramped up, the rider jumps onto the partner at pace 1.31, its slip rate leaping from 0.89 to 1.01, then climbs slowly; ramped back down, it stays on the partner to 1.30 and falls off at 1.295: a tread with memory. The tread is slanted and sits a little above the partner's frequency, so the flat tread written first missed, with a partner of quality 10 and again with one of quality 100. Sharpening the partner flattens the tread at every step (slope 0.66, 0.36, 0.27 and 0.25 as the damping falls from 0.1 to 0.01, RESONATOR-TREAD-b) and widens its memory, from 0.12 to 0.58 in slip rate: with the sharpest partner the rider, on the way down, stays on it from pace 1.30 to 1.085, slipping at 1.001 where alone it would slip at 0.42. The sharpest partner also catches the rider near half its frequency, by a jump at pace 1.10 onto a stretch a few percent fast, again missed by the flat bar. At twice its frequency the partner leaves the rider all but untouched: nothing in an overdamped rider's slip rings it there, and the paper's junctions carry an inertia the plain rider lacks.

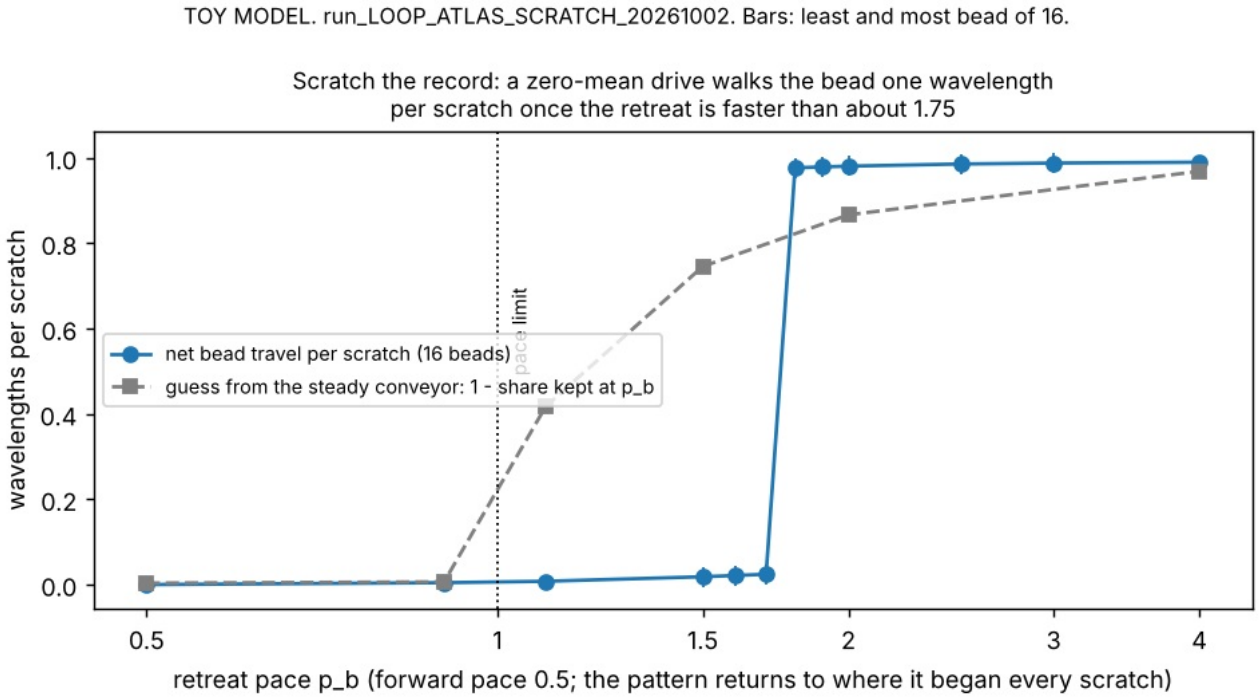


Figure 16. The scratch record: a zero-mean drive walks the bead one wavelength per scratch once the retreat is fast enough.

Bear's shuvit. One escapement keeps its memory. Bear drew a skateboard on two cylinders of different sizes: the board slides on the timing cylinder until it falls off at a cliff, then spins half a turn and lands on a new pad (his sketch of 4 October). In the comma toy of §4.1 that is a rider whose grip on the strong pattern changes sign each time it leaves a pad, a flip set off by the rider's own position and checked after every step (SHUVIT-1, written first). The flip turns the strong pattern into a rectifier. The board feels every tooth's pushing half and never its pulling half, as a full-wave rectifier passes both halves of an alternating current the same way round. So the slip becomes a forward step. At the golden ratio, with the weak pattern at 0.3, the board's drift was above the plain rider's at all 572 paces from 1.05 to 2.999, by as much as 0.644; inside the plain rider's first gear, where both ride the gear, by less than 0.001. With the same flips set half a turn the wrong way, the board ran backwards at every pace, at −0.531 to −0.590. The larger change is in the gears. The shuvit stretched first gear from paces 1.05–1.31 to 1.05–1.69, two and a half times as wide, and erased second gear: the board keeps pace with the weak pattern far past where the plain rider lets go. The lean written first, that the gears would count in half turns, missed, and its rival held: the only wide gear is one whole turn per beat, two shuvits, and none sits at a half turn per beat. For the engineer, a sign flip timed by the rider's own position is a commutator, the brushes of a DC motor that reverse the current every half turn so that the push always points the same way.

Timing advance. Motor builders move the commutator's switching point before or after its natural place, trading pull for top speed, and Bear asked whether the board has such a trade (SHUVIT-2, written first). With the cliff moved round the circle in steps of 15 degrees, advance and retard came out mirror images. Alone, the board drifts the same at an advance as at the equal retard, as the arithmetic says it must, within 8.6×10^{-4}; and at 90 degrees, where the flip falls in the middle of the pad, it drifts as the plain rider does, within 1.25×10^{-3}, a hair outside the bar of 10^{-3} written first, by a gap that shrinks as the pace rises, as a finite run's last part lap does. With the weak pattern, the natural point is best on both counts: first gear is widest there (188 paces) and the drift at the top pace highest (0.715), and both fall smoothly away on either side. The board beats the plain rider's mean drift out to between 90 and 105 degrees either way; near 120 its mean drift reaches zero, and beyond it the board runs backward. The lean written first, that the widest gear would sit off the natural point, missed, and its rival held: there is no crossover near the natural point. The one trace of a trade sits far out, at 120 degrees, where retard holds first gear to pace 1.149 and advance only to 1.111, at the same drift. The reason is the engineer's: a motor's advance pays because its coil current lags the switch, and this board has no lag, since its speed is its push.

A lag makes advance pay. That reason can be tested, and it held (SHUVIT-3, written first). Give the push a lag, as a coil gives its current one: after each flip the grip swings over to its new sign with a time constant of 0.3 instead of at once. Now advance pays. On every second pace of the same grid, the drift at the top pace rose from 0.446 at the natural point to 0.578 at 45 degrees of advance, and first gear was widest at 15 and 30 degrees (91 of the 286 paces, against 88 at the natural point). The envelope has its crossover: holding is best at a small advance and speed at a larger one, the bargain a motor builder strikes. The best angle sits near the coil's own phase lag, read after the data: through 30, 45 and 60 degrees the top-pace drift peaks near 41 degrees, and a lag of 0.3 at the board's slip rate there (2.42) delays a wave by 36. With a lag of 1.0 the best speed angle moved out to 60 degrees, the edge of the angles tried, where the same reckoning gives 70, and the widest first gear to 45 and 60. A lagging rider is another matter. Given weight 0.3 or 1.0, so that its speed answers its push late, the board kept its mirror: advance and retard came out alike within 0.006, and the natural point stayed best on both counts. The lean that weight would break the mirror missed. What has to lag is the push, not the rider; in motor terms, a heavy rotor gains nothing from advance, and a slow coil does.

The coil's rule, tested. On finer angles, every 2.5 degrees at the top pace, the best advance landed where the coil's delay puts it, as written first (SHUVIT-4). For a lag of 1.0 it landed at 70.0 degrees, where the reckoning gives 70.0; for 0.6 at 60.0, against 57.4; for 0.3 at 42.5, against 35.9. The rule fits best where the lag is largest. Weight, stepped from 0.1 to 2.0, narrowed first gear at the natural point at every step, from 94 of the 286 paces to 43, and kept the drift's mirror within 0.0061. But at light weights the gear itself was not quite a mirror image: at weights 0.1 to 0.5 the advance side held 2 to 4 paces longer, which missed the lean that the two sides would stay within 2, and the tilt was gone by weight 1.0. A tiny lag's cost grew, as written first, with the lag (0.005 to 0.04) and with the pace, but faster than the jump at the flip: the shift at 60 degrees was 2.2 times that at 30, where the jump's ratio is 1.73, and the lean, that the shift would follow the jump and so come out between 1.4 and 2.1 times, missed. Read after the data, the shift follows how steeply the drift falls with the angle there, whose ratio is 2.23. Retard pays more for a lag than advance gains: at a lag of 0.02, -0.028 at -60 degrees against $+0.012$ at $+60$.

4.11 What carries to radio hardware

The gearbox's laws read straight across to familiar hardware: the phase-locked loop and its loop filter, a synthesizer's divider chain, the opto-isolator, dither, and Bear's laser on a photoresistor. Stacked relays were then tried (the arrays of §6.13 set riders side by side on one drive, not in a stack): three gearboxes in a row, each fed through a loop filter. A stack needs no ladder of time constants: with the third stage 0.04 above its floor, it held its gear from the same second filter, 28.4, whether the first filter was 25, 100 or 400, so a filter need only be slow against the room of the stage it feeds, not slower than the filter below it (STACK-1). How slow is roughly one over that room while the room is small: filter time times room 0.75 to 1.39 at rooms 0.02 to 0.08, a slope of -1.35 across those five rooms on a grid of filters a quarter apart, with the ratio of stage two's kicks to stage three's teeth held at 1.25 (STACK-2). But it is not one law: written down first, a slope of -1 missed twice (-1.74, then -1.84 across all seven rooms to 0.16 with that ratio held), because below a room of 0.10 a pocket of misses opens between the quick filters, which hold, and the slow ones, and by 0.11 it has closed: the third stage then holds at every filter from about 1.5 up, and only the quickest few miss. Read on a grid of filters a tenth apart (STACK-3, STACK-4), the pocket's top edge is ragged: filter times room wanders from 0.83 to 1.06 at rooms 0.04 to 0.08 and climbs past 2 at rooms 0.01 and 0.015, so one over the room is only its rough size. Its bottom edge holds still at a filter of 3.45 from rooms 0.01 to 0.04, a fourfold range, then rises to meet the top edge. And the pocket is striped, not clean: at room 0.06, read on a grid five times finer, the filters from 4.1 to 5.5, 8.8 to 9.5 and 12.3 to 13.9 hold inside it. Kin within the program: *A Coincidence Has No Single Price* (Schulz, 2026), whose method appears here twice, as the relay's near-fraction hold and its three-beat rule, that near the watershed a relay run counts only if it lasts at least three beats of the relay's kicks against the next stage's teeth (the method is cited here, not the note's worked example).

4B. The record

4.12 The record of grooves

Bear asked for the tongue map drawn as a record, with a needle running from the outer ring to the centre (drive along the radius, one needle weight per record). Drawn that way, the record turned out to be the right picture (figure 18). The light needles' grooves are locked fractions, and they come in the order of the fraction tree. A groove's generation in the Stern–Brocot tree, the sum of its continued-fraction terms less one, orders it on the record. (The terms are the whole numbers met when the fraction is written as a ladder of one-over steps: 2/5 is 1/(2 + 1/2), so its terms are 0, 2 and 2, and it is born in generation 3.)

The fraction tree has an engineer's history. Achille Brocot, a Paris clockmaker, published it in 1861 as a way to design gear trains. To make one wheel turn at a ratio no simple pair of gears hits exactly, walk down the tree and stop at the first fraction close enough: the tree hands you the simplest pair of tooth counts that does the job. Moritz Stern had found the same tree a few years earlier, as mathematics. Each fraction on it is born between its two parents by adding the tops and adding the bottoms, 1/3 and 1/2 making 2/5. That child is called the mediant. A rider locked in a groove does what Brocot's clocks do: it turns a fixed whole number of times while the drive turns another. In the field's words the grooves are mode locks, the record is the map of Arnold tongues of Part 2 drawn as a disc, and the tree says which locks are simple and which sit deep (figure 17).

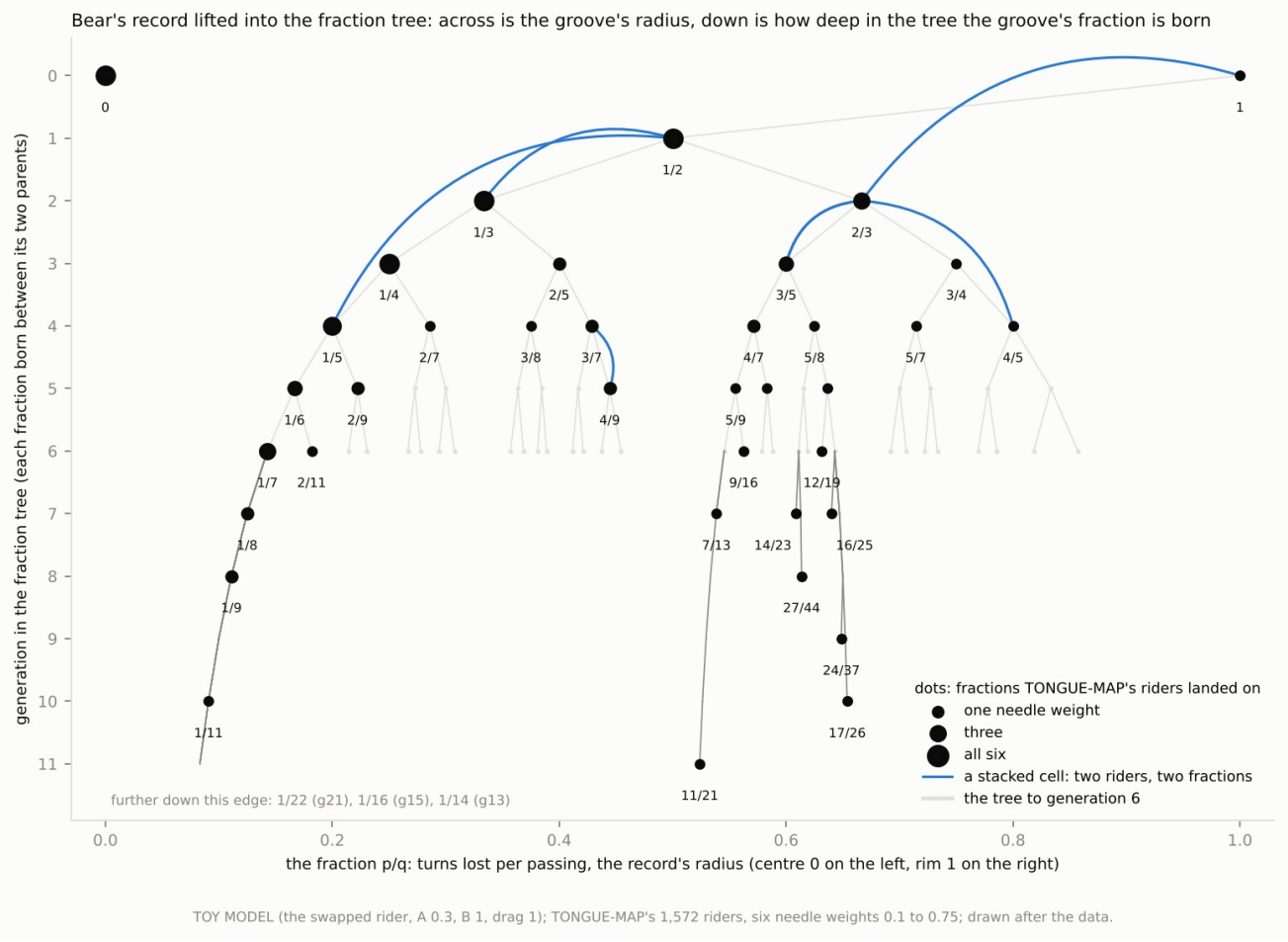


Figure 17. The record lifted into the fraction tree: across is the groove's radius, down is how deep in the tree its fraction is born.

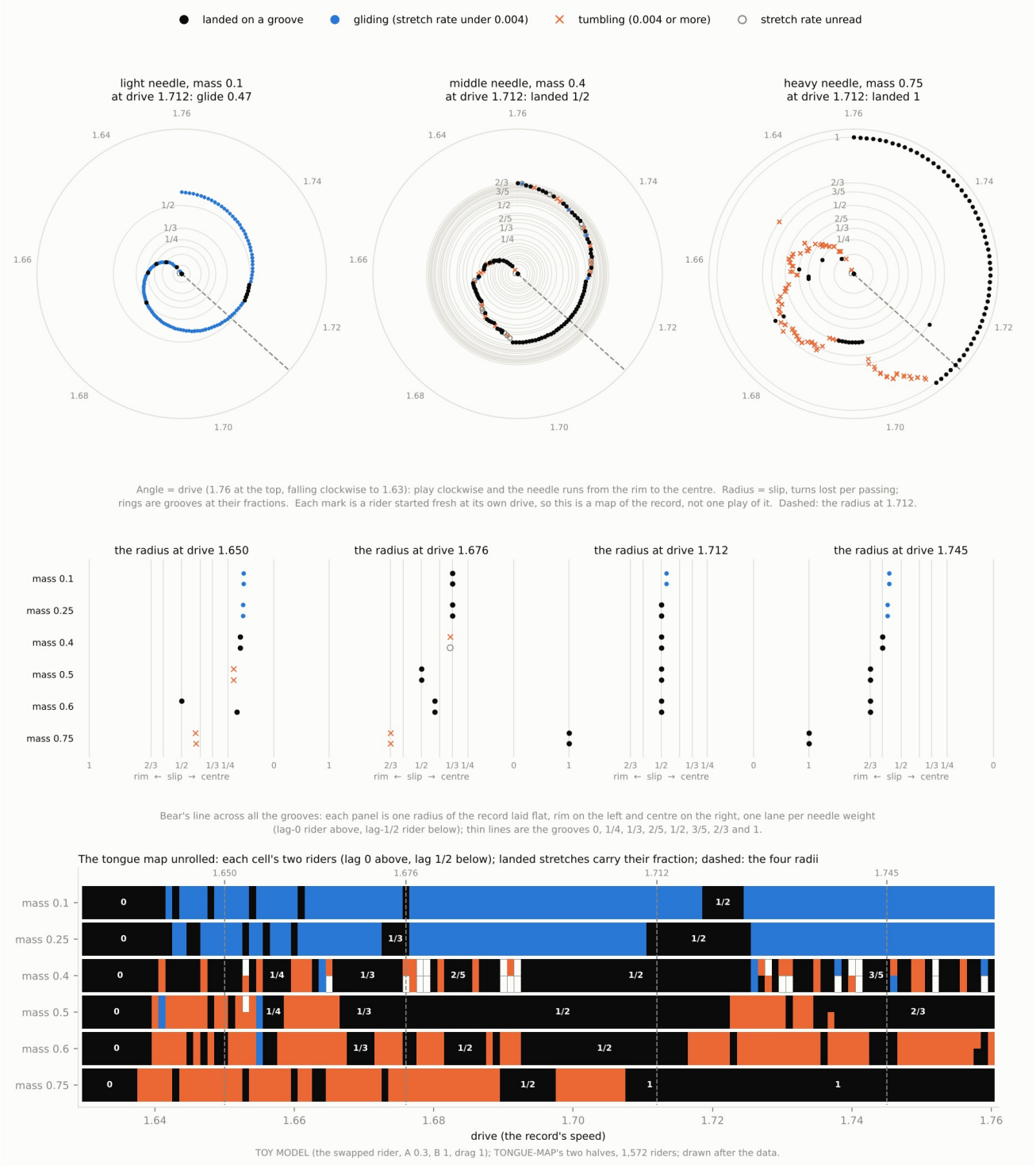


Figure 18. The record: three needles, each playing its grooves from the rim to the centre.

Bear's two needles. Bear asked what a needle does when it is played along the record rather than set down in one groove: does a light needle play one song both ways, and does a heavy one remember which way it came (RECORD, written first)? The rider was played down the record from 1.6770 to 1.6620 and back up, 2.5 millionths of drive per passing, at three weights, 0.25, 0.5 and 0.75. The light needle (0.25) played the same song both ways, to a window: in flight, with one groove, 1/3, at 1.67317 to 1.67648. The middle one (0.5), read after the data, held grooves most often, 1/2 at 1.6739 to 1.6769 and 1/3 below it, its two ways parting by about a thousandth of drive at those grooves' edges. The test written first also asked its four starts to agree within 0.01 in every window, and that bar sat under its instrument's floor: a window of 60 passings cannot read a slip finer than 1/60, and the starts differed by up to 0.0144. The cleaner test came next, a stretch rate measured with a shadow rider started 10^{-8} away (TUMBLE-1, written first). The light needle's flight is a glide, its rate inside the record's floor of about 0.002 (−0.0012 to +0.0010 per passing),

its slip climbing smoothly with the drive; the heavy needle (0.75) tumbles, at +0.40 to +0.64 per passing, doubling a small difference every passing or two. The heavy needle's two ways also disagreed, as written first, but only where a ramp reads a doubling late, not as a second state, so the ledger keeps that test as void. And at that speed it landed nowhere in the 2/3 island: it crosses the island in about 100 passings, where held riders took up to thousands to land. Played slower, the island alone at a quarter of a millionth per passing (RECORD-2, written first), the heavy needle landed there both ways, all four starts, and in the island's lower strip it read the dance its side gave it, 4 in 6 coming down and 8 in 12 going up: the rule of 'Playing the toggle' (§4.13) holds on a moving needle. At four times that speed only the four starts going up landed.

4.13 Grooves that share a drive are tree neighbours

Where a rider can land in either of two grooves depending on where it starts (a stack), the two fractions p1/q1 and p2/q2 satisfy $|p1\ q2 - p2\ q1| = 1$. In words: they are neighbours in the fraction tree. That is a clockmaker's test. Cross-multiply the two fractions, and the two products differ by exactly one. Then no simpler fraction fits between them, and the simplest one that does is their mediant. So a rider with two grooves to choose from never chooses between strangers; it chooses between next-door neighbours. To an engineer a rider with two possible landings is bistable, like a toggle that keeps whichever state it was pushed into. Here the two states always sit side by side on the tree. That rule was written down before the stacks were searched for, and it held: first on one pair, 3/5 and 2/3, found twice (STACKS, STACKS-3), then blind on fresh pages of the pad atlas, where 24 of 24 stacks were neighbours (PA-5c) and 15 of 15 were neighbours or the one pairing the rule allowed in advance, a whole gear beside a third (PA-5d). That is 39 of 39.

Playing the toggle. One fraction can also hold two dances at one drive, and that is a toggle of a second kind. At 1.6756 the rider loses two turns in every three passings either on a dance that repeats every six passings or on one that repeats every twelve (4 in 6 and 8 in 12, lesson 127); of 64 riders held there a long time, 12 sat on the six and 52 on the twelve. Bear asked whether a note played on the drive could choose the dance, as a bowed plate chooses where its sand lies (Chladni's figures). It can, and that was written first (SONG-1). Each rider's drive was slid in 50 passings to a note, held there 300 passings, slid back, and held at 1.6756 for 2,000. A note just inside the window (1.6757) moved no one. The three notes just below the window (1.6754, 1.6750 and 1.6748) sent all 64 riders to the twelve, whichever they started on, and the note at 1.6760, above it, sent all 64 to the six. The lean written first, that a note out in the tumbling would deal the riders afresh, missed: from 1.6750 every rider came home on the twelve. Only the two farthest notes mixed the riders, 1.6770 (26 of 64 ended on the six) and 1.6744 (19), and theirs were the fastest slides back into the window. So the redesign, written first, said that the side of the return decides and haste spoils it (SONG-2). Slid over 200 or 800 passings instead of 50, every rider at every note came home on its side's dance, all 64 on the twelve from 1.6744 and 1.6750 and all 64 on the six from 1.6760 and 1.6770; the riders that missed it fell from 19 and 38 to none. The fine scan registered for the coastline test below (COASTLINE: eight fresh starts at every drive 0.00002 apart) also shows the window's shape, a reading that was not registered. Between 1.67558 and 1.67584 fresh starts land on a dance of 24 passings at both edges, on the twelve inside them and on the six in the middle (1.67568 to 1.67574), with six and twelve side by side in between. A rider sliding back from above crosses that middle before it stops at 1.6756, and one from below does not. To an engineer this is a one-bit memory, written by the side the drive comes back from, so long as it comes back slowly.

Rocking the toggle. Bear's next picture was the plate itself: rock the drive about 1.6756 at a dance's own rhythm, and that dance should gather the riders (SONG-3, written first). It missed, and how it missed was the find. Rocked at six passings, the six-passing dance lost riders to the twelve, 5, 6 and 6 of its 12 at depths of 0.00002, 0.00005 and 0.0001 of drive. Rocked at seven passings no one left, and rocked at twelve, 7 left, only at the deepest. No twelve left under any rocking. A rocking at a dance's own rhythm throws riders off that dance, as a swing pushed in time swings until its rider falls off. Bear's plate was truer than the lean drawn from it: a bowed plate's sand is thrown off where the plate moves and settles on the lines that stand still, and here the lines that stand still are moments. Read after the data, which sixes fell was set exactly by where each sat in its six-passing cycle when the rocking began, and those that fell sat on a run of neighbouring steps of the dance: two steps at the shallowest rocking at six passings, three at the deeper two, and at twelve passings a run of three beginning two steps further on. SONG-4, written first on that reading, shifted the rocking's start one passing at a time. At every shift whole steps left or stayed together, and every step left at some shift and stayed at another, the open pair moving one step round the dance for each passing of shift. That held, but it could hardly have missed: shifting the rocking by a passing is the same, apart from its first and last few passings, as moving each dancer's start one step round its dance, and that rule applied to the first shift's row gives all six rows (SONG-SYMMETRY). Whether a deeper rocking opens a longer run of steps is open (Part 8).

The coastline. Bear also asked whether the drives where riders land have smooth edges, a few pads, or holes at every scale, coves inside coves, like a coastline (COASTLINE, written first). The fine scan above, eight fresh starts at every drive 0.00002 apart from 1.6740 to 1.6760, found 21 of 101 drives landing, in eight windows. The measure written first was a coastline's: how often two drives a given distance apart disagree, which falls in step with the distance for a smooth edge and more slowly for a ragged one. It fell far more slowly (slope 0.23, where a smooth edge gives 1), and the rival, smooth pads, failed. But seven of the eight windows were a single grid step wide, and windows

narrower than the ruler flatten the measure whether or not more hide below it, so the ledger keeps the test as void. The test that could tell was written down that day and run on 31 October: the same riders at ten times the magnification, over a stretch of 0.00034 where every rider at every coarse drive was in flight (COASTLINE-Z, written first). Two windows appeared between the coarse grid's points, all eight riders landing on 5/7 at 1.674568 and on 2/3 at 1.674636, a dance of 30 passings some 0.0009 below the 2/3 island, and both are a single step of the finer ruler too. The coves go on below each ruler tried.

4.14 The shaking has an address

Every record at once (32,080 riders at twenty weights, run on Bear's laptop) puts nearly all of the shaking in one place (figure 19). It lives in the room between gear 1's edge (about 1.636) and gear 2's door. Gear 1 and gear 2 are the record's two wide locks, told apart by the needle's speed: 0.618034 of the drive in gear 1 and 0.236068 of it in gear 2; they are not Part 4A's first and second gears. The door is a straight line in weight, drive = 2.0630 − 0.4719 × weight, so the room closes near weight 0.905, and the shaking all but stops there (figure 20; 36, 24 and 6 shaking riders at weights 0.88 to 0.90, none at 0.91 or 0.92; past the closing only a rare pair shakes, two riders at 0.93 that shake beside gear 2 while keeping the 1/2 groove's count, ROOM-1). Heavier needles shake more densely in what room is left.

Shaking is this book's word for chaos: motion that never repeats, and that two riders started a hair apart soon disagree about. Like a coin toss, it follows exact rules and still cannot be called for long, because the rules stretch a hair's difference into a whole turn (§2.6). Here the chaos has an address, the way a compressor's surge has a line on its operating map: a drawn boundary past which the machine stops behaving. For the rider the keep-out zone is the room. Its walls are known: gear 1's edge on one side, and on the other gear 2's door, which slides down as the needle gets heavier until the walls meet.

Past the closing, the door changes its law. Run again from weight 0.55 to 1.10 (9,648 riders, on Bear's laptop; DOOR-1), the door kept the straight line within 0.0015 from 0.55 to 0.90, and at every weight both surveys read, the two doors agreed to the last printed place: the same engine on a grid of drives and starts of its own, so a check that the door does not depend on the grid, not an independent confirmation. Past the closing it bent off the line, by +0.0018 at weight 0.95, +0.0049 at 1.00, +0.0060 at 1.05 and +0.0076 at 1.10, sliding down about 0.43 to 0.45 per unit of weight instead of 0.47.

There the room is gone and nothing shakes: from 0.95 up, every one of the 804 riders at each weight lands. Over a band of drives gear 1 and gear 2 are both open, and the rider's starting lag picks which (§4.17). The band is 0.019 wide at weight 0.95 and 0.039 at 1.00, and at least 0.04 above that, where gear 1 still lands at the highest drive read. Gear 1's edge hardly moves through this stretch, from 1.6370 at weight 0.80 to 1.6350 at 1.00, so the room's two walls cross near 0.905, as the line said, and past the crossing they overlap instead of leaving a room between them. The door is two laws, then: below the crossing it is the room's far wall, a straight line; above it, it is the low end of the overlap, with a gentler slope of its own.

For an engineer this is the trade a heavy, lightly damped part makes. The chaos goes, and in its place come two states side by side: hysteresis, which heavy junctions (a large Stewart–McCumber parameter) are known for. So past weight 0.905 the drive's history, not its value alone, says which gear the rider is in.

Two checks were made past the closing. Put on gear 2 and lowered a step of 0.001 at a time, a rider falls off where the survey's door sits, within two steps at weights 1.00 to 1.20 (DOOR-3), so the door there is gear 2's own edge, the low end of the hysteresis loop, and not merely the first drive at which the survey's four starts happened to reach it. And the edge keeps bending: at weight 1.20 it sits at 1.5115, 0.0148 above the straight line and 0.0033 above a straight line through the four heavy weights (DOOR-2, DOOR-3), its slope easing from about 0.45 to 0.40 per unit of weight. Its own law is a parabola in the weight. Carried on from the edges at 1.00, 1.10 and 1.20 as a curve whose slope eases by the same amount at each step, it was written down before the run as 1.4740 at weight 1.30 and 1.4390 at 1.40, and the edge came out there, to the drive step (DOOR-4); a straight line through 1.10 and 1.20 would have put it 0.0025 and 0.0075 lower. From 1.00 to 1.40 the slope per unit of weight runs −0.425, −0.400, −0.375 and −0.350.

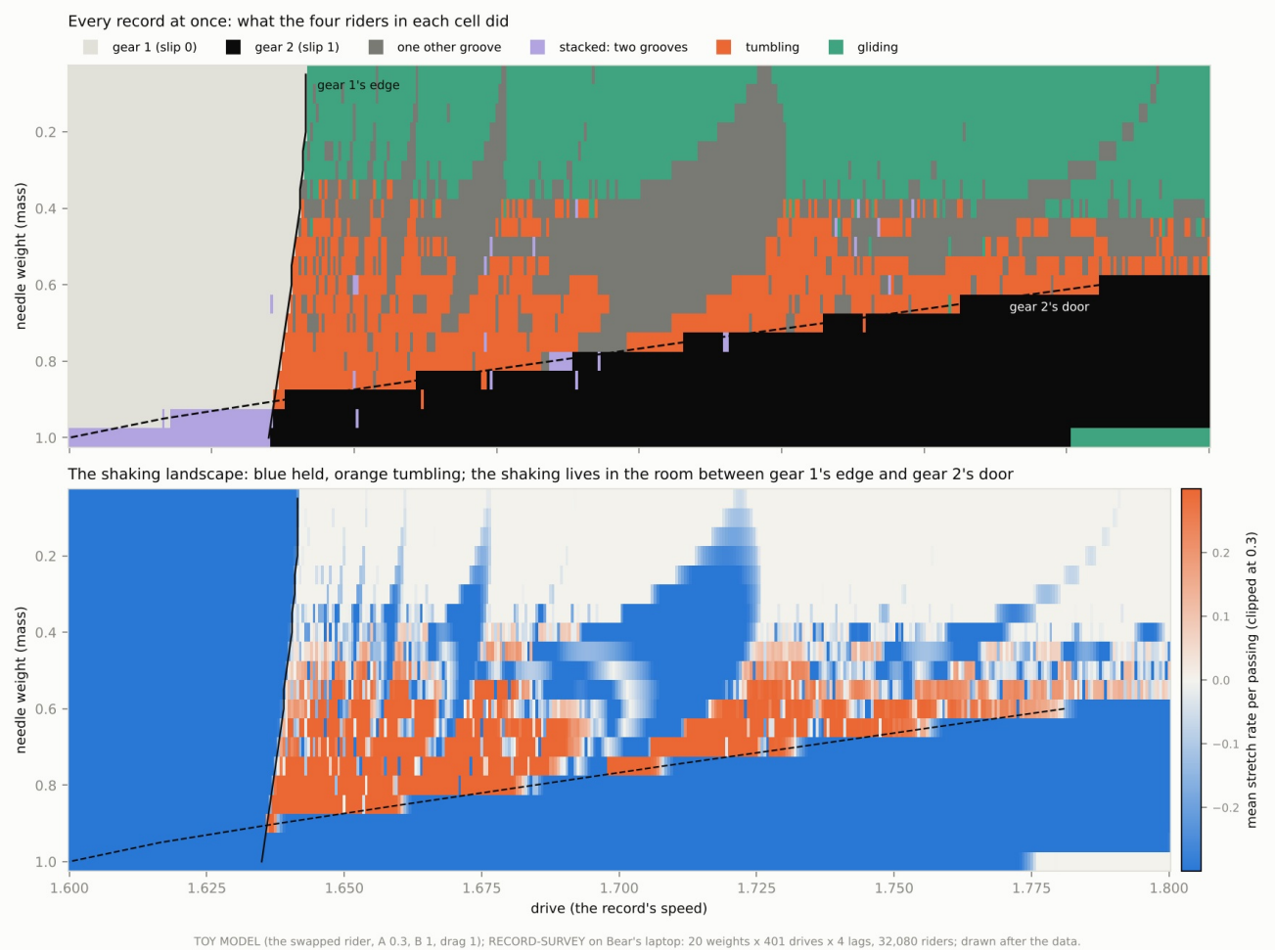


Figure 19. Every record at once: the shaking lives in the room between gear 1’s edge and gear 2’s door.

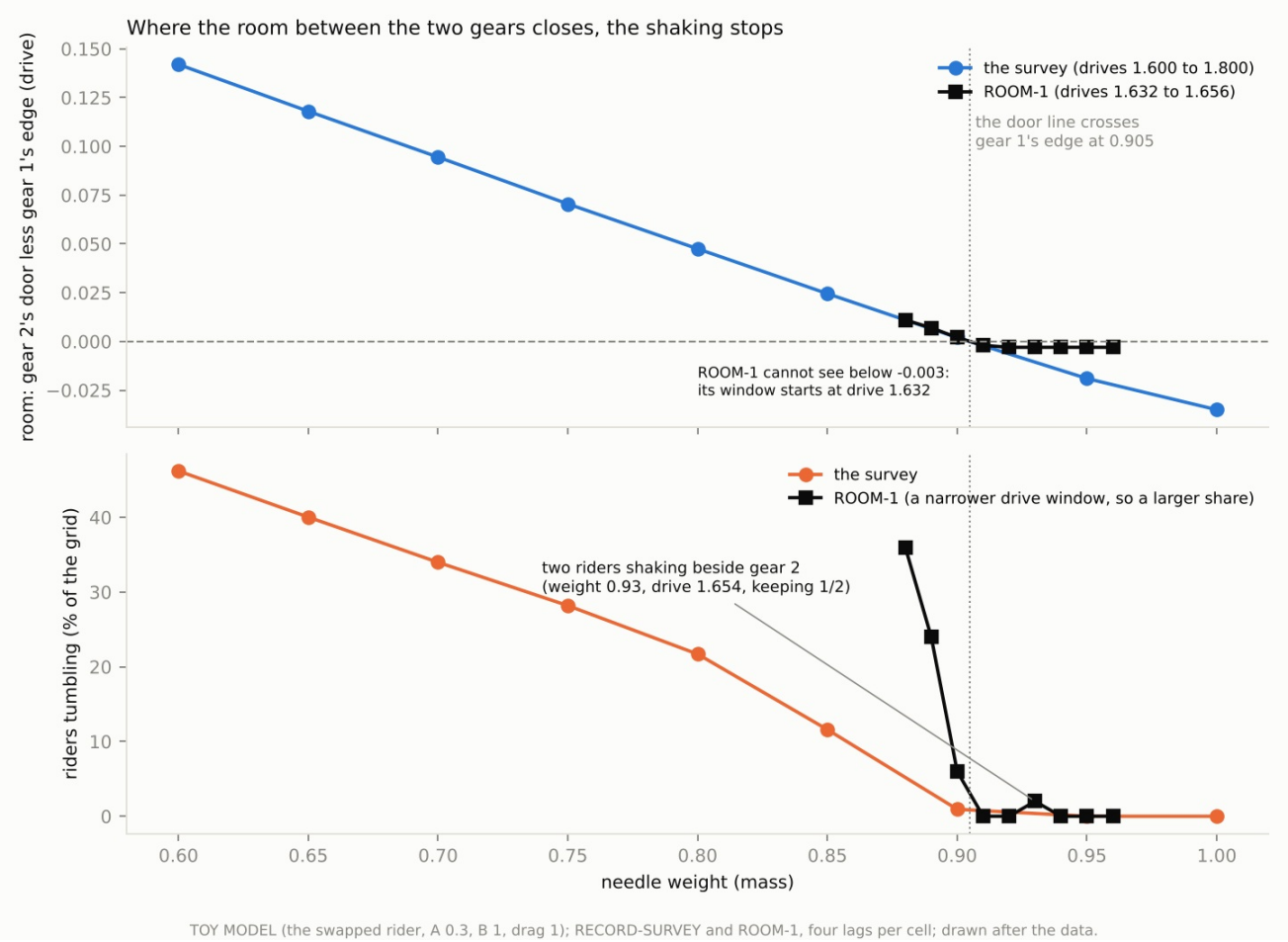


Figure 20. Where the room closes, the shaking stops.

4.15 The half groove’s ribbon

Beside gear 2, where the room is closed, a narrow ribbon of the 1/2 groove runs at every weight from 0.85 to 0.95. Going down the drive it plays period 2, then 4, then 8, then shaking that still keeps exactly half a turn per passing; held 20,000 passings, the count stays within a millionth of 1/2 (HOLD-1). The ribbon is the half groove’s own doubling ladder, followed in weight without a break out of the room (figure 21), as gear 2’s door slides past it at weight 0.80.

To an engineer’s eye the ribbon is a strange thing: the motion is chaotic, yet its count is locked. Over a long run the rider makes exactly half a turn per passing, while no two passings look the same. A watch can do this, keeping perfect time over a day while each tick wobbles; so can a phase-locked loop that holds its frequency exactly while its phase jitters. In the field’s words the rotation number stays locked while the orbit is chaotic.

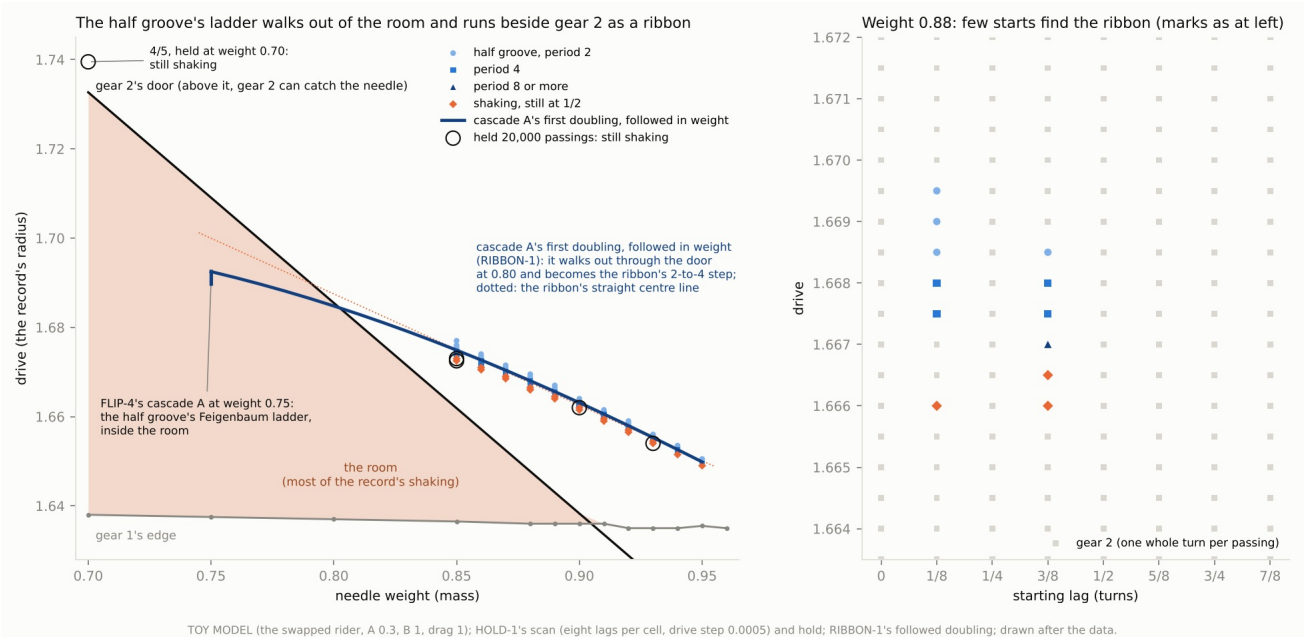


Figure 21. The half groove’s ladder walks out of the room and runs beside gear 2 as a ribbon.

4.16 The half-turn fold

A dance doubles when its multiplier reaches −1 (Bear’s half-turn fold), and the doubled dance is born at +1. The ladder of doublings shrinks onto Feigenbaum’s ratio: span ratios 4.46, 4.62 and 4.66 on the 1/2 groove at weight 0.75. On the 3/7 groove at weight 0.4 they climb toward it more slowly at first: 2.41, 3.14 and 3.94. Followed down to where the 896-passing dance itself doubles, eight doublings in all (LADDER-B-UP, on Bear’s laptop, registered before the run), they go on to 4.45, 4.62 and 4.66. Near the end the slow ladder closes on Feigenbaum’s ratio just as the fast one does, each gap to 4.669 about a quarter of the one before: 0.218, 0.050 and 0.012 on the 3/7 groove, beside 0.210, 0.054 and 0.012 on the 1/2 groove. The last gap is good to about 0.005, because the two deepest doublings are read to about 5e-11 of drive. The two ladders start differently and finish the same way.

In the field this is the flip bifurcation (figure 22), and the multiplier is the Floquet multiplier: the factor by which a small nudge grows or shrinks over one lap of the dance. At −1 a nudge comes back flipped and the same size, and the dance splits into one that takes two laps to repeat. Each doubling comes sooner than the last, and the gaps shrink by a ratio that is the same in many systems that double this way, from convecting fluids to driven circuits: Feigenbaum’s 4.669 (a driven circuit with a varactor diode agreed with its scaling and its rate of convergence, Linsay, 1981; checked against the abstract). Read by an engineer, a ladder like this is a warning light. Once two doublings have been seen, their spacing and Feigenbaum’s ratio predict where the shaking will begin, before the machine gets there. On the 3/7 groove that point is at drive 1.68830273, 2.6e-8 below the last doubling found (a reading, from the last gap and Feigenbaum’s ratio).

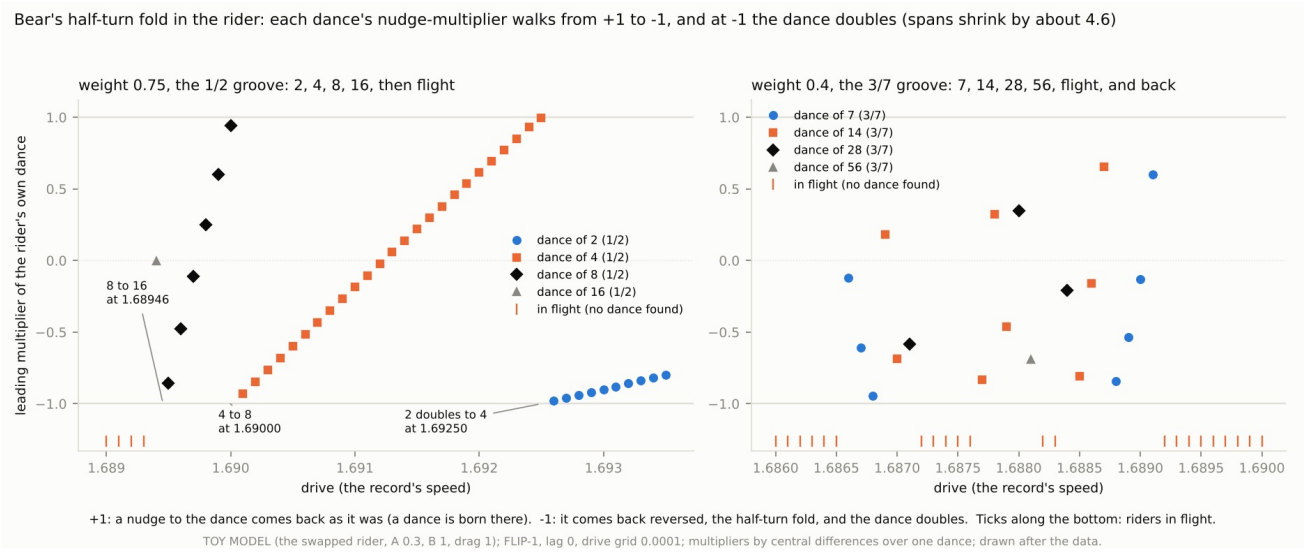


Figure 22. The half-turn fold: each dance's multiplier walks from +1 to −1, and the dance doubles.

4.17 Bear's doors

Where gear 1 and gear 2 overlap, both doors are open, and which one a rider takes depends on where it starts. The seam between the two basins is fractal (figure 23): its uncertainty exponent is 0.38 at drive 1.605 (BASIN-1, weight 1.00, the record's rider of §4.12). The uncertainty exponent says how much aiming a start more finely helps. For a smooth border it is 1: place a start ten times more finely and the chance of not knowing its door falls tenfold. At 0.38, ten times finer cuts that chance only about 2.4 times, the mark of a border crinkled at every scale.

A basin is the set of starts that end in one place, as every hillside in a river's basin drains to that one river. Here the line between two basins is not a clean ridge but a coastline, crinkled at every scale. The uncertainty exponent says how much better you must know the start to call the outcome. At 0.38, halving the share of starts you cannot call needs a start known about 6.2 times more finely. The field calls this final-state sensitivity (Grebogi, McDonald, Ott and Yorke, 1983; not read). To an engineer it means that near the seam no realistic tolerance on the starting state lets you promise which door the rider takes. Design away from the seam, or start the rider well inside one basin.

Bear's coin and its fins. At weight 0.40, with the ripple turned to run with the rider (the turned walkway), there is a band of drives, 1.434 to 1.443, where a rider lands on one of two faces depending only on where it starts: heads, the conveyor (§4.19), or tails, the 1/2 groove. Bear calls it the coin. Across the band the chance of heads rises in a straight line with the drive, and the edge between the faces is fractal, with an uncertainty exponent of 0.592 at drive 1.4380 (COIN-1, COIN-2, FRACTAL-1). Run again with each start's settling kept (SETTLE-1), the same 4,000 starts land exactly as before, and the seam decides fast: every start settles within 18 passings, half within 8, and the slowest sit on the crinkles, with a neighbour 1/40000 away on the other face for a fifth of the 200 slowest against one start in eighty overall. The seam does not hold riders; it only makes it impossible to say beforehand which way each one goes. The stretching while riders are still choosing is real, a positive 0.19 per passing for the 400 slowest, so something near the seam pulls neighbours apart; but this run cannot say whether it is chaos, since too few starts linger to measure how fast they leave (Part 8). The stretching is fastest before that: over the first five passings, which the registered rule left out, the same 400 stretch by 0.84 per passing on average, a factor of 67 in five passings (read afterwards from the same files, not registered). A second engine written from scratch (WITNESS-COIN-1), sharing only the equation, flips the coin the same way: with an adaptive eighth-order stepper in place of the fixed one, the face read from the slip instead of the strobe's period, and other settle and watch lengths, it put every one of the coin's 1,800 riders of record on the same face. On the map of start against drive Bear drew shark fins, and they are real. The main heads arc grows downward as the drive rises, its lower edge sliding from lag 0.632 to 0.272 across the band, and a thin heads stripe runs beside that edge like a shadow. Written down before the run, the stripe stayed 0.09 to 0.12 turn below the edge at every one of the 31 drives judged, read at 1,000 starts each, and a second, thinner stripe sat between them at every one (COIN-3). Each edge carries smaller copies of itself, which is what a fractal edge looks like when it is read one scale at a time. A small fin Bear drew near lag 0.79, which seemed to end at drive 1.4365, does not end: it is a stripe two to four thousandths of a turn wide that slides upward past the hundredths a coarser map sampled (lesson 14). At the top of the band, read on a dial 0.00002 of drive apart (COIN-4), the share of heads does not climb smoothly: it zigzags, up and down by as much as 0.04 between neighbouring drives. The small fin never joins the main arc. Tails is not eaten away; it ends all at once: at drive 1.44284 the last 52 tails starts turn heads together. Whether that was the 1/2 groove itself ending was then tested by carrying a tails rider up the drive on its own orbit (COIN-END). The lean, that the groove outlives the coin and only its catchment leaves the starts from rest,

missed, and its rival held: the carried rider leaves the groove for the conveyor between 1.44282 and 1.44284, its stretch rate climbing toward zero on the way, as at a fold. Read on a dial ten times finer, the fold sits between 1.442834 and 1.442836 (COIN-FOLD, a reading: its check was set against the wrong reference and failed).

The coin's third face. Just above the coin a few starts land on neither face: at drive 1.444, 16 of 100 starts dance the 2/3 groove, two turns lost in every three passings, danced twice over (COIN-2). Read on finer dials (THIRD-FACE-1, THIRD-FACE-2), the face is a window between drives 1.44346 and 1.44418 that opens and closes by the same road. Going up the drive: a lone 6/9 at 1.44346, shaking, the doubled form 4/6 at 1.44356, then 2/3 from 1.44358; then 4/6 again from 1.443990, 8/12 from 1.444100, 16/24 at 1.444115, and shaking from 1.444120 until every start lands heads from 1.444180. Written down before the runs, the closing cascade came in order, 3, 6, 12, 24, each stage one unbroken band of drives, and the window had a chaos of its own before it closed; the guess that it would open out of quiet water missed, since it opens out of shaking. The first ratio of band widths, 7.33, missed its registered band of 3 to 7, but the dial's step allowed anything from 5.25 to 11.50, so the miss decides nothing (lesson 1). Read later on dials up to ten times finer, with a settle three times as long (THIRD-FACE-3), the first doubling moved up: on the old dial a 2/3 still alternating as it settled had been read as 4/6. The 3 to 6 doubling sits between drives 1.4439900 and 1.4439925, where the 2/3 riders' stretch rate falls to zero, and the first ratio comes to 6.18, inside the 6.0 to 9.0 written down first (TF3-1b; the first finer dial stopped 0.0000025 short of the doubling and could not place it, a void). The next ratio, the 12 band over the 24 band, is about 4.0. The riders shaking on the third face all keep two turns in three passings, within 0.0004 of 2/3, and the dances that flicker inside that chaos (6/9, 24/36, 14/21) are 2/3 as well, so the face's chaos keeps a count (lesson 121). And where its 2/3 splits into 4/6, each stop splits the way Bear marked it on the picture: the lower stop into a near-level branch and a falling one, the upper into a near-equal pair, the middle one flat (read after the data). A window bounded by doubling at both ends is known in kind, as a remerging Feigenbaum tree (Bier and Bountis, 1984); the coin's tails opens the same way, through 2/4 below 1/2 (lesson 122). And the window does open by the same road it closes. On the finer dial, going up out of the shaking come 32/48, 16/24, 8/12 and 4/6, and then 2/3: a cascade run backwards, as written down first (THIRD-FACE-3). Below it, inside the shaking, sit the tripled dance 6/9 and its own doubling, 12/18. Why the first ratio sits so far above 4.669 while the next is near 4 is open (Part 8).

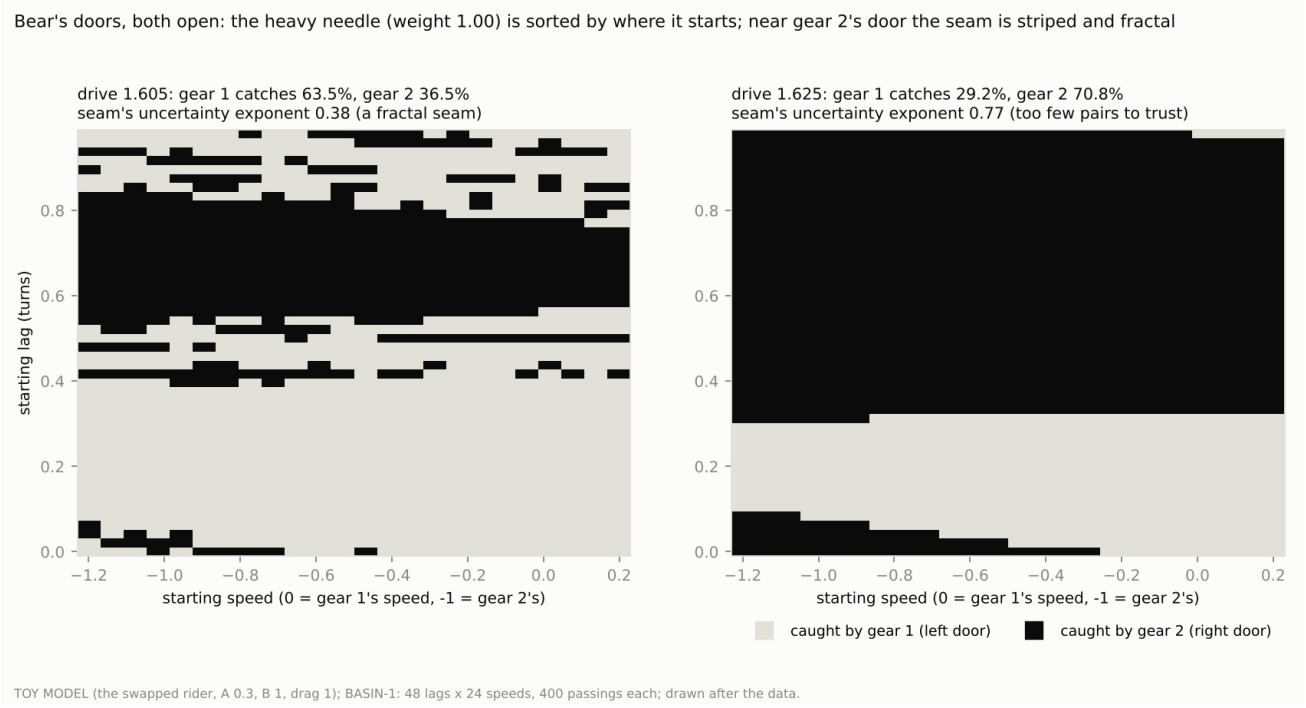


Figure 23. Bear's doors: the heavy needle sorted by where it starts; near gear 2's door the seam is striped and fractal.

4C. The atlas: the map in Bear's words

Part 4C is the fair's parking lot, where the skaters ride, drawn as a map. The half pipe, rocked in and never gone over, is the plain; the full pipe, ridden all the way round, is a fin, a whole-number slip; and the shuvit's half turn pictures the 1/2 groove of §4.19 (built as a toy, the shuvit became a commutator whose gears count whole turns, §4.10). Here those places get their coordinates, each named in Bear's words beside its classical match.

4.18 The landmarks, each beside its classical match

The pad atlas maps the whole plane of clock ratio and pull at weights from 0.05 to 10, the ripple against the ball. Each landmark is told in Bear’s words, then set beside the classical result it matches.

An engineer would call this map an operating map, the same kind of chart a pump or compressor maker prints: two knobs, one across and one up, and on the chart the regions where the machine does one thing or another, with their edges drawn. Physicists call it a phase diagram, or a bifurcation diagram. Bear named the landmarks in his own words, the way a hiker names a ridge by its shape, and the right-hand column sets each name beside the result in the literature it matches.

Bear’s word	what it is	the landmark it matches
the plain and its edge	rest, and the least pull at which the ball leaves it; across weights 0.40 to 0.74 the edge falls on a straight line	static pinning, the edge raised slightly by the ripple
fins	whole-number locks, 1, 2 and 3 ledges per tick at weight 2; their swept-back feet run along the edge	Shapiro-like steps
seams	the whole-number lines of the ball’s own pace	the same resonance condition, read across the plane
ramps	the level lines of one smooth field (ledges per tick), widened by locking; most numerous near weight 0.40	Arnold tongues and the devil’s staircase
seeds	the fractions that hug the edge; they slide toward the slower tick as the weight drops	the onset of rotation
the trickle	at weight 0.40, 1/n locks from 1/10 to 1/48; $n^2 \times \text{height} = 0.4895$ within 4% (TRICKLE-1b), the height being how far above the edge’s pull the 1/n lock first appears; two missing steps found blind exactly where predicted	the saddle-node ghost (square-root bottleneck) near a tongue’s edge
memory	from weight 0.70 up, a rolling ball keeps rolling below the pull where a still one can start; the gap grows to 0.61 at weight 10	the pendulum’s bistability (Stewart; McCumber); the switch at weight 0.7026
the weight makes the fractions	with no weight, both forcings lock on whole numbers only; at weight 0.05 an added drive locks on fractions as the moving ripple does	the overdamped junction, whose locks exist only at whole numbers
the ripple’s balance law	a fast ripple lifts the plain’s edge (as its strength squared), a slow one lowers it; the two balance at ripple speed 1/2 at every weight; the law predicted 45 to 47 of 51 columns blind (THEORY-1)	averaging of a fast force (Kapitza’s pendulum, 1951)
the fold facing the wave	the once-per-passing map folds at weight 0.33 to 0.40; only on folded pulls does the ball tumble	the circle map’s critical line
the stumble	with the ripple on, the edge’s crawl fades from weight 0.40 to 0.60 and collapses between 0.65 and 0.68	chaos in the rf-biased junction (Kautz and Monaco, 1985)
the fractions lose their room	the fractions live in a band just above the edge that closes from weight 0.60 on and is gone by 3.00	no named match found
the heavy ball’s plain	at weights 3 and 5 the memory floor equals the edge in every column and sits near the bare ball’s launch pull (the least pull that sets a ball rolling from rest half a turn along its ledge, with the ripple off)	the pendulum’s static pull, the ripple only delaying it

Two of the landmarks carry straight to hardware. Memory, the rolling ball that keeps rolling below the pull where a still ball can start, is the hysteresis of an underdamped Josephson junction: its current–voltage curve has two branches, and which one it sits on depends on where it came from. A two-valued curve like that is a one-bit memory, and early superconducting logic was built on junctions that latch this way (IBM's Josephson computer program of the 1970s and early 1980s; not read). The ripple's balance law is a plain cousin of Kapitza's pendulum, where a fast shake of the pivot holds a pendulum upside down: a fast force, averaged over its shaking, acts like a steady one that grows as the square of its strength.

The plain's edge under the ripple. The table's first row, the plain and its edge, holds a law of its own. In §2.4's figure 3 the estimate for the turned ripple's edge, written down before the run, came early, by 0.064 at weight 0.40, and missed (KN3-3). The runs that followed found why, and more besides.

The ripple's strength, at seven weights. Why did that estimate miss? The edge was found again later, exactly this time rather than on a grid: the drive at which the resting motion stops existing, a fold (§4.8), for seven weights from 0.05 to 5.00, seven ripple strengths from 0.05 to 0.60, and both directions (FOLD-DIR). Three things were written down first, and all three held. First, the ripple moves the edge by its strength squared, with a coefficient worked by hand from two effects: the shaking costs the ball some grip, the way a ball rocked about the top of a hill averages a little less than the top, and the ripple's push, in step with the drag, adds grip or takes it away depending on which way the ripple runs. At strength 0.05 that coefficient matched the exact edge within 0.19% at all fourteen settings. Second, what it leaves over grows as the strength to the fourth power: halving the strength shrank the remainder 14.0 to 16.6 times, where a fourth power gives 16. Third, the shaking is the same whichever way the ripple runs, and only the push turns over, so the weight drops out of the ratio of the two shifts. The ripple running against the ball raises the edge 0.1056 times as far as the turned ripple lowers it, at every weight from 0.05 to 5.00 (measured 0.10551 to 0.10584); on the golden ray that ratio is a golden number, $(\sqrt{5} - 2)/\sqrt{5}$. Put the other way round, the turned ripple moves the edge $\varphi^2 + \varphi^4 = 9.472$ times as far, φ being the golden ratio, 1.618 (figure 24). And the miss is explained. The estimate's formula was right, but it had been worked out at the shifted drive, where the turned ripple passes more slowly and shakes the ball harder. Worked at the unshifted drive, the same formula comes within 0.3% of the turned edge at weight 0.40, where the estimate had been 33% too far, and within 4.2% at every weight. In every row the exact method finished, a heavier ball's edge moves less; with the ripple turned around at the biggest strengths it lost its way at seven settings. Walking the strength up from a finished neighbour found three of them (weight 1.00 at strength 0.45, weights 2.00 and 5.00 at 0.60) and lost the other four, all with weights of 1 or less at strength 0.60 (FOLD-DIR-b). Mapping the whole branch lost them too (FOLD-DIR-c), so the edge was then read directly, the way the grid of §2.4's figure 3 reads it: a ball set down at rest, drive by drive, either stays on the plain or slips off (FOLD-DIR-d). Two things came out. At strength 0.60 all five weights of 1 or less, from 0.05 to 1.00, slip off at the same drive, between 0.88 and 0.89, where the shift worked to second order would have spread them widely; for the heavier balls, 2 and 5, the fold lies far higher, at 1.21 and 1.53, though their edges were not read this way. And at strength 0.45 a ball set down at rest slips off before the fold, by 0.05 of drive at weight 0.75 and 0.11 at weight 1: under a strong turned ripple the drive where the resting dance stops existing and the drive where a resting ball stops staying are no longer the same.

Why the ball slips off early. Read finer, by halving twelve times the gap between a drive where the ball stays on the plain and one where it slips off (FOLD-DIR-e), the five light weights at strength 0.60 leave the plain within 0.0023 of one another, between drives 0.8825 and 0.8848. The cause of the early slip at strength 0.45 came from raising the drive slowly under a ball already resting: its dance doubles, repeating every second passing instead of every one, first at drive 1.165 for weight 0.75 and 1.145 for weight 1, and a ball set down at rest slips off at 1.166 and 1.147, the same drives. The dance loses its rhythm before it loses its footing, and once it has doubled the ball is gone within a step. With that, the weight's part turns round. Under a gentle ripple (strength 0.30) a heavier ball's edge moves less, and the edge from rest sits on the fold within 0.0015 up to weight 1. Under a strong turned ripple a heavier ball slips off sooner: at strength 0.60 weight 2 leaves at 0.860 and weight 5 at 0.713, far below their folds at 1.21 and 1.53. For an engineer the knob reads the other way round under a strong turned ripple: weight does not hold a ball on the plain, it shakes it off.

Which door the ball leaves by. Raising the drive slowly under the heavy balls too (FX-2b) showed that the plain has more than one way out. At strength 0.45 the heavy balls leave the way the middling ones do: weight 2's dance doubles at drive 1.041 and the ball is gone at 1.046, weight 5's doubles and goes at 0.780. At strength 0.60 no ball doubles. The light balls leave where their dance stops existing, a fold, wherever that fold was found; at strength 0.60 it was not found for weights of 1 or less (FOLD-DIR-b and -c, above). The heavy balls leave by a fold too, but not of the dance walked up from a gentler ripple: the dance a ball set down at rest falls into ends at drive 0.860 for weight 2 and 0.713 for weight 5 (its multiplier climbing to +1 there), while the walked dance goes on existing to 1.21 and 1.53. The edge from rest is the fold of the dance a ball at rest falls into, within 0.0001: registered before the run, the fold came out at 0.859975 for weight 2 and 0.713061 for weight 5, against edges from rest of 0.85989 and 0.71310 (FX-2c). The walked dance is no second place to rest for weight 2, or only just. A ball set down a hundredth of a turn beside it fell off the plain at every drive tried from 0.90 to 1.20, against the lean written first (FX-2c, a miss), and a look after the data said why. A dance has two multipliers, and a nudge grows if either is larger than 1 in size; FX-2c's program had

kept only the larger of the two by sign. The drag fixes their product at e^(−T/m), e raised to minus the time of one passing over the weight, so the tiny one, under 0.0002 in size, has a partner of −7.8 at drive 1.20 and −69 at 0.90. The walked dance holds a ball only from its fold at 1.21082 down to 1.21040, where it doubles, and just below that the doubled dance holds a ball too (FX-2d). Weight 5's walked dance holds a ball far longer than the lean written first, that it would double within 0.01 of its fold: it rests from its fold at 1.528 all the way down to 1.465 (FX-2d, a miss). Below its doubling it hands the ball on: written first and held, a ball set beside it 0.001 and 0.005 of drive below the doubling settles into the doubled dance and stays (FX-2e), its two points 0.085 and 0.189 apart, growing as the square root of the distance, as a doubling born gently does. A guess that the width from doubling to fold follows the drag's product alone missed: at the gentler strength 0.525, weight 2's width came out 0.0019, 2.4 times the guess, and weight 5's 0.084 (FX-2e, a miss), so the ripple's strength sets the width as well as the weight. Read from how fast the dance's multipliers fall just below its fold, the width comes out within 10% and 23% at weight 2, where it is tiny, but only half the true width at weight 5, where it is a hundred times wider (FX-2f, a miss); there the width is measured, not foretold. Both walked dances sit far above the drives where a ball set down at rest leaves, 0.860 and 0.713. The drag's fixed product is also a free check, for an engineer, on any pair of multipliers: worked by one-sided differences the pair at drive 0.90 multiplied out 25% low, by central differences within 0.01% (FX-2d, written first). For an engineer there are three doors out of the plain, the fold, the doubling and the fold of another dance, and the door is worth naming before the number is trusted (Part 8).

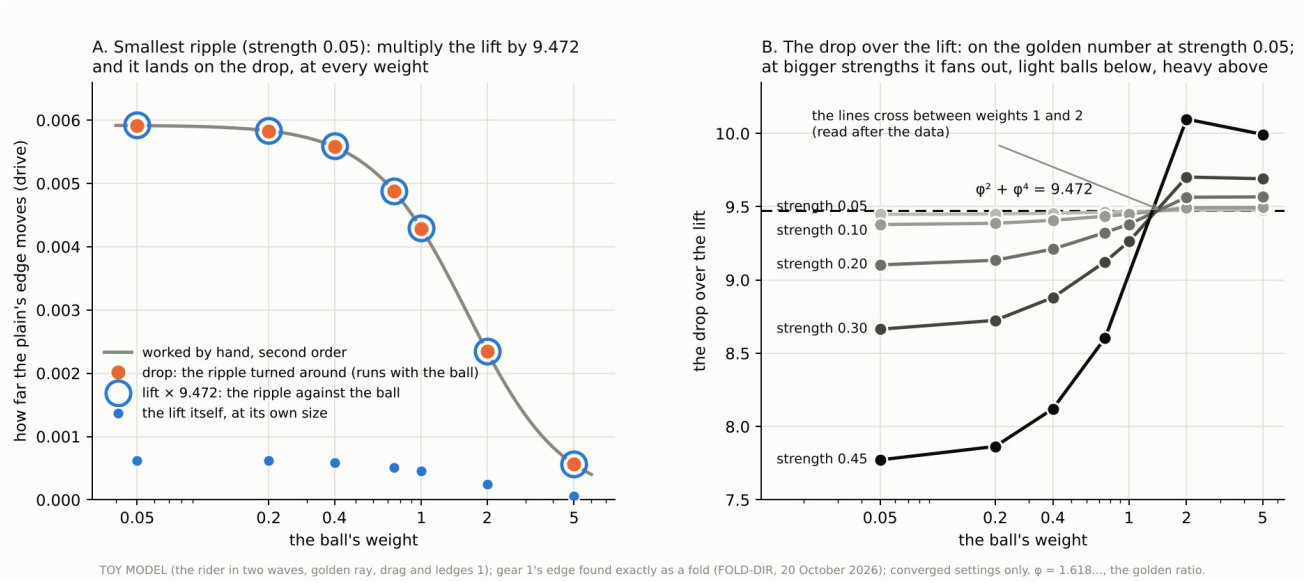


Figure 24. The ripple's two directions: turned around, it moves the plain's edge $\varphi^2 + \varphi^4 = 9.472$ times as far as forward, at every weight.

Reading the figure. In the left panel each dot is the exact edge for one weight at the smallest strength, 0.05. The orange dots are how far the edge drops when the ripple runs with the ball; the small blue dots near the bottom are how far it rises when the ripple runs against it, about a tenth as far. Multiply each blue dot by 9.472 and it becomes a blue ring, and every ring sits on its orange dot; the grey line is the shift worked out by hand. To the right the dots fall: a heavy ball shakes less, so its edge moves less. But the shaking enters both directions the same way, so it cancels from their ratio. The number itself comes from two sums. The shaking's loss is a quarter of the ledges' strength whichever way the ripple runs, and the push is half the ripple's passing speed, which on the golden ray is half of $1/\varphi$; against the ball the push nearly cancels the loss, leaving $(\sqrt{5} - 2)/4$, and with the ball the two add, to $\sqrt{5}/4$. The right panel shows the ratio at bigger strengths, where it fans out: light balls fall below the golden number and heavy ones rise above it. Seen first in this picture, and then registered and run at four more weights (FOLD-DIR-b, the run that also walked the strength up, above), the lines cross the golden number near weight 1.43 at every strength from 0.05 to 0.30 (1.456, 1.450, 1.427 and 1.397), not at φ. The crossing slides a little lighter as the ripple grows (a reading after the data, each crossing placed by a straight line between the weights run), so it is a narrow waist rather than a single point. It is the same kind of point as the pin of §4.26, where every weight's dance crosses at the weightless root: in both, a family of curves, one for each setting of a knob, crosses where the knob's first effect vanishes, on the curve the knob-free toy would draw, and in both the crossing is a narrow region whose width the next effect sets. Chemists call such a point isosbestic, and correlated-electron physics knows it from specific-heat curves that cross near one temperature (Vollhardt, 1997; Greger, Kollar and Vollhardt). The engineer's use of it is the zero-temperature-coefficient point of a transistor: bias there and a knob that drifts does nothing, to first order. The two numbers, the pin's drive 1.436 and the waist's weight near 1.43, agree only by chance; they belong to different knobs.

Fins and ramps, told apart. Figures 25 and 26 show one family at two weights. A ramp is a level line of one smooth field, how many ledges the falling ball drops per tick, and wherever the ball locks the line widens into an orange band. At weight 0.40 the ball locks on 133 different fractions, so the map is full of ramps. At weight 2 it locks on only 25, and most of its locked squares, 235 of 282, sit on the whole numbers 1, 2 and 3 ledges per tick (the pad atlas, run 5, read at lag 1/2): those whole-number ramps are the fins. The fins figure draws them in the same orange, and reads three rows across them to show the staircase they make.

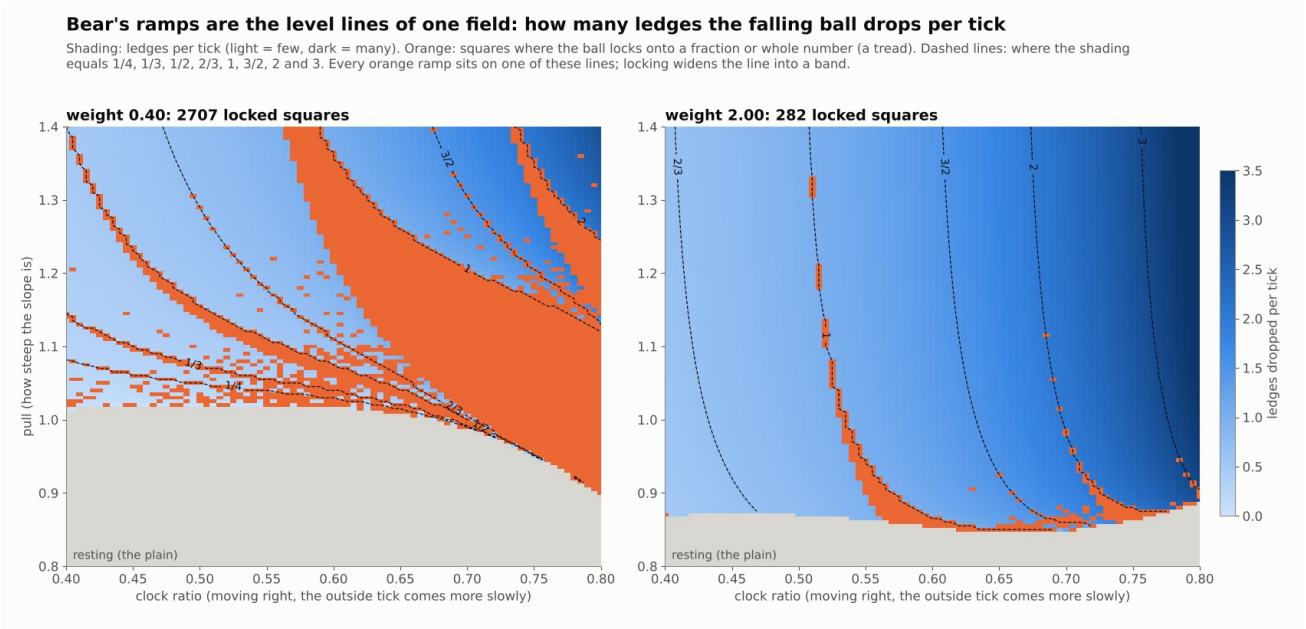


Figure 25. Bear's ramps are the level lines of one field: how many ledges the falling ball drops per tick.

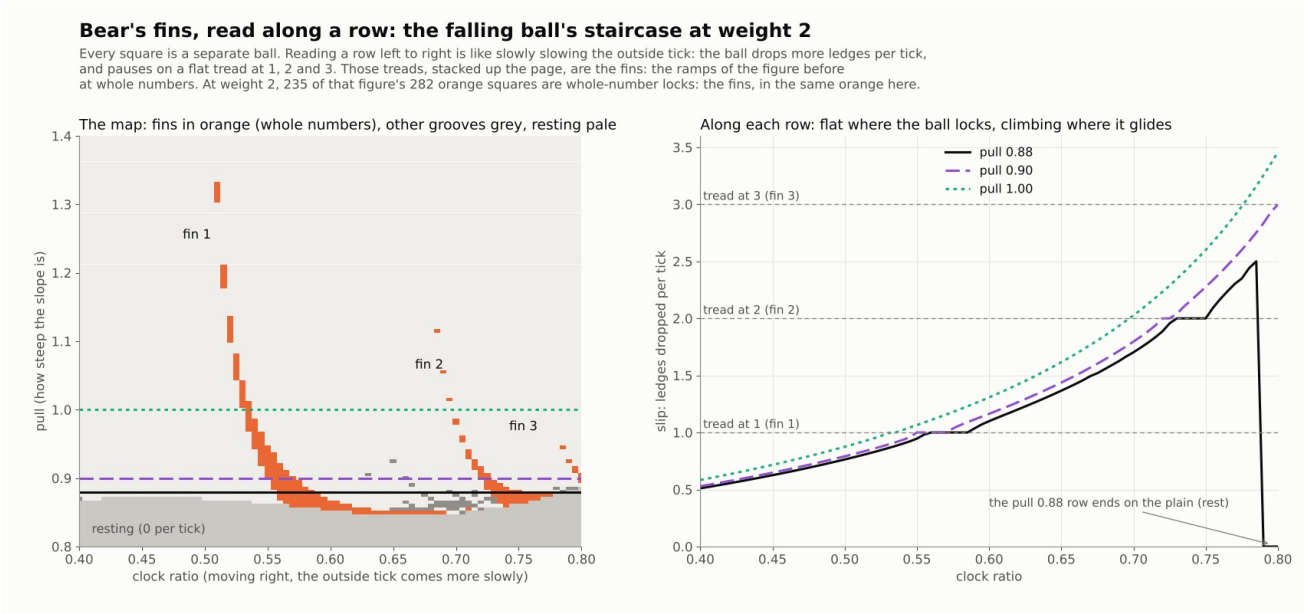


Figure 26. Bear's fins: the ramps at whole numbers, read along a row, the falling ball's staircase at weight 2.

4.19 The S, the foothold and the wrap (the turned walkway)

With the ripple turned to run with the ball, the 1/2 groove shows Bear's S: its lower stop plunges, strolls and falls. The stroll, the foothold, sits at lag 0.143 to 0.146 near drive 1.435 at every weight from 0.10 to 0.40, a fourfold change of weight. After the groove dies, other grooves' stops stroll on the same foothold (the second S). Demagnified, every weight's 1/2 groove traces nearly the same two stops, and its stops stay 0.713 to 0.730 of a turn apart (§4.24 says

why). At weights 0.24 to 0.34 the S's curve wraps once around the lag, through lag 0, and comes down into the birth ring of the conveyor (the 1/1 lock, where the ball surfs one ledge for every passing, as at the fair), where the upper stop's curve arrives straight.

In the words of §6.11, the foothold (figure 28) is the book's first weight-proof landmark, found by eye long before that section went looking for one: a fourfold change of weight moves it by no more than 0.003 in lag (read from the explorer's weight maps after SECOND-S-2). A climber's foothold is the same idea, a ledge that holds whatever the climber carries. The wrap is a winding (figure 27). The S's curve goes once around the lag, a whole turn, and a whole turn is a count, not a measurement: it cannot drift by a little. It is either there or it is not.

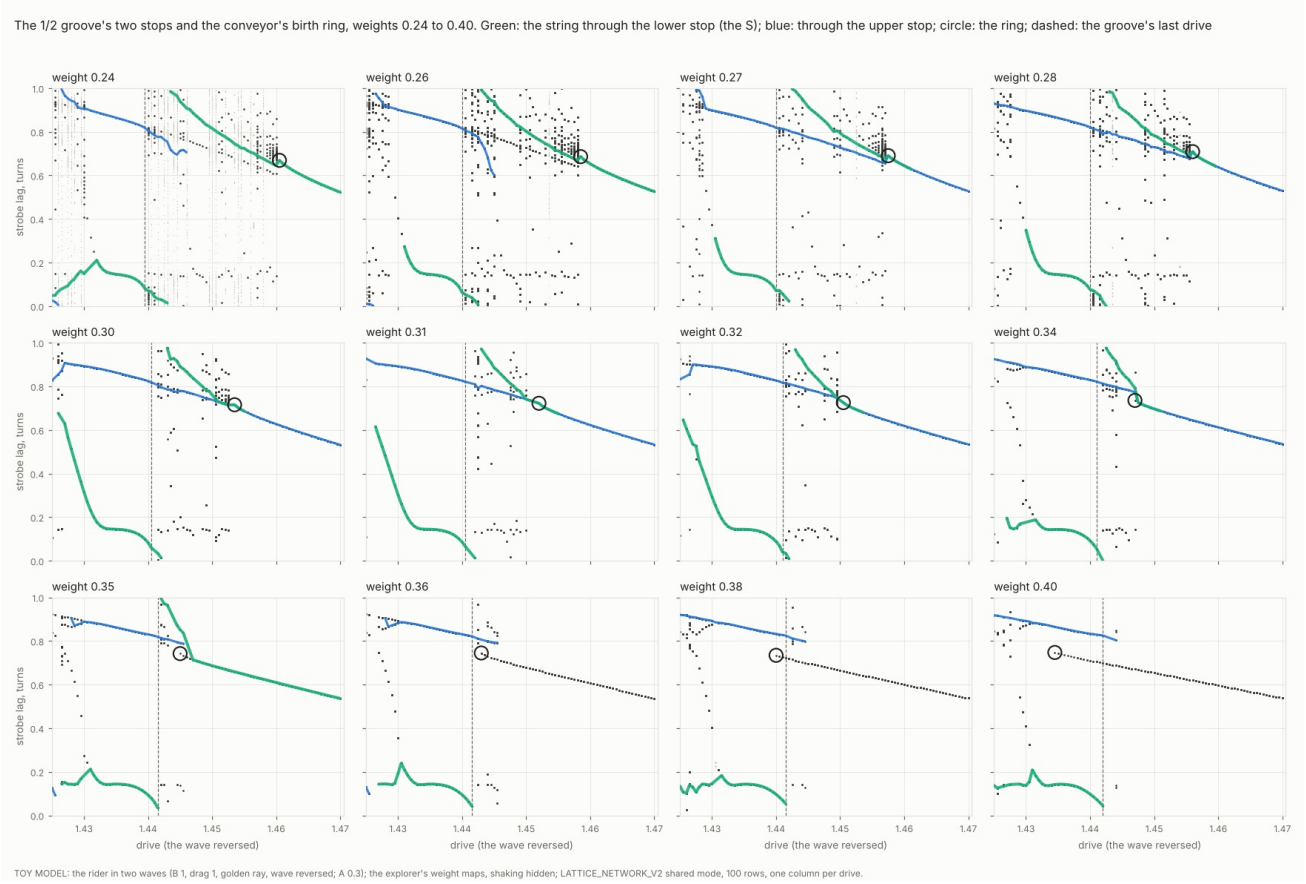


Figure 27. The wrap across weights: the S's string and the upper stop's string meeting at the ring.

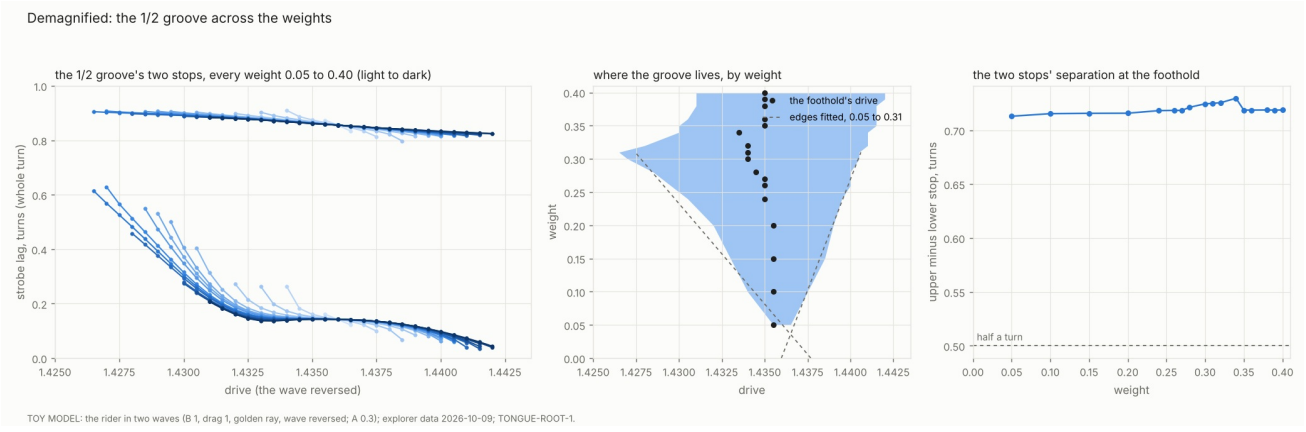


Figure 28. The foothold and the master curve: the 1/2 groove's two stops at many weights.

4D. The dancers

4.20 Where the order lives

Part 4D is the evening show in the dance hall, and it opens on a clue the atlas of Part 4C left behind. Beside the S's foothold the atlas found a pin. Read on a dial ten times finer, every weight's lower stop crosses at one and the same drive, 1.4360 (PIVOT-1), where the stops of all the weights lie within 0.0003 of a turn of one another: make the rider heavier or lighter, and that crossing stays put. Bear's saying, "bilateral symmetry is another stalking point", found what that drive is. It is where the 1/2 dance's two stops are mirror images of each other about lag 0 (the mirror drive, 1.43571 to 1.43596). In an engineer's words, the pin is an operating point that holds still while another knob, here the weight, is turned. The rest of Part 4D, §4.21 to §4.28, explains what the whole family of dances behind the pin obeys, and §4.26 why the pin does not care about weight.

4.21 The frozen balance point

Picture the strobe of Part 2, a light that flashes once per passing of the ripple, set to fire at the moment the ripple lines up with the ledges. In that flash the push on a ball sitting still is (1 + A) cos u − F: the ledges' pull, made stronger by the ripple's share A, less the steady push F. It is zero at two lags, ±arccos(F/1.3)/2π (with A = 0.3). Those are the frozen balance points: the knife edge on one side, where the smallest nudge tips the ball away, and the resting point on the other, where a nudge rolls it back. A ball sitting there is, for that instant, held still. In the field's words they are the two equilibria of the force frozen at that moment, one unstable and one stable: the snapshot a stroboscope takes. The dances of this chapter are read against them (figure 29).

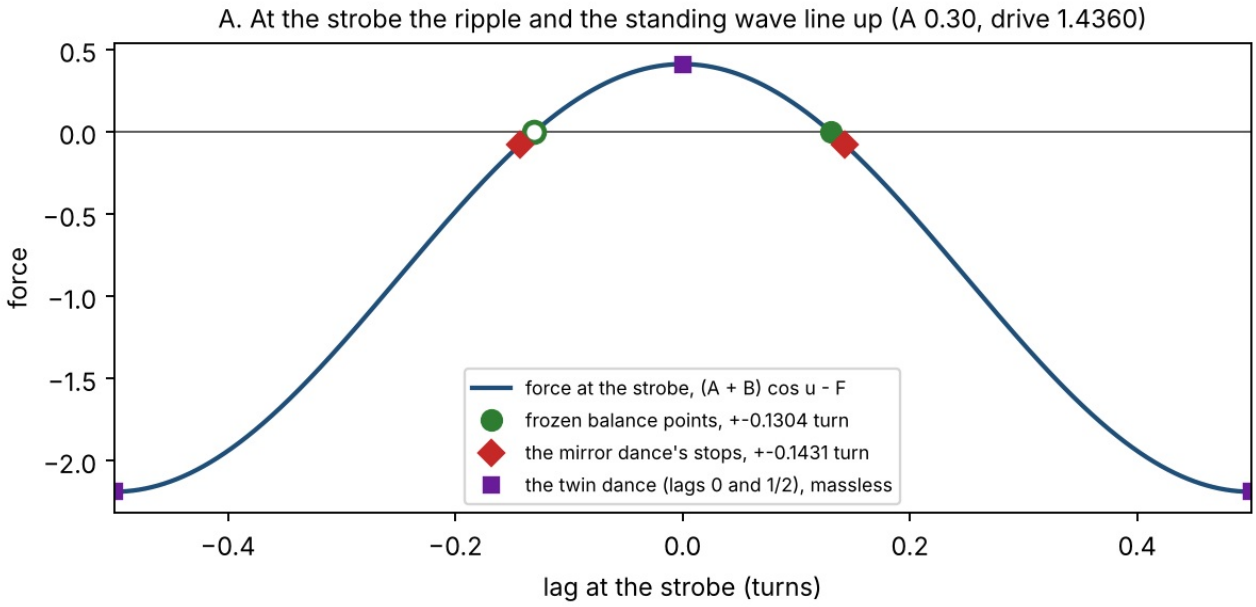


Figure 29. The push at the strobe, and its two frozen balance points.

4.22 At a root every start is a dance, and the mirror leaves two

With no weight, one passing carries the rider's lag through a Möbius map (§3.5), the same kind of map that carries a laser beam's shape through a lens (the ABCD law of optics; Kogelnik and Li, 1966). At special drives, the weightless roots, that map becomes a perfect gear: every start is a p/q dance, visiting q stops, moving on by p of them at each passing, and coming home after q passings. In the field's words, at a root the passing map is periodic, and every orbit is a periodic orbit. Run the film backwards and flip the lag, and the rider looks the same: the equation has a time mirror, which the field calls a reversing symmetry. That time mirror leaves exactly two of the dances symmetric, each its own reflection; the field's name for such a dance is a symmetric periodic orbit. We colour them red and purple (figure 30).

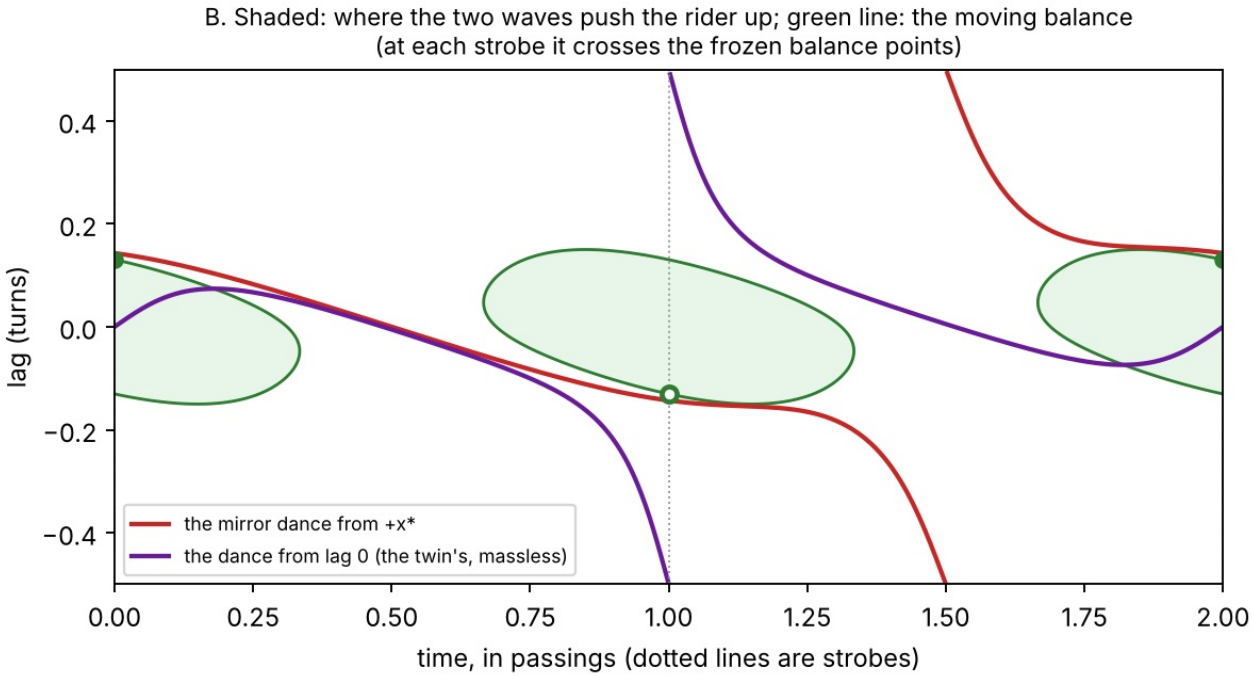


Figure 30. The moving balance line, and the two self-mirrored dances.

4.23 A touch of weight chooses the far-side leaper

Give the rider a little weight and the show narrows to two acts: only the two self-mirrored dances survive, and only one of them can stay. Other dances could survive too, in mirror-image pairs, but none was seen at any of these roots. One is stable and one is not. It is the vine swing over the pit at the afternoon stunt show, where whoever swings across the gap is the one who stays. The stable one, red, is always the dance whose big leap crosses the far side of the circle (lag 1/2) at half-time, the way a swinger is over the middle of the pit at the middle of the swing. That held at all 19 fractions of the toy on file when it was graded, and the newer rows were predicted before they ran. It has held under every single-tone drive run; a second tone strong enough to push a grip through zero breaks it (§4.28). The choice is the sign of the first-order grip M′ (§3.5, piece 5): how strongly a small weight first pulls a dance in or pushes it away. In the field's words it is a first-order perturbation result, the weight's first correction to each dance's stability deciding which one attracts. Most of the difference between the two dances, 69 to 83% wherever it was split (SELECT-1, PIN7-1h, PIN68-1, TONGUE-1), least near the middle of the circle and less there as the denominator grows, comes from one term: added up along the dance, how sharply the push bends across the lag (f_uu, its second rate of change) times how far the weight has already nudged the rider off its weightless path. In symbols it is ∫ f_uu δ; that δ is the nudge, not the excess of §4.28.

The two dances' grips are not equal, and how unequal they are comes in steps, like the gears of Part 4A. Red's M′ over purple's M′ depends only on the smaller of p and q − p. It is −2.80 to −3.02 when that number is 1, −2.34 to −2.52 when it is 2, and −2.18 to −2.22 when it is 3. This was found after the data, then predicted for four new fractions, and it held at all four (PIN68-1r). Its next step, written into the record as a test before 4/9 and 5/9 were run, put the ratio between −1.95 and −2.10 when the smaller number is 4; it came to −2.00 and −1.97 (TONGUE-1).

How wide each tongue opens. A small weight opens each fraction's tongue to a width in proportion to the weight. Per unit of weight, the width follows one rule across all nineteen fractions on file: it shrinks with the denominator as about q to the power −2.39, opens wider for fractions near the middle of the circle (as the sine of π p/q to the power 0.71), and a little wider on the upper side (by e to the power 0.49 p/q). Fitted before any ninths were run, those four numbers fit all nineteen to within 10%, where a power of q alone missed by up to 70%. Written down before the run, the rule placed 4/9 and 5/9 within 3% (0.000897 and 0.000950 per unit weight, against 0.000922 and 0.000974; TONGUE-1); q alone said 0.000716 for both.

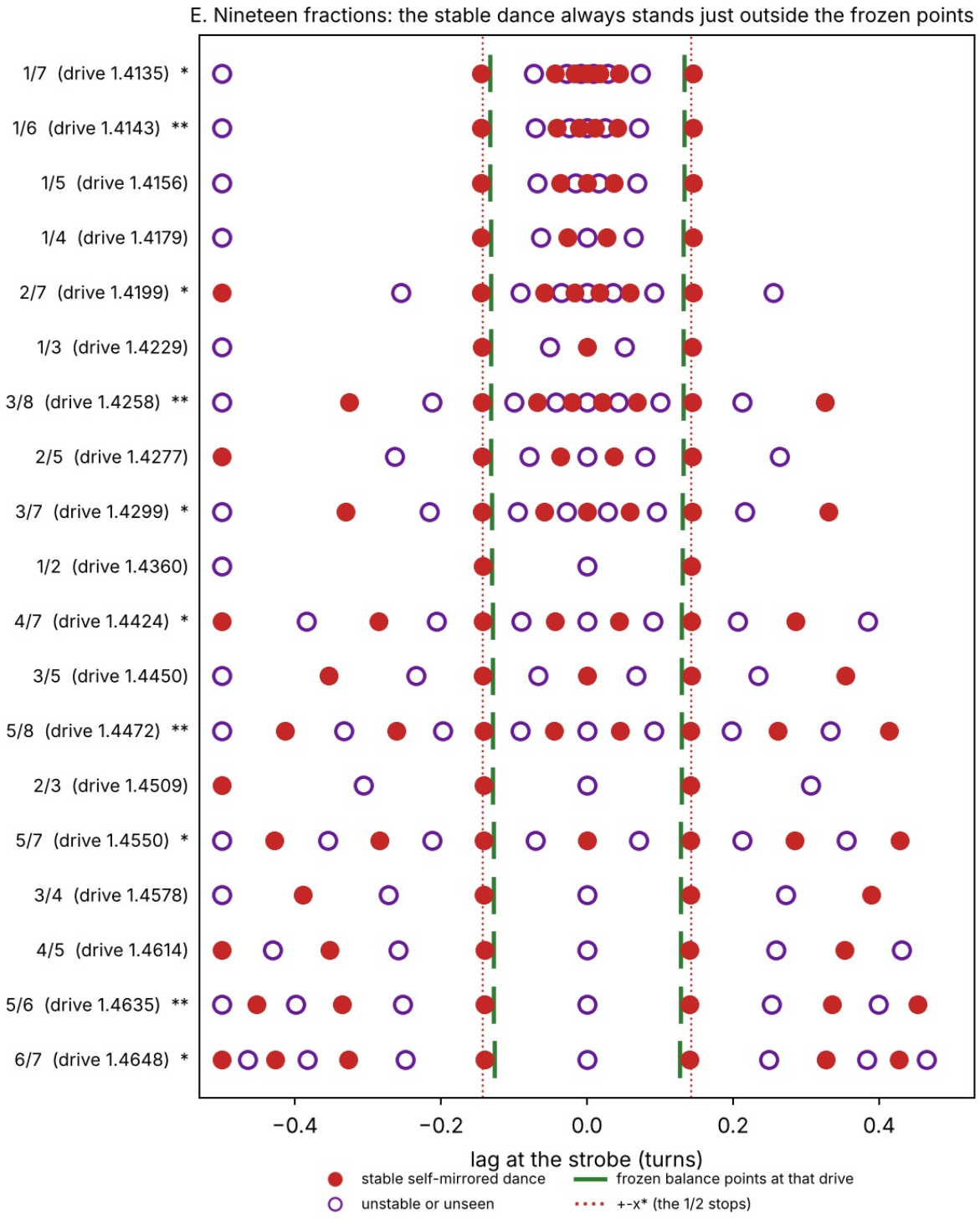


Figure 31. The self-mirrored dances of nineteen fractions at their weightless roots: red stable, purple not, green ticks the frozen points.

4.24 Launch and catch

The red dance's pair of stops is one passing: a launch just outside the knife-edge frozen point and a catch on the other side. It is a trapeze act: the flyer leaves from one fixed spot by the rig and is caught on the far side. Followed at every drive, the launch stays 0.0124 to 0.0131 of a turn outside the frozen point (CATCH-1), and it moves 44 to 128 times less than the other mirror step. At the 1/2 root its leap, 0.7138 of a turn, is the longest that one passing makes from any start, on the top of a smooth hill (SEP-1). Sliding the pair 0.018 of a turn either way, as weight 0.05 does, shortens the leap by only 0.0015 to 0.0018. A hilltop explains why: at the top of a hill a small step to either side lowers you only at second order, the same reason a quartz crystal's turnover and the weight-proof edge of §6.11 work. That is why the 1/2 groove's two stops stayed 0.713 to 0.730 apart at every weight in §4.19: the distance was the leap all along.

Which side of the line. On the turned walkway the launch never crosses the frozen point (figure 32). Followed from strength 0.25 to 1.00, it stays outside at every drive read: 364 drives at strengths 0.40 to 0.70, and 182 more at 0.80 and 1.00 written down before their run (CATCH-1, CATCH-2, CATCH-3). The gap grows with the drive, and at every drive it grows with the strength, but more slowly each time: at drive 1.00 it rose 0.00063 of a turn from strength 0.60 to 0.70, 0.00039 from 0.70 to 0.80 and 0.00014 per tenth from 0.80 to 1.00, as predicted. So on the walkway the side belongs to the setting, not to the strength; why the gap is the size it is remains a derivation (Part 8). Meanwhile the band of drives where the rider turns from resting to slipping collapses as the strength grows: 0.037 of drive wide at strength 0.40, 0.0076 at 0.50 and 0.00073 at 0.60, narrowing five times and then ten times per tenth of strength, faster than the three to four times guessed. Its logarithm is a parabola in the strength: drawn through those three and written down before the run, the parabola placed the window at 0.45, 0.55 and 0.65 to within 0.0005 of drive and gave its width to within a fifth (0.0181, 0.0027 and 0.00014 of drive), and inside each window the gap barely moves, by less than 0.00014 of a turn across 0.01 of drive (CATCH-4).

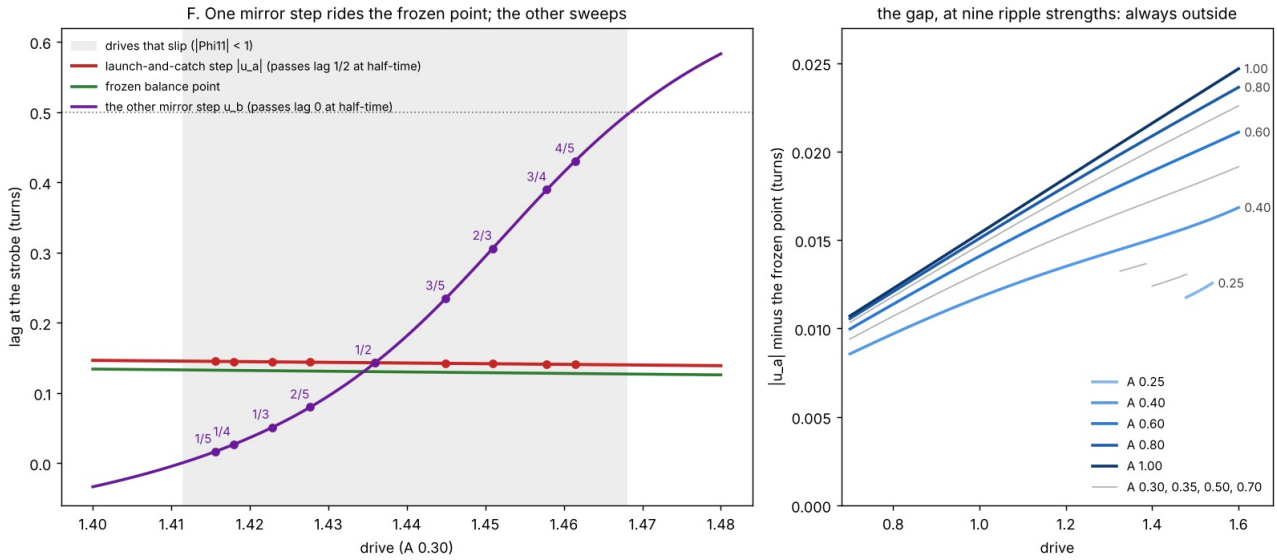


Figure 32. Launch and catch: the step rides the frozen point at every drive while the other mirror step sweeps across (left, strength 0.30). Right: the launch's step back from the frozen point at nine ripple strengths from 0.25 to 1.00 (CATCH-1, CATCH-2, CATCH-3); it grows with the drive and with the strength, and never crosses to the inside.

4.25 The counts, and the clock face

Bear counted figure 31 and found rules. A p/q dance has q stops. Each passing skips p places. Red and purple strictly alternate. Red has q − p − 1 stops between the frozen points and purple q − p. When q is odd, red passes lag 0 if p is odd and lag 1/2 if p is even. And the biggest red step grows across the rows, from 0.710 of a turn at 1/7 to 0.927 at 6/7 (PIN7-1).

Every count has the same reason. *A crooked clock face with its numbers bunched to one side. Straighten the face and every dance becomes a hand ticking evenly round it, the two dancers standing exactly half a tick apart. The leaper's mark is painted at the same hour on every clock. The only thing that changes from hall to hall is where the chalk line falls beside that mark, a hair before it or a hair after, and that one hair decides how many dancers stand between the lines.*

In numbers: stretch the lag by one number per row (§3.5, piece 6) and every passing becomes a plain turn of p/q. In the field's words, at a root the passing map is conjugate to a rigid rotation by p/q: a change of coordinates, the straightened clock face, turns it into an even tick. The stops sit evenly, the two dances half a gap apart, and the

leaper’s pair at exactly ±(q − p)/(2q). The window count is then one comparison. On the toy’s turned walkway the frozen points fall a hair short of the mark, so red has q − p − 1 stops between them. In the junction, and on the toy’s other side of the plane (§4.27), they fall a hair past it, so red has q − p + 1. All 45 rows tested agree. The count was found after the data, then written down for four fractions not yet run, 1/6, 5/6, 3/8 and 5/8: red 4, 0, 4 and 2 stops between the frozen points, purple 5, 1, 5 and 3. All four landed exactly (PIN68-1w).

When the chalk line falls more than half a tick past the mark, it sweeps over the next pair of purple stops, and the one comparison becomes a count: each dance has as many stops between the frozen points as it has points of its even lattice inside them. That happened on the toy’s other side at 5/8, 4/9 and 5/9 with a strong ripple, 1.05 to 1.19 half-ticks past; the lattice count agrees with every one of the 52 toy rows run (OFFAXIS-1, LATTICE_COUNT). Built on the toy, it was then registered as a prediction for the junction and read against every junction row on file, and against four new rows run after it was written down, at eighths and sevenths the junction had never run (3/8, 5/8, 2/7 and 5/7): 205 of 205 agree, with the pair inside the frozen points and outside them, and with second, fourth and other tones added (LATTICE-JJ-1). The junction’s counts are the same lattice count.

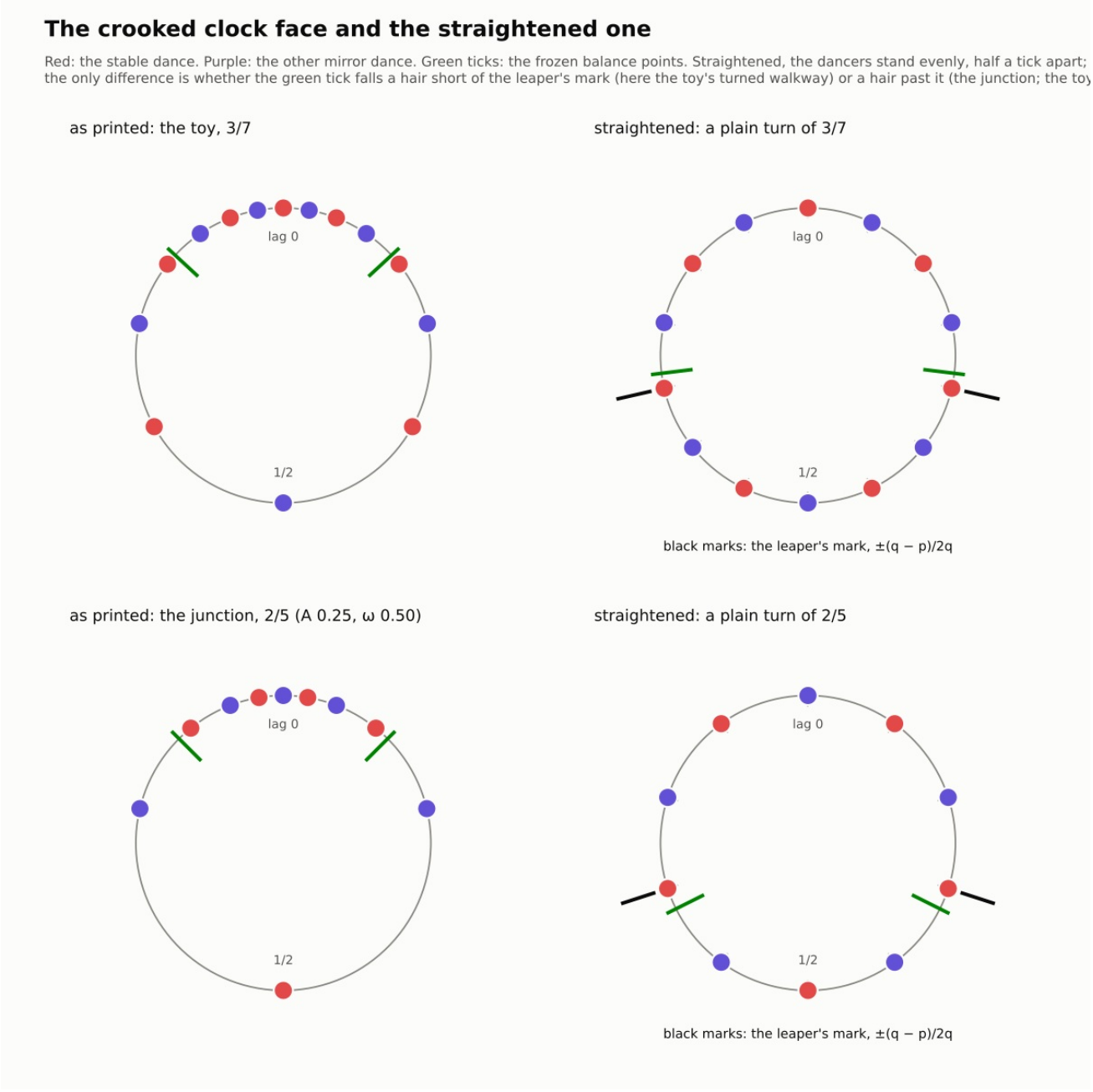


Figure 33. The crooked clock face and the straightened one: a toy dance and a junction dance, as their runs printed them and in the straightened lag.

4.26 Why the pin does not care about weight

This is the limbo of the evening show: the dancer holds the groove while the weight tilts them, and the bar does not move. At small weight, the weightless theory places the groove's mirror point exactly: drive D* = 1.4359681 and lag x* = 0.14309 at ripple strength 0.30, with the weighted riders' mirror drives agreeing to five decimals at weights 0.02 and 0.05, and at strengths 0.25 and 0.35 as well. As the weight grows, the mirror drive leaves D* slowly at first and then faster (by 0.0000081 at weight 0.10, 0.0002581 at 0.40). The pin is the mirror point of a dance that weight barely moves. In engineering words it is a point of first-order insensitivity, the same kind of point as the weight-proof edge of §6.11: there the weight's effect starts slowly and then grows faster than the weight does, about 32 times larger when the weight is four times larger, between weights 0.10 and 0.40. Its sharpness at strength 0.30 is partly luck, a sweet spot of the strength knob: at 0.25 and 0.35 the crossing is 7.5 and 5 times blunter, read between the dial's steps (SHARP-PIN-1a, below). A second run, on the same integrator with the ripple strength made a knob and other weights (0.05, 0.15 and 0.35), found the same at both strengths: the lightest weight's mirror drive on D* to five decimals, and a blunt crossing (PIVOT-2). Run at 0.30 with the same weights, the crossing is sharp again, within 0.00020 of a turn read between the dial's steps and 0.000014 from D*, against 0.00150 at 0.25 and 0.00100 at 0.35 read the same way (SHARP-PIN-1a; figure 35). Read only on the dial, the bluntness had looked up to twice as large.

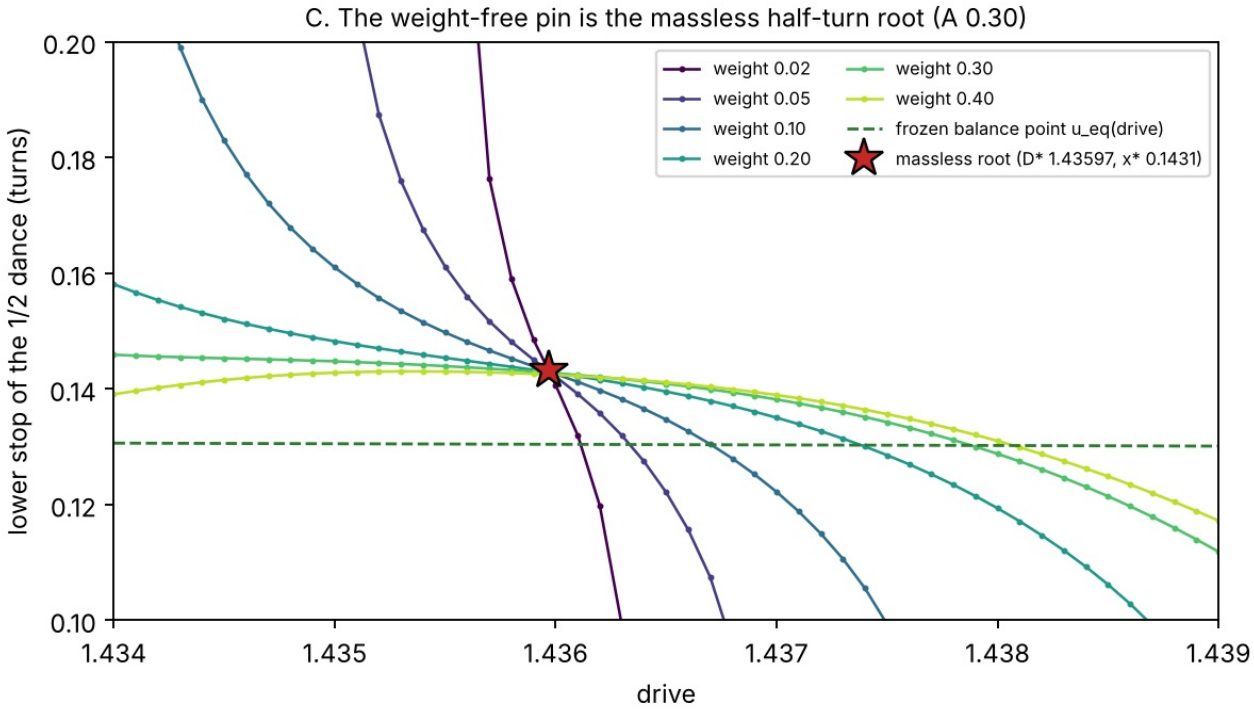


Figure 34. Every weight's lower stop against drive, through the weightless root.

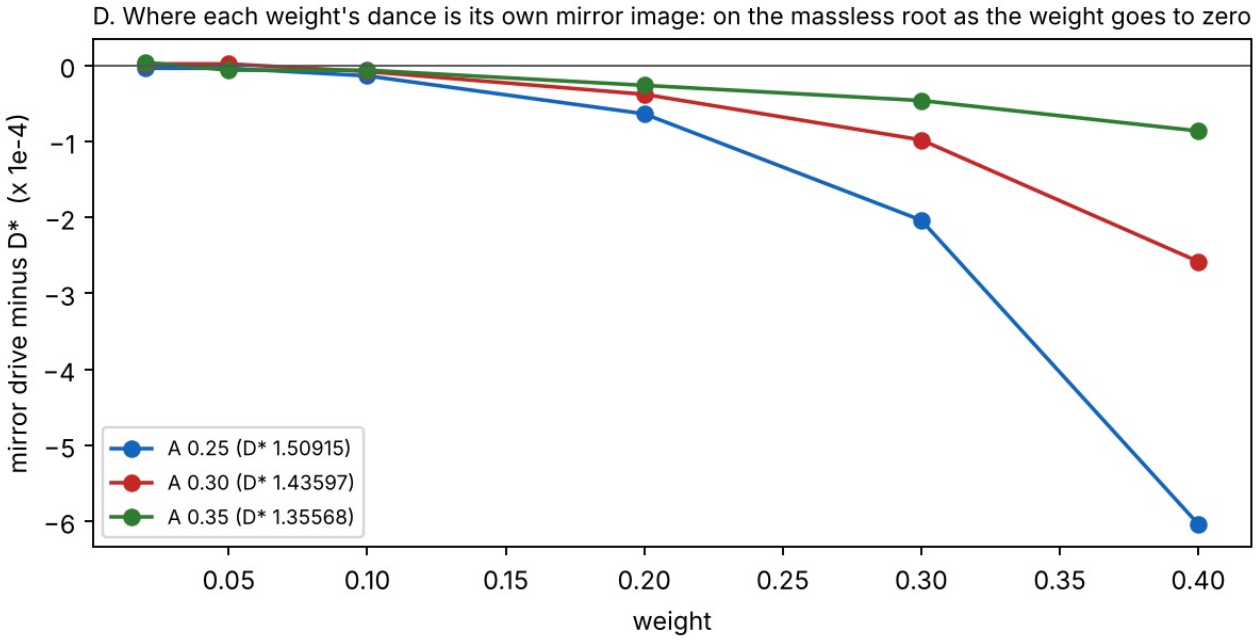


Figure 35. Each weight’s mirror drive minus the weightless root.

4.27 The other side of the plane

Everything in §4.20 to §4.26 was found with the ripple running with the ball. Does the shape endure when the ripple runs against it, on the atlas’s side of the plane? It is the same dance hall with the music turned around; in a travelling-wave tube’s terms, the wave runs with the beam on one side and against it on the other. The instrument was first made to reproduce the turned side’s seven roots exactly (TOY-SIDE-1). Then it was run on the atlas’s side at five ripple strengths, 0.20 to 0.40, with the fifths and sevenths added at 0.30: 35 rows in all (TOY-SIDE-1 to 4). At strength 0.30, the only one run on both sides when these rows were made (TOY-WALK-1 later ran the turned walkway at 0.10 and 0.20, §4.28), the roots sit 17 to 26% higher in drive than on the turned side.

The shape endures. At every one of the 35 roots weight keeps exactly the two mirror dances, and the stable one is again the far-side leaper.

The pair changes sides. The leaper’s pair sits inside the frozen points at all 35 rows, where on the turned side it sits outside. The counts follow the clock face either way (red $q - p + 1$ here). So which side of the chalk line the pair lands on is not set by the medium: the toy lands outside on one side of the plane and inside on the other, and the junction does both, depending on its drive (§5.2).

- The pair sits deeper inside as the ripple strengthens: on average 0.014 of a turn inside at strength 0.20, 0.028 at 0.30 and 0.035 at 0.40.
- Within one strength its depth drifts with the drive (read as the change from the lowest-drive root, 1/4, to the highest, 3/4), and the drift changes direction between 0.25 and 0.35, as was written down before those two strengths were run (TOY-SIDE-4f). At the weaker strengths the pair rises toward the frozen point as the drive grows; at the stronger ones it sinks deeper.
- Strength 0.30, where the first run happened to be made, sits close to where the drift vanishes. That is why its band looked so narrow, and why a fixed band failed at strength 0.20 (TOY-SIDE-3d).

The grip does not follow the junction’s knob here. A scaling law that carries in one medium need not carry in another, which is why every law in this book is tested in the medium it is claimed for. In the junction the grip ratio is set by strength over frequency squared (§5.2). The toy’s matching quantity would be strength over passing speed squared. It was tested with two in-between strengths whose rows overlap their neighbours on that scale, and it failed at all six overlapping rows, by 7 to 17% (TOY-SIDE-4e). What held instead, at all 14 rows and within 0.33%, is plainer: each fraction’s grip changes smoothly with the ripple’s strength (TOY-SIDE-4d). An earlier reading, that the grip’s steps “come back” when rows are matched on strength over passing speed squared, was made after the data and is withdrawn.

4.28 The grip multiplies

This is the longest section in the book, and it is one story told over many runs. Every tongue holds two dances: one that grips and a partner that wobbles. Their grips are opposite, and on the atlas's side of the plane nearly equal, like a seesaw that is almost level; the amount by which the gripping dance outweighs its partner is this section's hero, the excess. On the turned walkway the seesaw tips far over, the gripping dance holding about three times as hard (§4.23), and at strength 0.30 the excess reaches 1.89 (Scene 2). The story runs in six scenes. First, the excess multiplies: a dance that needs two steps the short way round gets the simplest dances' excess squared, and three steps cube it, as weavers knew for their satins. Second, the multiplying bends as the excess grows, and it bends the same way in the toy and in the junction, so one gearbox serves both. Third, a second tone at twice the frequency regears the box, and the section stops to ask how music reaches the dancers at all: the chalk line hears only the beat, the couples hear the whole bar, and their ear falls as one over the harmonic squared, a law then derived with nothing fitted. Fourth, pushed far enough, the second tone pinches a dance's grip shut and hands the hold across to the other dance through a walking pair; counting the drive's quanta says why one tone can never do that and two can. Fifth, the excess climbs down the Farey family tree, where a child's excess comes close to its two parents' multiplied, and the children of 1/2 carry an extra factor whose reason is still open. Last, real weight is put back in: the grips let go as the rider gets heavier, narrow tongues wander away from their weightless roots, and a dance's grip turns out to be the slope of its tongue's wrinkle, which is why a steady dance never comes without a wobbly partner; and the trace that sorts the motion turns out to draw a light cone, which weight gives to every step. Each paragraph below starts where the one before it stopped, so they read best in order.

Scene 1: the excess multiplies

Write δ for the stable dance's grip in excess of a plain mirror: the grip ratio is $-(1 + \delta)$. A plain mirror, where the two dances hold equally hard in opposite directions, has $\delta = 0$. On the atlas's side the simplest dances (the short way round takes one step) have δ of about a third at strength 0.30: 0.33 to 0.35 for 1/2, 1/3 and 1/4.

- **Two steps square it.** A dance whose short way round takes two steps has δ equal to the simplest dances' δ squared, compared at the same place on the drive axis. The exponents read 1.97 to 2.14 at all twelve such rows over five strengths.
- **Three steps cube it.** At 3/7 and 4/7 the exponents read 3.02 and 3.05.
- **How it was found.** It was seen after the data at strength 0.30 (TOY-SIDE-2), written down as a prediction before the next runs, and then held blind at all eight rows at the four other strengths (TOY-SIDE-3g, 4g).

So the gearing does not add the stable dance's extra grip, δ, up step by step. It multiplies it, as if every extra step on the short side had to pass through the same narrow door.

Twills and satins. Weavers drew the same line long ago. A satin moves its interlacing point a fixed count of threads at each pass, and the count must share no factor with the number of threads and must be neither 1 nor one less than that number, because those two give a twill ("Satin", The Icon Analyst 59, 2000, which collects the historical sources; checked 13 October). A dance p/q moves p of its q stops at each passing, with p/q in lowest terms, so the weaver's rule picks out exactly the dances whose short way round takes two steps or more. The simplest dances are the twills; every other dance is a satin, and its extra grip is the twill's, geared once for each extra step.

Scene 2: the multiplying bends, the same way in both media

The junction does the same where δ is small. At the junction's settings of §5.2 the simplest dances' δ is near or above 1, and there the next step shrinks it only by a factor of 0.57 to 0.78; the power form does not hold. But run at weak strength over frequency squared, where its δ is 0.16 to 0.32, the junction squares it too: exponents 1.97, 2.03, 2.04 and 2.05, predicted before the run (JJ-SMALL-1c). One law in both media where the excess is small, bending away from it as the excess nears 1 (figure 36).

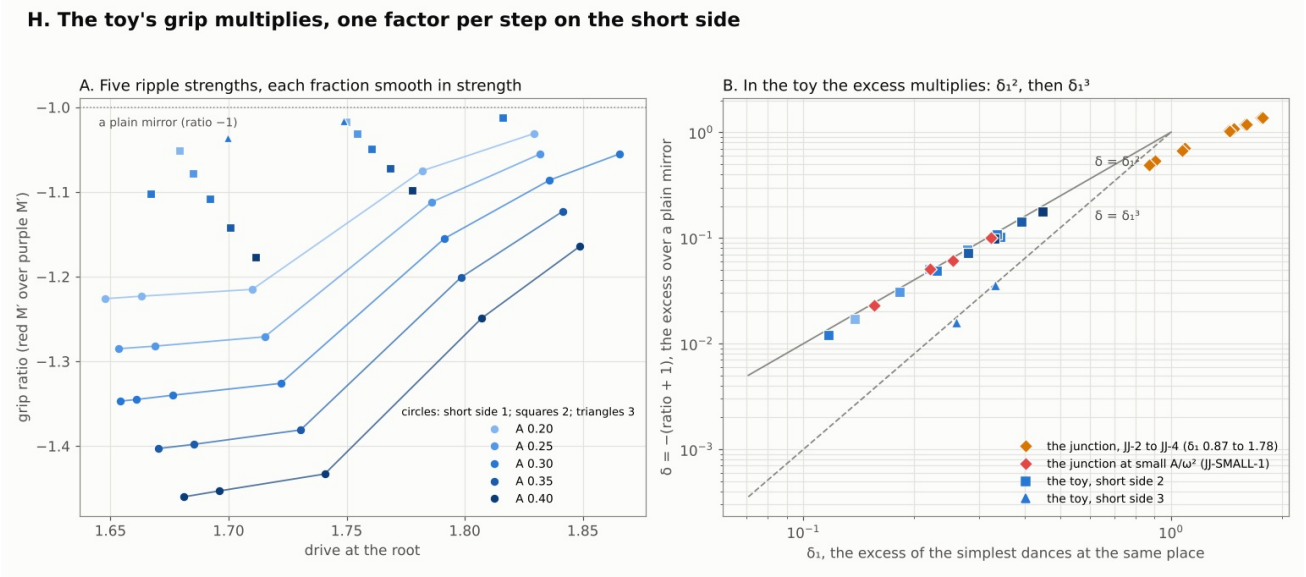


Figure 36. The toy's grip multiplies. A: the grip ratio against drive at five ripple strengths on the atlas's side; circles are dances whose short way round takes one step, squares two, triangles three. B: the excess δ of each two- and three-step dance against the simplest dances' excess at the same place, on logarithmic scales, with the lines $\delta = \delta_1^2$ and δ_1^3; the junction's rows are the diamonds, on the line where δ_1 is small (red) and below it where δ_1 nears 1 (orange).

Not bigger, not smaller. Two off-axis runs asked whether the power is really the number of steps (OFFAXIS-1). Four-step dances, never run before, came out at exponents 4.26 and 4.26 in the toy and 4.18 and 4.04 in the junction; three-step dances on a new denominator, 3.20 and 3.35 in the toy, and 3.04 and 3.06 in the junction. The power is the number of steps, leaning a shade above it (1 to 12%) where δ_1 is larger. So it is the small-excess limit of something that bends as δ_1 grows, which is what the junction showed near $\delta_1 = 1$.

One gearbox for both media. The multiplying is exact only while δ_1 is small. Measured as the step a second step takes, δ_2 divided by δ_1, all 28 two-step rows of both media lay on one smooth curve. The step equals δ_1 at small δ_1, bends over as δ_1 grows, and levels off toward about 0.92. Fitted before the runs, that curve then predicted the stretch no run had visited (δ_1 0.43 to 0.73) blind, in both media, within 0.017 (GAP-1). So each medium sets its own first gear: the ripple's strength in the toy (δ_1 about 1.1 times it at moderate strength), strength over frequency squared in the junction. Every step after the first is geared the same way in both.

The bend belongs to the grip, not to the medium. Until then the toy's two-step rows had sat only on the curve's straight start, below $\delta_1 = 0.45$, and every row out on its bend was the junction's. So the toy was pushed to strength 0.60 and 0.70, where its own δ_1 reaches 0.51 to 0.70, into the stretch where the junction had already bent (TOY-BEND-1). Its four two-step rows landed on the curve blind, within 0.0073; the plain power law missed them by 0.10 to 0.18. The toy bends exactly where the junction bends, so the bend is a property of how big the first grip is, not of which medium is gripping. One miss came with it: the toy's first gear is not a fixed 1.1 times its strength. δ_1 over the strength, at the 1/2 dance, rises from 1.07 at strength 0.20 to 1.09 near 0.35 and then falls, to 0.96 at 0.70, so the band of 1.0 to 1.2 that had been registered from the weaker runs failed at the strongest. The first gear bends too. That fall was written down as the next test, and run later at strength 0.80 it went on: the 1/2 dance's δ_1 over the strength came to 0.90, under the 0.93 the test asked for, each step down larger than the last (TOY-BEND-2). The toy's two two-step rows there sat on the gearbox's curve again, within 0.009, at δ_1 up to 0.75, while the plain power law missed them by 0.15 and 0.21.

The toy's other side covers the whole bend. The toy has a second side, the turned walkway, where the ripple runs with the ball instead of against it, and there its first gear is far steeper: δ_1 at the 1/2 dance is 4.4, 5.2 and 6.0 times the strength at strengths 0.10, 0.20 and 0.30, against about 1.1 on the atlas's side. So weak ripple on the walkway sweeps δ_1 from 0.34 up through 1, and the two-step rows already on file at strength 0.30 reach δ_1 1.89 (the 1/2 dance's own is 1.80). Run at 0.10 and 0.20 (TOY-WALK-1, a run waiting since 13 October), the walkway's four two-step rows landed on the curve blind, within 0.019, at δ_1 0.34 to 1.06; the two rows on file at 0.30, never used in the fit, sit within 0.024 at δ_1 1.86 and 1.89. The δ_1 of the 1/2 dance passed through 1 between the two strengths, as guessed, and the pair stayed outside the frozen points at every row, as it was at 0.30. With both of its sides, the toy now runs through every stretch of the gearbox the junction does, in its own medium.

Every extra step through the same gear. If the gearbox is one machine, a dance with three or four steps on its short side should be geared at every step, not just the second. Three- and four-step dances were run inside the middle stretch, two in the toy at strength 0.50 and four in the junction at strength 0.25 and frequency 0.55 (DANCE-3). The plain power law missed all six, by 24 to 56%: every step bends, not only the second. The rule registered for how the steps combine, the two-step gear raised to half the number of steps, held at four rows and missed the junction's four-step dances by 17 and 19%, so it failed as registered. Read afterwards across all eighteen three- and four-step rows on file, a simpler rule carries them: each extra step multiplies the excess by the same factor as the second step did. The measured factor is 0.97 to 1.11 times the second step's, from δ_1 0.26 to 1.09 in both media, and within 2% at the two largest. Found after the data, it was then written down first and tested in the junction's far bend, at two settings where the first gear is larger than at any row it was found on (δ_1 1.42 to 1.47, against 0.26 to 1.09): it held at all eight new rows, each extra step 1.01 to 1.04 times the second step's factor, while the combining rule registered for DANCE-3 missed every one, by 26 to 45% (DANCE-4). Figure 37, C shows the eighteen rows read before it.

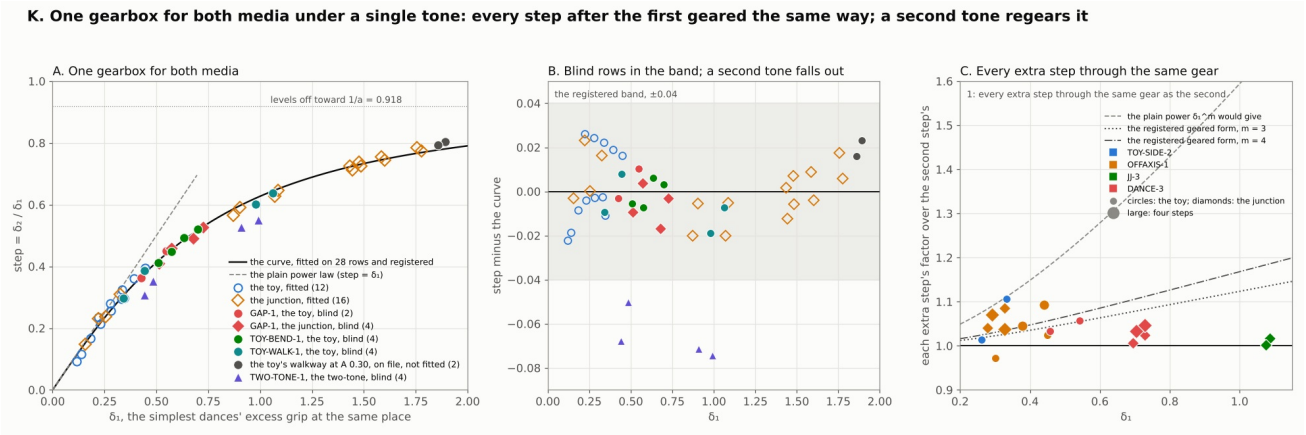


Figure 37. One gearbox for both media under a single tone. A: the step δ_2/δ_1 of every two-step dance against δ_1, toy (circles) and junction (diamonds); hollow, the 28 rows the curve was fitted on; filled, rows run after it was registered, each blind: the empty stretch, the toy's bend and the toy's turned walkway on the curve, the junction with a second tone (triangles) below it; grey, the walkway's two rows at strength 0.30, on file but never fitted. B: each row's distance from the curve, with the registered band of 0.04. C: for every three- and four-step dance on file, the factor each extra step multiplies by, over the second step's; the lines are what the plain power law and the registered combining rule would give.

Scene 3: a second tone regears the box, and how the music reaches the dancers

A second tone regears it. Event time says the junction's moves should not care what shape its radio wave has, so long as the wave keeps the time mirror. So the junction was given a second tone at twice the frequency, a twentieth of a unit either way, at a setting where the plain wave sits on the curve (TWO-TONE-1). With the second tone set to zero, the new engine reprinted the old run's 28 row lines byte for byte, four for each of its seven fractions (the fraction, its two dances and its checks). With either sign the moves were unchanged: every check held, and the far-side leaper was the stable dance at all fourteen rows, the seven fractions under each sign, with the pair inside the frozen points and red's count as before. The gearbox did change. The second tone moved the first gear a long way (δ_1 at 2/5 went from 0.73 to 0.99 with one sign and 0.48 with the other), and both signs pulled the second step below the curve, by 0.05 to 0.075. The prediction that the curve would carry them failed at all four rows; the rival, that the music's shape regears the steps, held. So the curve in δ_1 alone describes one shape of music, a single tone, which both media were driven by in every earlier run. The dance moves travel between kinds of music. The gearing is set partly by how hard the first dance grips and partly by the shape of the wave. With the second tone pulling one way, the second step's share of a plain square (δ_2 over δ_1 squared, which is 1 when the second step simply squares the first grip; it is not figure 37's step δ_2/δ_1) read 0.70 and 0.73, against 0.72 with the single tone, although δ_1 had fallen by a third: as if the step remembered the music rather than the first grip.

How the tone regears it. Run again at half the tone, ±0.025 (TWO-TONE-2), the two effects of the second tone came apart. The pair's distance from its frozen point is linear and odd in the tone, moving in proportion to the tone and the other way when the tone's sign flips: at every row it changes in step with the tone, half as much at half the tone (0.49 to 0.51 of the change at ±0.05), on one straight line through all five runs to within 0.0002 of a turn. Most of that change is the chalk line moving, not the couple. The frozen point sits where the push at the strobe is zero, and the second tone changes that push directly, so the frozen point slides 0.21 to 0.22 of a turn per unit of the tone. The pair itself drifts the same way, a tenth to a sixth as far (0.021 to 0.037 per unit, from 1/4 to 3/4), on a straight line to within 0.00003 of a turn. The gearing is neither: it does not move in proportion to the tone, and flipping the tone's sign does not simply turn it round. The guess was that it falls the same way at both signs, about a quarter as far at half the tone; that held on the plus side and failed on the minus side, where the step rose a little instead. Read across all five runs,

the gearing is a hump, highest a hair below no second tone; the guess had put the top exactly at zero. Four more settings, run to test that (HUMP-1), confirmed it blind: across all nine the top sits at a second tone of about −0.005 to −0.009, and the couples' own drift stayed on its straight line to within 0.00003 of a turn. They also showed the hump is not a parabola. On the plus side the gearing falls steadily; on the minus side it holds a shelf down to about −0.025 and then drops, most steeply between −0.0375 and −0.05, so the parabola fitted on five settings missed one of the eight new values by 0.012 (figure 38). Run closer to where the drive's crest goes flat, which is at a second tone of −A/4, −0.0625, with tones down to −0.055, the drop steepened as written down first: its fall per unit of tone nearly doubled between −0.0375 and −0.055 at both dances, while on the plus side the fall kept the same pace to within 5% (HUMP-2), and a parabola through all nine earlier settings missed the new values by up to 0.028.

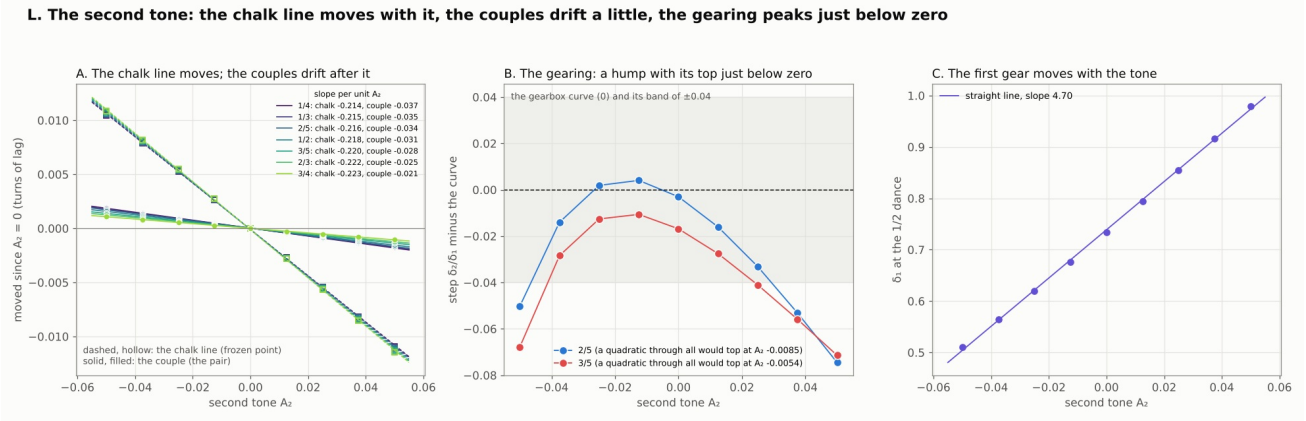


Figure 38. The second tone at the junction (strength 0.25, frequency 0.55), at nine settings from −0.05 to +0.05. A: how far each of seven dances' frozen point (dashed) and pair (solid) has moved since no second tone, with straight lines: the chalk line does most of the moving, and the couple drifts after it a tenth to a sixth as far. B: the two-step dances' step minus the gearbox curve, joined point to point: a hump whose top sits just below no second tone, with a shelf and then a drop on the minus side and a steady fall on the plus side. C: the simplest dances' excess grip at the 1/2 dance, which moves almost in a straight line with the tone.

The chalk line does not drag the couples. Both move the same way when the second tone changes, so it is fair to ask whether the couples simply follow their chalk line. Two kinds of music settled it (DRAG-1). A fourth harmonic was chosen to change the drive at the strobe instant exactly as the second tone at −0.025 does, so the chalk line moved almost the same (1.05 to 1.08 times as far); the couples moved only half to three quarters as far as with the second tone. Then the second and fourth tones were played together, chosen so that the drive at the strobe instant did not change at all. The chalk line barely moved, and slightly one way; the couples moved the other way, by exactly what the two tones do separately, added (within 0.00001 of a turn at every dance). So the chalk line answers the music at one instant, the strobe beat, and the couples answer every tone over the whole cycle, adding their effects. They move together only because they hear the same music.

The couples' ear. If the couples answer the whole cycle, how strongly do they answer each kind of tone? Six kinds of music were played, each sized to lower the drive at the strobe beat by the same amount, so the chalk line jumped the same way in all of them (EAR-1): the fundamental (a slightly stronger radio wave) and harmonics 2 to 6. The couples' answer falls as the harmonic gets faster. Through the third harmonic's answer at each dance, the line one over n squared carries the fourth, fifth and sixth within 9%, and even the fundamental within 16%. The second harmonic is the one exception, falling 50 to 69% below the line, and it is also the one tone that moves the tongues the other way (the root bias rises with it and falls with the others), as if it beat against the fundamental. The guess had been only that the answer falls at every step; it missed by the printing's last digit, where the second and third harmonics tie at the 1/2 dance. The one-over-n-squared ear was seen after the data, on five harmonics; the seventh and eighth were then run as its test, with their answers written down first, and both landed on the line at every dance, within 0.00002 of a turn (EAR-2, figure 39). The chalk line, meanwhile, moved the same distance for every harmonic from the third to the eighth, 0.00555 to 0.00594 of a turn depending on the dance (the fundamental and the second harmonic also move the tongues themselves, figure 39 B), as the design makes it: each tone was sized to change the push at the flash by the same amount, and the chalk line moves only where that push moves. It hears the drive only at the strobe beat, so the pitch means nothing to it. With the couples falling as one over n squared beneath it, the chalk line outruns them by n squared: at the 1/2 dance it moved 7.3 times as far as the couples at the third harmonic and 47.6 times as far at the eighth (figure 39, C). Only with the fundamental do the two move nearly together, the chalk line 1.2 times as far.

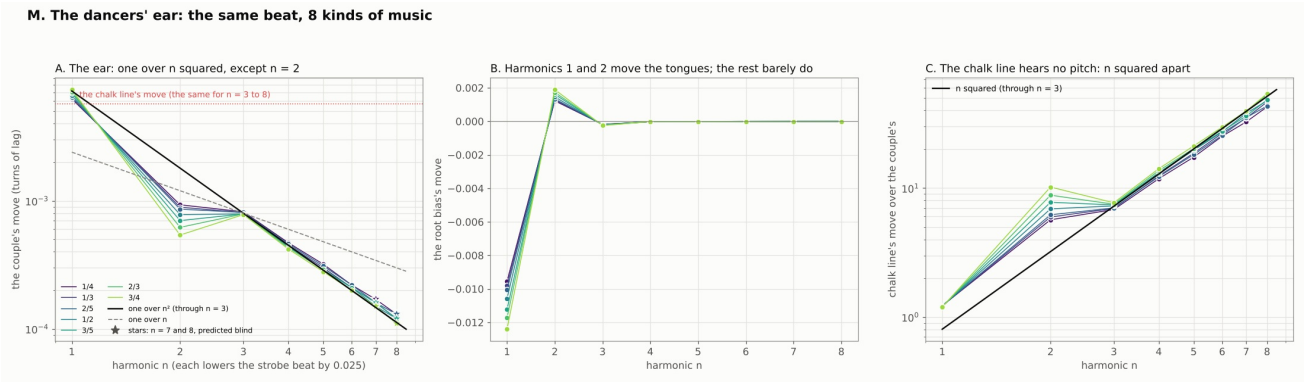


Figure 39. The dancers' ear at strength 0.25, frequency 0.55. A: how far the couples move (the pair's lag) when one extra tone, harmonic 1 to 8, lowers the strobe beat by the same amount, at seven dances, on logarithmic scales (stars: the seventh and eighth, predicted before they ran); the solid line falls as one over n squared through the third harmonic, the dashed as one over n; the dotted line is the chalk line's own move, the same from the third harmonic to the eighth. B: how far each tone moves the dances' roots: the fundamental and the second harmonic move the tongues, in opposite directions, and the rest barely do. C: the chalk line's move over the couple's at each dance: the chalk line hears no pitch, so it outruns the couples as n squared (the solid line, through the third harmonic).

Why one over n squared. The couples' place is not a point the music pushes around. At a root every weightless path closes on itself, and the pair is set by the shape of the one-beat passing matrix, through the straightened lag (§3.5, piece 6). An extra tone changes that matrix by adding up, over one whole beat, the tone times a smooth response. Over whole beats the first part of that sum cancels. What is left is set by the corner the response turns at the strobe, where each beat starts and ends, and a corner once a beat gives one over n squared, the way a triangle wave's harmonics fall. Worked through, the corner does not depend on the bias or the strength at all, and the couples' move comes out as $a \cdot \sin(2\pi x)/(2\pi n^2\omega^2)$: a the tone's size, x the couple's lag, ω the radio wave's frequency, and nothing to fit. The chalk line, which answers only the beat itself, moves about $a/(2\pi \sin 2\pi x_e)$, the tone's size divided by 2π and by the sine of the chalk line's own place x_e, whatever the harmonic, so the gap between them grows as n^2 (figure 39, C). Computed in full, the same first-order theory reproduced all 56 measured moves, harmonics 1 to 8 at seven dances, to within 0.00004 of a turn as the runs print, the second harmonic's departure included (KERNEL-1, figure 40). The closed form is its large-n end: within 2.6% at the eighth harmonic and 0.2% at the thirtieth, the gap shrinking as one over n squared. At first order the faster tones do not move the tongues at all. The small moves the runs show are second order in the tone, nearly the same at every dance and falling as one over $(n\omega)^2$, as when a fast tone shakes the junction's phase and slightly weakens its pull. The tenth and twelfth harmonics were then run with their answers written down first from the closed form, and all fourteen landed within 0.000015 of a turn, as close as the runs print (EAR-3, the stars in figure 40). A second engine, written from scratch and sharing only the equation, then heard the ear again (WITNESS-EAR-1). It finds each root by closing the flow itself rather than through the passing matrix, finds the couples as the flow's own mirror steps, and tells the stable dance by giving the rider real weight. It found every root and every couple of the 77 rows on file to the runs' last printed digit, and with its finer reading the theory's widest miss, the fundamental at the 3/4 dance, comes to 0.000042 of a turn rather than the printed 0.000037: a second-order part, not the printing. It then carried the ear on to the sixteenth, twentieth and thirtieth harmonics, which no run had reached beyond first order, and every move landed within 0.3% of the theory, as written down first. The tongues' second-order slide makes a small lesson in a ruler's grain. Read from the fourth harmonic to the eighth, it spread from about 0.22 to 0.24 times $(a/n\omega)^2$ and was taken for one number with some scatter, which is why it was first told as the same at every dance. Carried out to the thirtieth harmonic and read to a billionth of a turn, the scatter sorts itself into two tidy parts: within each dance the slide settles as the harmonic rises, and from dance to dance it eases steadily, 0.237 at 1/4 down to 0.217 at 3/4. And with real weight the couple sat in the stable dance at all 98 settings.

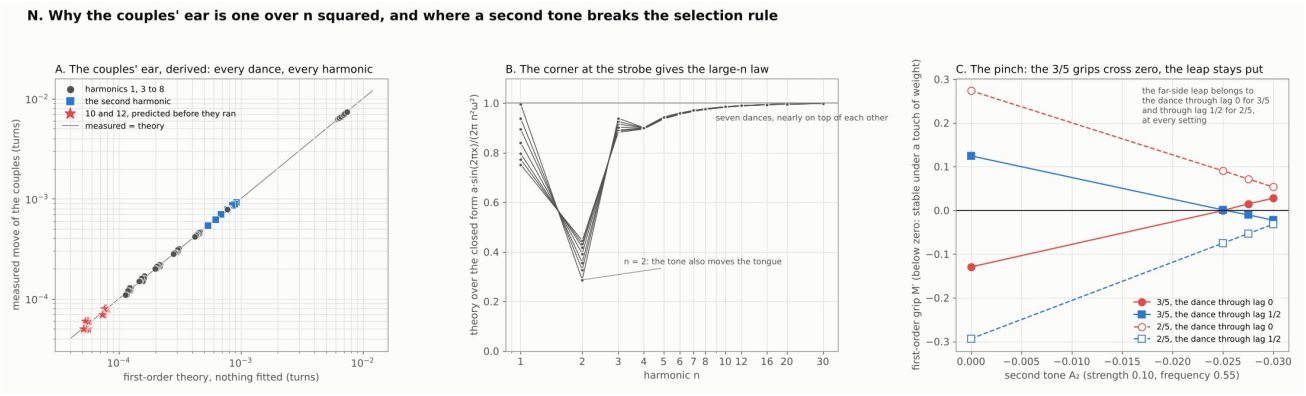


Figure 40. Why the couples' ear is one over n squared, and where a second tone breaks the selection rule. A: every measured move of the couples (harmonics 1 to 8 at seven dances, strength 0.25, frequency 0.55) against the first-order theory on the passing matrix, with nothing fitted; squares, the second harmonic; stars, the tenth and twelfth, whose answers were written down before they ran. B: the theory over the closed form $a \cdot \sin(2\pi x)/(2\pi\, n^2\omega^2)$ at each dance, reaching 1 as the harmonic grows, with the second harmonic far below. C: the 3/5 dances' first-order grip against the second tone at strength 0.10; the grips cross zero between −0.025 and −0.0275, and the far-side leap stays with the dance through lag 0, so past the crossing the stable dance no longer holds it; the 2/5 dances (dashed, hollow) are heading the same way.

Scene 4: the pinch, the swap and the hand-off

The second tone steers the tongues. In the usual map, with bias running across and the radio wave's strength running up, each dance's tongue hangs like an icicle. A stronger wave slides every tongue toward lower bias. The second tone is a third knob, and along it the tongues lean the other way: at the same change at the beat it slides them about a seventh as far as the fundamental does, and harmonics three and up hardly move them. The guess for why was that the junction bends the fundamental into a ripple at twice its frequency, of a size near the square of its strength, which the second tone lines up with. Run at strengths 0.20 and 0.30 (PUSH-1), the push did grow faster than the strength, by 2.06 to 1.93 times at the four lower dances, powers of 1.78 to 1.62 of the strength where a square would give 2.25 times, and only 1.84 to 1.67 times, powers of 1.50 to 1.26, at the three upper ones. That looked like a square at weak drive that bends as the drive grows, so the weaker strengths 0.10 and 0.15 were run to test it (PUSH-2). They broke the square the other way. Going weaker, the push fades faster than a square: from strength 0.15 to 0.10 it falls 3.25 times at the 1/4 dance, close to a cube, and 2.18 times at 3/4. No single power fits any dance. At every one the power falls steadily as the wave strengthens, from 2.9 to 1.7 at 1/4 and from 1.9 to 1.2 at 3/4, across strengths 0.10 to 0.30. Weaker still, with a gentler second tone of −0.010 (the method needs it under a quarter of the drive), the power keeps rising: at the 1/4 dance 2.7 between strengths 0.07 and 0.10 and 4.7 between 0.05 and 0.07, at 3/4 2.0 and 2.5 (PUSH-3). A guess that the 1/4 power would already pass 3 between 0.07 and 0.10 missed: it passes 3 only below 0.07. The push is not straight in the second tone's size either: at strength 0.10 a second tone 0.4 times as large pushes 0.48 to 0.64 times as far. Split into a part straight in the second tone and a part bent by it, from two gentle tones at each strength, the push comes apart cleanly (PUSH-4). The straight part grows as the first tone's strength squared, with powers of 1.79 to 1.95 between strengths 0.05 and 0.10 at the seven dances, as frequency mixing says: the tone and its octave make a steady push only together, at third order (for the drift, Hänggi and Marchesoni's Eq. 17; for the edges of a weightless junction's zero-voltage step, as the first tone's strength squared times the second's, Scheer and colleagues' Eq. 12). The bent part, the second tone acting on its own, is −0.21 to −0.23 times the second tone squared, the same at every strength to 2%, and the two parts together give the strong tone's push at strength 0.10 within a millionth. Both were written down before the runs. The single powers of 1.2 to 4.7 found before were a shrinking square less a steady bend. So the third knob's markings are stretched by the radio wave's strength: the louder the radio wave, the farther each turn of the knob slides the tongues, and the slow dances feel the band most. The couples hardly feel it. The same second tone moves them 0.00092 of a turn at the 1/2 dance at strength 0.10 and 0.00075 at 0.30, while its push on that tongue grows 9.5 times.

The second tone can swap which dance holds. At strength 0.10 the second tone did something no earlier run had shown: it all but erased the grip of the 3/5 dance. Its two dances' M′, −0.13 and +0.12 without the tone, fell to −0.0005 and +0.0010, smaller than the engine's spread of 0.004, so their signs cannot be told apart, and the selector M lay near zero all the way round. The same tone cut the 2/5 dance's grip to under a third. At strength 0.15 the tone cut the 3/5 grip by 42 to 47%, and at 0.20 by 22 to 28%. It is as if the tone pinches the 3/5 tongue nearly shut at this one setting. Pushed a little further, to −0.0275 and −0.030, the grips crossed zero, as predicted (PINCH-1): the dance through lag 1/2 became the stable one (M′ −0.0102, then −0.0225) and the dance through lag 0 the unstable one (+0.0148, then +0.0277). The far-side leap did not move with them. It stayed with the dance through lag 0, now the unstable one, as the straightened lag says it must: there the far-side step starts at $\pm(q - p)/(2q)$, a stop that only the dance through lag 0 has at 3/5. So the selection rule, the stable dance is the far-side leaper, which held at every root

of every run in both media under every music tried before, is not a law of the geometry. Under a single tone the grip's sign has always favoured the far-side leaper; a second tone can push it the other way. The 2/5 dance's grips are shrinking toward the same crossing, near −0.034 to −0.037 if they keep their pace.

Under real weight. A second engine then followed the 3/5 tongue with real weight in it (PINCH-2). For each of 200 starting lags it asks at what bias an orbit starting there closes after five beats. Plotted round the circle, the answer is the tongue's wrinkle: five humps, whose spread is the tongue's width and whose lean decides which dance holds. Away from the pinch the width grows in step with the weight, as first-order theory says. At the pinch it collapses, 10 to 50 times narrower, and what is left grows as the square of the weight: the first-order wrinkle really goes flat, and only weight's next order holds the tongue ajar. The swap survives real weight. Past a second tone of −0.0246, the same to four figures at weights 0.005, 0.01 and 0.03, the dance through lag 1/2 is the stable one, and the far-side leap still belongs to the dance through lag 0. The 2/5 dance swaps through a window of its own: at weight 0.01 the dance through lag 1/2 lets go at −0.0336 and the dance through lag 0 takes hold at −0.0372, both read again on tones 0.0001 apart (PINCH-3), and at its pinch the wrinkle's five humps vanish and a finer ten-hump wrinkle shows through (figure 41). The pinch has a name in the circle-map literature: a tongue pinches where q steps of the map add up to a plain turn (§2.5). In the weightless junction that happens at every fractional root, which is why the junction without capacitance shows only whole-number Shapiro steps, and weight is what opens the fractional ones. The second tone adds a second, rarer pinch, where even weight's first-order opening vanishes.

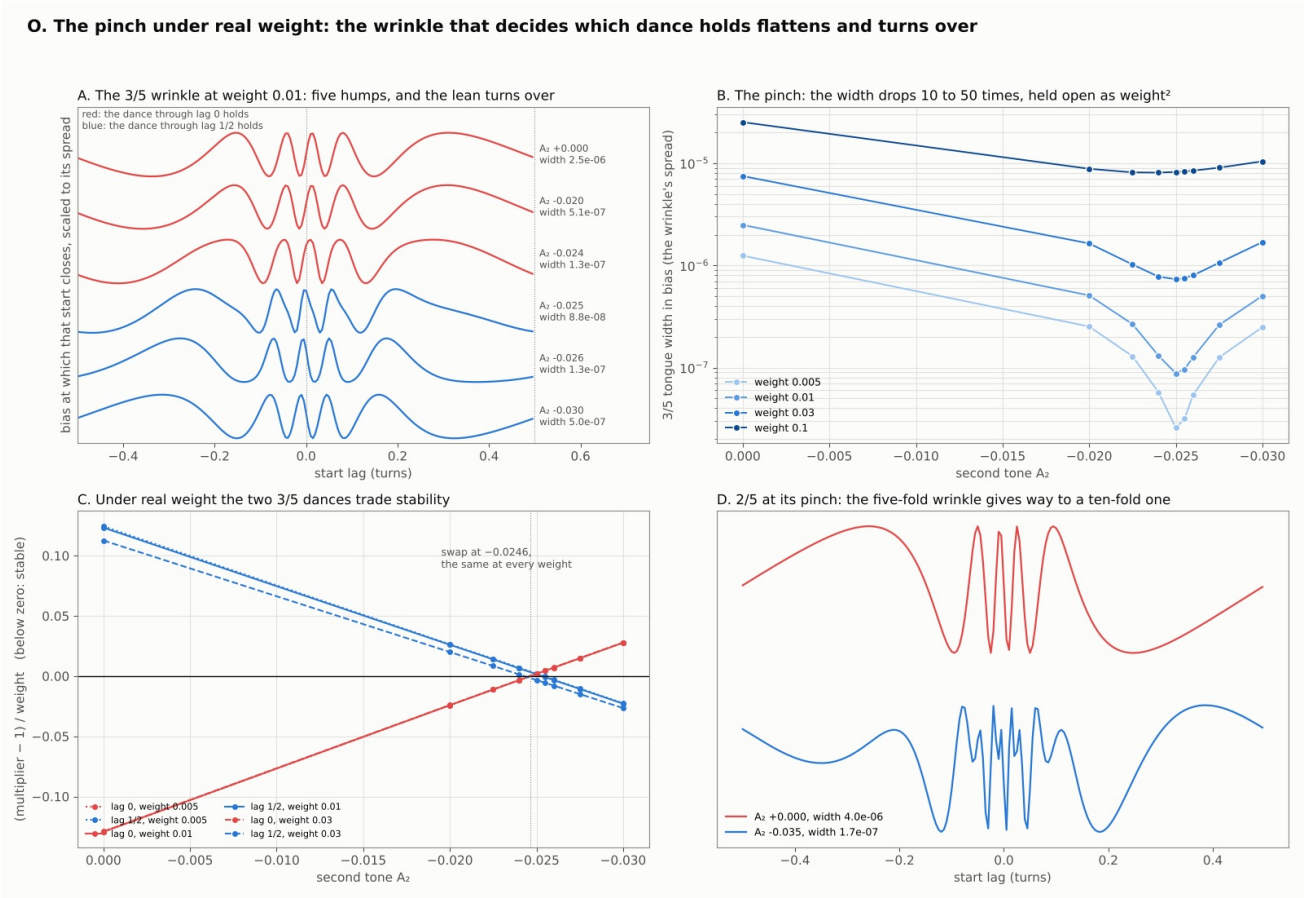


Figure 41. The pinch under real weight, at strength 0.10. A: the 3/5 wrinkle at weight 0.01 (the bias at which each starting lag closes after five beats), each scaled to its own spread, at six second tones: five humps whose lean turns over between −0.024 and −0.026. B: the 3/5 tongue's width against the second tone at four weights: a deep dip at the pinch at the light weights, washed out by weight 0.1. C: each 3/5 dance's multiplier minus 1, over the weight; the curves for three weights lie on top of each other and cross at −0.0246. D: the 2/5 wrinkle at weight 0.01, at no second tone and at −0.035, the setting run nearest its pinch (the dance through lag 1/2 lets go at −0.0336, PINCH-2, read again by PINCH-3), where the five humps give way to ten.

The long jumper's habit. Why, under a single tone, is the far-side leaper always the stable dance, and why can a second tone change that? Count what it takes to grip a dance. A dance of p turns in q beats is held by the q-fold part of weight's leftover, and to keep in step with that dance the leftover needs p quanta of the drive's beat. One tone of strength A supplies them one way only, as A to the power p. So the grip should grow as A^p at weak drive and keep its sign however the tone's strength changes. A second tone at twice the frequency gives a second way, trading two

single quanta for one double one, and where the two ways cancel the dance pinches, at a second tone in proportion to A^2. Read after the data, the stable grip's power of A at strength 0.10 was 1.1 for the one-turn dances, 1.84 and 1.92 for the two-turn dances, and 2.41 and 2.63 for the three-turn dances. Then it was tested blind (FAR-1). Between strengths 0.05 and 0.075 the powers came out 1.06 to 1.07, 1.95 to 1.98, and 2.78 to 2.89, closing on 1, 2 and 3. The pinch moved as the strength squared: the swap tone over A^2 stayed within 3 to 6% (3/5: −2.465, −2.434, −2.394; 2/5: −3.363, −3.270, −3.164 at strengths 0.10, 0.125, 0.15; that each ratio slides steadily with the strength is only a reading from these three strengths). At strengths 0.05 and 0.075 the far-side leaper was again the stable dance at every row. So a single tone cannot flip the habit, because it has only one way to build the grip; the second tone can, because it brings a second way with the other sign. Which sign the single way carries, and so why it is the far-side leaper that one tone favours, is still to be derived.

The hand-off. The swap is not a jump. At strength 0.15 and weight 0.005 a fine scan of the second tone (WALK-1) found a window, from −0.0539 to −0.0585, in which neither self-mirrored dance holds. Inside it the hold belongs to a pair of new dances, mirror images of each other, neither through lag 0 nor through lag 1/2. As the tone changes they walk from the stops of the one dance to the stops of the other: the stop nearest lag 0 slides from −0.007 to −0.025, about half a gap. At the window's middle the wrinkle shows ten humps instead of five, and both members of the pair hold at once. Bear had drawn the wrinkle as an N that "pinches twice" (§4.29); those two pinches are the window's two edges. At the first, the dance through lag 0 lets go; at the second, the dance through lag 1/2 takes hold; between them the walking pair carries the baton (figure 42). §4.23 had said that weight keeps only the two self-mirrored dances, and that others could appear in mirror pairs but none had been seen. Here they are, at a pinch.

The window belongs to light weight. At strength 0.10 the hand-off was read again at nine weights from 0.005 to 0.05 (WINDOW-WEIGHT). The tone at which the dance through lag 0 lets go hardly moves, −0.02464 to −0.02457, but the tone at which the dance through lag 1/2 takes hold walks toward gentler tones as the weight grows, and the window narrows: 0.00077 wide at weight 0.005 (0.00076 on WALK-3's finer tones), 0.00030 at 0.02, shut at about 0.025, where it was written down first to shut (between 0.02 and 0.03, near 0.026). It narrows as the square of the weight: one square law fits all nine weights, 0.005 to 0.05, within 0.00004. Past that weight the window turns over. The dance through lag 1/2 takes hold before the dance through lag 0 lets go, so over a stretch of the second tone both self-mirrored dances are stable, 0.00033 of tone at weight 0.03 and 0.0025 at 0.05. Heavier still, the overlap outruns the square law: 0.0042 at weight 0.06, 0.0064 at 0.07, 0.0094 at 0.08 where the law carried on says 0.0076, and about 0.020 at 0.10, where the dance through lag 1/2 already takes hold at a second tone of −0.004 (WINDOW-WEIGHT-2, WINDOW-WEIGHT-3). Under a single tone the dance through lag 0 is still the stable one there. The two sit at biases about a tongue's width apart, so whether a rider there keeps whichever dance it had, giving the hand-off a memory, is not yet known (Part 8).

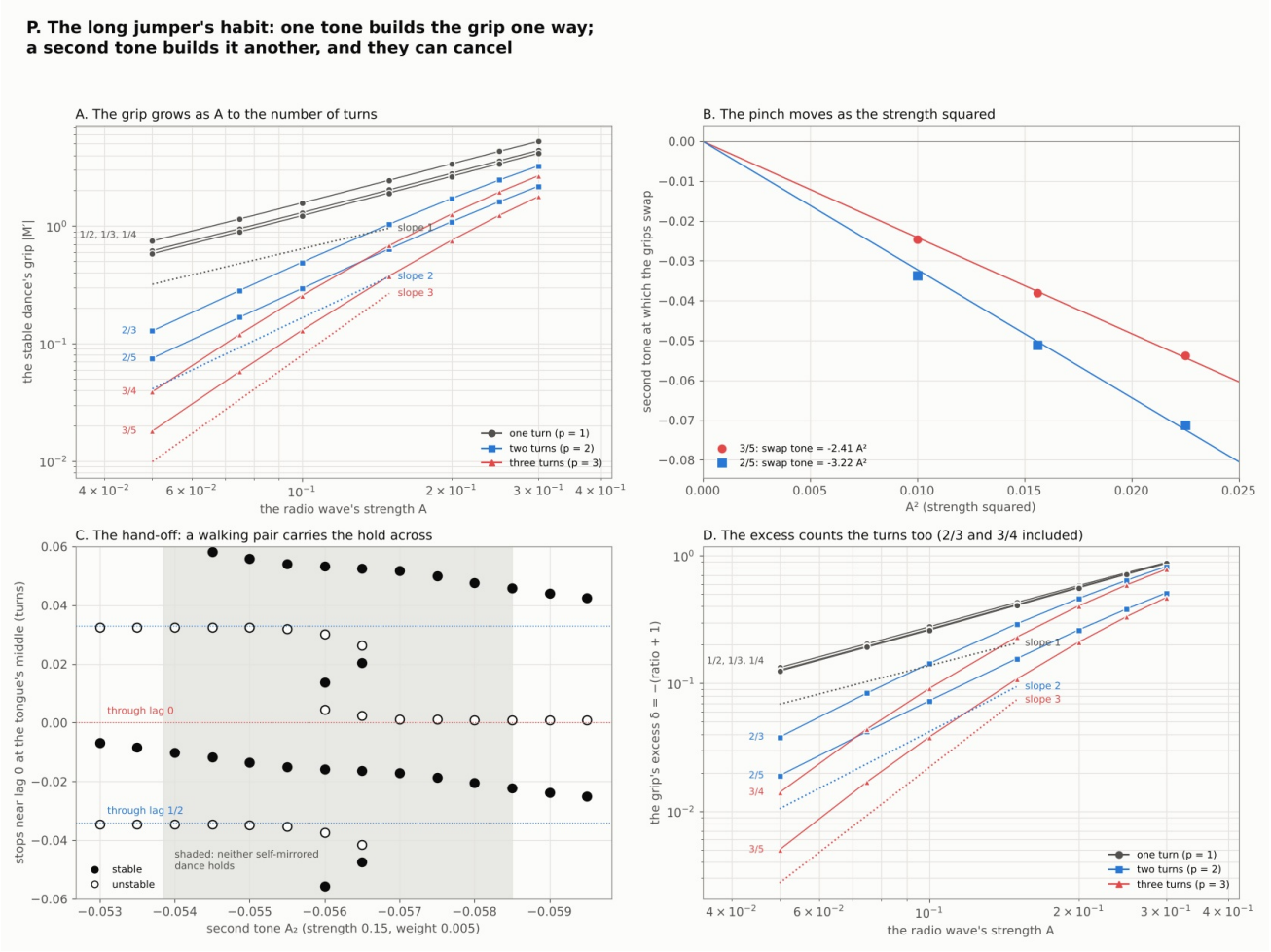


Figure 42. The long jumper's habit, at frequency 0.55. A: the stable dance's grip against the radio wave's strength, seven fractions, on logarithmic scales; the guides have slopes 1, 2 and 3, and each dance follows the slope of its number of turns as the drive weakens. B: the second tone at which the 3/5 and 2/5 grips swap, against the strength squared: straight lines through the origin. C: the hand-off at strength 0.15: the 3/5 stops near lag 0 at the tongue's middle bias against the second tone, stable filled and unstable hollow; in the shaded window neither self-mirrored dance holds, and the stable stops walk from one dance's places to the other's. D: the grip's excess δ against the strength, on logarithmic scales: it too follows the slope of each dance's number of turns, 2/3 and 3/4 included, though they take one step the short way.

Scene 5: down the Farey family tree

What the excess counts. This section began with a law about the excess δ, the lopsidedness between the stable dance's grip and its partner's: a dance two steps round the short way has the simplest dances' δ squared. The long jumper offers another count, turns instead of short-way steps, and the two disagree at 2/3 and 3/4, which take one step the short way but two and three turns. Before the weakest runs were read, the short-way count was registered for the excess and the turn count as its rival (DELTA-WEAK-1). The rival held. Between strengths 0.05 and 0.075 the excess grows as A to the power 1.06 at the one-turn dances, 1.96 at 2/5 and 2/3, 3.02 at 3/5 and 2.82 at 3/4; at strength 0.05, 2/3's excess is 0.038 and 3/4's 0.014, against 0.125 to 0.133 for the one-turn dances (figure 42, D). The old law still held as registered at the two-step dances, exponents 1.94 to 2.08, because it reads δ_1 off the neighbouring dances in bias, and 3/5's neighbours are 1/2 and 2/3, one of which already carries two turns. Counting quanta predicted the rival and should have been the bet: the excess weighs the leftover's 2q-fold part against its q-fold part, the first needs twice the quanta of the second, so their ratio grows as A^p. One more thing was seen after the data and priced, not tested. Each middle dance is the Farey mediant of its two neighbours, and its turns are the sum of theirs (2/5 from 1/3 and 1/2; 3/5 from 1/2 and 2/3). At weak drive its excess is close to the product of theirs: 0.96 to 1.13 of it for 2/5 and 3/5 at strengths 0.05 to 0.10, falling as the excess grows. That is what one would expect if a mediant dance is its parents' dances played one after the other, which an earlier run found for one groove near the conveyor's edge (MEDIANT-1), with the costs multiplied rather than added.

Down the family tree. Four new fractions, one level further down the tree, then tested the product blind at strength 0.10 (FAREY-1): 3/7 from 2/5 and 1/2, 4/7 from 1/2 and 3/5, 3/8 from 1/3 and 2/5, 5/8 from 3/5 and 2/3. Each child's excess came out within 6% of its parents' product: 1.037, 1.024, 0.952 and 0.942 times it (figure 43). The tighter guess, within 5% everywhere, missed at 5/8. The old law was registered as the rival, and it held too, within 11%, because at this strength its straight line in bias and the product differ by 5.2% at most. So the test could not tell them apart. The next one is designed to: at strength 0.05, where 2/3's excess is 0.29 of 1/2's instead of the 0.52 it has at 0.10 (both at frequency 0.55), the two predictions should part, and both were worked out before the test was registered (FAREY-2).

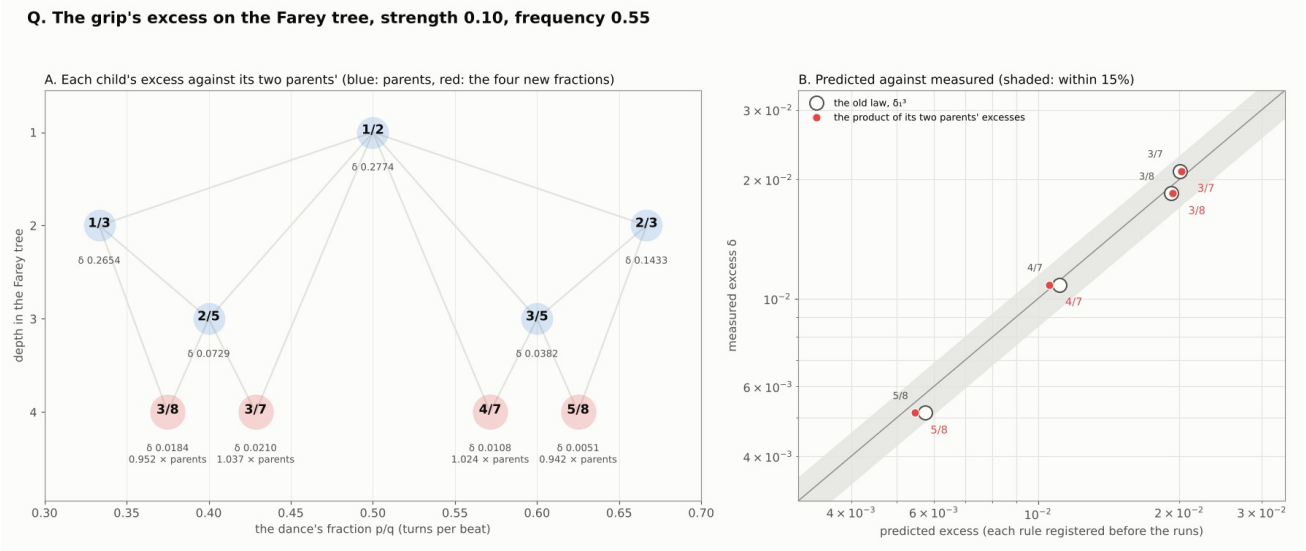


Figure 43. The grip's excess on the Farey tree, at strength 0.10 and frequency 0.55. A: the dances from 1/3 to 2/3, each placed at its fraction and its depth in the tree and joined to its two parents, with its excess δ; under each of the four new fractions, its excess over the product of its parents'. B: each new fraction's measured excess against the parents' product (red) and against the old law $\delta_1{}^3$ (hollow), with the line of equality and a band of 15%: both laws land inside, a few per cent apart, so this strength cannot tell them apart.

A ruler too coarse, and a finer one. FAREY-2 was built to part the two laws at strength 0.05. The parents and the children's places were found first and both predictions computed from them, and only one child, 5/8, had predictions far enough apart to be a fair test: its parents' product said 2.08 ten-thousandths and the old law 2.64. It came out at 1.96, and an earlier draft of this section called that a win for the product. It was not a test at all. The engine finds each dance's grip by a finite difference, and that route loses digits as the grip shrinks. Every row prints how far its two routes to the grip disagree, and at 5/8 that put the excess's uncertainty at about 1.0 ten-thousandth: half the excess, and nearly twice the gap between the two predictions. The reader had scaled the uncertainty down by the excess itself, so its precision check passed when it should have failed. The verdict is void. The same ruler left most of FAREY-3's children at strength 0.03 as noise, and three of them read negative. A second route to the grip, an integral along the same orbit with no difference to take, keeps its digits however small the grip gets (JJ_11). With it every row at both strengths was resolved, down to 5/9's excess of a hundred-thousandth, known to 9%. Measured again (FAREY-2b), 5/8 came out at 0.9962 of its parents' product and 0.7856 of the old law: the finite difference had been right, within its noise. At this strength every child but 4/7 sat nearer its parents' product than the old law; 4/7, registered to do so, came out 1.1138 of the product and 0.9284 of the old law, a miss, and the first sign of the factor that follows. At strength 0.03 (FAREY-3b), the three graded rows sat at 1.1472, 1.0091 and 0.9536 of their parents' product against 0.7411, 0.5446 and 0.6876 of the old law, and all seven children were nearer the product. There the old law misses the seven by 17 to 67%; the product by 1 to 18%.

The parent 1/2 carries a factor. The product's misses are not scattered. A child with 1/2 for a parent sits above its parents' product, and the others sit near it. At strength 0.05, after FAREY-2b was read, the children of 1/2 averaged 1.1078 of their product (2/5, 3/5, 3/7, 4/7) and the others 0.9943 (3/8, 5/8, 5/7). So before FAREY-3b's children were read, a factor κ was written down for 1/2 (two of the rows, 3/7 and 3/8, had been seen roughly in FAREY-3's finite-difference run, so the test was blind at the other five), read from the same run's first generation (2/5 and 3/5 over their parents' product, averaged): every child of 1/2 in the next generation should lie within 0.08 of κ over its product, and every other child within 0.08 of 1 (X-d). At strength 0.03, κ came out 1.1355, and the children of 1/2 landed at 1.1788 (3/7), 1.1472 (4/7), 1.1786 (4/9) and 1.1559 (5/9), all within 0.043 of it; the others at 1.0762, 1.0091 and 0.9536. The rival, children of 1/2 near 1, failed.

A factor that grows, and goes with the parent. κ is not a fixed number. It is 0.9754 at strength 0.10, 1.0886 at 0.05 and 1.1355 at 0.03, growing as the drive weakens (figure 44). At 0.10 a factor that size cannot be told from the plain product, which is why FAREY-1 saw neither. Read after the data, those three values lie on a straight line in the strength to within 0.001, which put κ at 1.1578 at strength 0.02. Run there with that number written down first (KAPPA-1), κ came out 1.1589, and the children of 1/2 sat near it again. One row broke the rest of the rule. 3/8, whose parents are 1/3 and 2/5, has climbed at every strength (0.9522, 1.0426, 1.0762, 1.0920 of its parents' product), and at 0.02 it sat 0.092 above 1, past the band of 0.08. Its parent 2/5 is itself a child of 1/2, so 1/2 is its grandparent. But 1/2 is a grandparent of 5/7 too, through 2/3, and of 5/8 through both its parents, and those two stayed near 1 (5/7 at 0.9669 at strength 0.02, KAPPA-1; 5/8 at 1.0091 at strength 0.03), so having 1/2 as a grandparent does not carry the factor by itself. Two readings fit those numbers. In one, the factor goes with having 1/2 as a parent. In the other, it is a hump in the child's place p/q, highest near 1/2 and higher on the left than at the mirror place on the right (at 0.03, 3/8 sits at 1.0762 against 5/8's 1.0091, and 2/5 at 1.1743 against 3/5's 1.0967). 5/12 tells them apart: 1/2 is not its parent (its parents are 2/5 and 3/7), but it sits between 2/5 and 3/7. Run at strength 0.05 with the place reading written down first as the lean (PLACE-1), 5/12 came out 1.0283 of its parents' product, not the 1.1308 its place predicted. With it, 5/13 (1.0285) and 4/11 (1.0426) sit level with 3/8 (1.0426): the rows on the left that 1/2 is not a parent of stand together a few per cent above the plain product, wherever they sit, and the children of 1/2 stand at κ. The factor goes with the parent, not the place. Then every child of 1/2 that resolves at strength 0.05 was run with that written down first (PARENT-1): 4/9 came out 1.1443 and 5/11 1.1450, beside 3/7's 1.1401 on the same side, and 5/9 1.1307 beside 4/7's 1.1138 (read again below with a finer ruler, 5/11 is 1.1453 and 5/9 1.1303). Two far-left rows that 1/2 is not a parent of, 2/7 and 3/10, came out 1.0211 and 1.0442. So at this strength the left side holds two plateaus: the children of 1/2 at 1.12 to 1.145, everyone else at 1.02 to 1.04. Along the ladder of children that walks toward 1/2, 2/5, 3/7, 4/9 and 5/11, each a child of 1/2 and of the one before, the factor settles: 1.1179, 1.1401, 1.1443, 1.1450, each step about a fifth of the one before.

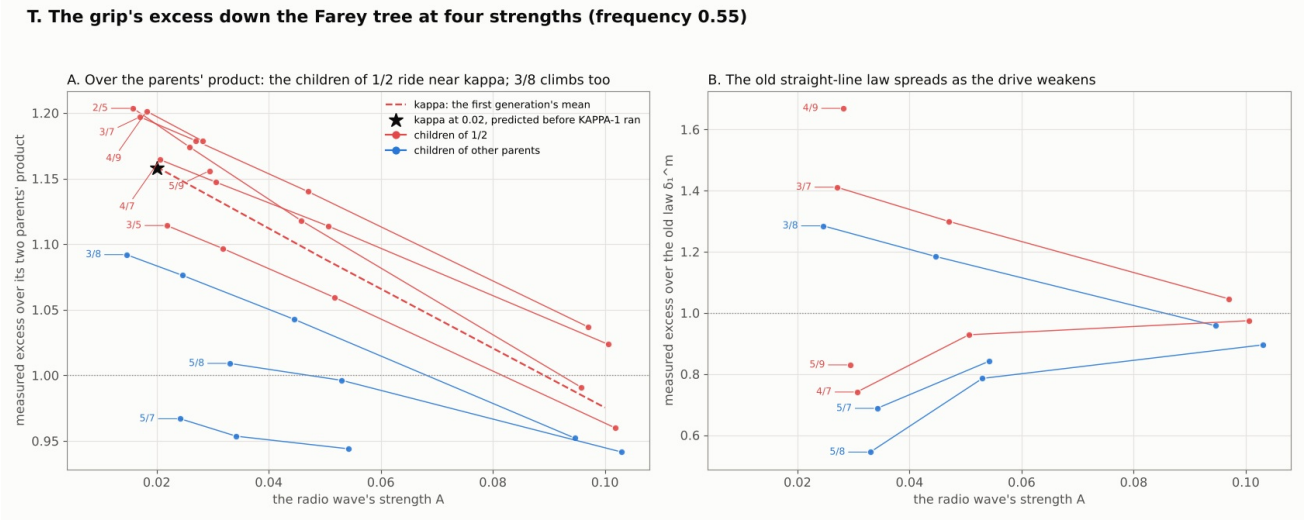


Figure 44. The grip's excess down the Farey tree at four strengths (frequency 0.55): FAREY-1 at 0.10, FAREY-2b at 0.05, FAREY-3b at 0.03 and KAPPA-1 at 0.02, the last three by the integral route (rows the route could not resolve at 0.02 left out). A: each child's excess over its two parents' product; red, the children of 1/2, blue, the others; dashed, κ, the first generation's mean, which grows as the drive weakens on a straight line in the strength; the star, κ at 0.02 as predicted before KAPPA-1 ran. The red rows ride near κ; of the blue, 5/7 and 5/8 stay near 1, and 3/8, whose grandparent is 1/2, climbs. B: the same children over the old straight-line law, which spreads from within 11% at 0.10 to 17 to 67% at 0.03.

Two arms, read with a finer ruler. The next rungs needed a finer ruler, and finding it took three tries. A deep rung's excess is the small difference between two grips that agree to five or six figures, and each grip is itself the small remainder of two much larger sums along the orbit. The first try (LADDER-1) read each grip on its own orbit, which works only for dances of an odd number of beats: at 1/2 the two starts lie on one orbit. Its error for each grip was assumed, not measured, and the two deepest rungs fell below it. The second (LADDER-1b) halved the step twice. The difference moved by 1e-11 to 5e-11 each time, in no steady direction, and did not shrink. That is the sign of bits lost to rounding along the orbit, not of the step being too coarse, and it was larger than the deepest rungs' whole excess. Carried in extended precision (LADDER-1c), the wander fell a thousandfold. 6/13 came out at 1.1459 of its parents' product and 6/11 at 1.1373, both inside the bands written down before the runs, and 7/13 at 1.1403; 7/15 was still too small to read. So the two arms are not mirror images. At the sixth rung the right one stands 0.0087 below the left, as the registered numbers said, and it climbs more slowly: 1.0592, 1.1138, 1.1303, 1.1373, 1.1403, each step 0.30, 0.42 and then 0.44 of the one before. The left arm's last steps are small, 0.0011 and 0.0006 with 5/11 and 5/9 also read in extended precision, but they no longer shrink by a fifth. Read after the data, pairing the arms by period

(2/5 with 3/5, 3/7 with 4/7, and on), the gap between them is 0.0587, 0.0262, 0.0140, 0.0081 and 0.0056, falling about as the period to the power −2.45 (figure 45, A and C). If it keeps falling that way, the arms meet at one height at 1/2: one arch with a steep left shoulder and a slow right one. At strength 0.05 the ruler cannot go deeper: the predicted 8/15 would carry about 10% uncertainty. So the question was carried to strength 0.10, where the grips are larger and every rung to period 17 resolves (LADDER-2). There too the left arm stands above the right at every period. The law read at 0.05, carried over unchanged, missed: from period 9 on, the gap sits 16 to 31% above it, closing more slowly at first and then faster. But the gap does close, from 0.0308 at period 5 to 0.0019 at 17, under a quarter of its period-9 value as the meeting reading said, and the rival in which it levels off failed. Read after the data, each arm's steps extrapolated to their end put the left arm's top at 1.0611 and the right's at 1.0608. That is one arch, with the two curves crossing at 1/2.

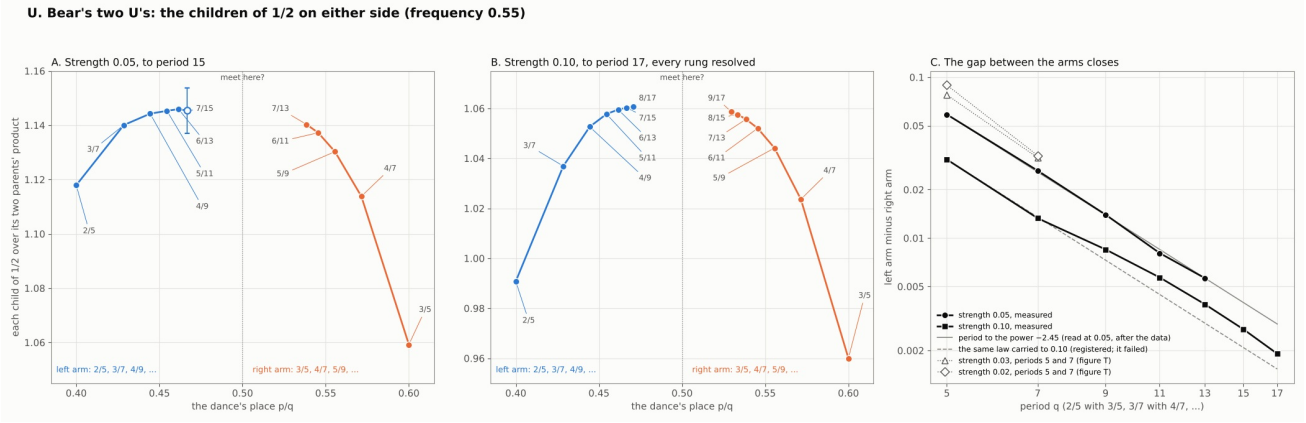


Figure 45. Bear's two U's: the children of 1/2 on either side, at frequency 0.55. A: at strength 0.05, each child of 1/2 over its two parents' product, against its place p/q. The left arm (blue: 2/5, 3/7, 4/9, 5/11, 6/13, 7/15, each a child of 1/2 and of the one before) climbs fast and levels near 1.146; the right arm (orange: 3/5, 4/7, 5/9, 6/11, 7/13) stands lower and climbs more slowly; 7/15 carries its uncertainty. B: the same at strength 0.10, every rung to period 17 resolved, the two arms closing on about 1.061. C: the gap between the arms at each period, left minus right, on logarithmic scales, at both strengths; the grey line is the period to the power −2.45 read at 0.05 after the data, and the dashed line is that law carried to 0.10, as registered, which the measured gap leaves from period 9; open markers, the gaps at periods 5 and 7 at strengths 0.03 and 0.02. Values from FAREY-2b, PARENT-1, LADDER-1c and LADDER-2.

What is left. Four things are left. Why 1/2: in these trees it is the one parent that is its own mirror, since the trees pair each fraction with one minus it (2/5 with 3/5, 3/8 with 5/8) and 1/2 is the only fraction equal to one minus itself; whether that is the reason is not known. The question has narrowed. Two other candidates fail on numbers already on file. One is the size of the parent's own excess: 1/3's (0.1267) is almost 1/2's (0.1329), yet its children that 1/2 is not a parent of sit at 1.02 to 1.04. The other is an even period: the children of 1/4, 3/4 and 3/8 carry no such factor (2/7 1.0211, 5/7 0.9440, 4/11 1.0426, 5/13 1.0285). What is left is that 1/2 is the one fraction whose two parents are whole slips, and the middle of its interval. One interval up, 3/2 shares both properties, so its children should show whether the factor follows them. But at the same drive the dances there grip hundreds of times more weakly (3/2's excess 0.00054 against 1/2's 0.1329, 5/3's 0.0000043), and their grandchildren sit below what the ruler can read. That test waits for a stronger drive and an engine that is both exact enough and able to pick each dance by its slip. And the second generation's children of 1/2 sit a little above κ at every strength, by 0.029 to 0.055 on average, the margin shrinking as the drive weakens, so the factor is not yet the whole story. And the rows 1/2 is not a parent of are not quite at 1 either: on the left they sit at 1.03 to 1.04 at strength 0.05, while 5/7 on the right sits at 0.94. And how the arms meet. At 0.10 they close toward one top, but by no single power of the period: the local power climbs from 1.8 to 2.8, and the gap's ratio over each two periods creeps from 0.64 to 0.71. At 0.10 the next rungs, 9/19 and 10/19, cannot say whether that closing settles into a fixed ratio: their gaps would sit so near the ruler's floor that the ratio over two periods would carry about 21 times the difference between a closing that settles and one that keeps creeping. The question waits for a ladder at a stronger drive, where the grips are larger (Part 8).

Scene 6: real weight put back in

Heavier, the grips let go. Under real weight, at strength 0.10 and frequency 0.55, the 3/7 dances behave as first order says, the one through lag 0 stable, within 0.6% of their grips at weights 0.005 and 0.01 (NEWTON-FREE-1). Heavier, they leave their grips, the stable dance first, as predicted (NEWTON-FREE-2): at weight 0.03 the stable dance's multiplier sits 4.6% from its first-order grip and the unstable one's 1.5%; at weight 0.1, 46.5% and about 20%. At 3/5, on the same drive, it had been the unstable dance that left first (PINCH-2, read after its data).

Why narrow tongues wander. Solving for the bias at which each start closes also shows how a tongue moves under weight. Its width grows in step with the weight, doubling from weight 0.005 to 0.01 at 3/7 and at 3/5. Its centre slides away from the weightless root four times as far over the same doubling, and nine times as far from 0.01 to 0.03 at 3/5: the slide grows as the weight squared. Read after the data, the reason looks simple. Weight's first-order leftover is mirror-odd and averages to zero round the circle, so it can tilt the tongue's wrinkle but cannot shift its centre; only the second order can. A narrow tongue therefore leaves its own weightless root at a small weight, about 0.003 at 3/7 and 0.006 at 3/5 at strength 0.10, which is why a search held at the root finds nothing there. Then it was tested blind at three more fractions, 2/5, 3/7 and 4/7 (SLIDE-1). The widths doubled at every doubling of the weight, 2.000 to 2.001. The slides had been registered against the weightless root, which is printed to seven decimals, and at the lightest weight they are only a few hundred-millionths, so that grade failed on resolution (ratios 3.86 to 5.67). Read again after the data without the root, from the change of the centre between neighbouring weights, the ratios are 4.0011, 4.0008 and 4.0010; a slide growing as the weight squared gives exactly 4. The weight at which the root leaves its tongue, worked out from the two lighter weights, said the root would still be inside at weight 0.01 at 2/5 (it leaves at 0.029) and already outside at 3/7 (0.0028) and 4/7 (0.0011). It was, at all three. Written down first and asked again at strength 0.25, the same three dances gave ratios of 4.0024, 4.0006 and 4.0028, and their widths doubled with each doubling of the weight, to within 0.0005 (SLIDE-2): at the stronger drive too the slide has no first-order part, as the mirror reading says it cannot.

The grip is the wrinkle's slope. Each start on the circle closes after q beats at its own bias, and plotted round the circle those biases make the tongue's wrinkle. Because every start closes, the multiplier of the dance through a start is tied to the wrinkle's slope there: the dance is stable where the wrinkle climbs and unstable where it falls, and the two meet at its top and bottom, the tongue's edges, where the pair is born and dies. Bear's picture for it is partners who hold each other on their patch. In the raw lag the sign of that rule held at 99.7% of 10,584 starts, but its size did not fit one gear, because the raw lag is a crooked clock face and the wrinkle falls off cliffs where the starts bunch (SLOPE-1). Read in the straightened lag, where the weightless passing is a plain rotation, the multiplier minus 1 is the wrinkle's slope times one gear: q times how fast the rotation per beat grows with the bias, which the weightless passing matrix gives with nothing fitted. In all 53 tongues tested, at strengths 0.10 and 0.30, inside the hand-off window and outside it, the gear fitted to the data came within 1 to 2% of that prediction (SLOPE-2), and most of the gap is the reader's own finite differences. So a dance's grip and its tongue's shape are one object.

Partners come in pairs, and the trace sorts the motion. With weight, the steady dance is a sink: both of its multipliers are below 1, the near-1 one by its grip and the other, set by the damping, almost 0. The wobbly dance is a saddle: its near-1 multiplier is above 1, so it pushes its neighbours away along the slip while the damping still pulls them in across it. Counted the way topologists count fixed points, a sink is +1 and a saddle −1, and on a ring the counts must add up to the ring's own number, which is zero. So a steady dance cannot exist without a wobbly partner. On the wrinkle the same fact is plainer still: a wave crosses any level as often climbing as falling, so at every bias inside a tongue there are as many steady stops as wobbly ones, alternating round the circle, as the engines' own checks find at every root of every run. The pair is born together at one edge of the tongue and vanishes together at the other, the way a pentagon and a heptagon appear together in a hexagon lattice as one dislocation. Bear drew it in an old notebook as a +1 well beside a −1 saddle. The weightless junction carries a deeper version of the same sorting. Its passing matrix has determinant 1, so it belongs to SL(2,R), the group of rigid motions of the curvature −1 plane, acting on the circle of phases at that plane's rim. Its trace sorts the motion, as the same trace sorted the islands of a light star around an eccentric binary in the team's three-body work, *Held, Rocking, Locked*. Where Φ11, half the trace, lies between −1 and 1, the map is a rotation by acos(Φ11)/π of a turn per beat. Where it lies beyond ±1, the map slides along a line with one attracting and one repelling point on the rim, and the junction locks to a whole number of slips per beat: an even number where Φ11 is above 1, an odd one where it is below −1. Tested over 1,401 biases at strengths 0.10 and 0.30, with the slip measured by direct integration and not from the matrix, every hyperbolic bias locked to a whole slip and every elliptic bias turned by acos(Φ11)/π, to within 0.002 of a turn (HYPER-1). The fractional dances live in the elliptic stretches, and weight turns each of their weightless rotations into a sink and a saddle.

The light cone. The same group has a second life, and Poincaré's reading of relativity points to it: he saw a Lorentz transformation as a rotation of four-dimensional space about its origin ("Sur la dynamique de l'électron", Rendiconti del Circolo Matematico di Palermo 21, 1906, §9; checked 29 October). SL(2,R) stands, two to one, for the Lorentz group of a world with two directions of space and one of time. In that world a passing with Φ11 between −1 and 1 is a rotation; one with Φ11 beyond ±1 is a boost, the change of a traveller's speed in special relativity; and |Φ11| = 1, half the trace equal to plus or minus one, is the light cone between them. A boost has a rapidity, arccosh|Φ11|, the number whose hyperbolic cosine is |Φ11|, which plays the part for a boost that an angle plays for a rotation. Inside a whole-slip step the rapidity is the step's grip: the locked orbit's multiplier, the share of a nudge left after one passing, is e to the minus twice the rapidity. The two-to-one is not idle. A matrix and its negative make the same Lorentz motion, and the sign that picture throws away is the one that says whether the whole slip is even or odd. All of this is known mathematics, checked here on the book's own engines rather than predicted (LIGHTCONE). At strengths 0.10 and 0.30 the steps' edges found from the cone and found by integrating the phase alone agree within 8×10^{-14}, and at 32 biases inside steps the multiplier matches twice the rapidity within 4×10^{-12}, at rapidities from 0.27 to 4.05. Over the

whole plane of bias 0 to 2 and strength 0 to 1.2 at frequency 0.55, 19,481 settings, the steps read from the slip alone and the cone drawn from the matrix agree at 19,477; the other four lie within 2×10^{-5} of the cone, where the turn per beat is too slow for a run of 150 beats to tell from a step (figure 46). One of the four sits exactly on the cone, for a simple reason. At strength 1 and a bias equal to the frequency, the phase can turn in lockstep with the radio wave, a quarter turn ahead of it (in §5.2's ψ, the phase less a quarter turn, simply $\psi = \omega t$), because then the wave's push and the junction's own pull cancel at every instant and the bias alone carries the phase round once a beat; a nudge to that motion is neither held nor shed over a beat, its multiplier exactly 1, so that rider walks the cone itself, at the lower edge of the one-slip step. In the junction's circuit the supercurrent then carries the whole radio wave and the shunt resistor beside it carries only the steady bias, so the voltage across the junction is pure DC: flat to 1.1×10^{-9} over 50 beats, against a ripple of 1.65 just inside the step (SWING-AND-RIPPLE). Every point on the cone is balanced just as tightly; this one is special only because a pencil can write its motion down. It is the simplest of a family of such solutions: Buchstaber and Tertychnyi built explicit solutions of this equation from polynomial solutions of a linear equation tied to it (2013, checked against its record), and Buchstaber and Glutsyuk proved that those polynomial solutions occur exactly where a bias that is a whole multiple of the frequency meets a step's edge, away from the points where two steps touch (2017, their Theorem 1.15, read through a tool's summary). At the first multiple and strength 1 the polynomial is a constant, the case this rider is (Buchstaber and Tertychnyi, 2015, Appendix B, read in part), though the rider's formula itself was not found written out in what was read. The tongues' tips sit where a free rider, with no radio wave, slips round at one, two or three times the drive's frequency. Its own rate, at a bias B (the steady push of §5.2's equation), is the square root of B squared less 1, so the tips stand at biases of the square root of 1 plus (n times the frequency) squared: 1.1413, 1.4866 and 1.9294.

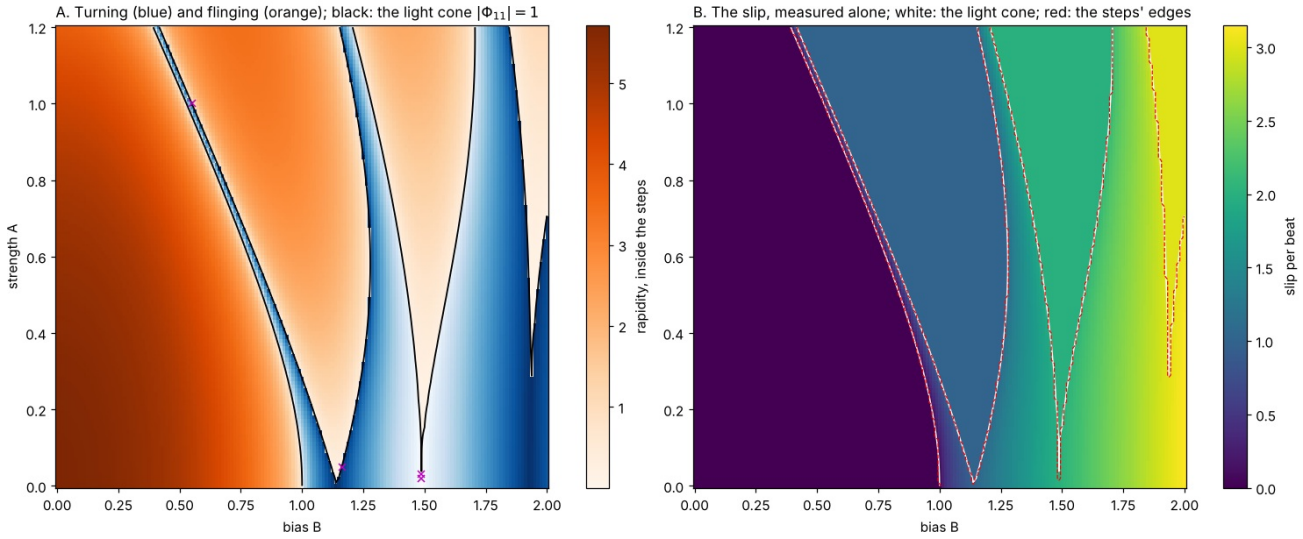


Figure 46. The weightless junction's light cone at frequency 0.55, over bias 0 to 2 and strength 0 to 1.2 (19,481 settings). A: where one passing is a rotation (blue, the turn per beat, acos(Φ11)/π of a turn) and where it is a boost (orange, the rapidity arccosh|Φ11|, which inside a step is its grip); black, the light cone |Φ11| = 1; the four crosses are the only settings where the step read from the slip disagrees with the cone, all within 2×10^{-5} of it. B: the slip per beat by direct integration, with nothing taken from the matrix; white, the light cone; red dashed, the steps' edges read from the slip alone. The two panels are one picture drawn twice. From LIGHTCONE.

Weight gives every step its own cone. Weight takes the passing out of SL(2,R). The damping shrinks every patch of starts by the same factor each passing, e to the minus the passing's time over the weight (Liouville's formula, §3.6), so the ruler becomes τ, half the trace divided by the square root of the determinant. Where τ is above 1 the step's two multipliers are real and positive, and a nudged rider settles straight back; where τ is below −1 they are real and negative, and the rider flips side every beat as it settles; where it is below 1 they are a complex pair, and the rider rings as it settles. Five things were written down before weight was switched on (WEIGHT-CONE), and all five held. Over the plane at weight 0.05 (figure 47, A), whole steps stayed whole: deep inside the weightless boosts the same whole slip held at all 13,889 settings. Every half and third step that weight opened, 110 settings, lay in the weightless rotations, each on the line where the weightless rider already slipped by that same fraction, within 0.0136 of it. Weight does not make fractional dances from nothing; it catches a drift the rider already had, turning a slow rotation into a sink and a saddle, over the whole plane. On one step, the zero-slip step at strength 0.30 and bias 0.70 (figure 47, B), the step's own cone was predicted to be crossed, from settling straight to ringing, within 25% of the weight at which a plain damped spring starts to ring: one over four times its stiffness, with the step's stiffness averaged over the beat (0.6728, so 0.3716). The first crossing came at weight 0.3064, 0.82 of that, and the prediction held; once the step rings, each multiplier's size matches Liouville's within 8×10^{-12}. But that crossing is a hairline, 7×10^{-5} wide. Past it both multipliers are real and negative, and the rider kicks a nudge to the other side every beat, smaller each time, out

to weight 0.413; then comes a second short window of ringing, a long stretch of settling straight, and the ring that lasts only from weight 1.669, 4.49 times the guess. At the guessed weight itself the step is far from ringing: τ is −30.2 there, where the plain spring has exactly +1. So the averaged stiffness lands near the first crossing, and whether it is the reason is not shown; a stiffness that the drive pumps sets the rest. τ moves smoothly with the weight, so a change of sign between two samples always hides a window of ringing, and every crossing was pinned by bisection.

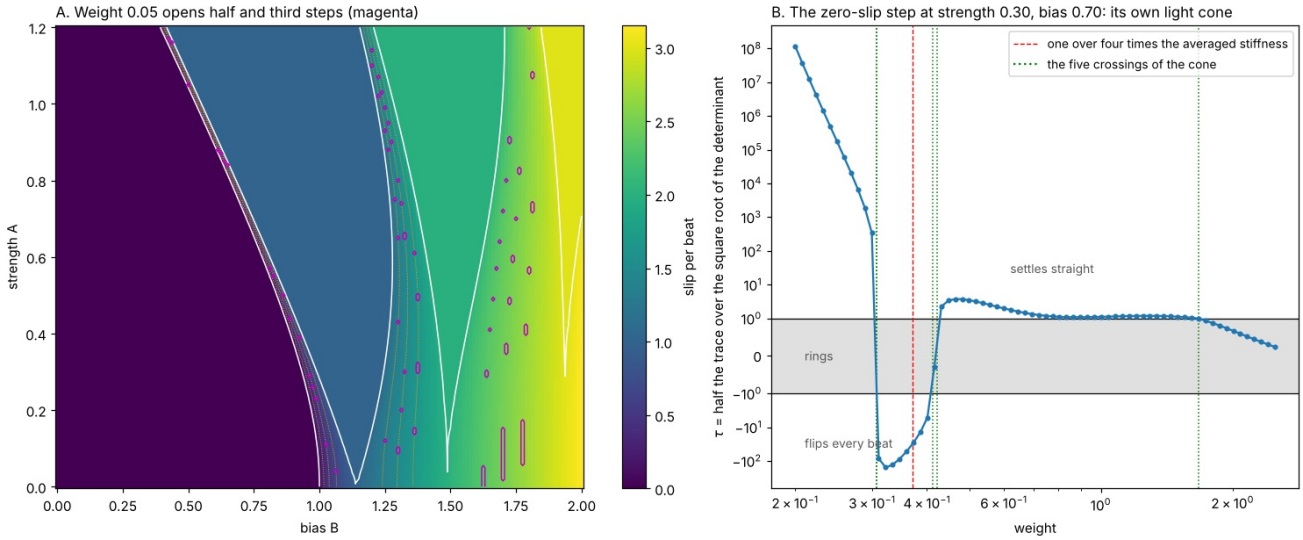


Figure 47. What weight does to the light cone, at frequency 0.55. A: the slip per beat at weight 0.05 over the plane of figure 46; white, the weightless light cone; magenta, the half and third steps that weight opened, every one inside the weightless rotations; orange dotted, the lines where the weightless rider slips a half, a third or two thirds of a turn per beat beyond a whole number, along which they lie. B: the zero-slip step at strength 0.30 and bias 0.70: τ, half the trace over the square root of the determinant, against the weight, on a scale that is even between −1 and 1 and logarithmic outside; grey, a complex pair, where a nudge rings; above it, two real multipliers, where it settles straight; below it, two real negative ones, where it flips side every beat. Green dotted, the five crossings of the cone, found by bisection; red dashed, one over four times the averaged stiffness, the weight registered for the first crossing. From WEIGHT-CONE.

How wide the hand-off is. At strength 0.15 the hand-off window was 0.0046 of a tone wide (WALK-1). At the same weight, 0.005, it is 0.00210 wide at strength 0.125 and 0.00076 at 0.10 (WALK-2, read again on tones ten times finer by WALK-3): the first about 2% from the width guessed beforehand from a single power through 0.10 and 0.15, the second 2.5% under the coarser scan's 0.000783 (WALK-2). The width grows about as the strength to the power 4.4 to 4.5. Weight closes it, as the weight squared. At strength 0.15 the window is 0.00469 wide at weight 0.0025, 0.00435 at 0.01 and 0.0032 at 0.02 (WINDOW-2): 0.0047 less 3.8 times the weight squared. A guess that it would keep its width out to weight 0.01 missed by 6%. Nearly all the narrowing happens on one side. The dance through lag 0 lets go at the same tone at every weight, to within 0.00005; the dance through lag 1/2 takes hold later as the weight grows. At strength 0.30 the window is 0.056 wide at weight 0.01 and 0.033 at 0.03 (LAB-MAP-2). Through those two weights the same rule predicted the window at weights 0.005 and 0.02 before they ran, 0.0577 and 0.0473; they came out 0.0576 and 0.0476 (WINDOW-3). So the window has a width without weight, and weight's second order trims it from the side of the dance that takes over. The 2/5 window does the opposite. At strength 0.10 it widens with the weight, on both sides, as the weight squared: written first from weights 0.01 and 0.03 as 0.003524 plus 0.859 times the weight squared, it came out 0.003550 wide at weight 0.005 and 0.003869 at 0.02, against 0.003546 and 0.003868 (WINDOW-4). The width without weight is set where each self-mirrored dance's first-order grip crosses zero. Read on each dance's own orbit, the lag-0 grip crosses zero at a second tone of −0.053855 at strength 0.15 and the lag-1/2 grip at −0.058585, a window 0.004730 wide, where the weighted runs, carried to no weight, had put the edges at −0.053862 and −0.058589. At strength 0.30 the crossings are −0.181107 and −0.239381, against −0.181110 and −0.239527 (WINDOW-0b). Between the two crossings both grips are positive, both self-mirrored dances wobble, and the selector has twice its usual number of zeros: the walking pair is already there without weight. A first attempt read the grips with an engine that hands every zero to the nearest self-mirrored dance; inside the window it averaged the walking pair into them, and every grade failed (WINDOW-0). The ten-humped wrinkle, where both walking dances hold at once, showed inside the window at 0.125 and nowhere outside it, as predicted. At 0.10 it did not show at all, not even on a scan whose steps were a twentieth of the window (WALK-2c). Either it shrinks faster than the window or it is absent at weak drive. It closes with weight too: at strength 0.15 it is gone by weight 0.01, and at 0.30 it spans eight of the scanned tones at weight 0.005 and three at 0.02.

4.29 What the dances are like: pictures, graded

Section 4.28 told the dancers' story in runs. This section tells it in pictures, because that is how most of it was first seen. The story in one breath: the grip multiplies step by step, through one gearbox in both media; a second tone regears it without changing the moves; the chalk line answers the beat while the couples answer the whole bar, so their ear falls as one over n squared, a law now derived; pushed far enough, the second tone pinches a dance's grip through zero, and there the stable dance and the far-side leap part company, under real weight as well as on paper; one tone never flips that choice, because it can build the grip only one way, while a second tone brings another way, and at the pinch the hold is handed across by a walking pair.

Along the way Bear and Claude reached for pictures: a pendulum, a comma from music tuning, a long jumper, trapeze artists at the catch, a downhill skier, a skater with four blades. A picture earns its keep when it points at something a run can check, so each one is graded here. *Exact* means the picture is the thing itself in other clothes. *Same structure* means the picture and the dance share their bones, the same count or the same shape, though the stuff they are made of differs. *A picture with one true point* gets one thing right and is only a picture otherwise. *A picture*, plain, helped someone think and is kept for that, but the book does not lean on it. Read the table as a field guide. The second column says whose picture it was, the third what it gets right, and the fourth where it stops. The fourth is often the most useful column: the place where a good picture breaks is the place where the physics is doing something the picture cannot.

picture	whose	what it gets right	where it stops	grade
a driven, weighted pendulum	Bear	the toy and the junction are driven, damped pendulums; weight is the toy's mass and the junction's capacitance	nothing	exact
the comma of music tuning	Bear	a cycle that almost closes, and its leftover. Without weight a q-beat dance closes exactly from every start; weight leaves a q-fold leftover whose lean decides where it holds, and the pinch is where that lean goes flat	a tuning comma is the same gap everywhere and nothing locks; our leftover varies round the circle, and the variation is what locks	same structure
a winding residual, a holonomy between frames	Bear	the leftover is what remains after carrying a start once round the closed q-beat loop with weight switched on: a return defect, first order in weight (the subharmonic Melnikov function)	there is no curvature here beyond the flow itself	same structure
an N-shaped wrinkle that pinches twice	Bear	the N: each hump of the wrinkle crosses its mean twice, once at the stable dance and once at the unstable one; the two pinches: the hand-off window's two edges, where first the dance through lag 0 lets go and then the dance through lag 1/2 takes hold (§4.28, Scene 4); at 2/5's pinch the five humps give way to ten	nothing found yet	exact
a triangle wave's corner	Claude	the couples' answer to a tone turns a corner once a beat, so its harmonics fall as one over n squared	nothing	exact

picture	whose	what it gets right	where it stops	grade
a sample-and-hold beside a filter	Claude	the chalk line reads the drive at one instant; the couples carry the whole bar; they part as n squared	real samplers hold a value; the chalk line is a balance, not a memory	same structure
a capacitor baseline with a triac gateway	Bear	the capacitor: the junction’s capacitance is the weight that opens the fractional dances	a triac switches on a trigger; which dance holds is set continuously by the leftover’s lean	capacitor exact, triac a picture
gears on one shaft (figure 40, A)	Bear	each harmonic’s seven dances sit together, and the clusters step down by $(n/(n+1))^2$, fixed by the harmonic number alone, as a gear train’s ratios are fixed by its teeth	it is not §4.28’s gearbox of the grip, which multiplies steps of one dance	a picture with one true point
a damped pendulum (figure 41, D)	Bear	the system is damped	the panel’s look of a ringing that dies away comes from the crooked clock face: in the raw lag the stops bunch near the frozen points; straightened, the humps are even	a picture with one true point
a long jumper	Bear	the far-side leaper is the dance whose big step crosses the far side of the circle at half-time	it is the stable one only while the music’s leftover leans its way	a picture
trapeze artists at the catch	Bear	in the hand-off window a pair of mirror-image dances carries the hold from one self-mirrored dance to the other, and at the window’s middle both are stable at once, as flyer and catcher both grip at the catch	a trapeze catch passes one body between two hands; here two dances, each the other’s mirror image, hold side by side, and neither is the one that let go	a picture with one true point
a latch: a number you cannot cut	Bear	a dance’s p/q is a whole-number winding; it holds across the whole width of its tongue and changes only at the tongue’s edges, where the smooth motion can no longer carry it	the edges are not cuts in space but places where the dance stops existing	same structure
partners who hold each other on their patch	Bear	the stable dance and its unstable partner are two points on one wrinkle, born together at one edge of the tongue and gone together at the other; between, the stable one sits where the wrinkle rises and the unstable one where it falls, so neither exists without the other (the sign held at 10,555 of 10,584 starts, read after the data)	they exert no pull on each other; they are two halves of one hill. In the raw lag one gear does not fit the sizes of their grips (SLOPE-1); in the straightened lag it does, within 1 to 2%, with the gear predicted (SLOPE-2), and that 1 to 2% was most likely the reading’s grid: at the crossings the gear holds within about 0.4% (SHAPE-1c)	same structure
the chaperone: “leave	Bear	the stops of the dance	the finite difference’s	same structure for the

room”		that holds and of the one that does not alternate round the circle, so between any two stable stops stands an unstable one, and starts on either side of it go to different stable stops	nudge, which the picture first answered, was the ruler’s own poke, not anything in the dance; no rule is enforced, the spacing is geometry	alternation; the nudge a picture
adding up the residuals	Bear	the integral route adds, step by step along the whole orbit, how much the drive stretches and bends a neighbouring path, with nothing nudged; the grip is that sum	a residual is usually what a fit leaves over; these are the pulls themselves, every one of them counted	same structure
mirror dipoles: complementary, sorted partners	Bear	the two dances’ grips are nearly equal and opposite, a ratio of −(1 + δ), like a dipole whose small mismatch δ is the lopsidedness; they come as a pair and go as a pair	they are not each other’s mirror images (each is its own mirror in time), and nothing passes between them	same structure
a +1 well and a −1 saddle	Bear (notebook, and the saddles-and-wobblers diagram, a wavy line with the steady dancers drawn on its crests; lesson 103)	with weight the steady dance is a sink (index +1, a well) and the wobbly one a saddle (index −1); on a ring the indices add to zero, so they come in equal numbers and are made and unmade together at the tongue’s edges, like a pentagon–heptagon pair; a second tone can turn the wrinkle’s lean over, so wells and ridges trade places (the swap), which answers his “does tone reset the dancers?”: yes	the diagram puts the steady dancers on crests and the wobblers in a trough; on the wrinkle they stand on its climbing and falling sides and meet at its crests and troughs, and on the landscape whose slope is the wrinkle less the bias, the steady sit in the wells and the wobbly on the ridges; nothing here is a curved space, the ±1 is a count	same structure
keep the ratio, or be pulled off one way or another	Bear	while the two grips keep their ratio near −1, one steady and one wobbly, the pair holds its place; when a second tone turns the ratio over, both self-mirrored dances are wobbly at once (both multipliers above 1 inside the hand-off window), and the lead passes to the walking pair, which steps off the self-mirrored places both ways at once, as mirror images	the ratio does not pull anything; it is a reading of two slopes of one wrinkle, and what moves them is the music	same structure
the −1 curvature plane	Bear, with the trace sorting of the team’s three-body work	the weightless passing matrix is in SL(2,R), the rigid motions of the curvature −1 plane; with Φ11 between −1 and 1 it rotates (the fractional dances’ stretches), with Φ11 beyond ±1 it locks to a whole slip, even or	with weight the map is no longer in SL(2,R): the damping shrinks area, and that is what turns a rotation into a sink and a saddle	exact for the weightless junction

		odd by its sign (HYPER-1, 1,401 biases at two strengths)		
a light cone: a Lorentz transformation as a rotation of spacetime (Poincaré)	Bear	SL(2,R) stands, two to one, for the Lorentz group of two directions of space and one of time; a weightless passing is a rotation inside the cone and a boost outside it, the cone is Φ11 = ±1, and inside a step the boost's rapidity is the step's grip (LIGHTCONE, 19,477 of 19,481 settings; figure 46)	nothing here moves at the speed of light; the sign of Φ11, which the Lorentz picture loses, says whether the whole slip is even or odd; with weight the passing leaves the group and each step gets a cone of its own, which one step crosses five times as its weight grows (WEIGHT-CONE, figure 47)	exact for the weightless junction
two U's offset side by side and intersecting; two blades cutting the skater's ice	Bear	the children of 1/2 climb toward 1/2 on two arms, an upside-down U on each side, and the arms are offset: at every period the right one stands lower (0.0587 lower at period 5, 0.0056 at 13), and it climbs more slowly; the gap closes about as the period to the power −2.45, and at strength 0.10, read to period 17, the gap closes to 0.0019 and each arm's extrapolated top lands within 0.0004 of the other's, so the two curves meet, intersecting, at 1/2 (LADDER-1c, LADDER-2; figure 45)	each curve is a row of separate dances, not a smooth floor; the closing follows no single power of the period, and the tops are extrapolated, read after the data	same structure; the meeting held as registered at strength 0.10
a four-legged skater, each pair of blades carving one lip of the groove, the dancers holding and bracketing it	Bear	the groove is a dance's tongue in bias, and its two lips are the tongue's edges. At each lip a steady stop and a wobbly stop meet and vanish together: their multipliers meet at 1 where the wrinkle's slope is zero (SLOPE-2). So each lip is carved by a pair of blades, two lips and four blades. As the bias crosses the groove each steady stop slides from one lip to the other; beyond the lips no dancer exists and the slip resumes, so the dancers bracket the groove exactly	the steady stop that closes the top lip meets the wobbly stop on one side of it, and at the bottom lip the wobbly stop on its other side, so the four blades are two dances, a skating couple, not four skaters; nothing holds the lips up, they are where the pair runs out. Read on the tables' grid the dancers seemed to give the width 3 to 6% short (GROOVE-1); that was the grid. In the straightened lag the humps are sines, and read cleanly the dancers give the width (SHAPE-1b, 1c)	exact
a downhill skier, low at the back, whose poles steer and can pole-vault a gap	Bear	the steady dance sits low, in a well, and stays where it is; at a pinch the hold has to cross a gap, from the dance	nothing is planted or pushed: the walking pair exists only inside the window, and the vault is made by the music	a picture with one true point: the hold is carried across the gap by a pair

		through lag 0 to the dance through lag 1/2, and inside the hand-off window it is carried across by the walking pair, two mirror-image dances like a pair of poles, each walking half a gap while both self-mirrored dances wobble (WALK-1, WINDOW-0b)	changing, not by the skier	
drive quanta	Claude	a dance of p turns in q beats needs p quanta of the drive's frequency to grip it; one tone supplies them one way, a second tone a second way, and where the two cancel the dance pinches	the quanta are orders of a Fourier expansion, not particles	same structure; tested blind and held (FAR-1); the excess counts the same way (DELTA-WEAK-1)

The drive-quanta picture is the one this section leans on hardest. It is ordinary bookkeeping of a Fourier expansion, the same counting that tells how many photons a multiphoton resonance needs, and here it predicted two things blind: the grip's power p, and a pinch that moves as A^2. Bear pictured the dances as a track meet, with long jumpers, relays, high jumpers and pole vaulters, and one event fits closely. The far-side leaper is the long jumper, and the swap at a pinch is a relay: the hold is passed through a pair of walking dances inside a window (§4.28, figure 42). The other events stay pictures. A whole-number count is here too: the power of the grip is the number of turns the dance makes.

The groove held by its dancers. Bear's four-legged skater asked for one more test: can the dancers alone give the groove's width? Every starting lag closes at its own bias, and the dancers standing at the groove's middle bias lean by an amount set by how steeply that bias changes there (SLOPE-2). If each hump of the wrinkle is a sine in the straightened lag, the dancers' lean gives the hump's height, and that height is the groove's width. Read that way on tables already on file (GROOVE-1), the dancers came up 3 to 6% short at strength 0.10. The first explanation was the hump's shape. It was wrong. Computed on starts spaced evenly in the straightened lag, the weightless selector's humps are sines to within 0.14% (SHAPE-1b). The shortfall came from the reading. The tables' starts are spaced evenly in the raw lag, which is coarse in the straightened lag just where most dancers stand, and a straight line between two starts undercounts a lean at its peak. Replaying that same reading on the selector, with no table, no weight and no gear, reproduces GROOVE-1's numbers within 0.4% at all nine cells (SHAPE-1c). Read cleanly, the dancers give their groove's width, and the weightless gear holds within about 0.4%; SLOPE-2's 1 to 2% excess was most likely the same grid. The selector also shows Bear's skating couple: its middle level is crossed twenty times instead of ten exactly between second tones -0.182 and -0.238 at strength 0.30, the edges of the weightless hand-off window found earlier from the grips alone (-0.1811 and -0.2394). At weight 0.005 the tables show the doubled count only from -0.200 to -0.228. Near the window's edges, where the humps change from five to ten, that weight is already too large for first order.

Part 5 — The substrates it drives through

> *a single tune whistled in a dozen rooms: a long glass tube where a beam of sparks hums it, a dish of tiny beads drifting over bars of light, a country's power lines turning in step, a crystal humming under a radio, a cluster of heart cells keeping the beat. Each room has its own furniture and its own echo, but the same tune comes through. In only one room has anyone left the windows open, so that you can record it yourself.*
>
> — no-physics read 122

5.1 Media that share the equation

Some real media obey the rider's equation, or its close cousin, and can test the ruler directly. A survey on 11 October listed twelve; seven of them are shown here. The ones marked "checked" were read against their own pages; the rest come from a literature search run with Claude, kept as reported until checked.

medium	what plays the ball's lag, the pull and the outside tick	measured landmarks	the cheapest ruler test
travelling-wave tube (Doveil, Macor and Elskens 2006; checked)	an electron's position; two waves; no drag	a devil's staircase; the onset of chaos	digitize the staircase and compare with our toy at very low drag
colloids on an optical washboard (Juniper et al. 2015; checked)	a particle's position; a steady and an alternating drive; nearly weightless	whole-number steps; Arnold tongues described	test the square-root onset and step widths; memory must be absent, a control
the power grid (Bonneville events, Ferraro 2021; checked)	a rotor's angle; mechanical power; electrical coupling; inertia and damping	the swing equation fitted from real events	done: §5.3
grid frequency, Continental Europe (from Jumar and colleagues' database of twelve synchronous areas, a 2020 preprint; checked against its arXiv record)	system frequency	none yet	background for the grid test
charge-density-wave crystals (Brown, Mozurkewich and Grüner 1984; abstract checked)	the wave's phase; steady field; pinning; radio field	subharmonic steps, a devil's staircase	digitize the step positions; test the trickle's $n^2 \times$ height
Josephson junctions (Kautz and Monaco 1985; abstract checked)	the phase difference; the bias current; the critical current; the microwave	chaos in the rf-biased junction	run on our own bench: §5.2
phase-locked loop hardware (from the search)	the phase error; frequency offset; detector; loop filter	lock range against capture range (memory)	place the loop on the memory curve

The weight decides which medium is our true twin. With no weight, an added drive and a moving ripple both lock only on whole numbers. At weight 0.05 the added drive locks on fractions just as the moving ripple does (ADDED-1). So a medium must be matched on weight before it can be called a twin.

5.2 The Josephson junction, run on our bench

The junction is the first real device whose standard equation shares the rider's time mirror. We ran that equation through the same instrument, with its capacitance playing the weight. In symbols, $\beta\,\psi'' + \psi' = B + A\cos\omega t - \cos\psi$, where ψ is the junction's phase less a quarter turn, B the bias (a new B: the steady push, which plays the toy's pull; the ledges' strength, called B in §2.1, is here the 1 in front of $\cos\psi$), A and ω the radio wave's strength and frequency, and β the capacitance. In words: the same balance of weight, drag, steady push and a push that comes and goes, except that the junction's radio push does not depend on where the phase sits.

What carried over. At every root, weight keeps exactly the two self-mirrored dances, and the stable one is again the far-side leaper. That held at 44 roots, across nine fractions and six settings of the radio wave (JJ-1 to JJ-4).

What does not carry over unchanged. At the settings first run, the leaper's pair sits inside the frozen points, not outside as on the toy's turned walkway, so red has $q - p + 1$ stops between them instead of $q - p - 1$ (the clock face of §4.25 says why). The guess that the pair would sit outside, as in the toy, failed at all seven roots of the first run; its redesign turned it into a rule for those settings. But it is not the junction's rule everywhere. At weak, fast drives the pair sits outside, exactly as on the toy's turned walkway, and red's count follows (JJ-SMALL-1, SIDE-MAP-1). Mapped over 48 more settings, the plane falls into three zones, each set by strength over frequency squared, A/ω^2 (figure 48):

- below about 0.12 the strobe has no frozen points at all, because even the least current is above the critical one;

- from there to about 0.33 to 0.42 (the flip drifts a little with the frequency, a reading from a straight line between neighbouring settings) the pair sits outside;
- above that it sits inside, at every grid and corner setting run.

The far-side leaper is the stable dance in all three zones, at all 48 settings, including the eight with no frozen points to land beside.

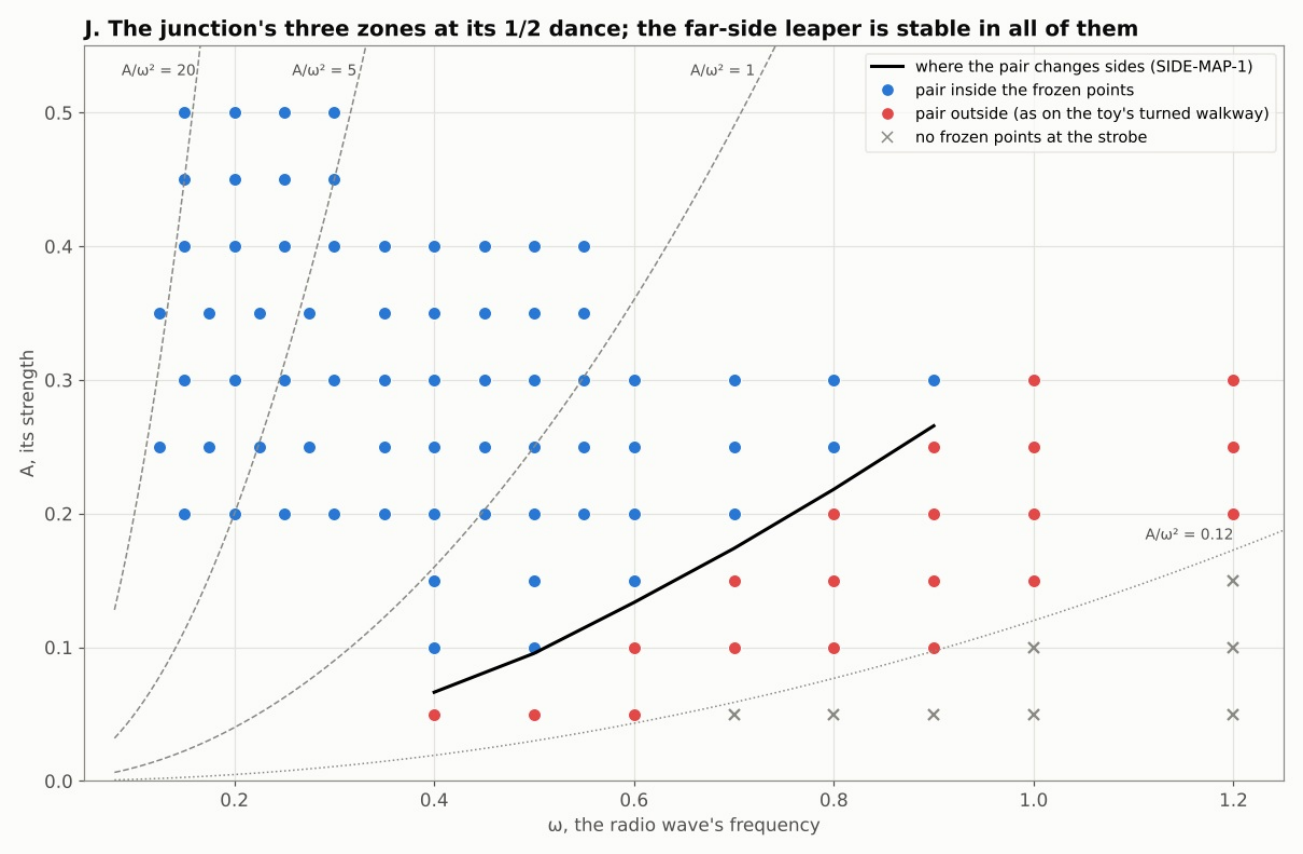


Figure 48. The junction’s three zones over its strength and frequency, at the 1/2 dance: the pair inside the frozen points (blue), outside them (red), or no frozen points at the strobe (crosses). The black line is where the pair changes sides; the dashed curves are equal A/ω².

One knob for the grip. The grip ratio (red’s M’ over purple’s) steps with the smaller of p and q − p, as in the toy, keeping 0.81 to 0.87 of itself at each step. Its size is set by one knob alone: the radio wave’s strength over its frequency squared, A/ω².

- The guess that strength over frequency set it failed at a fourth setting (JJ-3e).
- The four settings then lay on one line against A/ω², and that line predicted two new settings blind, chosen so that the two guesses disagreed (JJ-4a).
- Over a plane of 25 settings, six of them already on record, it held within 0.063 at the ten new settings inside the line’s range (JJ-GRID-1). Three very different settings with the same A/ω² agree within 0.028 (the same run).
- The line keeps working from 0.66 to 2.5 and bends only in the strong, slow corner (figure 49).
- Why A/ω²: at small strength the excess grip is first order in the strength, with a coefficient $-0.85/\omega^2 + 0.16/\omega$: a part that grows as one over the frequency squared, less a smaller part, of the other sign, that grows as one over the frequency. That was measured at five frequencies and confirmed blind at three more within 1.2% (SMALL-A-1, 2). The first term dominates at slow drives, which is why one knob served across the grid. Why the two numbers are what they are is not derived.

By contrast, where the pair sits and how fast it crosses the strobe answer to both dials, and follow no simple law.

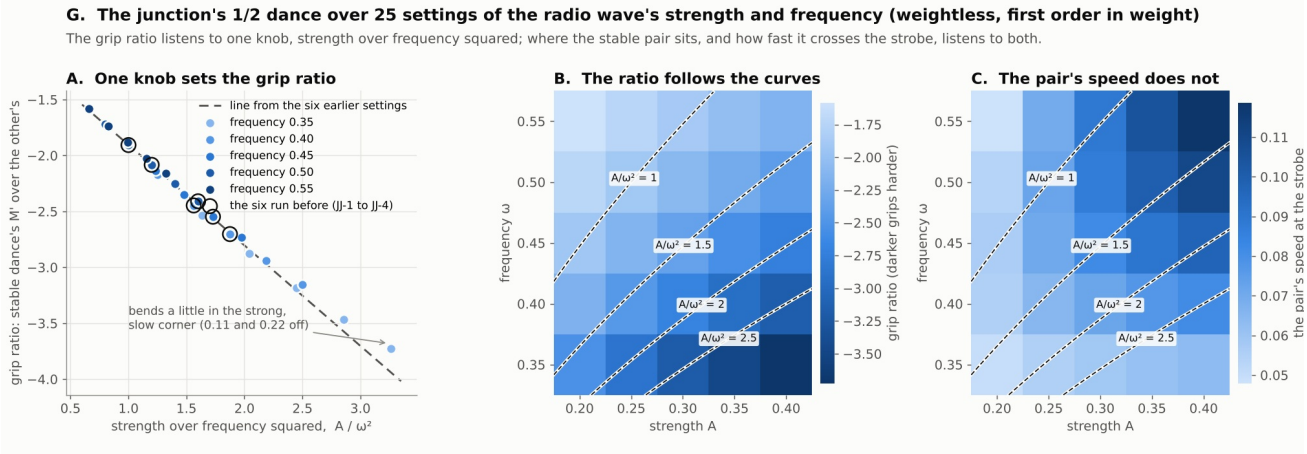


Figure 49. The junction's 1/2 dance over 25 settings of the radio wave's strength and frequency. A: the grip ratio against A/ω², with the line from the six earlier settings. B: the ratio over the plane, with curves of equal A/ω². C: the pair's speed, which does not follow those curves.

Into the strong, slow corner. The straight line bends where the drive is strong and slow, so the corner was run (figure 50), at strengths 0.20 to 0.50 and frequencies down to 0.10 (JJ-CORNER-1, 2):

- **The rules hold.** The selection rule and the pair inside held at every setting where the instrument could be trusted.
- **The pair never crosses there.** It closes on the frozen point roughly as the frequency squared (its offset over ω² stays between −0.070 and −0.106), so the margin shrinks toward zero without changing sign.
- **The knob survives where the line does not.** One smooth curve in A/ω² (a cubic in its logarithm) carries all 45 settings of the grid and the corner within 4.4%, out to A/ω² = 22, where the line is off by 46%. That curve then predicted eight new settings blind, within 4.5%.
- **The dance hides as the drive slows.** The 1/2 dance lives in a window of bias whose width falls exponentially in 1/ω, from 0.07 at the fast, weak edge to about 3 × 10^{-7} at strength 0.50 and frequency 0.15 (WINDOW-1, the run that measured this window of bias, not the hand-off window of §4.28; every residual under 0.04 in the logarithm). At frequency 0.10 the window is too narrow for our double-precision instrument to land in, so those settings are left out. The exponent grows with strength. The estimate tried after the data takes it from the stretch of each cycle where the current is below its threshold: half the sum, over that stretch, of the square root of one minus the current squared. That does not give it, so it is measured, not explained.

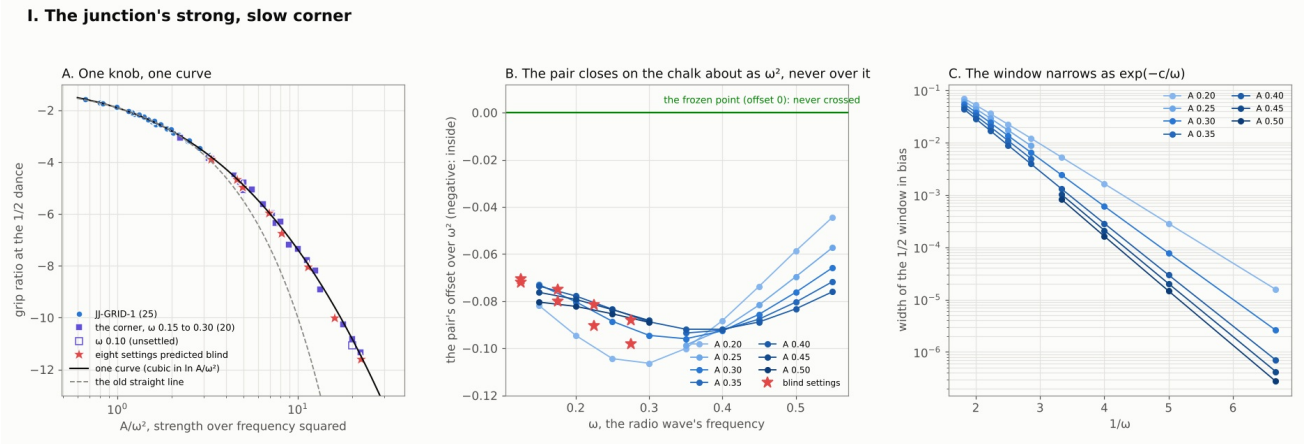


Figure 50. The junction's strong, slow corner. A: the grip ratio against A/ω², with the grid, the corner and eight settings predicted blind; the old straight line fails and one curve holds. B: the pair's offset over ω², which stays negative: the pair closes on the frozen point but never crosses. C: the 1/2 window's width in bias against 1/ω, a straight line on a logarithmic scale at every strength.

A second tone. Add a tone at twice the frequency, small enough that the strobe still falls at the least current, and the wave keeps its time mirror. At strength 0.25 and frequency 0.55, with the second tone at ±0.05, every rule above held at all fourteen rows: the far-side leaper, the pair inside, red's count. The frozen point slid toward the pair when the second tone added to the least current and away from it when the tone took away, and the pair drifted the same way a tenth to a sixth as far; both moves are straight lines in the tone (TWO-TONE-2). At strength 0.10, a second tone of

−0.0275 or −0.030 pushed the 3/5 grips through zero, and there the far-side leaper became the unstable dance (PINCH-1, §4.28). What changed was the gearing between dances: a hump in the tone with its top just below zero, falling steadily on the plus side and holding a shelf on the minus side before it drops, not a parabola (§4.28, TWO-TONE-1, 2, HUMP-1; figure 38).

The counts under weight. The count rule is a weightless rule, and weight is the term that breaks the mirror, so each dance was carried up in weight from 0.01 to 1 at two settings, one with the pair inside the frozen points (strength 0.25, frequency 0.55) and one with it outside (strength 0.10, frequency 0.60). A dance with weight lives in a tongue of bias, not at a single root, and where in its tongue the bias sits matters. Held at the weightless root (WEIGHT-WALK-1), the dances seemed to slide sideways and their counts to change somewhere between weight 0.01 and 0.2. With the bias following each tongue and the count read at its middle (WEIGHT-WALK-2), most of that slide disappeared where the pair sits inside: every count held at every weight the instrument could follow, 1/2 and 1/3 all the way to weight 1. Where the pair sits outside, the middles do slide: 1/2 and 1/3 each gained a stop by weight 0.3, and 2/5, whose outer stops sit only 0.006 of a turn outside their lines and move fast with weight, had changed by 0.01. At a tongue's edges, where the stable dance meets the unstable one, the count is one different at every weight. So with the pair inside, the rule holds at the middle of the tongue far beyond small weight, and with it outside it gives way between weight 0.1 and 0.3 (by 0.01 for the narrow 2/5). The narrowest tongues still slip away between rungs of the weight ladder, so the higher fractions were followed only part of the way, and by weight 1 the stable dance itself has become unstable. On a finer ladder, a sixth of a decade per rung, with each rung begun where the last two rungs put the middle (WEIGHT-WALK-3), the picture sharpens. Where the pair sits outside, 1/2, 1/3 and 2/3 all change their count at the middle on the same rung, weight 0.32, while each dance is still stable; for 1/2 and 1/3 that rung was called before the run from how far each stop sat from its line with no weight and how fast it moved at weight 0.01. Where the pair sits inside, no count changed while its dance was stable: 1/2 kept its count to weight 1, and the two that changed (1/3 at weight 0.68, 2/5 at 0.46) did so only after the dance being followed had turned unstable, its multiplier past 15. The same margin rule missed both narrow dances there, which held their counts past the rungs it named. The narrow tongues were followed further than before but not all the way: 2/3 at the inside setting was lost at weight 0.32 again, where its stops jump farther between rungs than the engine allows, a miss. The guess for the middle missed by more than a half-width twice, a miss; at the narrow tongues the edges the engine reports sit at its own probe points, so the widths there measure its steps more than the tongues; and the narrowest dance (2/5 at the outside setting) was followed one rung further on one computer than on the other, so its last rungs were not yet repeatable.

The counts under weight, followed further. An edge finder that walks out in small steps, halving its weight step where a dance is lost (WEIGHT-WALK-4), took these up: the two computers now reach the same weights with the same numbers, both 2/3 dances were followed to weight 1, and the outside 2/5 to about 0.079. At the outside setting the narrow tongues widen in step with the weight, by 1.47 a rung on a ladder whose rungs are 1.47 apart, at 2/5 on every rung, and at 2/3 until its dance turns unstable past weight 0.32, where that steady growth missed. With the pair inside, no count changed while the dance was stable at the two dances not run before, 3/5 and 3/7. Outside, the margin rule missed again, at 3/7, whose count changed at weight 0.22 where the rule called 0.68: its premise, that each stop keeps moving at its rate at weight 0.01, gives way. The guess for the middle from the last two weights missed by more than a half-width again, worst after a dance had turned unstable, as called, but also where the dance was still stable, against the call; there the miss grows about threefold a rung, near the rung's ratio cubed, pointing to a term past the weight squared that a guess from two weights leaves out. The outside 3/5 dance was lost early, near weight 0.0032, before the margin rule could be tried on it, a void. A search at six weights from there to 0.01 found it stable at every one, sliding down in bias as the weight grows, so the dance had not ended: the follower had turned it away, its stops having moved farther between weights than the follower's rule allows (LOST-3/5). With the lag kept within one turn, the stretch rate at all six weights and 81 biases stays within 2×10^{-4} a passing of zero, about what the dances' own grip allows and far below the 10^{-3} set before the run as the mark of chaos (LYAP-2); a first reading that crossed that mark came from rounding in a lag left to grow, and was ruled void.

What an experiment could check. A junction with a little capacitance, its fractional steps measured with their phase taken at the low point of the drive, would show the far-side rule directly. The Loop Atlas's pulsed needle, this same equation at weight 1 and frequency 1, showed the same placement first: its slow stops sit inside the frozen points and drift back, as here (its STRETCH_READS run). Its orbits are not the self-mirrored dances the count rule is about, so it says nothing of where that rule stops; where the rule stops as weight grows is open (Part 8).

Whose house this is. For the weightless junction, Buchstaber, Karpov and Tertychnyi (2010) and Ilyashenko, Ryzhov and Filimonov (2011) prove that it locks only at whole-number rotation numbers, and Glutsyuk has carried the work on since. Both papers were checked against their records on Math-Net.Ru, the Steklov Institute's journal portal, on 13 October and again on 31 October; Glutsyuk's later work was read only through a tool's summary. Ilyashenko, Ryzhov and Filimonov add, in their abstract (also read through a tool's summary), that the whole-number rule belongs to that exact equation: perturbing it in general opens locking at other rotation numbers. A fractional dance at zero weight therefore lives only at an isolated point, which is what our weightless roots are, and weight is one such perturbation: it opens each root into a tongue. If that reading holds, they proved the backbone of what we measured.

5.3 The power grid is the toy, exactly

A generator's swing equation is our rider with no ripple, exactly, not by analogy. Two integrations agree to 10^{-14} radian, which confirms the rescaling's algebra (MAP-1). The toy's weight is the machine's inertia times its peak power over its damping squared, and its pull is the mechanical power over the peak. A small swing's damping ratio ζ, its damping as a share of the damping that would just stop it ringing, obeys $4\,\zeta^2 \times \text{weight} \times \cos(\text{standing angle}) = 1$: four times the damping ratio squared, times the weight, times the cosine of the angle the machine stands at, makes one. Because the damping enters squared, a grid's place on the map is set mostly by how well its swings die out. At the memory switch (weight 0.7026) that damping ratio is about 0.60. Any machine whose swings die out more slowly sits on the memory side.

We fitted real events: Ferraro's thirteen Bonneville Power Administration recordings, ten events and three Chief Joseph brake tests (three meters, 60 readings a second). One event's file is damaged and one has no swing to fit. Of the other eleven, ten give clean ringdowns, and the third brake test's two good-shaped fits (0.394 and 0.397 Hz) agree with the first two brakes but fall short of the fit-quality bar (GRID-1c). The ten have damping ratios 0.039 to 0.166, at frequencies 0.375 to 0.416 Hz in nine of them. In the toy's units the grid's swing has a weight of 10 to 191, and it sits a quarter to a sixteenth of the way to the switch: deep on the memory side, as the textbooks expect of lightly damped power swings. This is a ruler check on real data, not a discovery.

5.4 Borrowed rulers

Three bodies, seismic rings, the Gaia loops and the integers do not share the equation: the islands of a light star around an eccentric binary (*Held, Rocking, Locked*), the loops walked round triangles of GPS stations over the Cascadia subduction zone (*Loop Walking Cascadia*) the loops read on the Gaia star catalogue (*Reading the Sky Through Loops*) and the loop walk through the integers (*Taking a Loop Walk Through the Integer Forest*), all the team's other work, listed under Sources checked. For them the pendulum certifies the ruler, not the mapping, and each needs its own check that its loop behaves like this one. Bear's three-body pictures taught Part 3's reading method: zoomed out, a slow rhythm appears that a short run cannot hold, and the pictures are mirror-symmetric. They stay pictures and lessons here, not tests; the tests belong to the three-body work.

5.5 The re-aim at a real medium

The toy pushes on nothing it carries (§1.2). Its next aim, which Bear ruled on 10 October belongs in this book, is a toy that pushes on something real, starting with a tether in orbit, with the gravity gradient as its medium. That toy has not been built yet. When it is, it will go round the same loop: a picture, a written guess, a reader first, a run, every line read, and every miss redesigned.

5.6 Where else the dancers' rules should be tried

Every medium in §5.1 that has inertia and a periodic drive can be asked the dancers' questions. The rule for this list is simple: what we can run ourselves, we run, and only what we cannot is left for others. The cheapest are listed first. Each says what plays the weight, what to change, what our rules predict, and who can try it.

medium	the weight	what to change	what our rules predict	can it be tried
Josephson junctions with capacitance (§5.2)	the capacitance	add a second microwave tone at twice the frequency	fractional Shapiro steps of numerator 2 and 3 shrink to almost nothing at a second tone growing as the square of the first, and past it the other branch of the step is the stable one; steps of numerator 1 change by only about a tenth (at strength 0.10, a second tone of −0.030 cut those grips by 11 to 13%)	ours, in the model, done: the pinch moves as the square of the first tone (FAR-1), the swap survives real weight (PINCH-2), the hold is handed across by a walking pair (WALK-1). The dip fills in as the capacitance grows: at strength 0.10 the 3/5 step narrows 48 times at the pinch at weight 0.005, 10 times at 0.03 and 3 times at 0.1, and at that strength the steps are a millionth of the bias wide, far too narrow for a lab. Mapped further, at strengths 0.20 and 0.30 (LAB-MAP-1 and 1b): the dip's depth is about 0.2 to 0.33 divided by the weight at every strength, 7 to 10 at weight 0.03 and about 3.3 at 0.1, and at weight 0.3 no swap was found in the scanned tones, the dance through lag 0 stable throughout. Predicted from that law before they ran, weights 0.15 and 0.2 at strength 0.20 gave depths 2.19 and 1.66, about a third over the weight each time; the swap moved outward with the weight, from −0.092 at weight 0.03 to −0.114 at 0.2, and at 0.3 it is gone (LAB-MAP-2). Carried tone by tone to −0.30, every start settling (LAB-MAP-3), the multiplier of the dance through lag 0 rises to 0.9625 near −0.17 and turns back down, to 0.742 at −0.30, while the dance through lag 1/2 stays unstable throughout: the hold never passes across. That was written down before the run, from the turn-over the earlier scan had begun to show. At weight 0.1 the 3/5

				step is 1.0 to 1.6 ten-thousandths of the critical current wide, a third of that at the pinch. So the lab test needs a junction whose weight (its Stewart–McCumber parameter) is 0.1 or less (junctions shunted from within, made for electronics, are reviewed by Belogolovskii and colleagues, 2017; whether theirs reach 0.1 is not checked here) and a measurement that resolves steps near a ten-thousandth of the critical current; whether that is within reach, and with what noise, is for others, in a lab with a two-tone source
phase-locked loops and injection-locked oscillators	the loop filter's lag	give the reference's frequency a tone and its octave	sub-harmonic lock ranges p/q with p of 2 or more pinch; the stable phase of the lock jumps to the other branch	ours for the loop, done in a behavioural circuit model with a real multiplier detector and an RC filter, the carrier's ripple included (PLL-2): at strength 0.10 and weight 0.03 the 3/5 lock pinches and swaps as in the junction. The detector's ripple pushes the whole lock down in bias, by 0.043 with a carrier ten times the tone and by 4×10^{-7} at a thousand times, the size a one-line estimate of the ripple's steady push gives, but moves the swap by at most 0.002, and by under 0.0005 once the carrier is fifty times the tone. The injection-locked oscillator is not yet tried (Part 8); in hardware by others
charge-density-wave crystals	the wave's inertia	two radio tones	subharmonic steps pinch the same way, if the crystal's pull is close to a single sine	by others; from the literature search of §5.1, its abstract checked
colloids on an optical washboard	almost none	a second drive tone	with no inertia and a purely sinusoidal trap, fractional steps should be absent (every weightless root is pinched); any fractional step seen comes from the trap's own harmonics or a little inertia, and a two-tone drive tells those apart	by others
the power grid's swing equation (§5.3)	the rotors' inertia	none in practice	real grids sit on whole-number locks, so the pinch should not matter there	the least likely to show anything
tuning and music	none	none	the comma is a picture, not a medium: nothing locks	no

5.7 A rider moved by two waves

The junction can also be steered. Give it a second radio tone at twice the frequency and it slips one way more easily than the other, with the direction set by the timing between the two tones. That result, the toy's first sight of it, and a manual for every knob are the whole of Part 6.

Part 6 — The move: a rider moved by two waves, and how to drive it

> *the idea in my head was the phase offset or quadrature between the blades/pads doing work*
>
> — Bear, 15 October 2026

Everything before this part says what a rider does. This part says how to make one go where you want: which knobs to turn, how far, and what each one costs. It is written as a manual. First comes the question and the wall it has to respect, then what the bench measured, then every knob one at a time, and last a set of instructions you could follow at a workbench. The rider is the Josephson junction of §5.2, run on our bench, because it has the toy's weight, honey and dips, and because a laboratory can test it.

A few words first. **The move** is slipping one way more easily than the other. **The step** is the band of steady pushes over which the rider stays locked and does not slip at all. Its two **edges** are the pushes at which it starts to slip, forward (B+) or backward (B−, a negative number). **The lopsidedness** is B+ + B−, the forward edge less the size of the backward one: zero for a fair step, negative when forward is easier. **Quadrature** is a quarter turn between two tones. **The zero** is the timing between the two tones at which the step is fair again.

6.1 The question, and the wall

Bear asked whether two waves with a phase between them could move a rider one way: a skater whose two blades, or a vehicle whose two pads, work in step but out of phase. They can, and the effect is known and used. A force made of a tone and its second harmonic, averaging to nothing, moves cold atoms through a symmetric lattice of light in a direction set by the relative phase (Schiavoni, Sanchez-Palencia, Renzoni and Grynberg); Flach, Yevtushenko and Zolotaryuk (2000) worked out the symmetry reason. A travelling-wave ultrasonic motor adds two standing waves 90 degrees apart in space and time, so the stator's surface points travel in ellipses and push the rotor along. In each, the push goes into something held, the lattice or the stator: nothing here moves a closed box. That is the wall of §1.2 again, and this part keeps to it. The rider below pushes on its floor, and the floor is held by whatever holds the laboratory down.

The toy saw it first. Bear had drawn this on 4 October, as a wheel pulled by two chains in turn, each taking a quarter turn and handing off, always 90 degrees apart. The toy's relay was then fed only a tone and its octave, with the octave's phase turned in 72 steps (the toy's two-tone run of 4 October, tests TT1 and TT3). With no weight, the second stage was left unharmed over about half the circle of phases, and that half was centred within 15 degrees of 90 at all 17 readable points, 13 of them at exactly 90. Given weight with its drag held fixed, the half-circle slid: by a median of 5 degrees at weight 0.1 and 17.5 at weight 0.3. That is the same shape the junction shows, eleven days later and in another medium: a timing pinned on a quarter turn when the rider has no weight, and slid off it by weight.

6.2 The lopsided step

In symbols, with ψ the junction's phase less a quarter turn:

$$\beta\,\psi'' + \psi' = B + A\cos\omega t + A_2\cos(2\omega t + \varphi) - \cos\psi.$$

In words: it is the junction's equation of §5.2 with a second radio tone added. That tone runs at exactly twice the frequency, has strength A_2, and is turned ahead of the first by an angle φ, the timing (in Part 6, φ always means this timing, not the golden ratio). As before, β is the weight, B the steady push (the bias current), and A and ω the first tone's strength and frequency. Every push and strength is measured as a fraction of the junction's critical current. With the second tone turned off the step is fair: both edges sit at 0.9331, and the lopsidedness is 4×10^{-16} (RATCHET-1b's check run with no second tone), which is zero to the computer's last digit.

The junction shows the move exactly (RATCHET-1b, figure 51). With a second tone at a relative phase φ, the zero-slip step stops being symmetric. At strength 0.30 and a second tone of 0.10, in phase, it reaches 0.9241 forward and 0.9378 backward, so a push that starts it slipping one way is too small to start it the other way. The lopsidedness follows $\cos\varphi$ to within 0.03% and vanishes exactly in quadrature, as the equation's own symmetries require when the rider has no weight. The same law is derived for the device itself, read as a diode: at small drive the lopsidedness of the junction's zero-voltage step goes as the cosine of the timing (Scheer, Seoane Souto, Hassler and Danon, 2025; §6.9). The second tone used throughout this book, in phase, is the most lopsided setting. With weight that symmetry breaks, and the zero moves off quadrature, as the next paragraph measures. The first attempt read the lock wrongly at phases other than 0 and π, and the exact mirror symmetry registered beforehand caught the error at once (RATCHET-1).

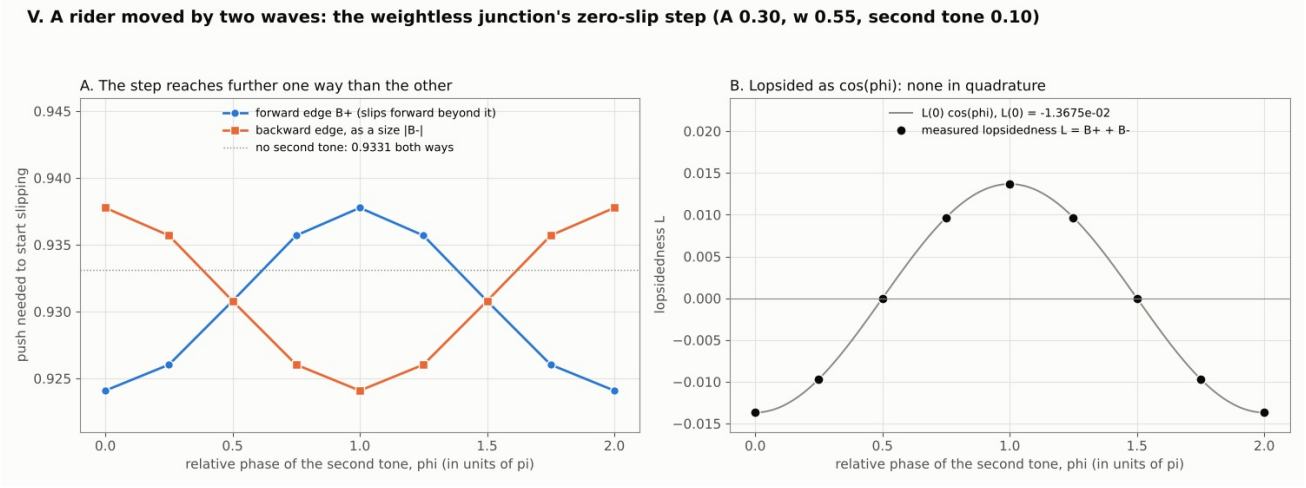


Figure 51. A rider moved by two waves: the weightless junction's zero-slip step at strength 0.30, frequency 0.55, second tone 0.10, against the second tone's relative phase φ. A: the push needed to start slipping forward (B+, blue) and backward (the size of B−, orange); dotted, the symmetric step with no second tone. B: the lopsidedness B+ + B−, and its value at φ = 0 times cos φ (grey). In phase the rider slips forward more easily; in quadrature the step is symmetric. From RATCHET-1b.

6.3 Weight slides the zero

With weight, the zero slides. Give the rider weight and the step's lopsidedness is still one clean cosine of φ, but the whole wave slides along the circle of phases (RATCHET-2, figure 52). Two exact symmetries of the equation say how it must move. Flipping the slope and shifting the second tone by half a turn gives the mirror: the lopsidedness at φ + π is exactly minus that at φ, at every weight. Running time backwards and turning the lag over changes the weight's sign, so the part that goes as cos φ can change only as the weight squared, while the part that goes as sin φ, the part that moves the zero, must grow in proportion to the weight. Both held, each registered before the run: doubling a light weight doubled the sin φ part (a ratio of 2.000 to within two parts in ten thousand, at three phases) and quadrupled the change in the cos φ part (a ratio of 4.000); the mirror held to every digit; and at every weight up to 0.32 one cosine fits all sixteen phases to within two parts in ten thousand of its height. At light weights the zero slides about 59 degrees of φ per unit of weight: 0.147 degrees at weight 0.0025 and 2.34 at 0.04. By weight 0.32 it has slid 16.9 degrees and by 0.64 it has slid 25.2, so the slide grows more slowly than the weight once the rider is heavy. The rectifier hardly weakens on the way: the cosine's height is 0.01367 with no weight and 0.01342 at weight 0.32. Weight turns this rectifier rather than weakening it.

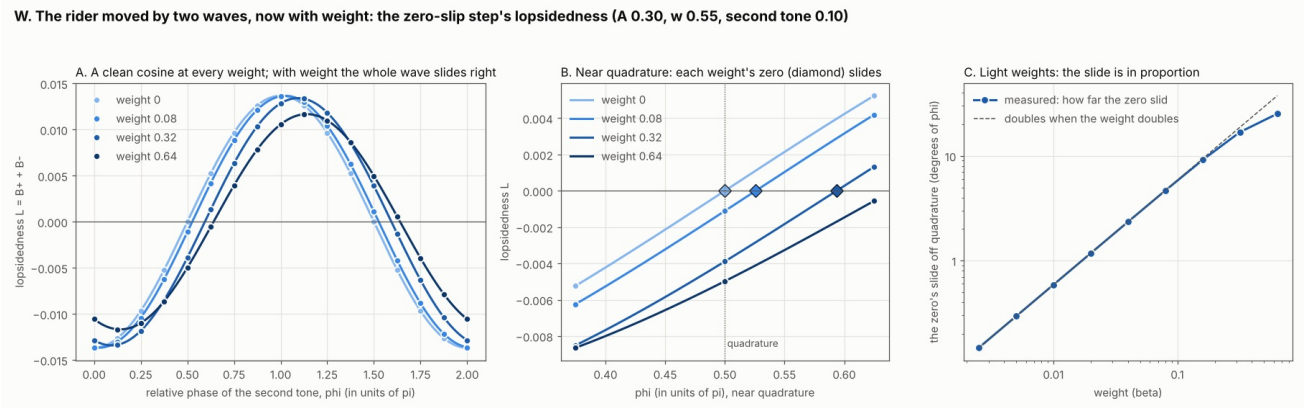


Figure 52. The rider moved by two waves, now with weight: the junction's zero-slip step at strength 0.30, frequency 0.55, second tone 0.10. A: the lopsidedness B+ + B− around the circle of the second tone's phase φ, at weights 0, 0.08, 0.32 and 0.64 (light blue to dark); dots are the sixteen measured phases, lines the one cosine fitted to them. B: the same near quadrature, magnified, with each weight's zero marked by a diamond (weight 0.64's zero lies just beyond the panel, at 0.64 π). C: how far the zero has slid from quadrature against the weight, both scales logarithmic; the dashed guide doubles when the weight doubles. From RATCHET-2.

6.4 Out and back

The cold atoms tell the same symmetry story from the other side: there, dissipation breaks the symmetry and lets a current flow (Gommers, Bergamini and Renzoni, 2005). Here the weightless rider, all honey, is held symmetric, and the equation says a rider with weight and no honey at all would be held symmetric too, because then time runs backwards exactly. The zero moves only when weight and honey are both present. So somewhere in between the slide must reach a largest value, and it does (RATCHET-3, figure 53). Followed in small steps from no weight to weight 20, the zero slides out, peaks at 27.4 degrees near weight 1.1, and comes back home: 3.4 degrees by weight 20. Both tails are clean. At light weights the slide grows in proportion to the weight; at heavy weights it falls as one over the weight, slide times weight settling near 69 degrees. The simplest hump with both tails, drawn from the two end weights alone, peaks at weight 1.09 and 29.1 degrees, close to the measured peak but a little too tall. On the heavy side the rectifier also fades, its height falling nearly as the weight to the minus fourth: by weight 20 it is a sixteen-thousandth of its weightless height, and its zero still comes home on schedule. Written down first from the steps before it, each about 0.69 the size of the one before, so that the powers close on −4 from above, the next rung's power was to be −3.977; at weight 24.5 it came −3.9771, and at all seven heavy weights from 8.2 to 24.5 the height stays in proportion to the second tone's strength to one part in ten thousand (RATCHET-80): a minus-fourth law, reached from above, with the second tone entering once. One would expect that fading if a heavy rider, slow to respond, feels the drive as fast and averages the two tones away; that reading is not yet tested. So the timing between the two tones that gives no drift depends on how heavy the rider is against its honey, it departs most from quadrature when the two are matched, and it returns to quadrature at both extremes. As before, every push goes into whatever holds the floor still.

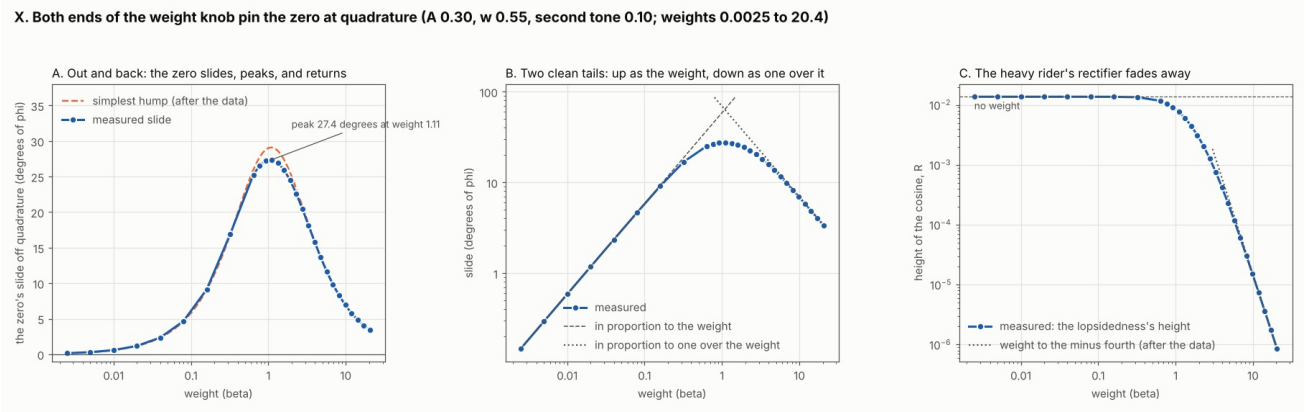


Figure 53. Both ends of the weight knob pin the zero at quadrature: the junction at strength 0.30, frequency 0.55, second tone 0.10, followed from weight 0.0025 to 20.4 in steps of a factor of 1.2 above 0.64. A: the zero's slide off quadrature against the weight (log scale): out and back, peaking at 27.4 degrees near weight 1.1; dashed orange, the simplest hump with both tails, read after the data. B: the same on log-log scales, with guides in proportion to the weight and to one over it. C: the height of the lopsidedness's cosine, flat to about weight 0.3 and then falling nearly as the weight to the minus fourth (guide read after the data). From RATCHET-3.

6.5 Two numbers set the zero

What sets the peak. Run at two other drive frequencies, the out-and-back hump is there every time, and the mirror holds exactly at every weight (RATCHET-4, figure 54). Far on the heavy side all three frequencies follow one law: the slide times the weight times the drive's frequency settles near 38 degrees (37.96, 38.07 and 38.11 at weight 20.4), so a heavy rider's zero depends only on how its settling time compares with the drive's period. The peak follows the same rule where the drive is fast: the peak's weight times the frequency, registered beforehand, matched within half a percent at frequency 0.70. At frequency 0.40 it came 14% high, outside the registered band, and the peak is lower, 22.8 degrees against 27.4. At that frequency the drive is also stronger compared with its frequency, and the light end follows that strength, not the frequency alone. So the run was repeated at frequency 0.40 with the drive weakened to the same ratio as at 0.55 (RATCHET-5, orange in figure 54). Then the peak's weight times the frequency came to 0.5784 against 0.5782, the peak to 27.1 degrees against 27.4, and the whole curve, drawn against the frequency times the weight, lies within half a degree of the curve at 0.55 at every weight, a tenth of a degree at the median. The miss belonged to the drive's strength. The rectifier's zero is set by two numbers: the weight times the frequency, which compares the rider's settling time with the drive's beat, and the drive's strength over the frequency squared. The two rivals, a resonance with the rider's own swing and a peak that does not move, both failed.

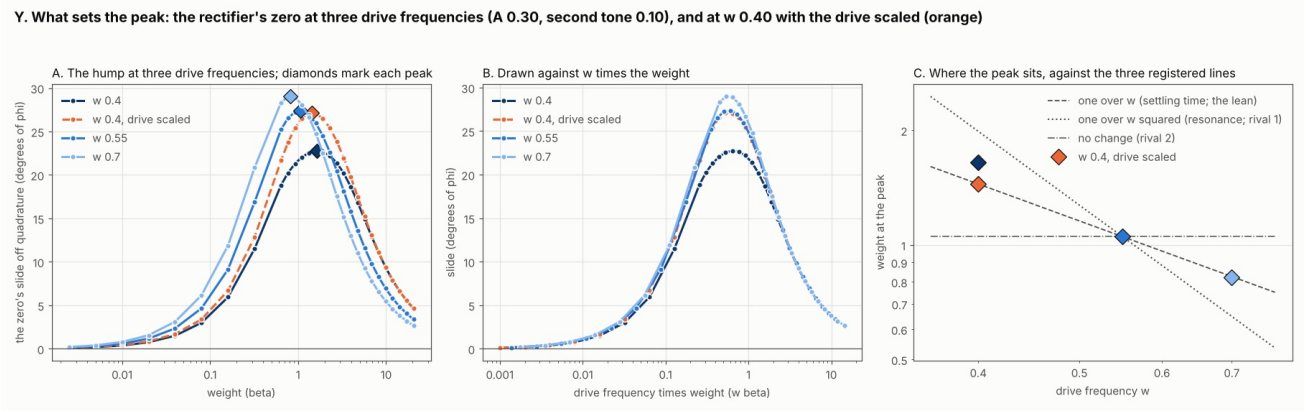


Figure 54. What sets the peak: the junction's rectifier at drive frequencies 0.40, 0.55 and 0.70 (dark to light), strength 0.30, second tone 0.10. A: the zero's slide against the weight, each peak marked by a diamond. B: the same against the frequency times the weight; the heavy sides fall onto one curve. C: the peak's weight against the frequency, with the three lines registered before the runs: one over the frequency (dashed), one over its square (dotted) and no change (dash-dot). Orange, dashed: frequency 0.40 with the drive scaled to the same strength-to-frequency-squared ratio as at 0.55; its curve lies on 0.55's in B and its peak on the one-over-frequency line in C. From RATCHET-4 and RATCHET-5.

Two numbers, tested from the other side. If the zero depends only on the weight times the frequency and on A/ω^2, then raising the drive at frequency 0.55 to the ratio that frequency 0.40 had at full strength (1.875) should move 0.55's curve onto 0.40's. It did, within the bands written down before the run (RATCHET-6, figure 55). The peak's weight times the frequency came to 0.634 against 0.658, the peak to 23.6 degrees against 22.8, and the whole curve, drawn against the frequency times the weight, lies within a degree of 0.40's (0.19 at the median); the light end and the heavy tail landed too. The rival, a peak whose place is set by the frequency times the weight alone, failed. The fit is looser than on the first ratio, where it was half a degree, and on both ratios the faster run peaks a little higher and a little earlier, so the frequency keeps a small say of its own. A third frequency on the first ratio, 0.70 with its drive raised, then landed within half a degree of 0.55's curve and peaked 0.35 degree higher, both as written down before it ran (RATCHET-7). At that ratio the peak climbs steadily with the frequency: 27.1, 27.4 and 27.7 degrees at 0.40, 0.55 and 0.70. One guess failed: that the rectifier's size grows in proportion to the drive along a ratio line. Scaled down by 0.53 its height went to 0.54 of itself; scaled up by 1.89 it went to 1.82: a little slower than the drive, and more so as the drive grows. Registered that way, it held at the third frequency: scaled by 1.62, the height went to 1.58. So the timing that gives no drift is set, very nearly, by two numbers; how strong the drift is when the timing is off depends a little on the frequency as well. Along the ratio itself, at frequency 0.55, the peak falls at every step run, from 28.9 degrees at ratio 0.612 to 21.2 at 2.75, while its place moves from weight times frequency 0.575 to 0.711 (RATCHET-83). The fall is steepest near ratio 1.3 to 1.5, about 4.5 degrees per unit of ratio, and slows past it, to 2.4 between 2.25 and 2.75. Written first, a curve through three ratios placed 1.30 and 1.50 within a quarter of a degree, but it said the fall would steepen past 1.875, and it slowed instead. The last ratio, 3.25, rerun with a guard that keeps each edge of the step on its own side (R83-e; a first run had let one edge jump onto the other in two of its sixteen timings), peaks at 20.4 degrees at weight times frequency 0.734: the fall goes on at every step, slowing again, to 1.6 degrees per unit of ratio, and the peak lands within 0.04 degree of the 20.43 that a curve through ratios 1.50 to 2.75 put there before the run; the straight line's 20.01 missed. Run past it at ratios 3.75, 4.0 and 4.25 (R83-F), the peak goes on falling, to 19.91, 19.77 and 19.71 degrees, each step's fall per unit of ratio about half the last (0.98, 0.54, 0.25): the curve written first that had the fall level off near 4.0 missed, and the steeper one held, within 0.1 degree. At 4.5 the fall ends: the peak comes back up to 19.73 degrees, 0.05 above where a fall that went on halving would have put it, so the lowest peak lies between ratios 4.25 and 4.5, while the peak's place, weight times frequency, which rose to 0.734 at ratio 3.25, has turned and falls, to 0.685 (R83-G). Between them, at 4.375, three forecasts written first held: the peak is 19.707 degrees, 0.0005 below a parabola through the peaks at 4.0, 4.25 and 4.5; its place, weight times frequency, is 0.698; and the window's weightless height sits on the straight line drawn for it, within 0.0001. So the fall bottoms out near ratio 4.32, at about 19.706 degrees (R83-H2; the first run of 4.375, R83-H, was lost to a stop rule of ours stricter than the reader, and was run once more). At 5.0 the engine could not follow the window's edges past a weight of about 1.6 to 1.9 at seven of its sixteen timings (R83-G). The guess written first was that the window closes there. Followed on a finer ladder of weights, it is open instead, 1.6 to 1.9 times its weightless width and still widening, so what stops the engine is its own limit on how far an edge may move in one step (R83-H). The peak at 5.0 lies past the weights followed at every timing, and that ratio is still not read.

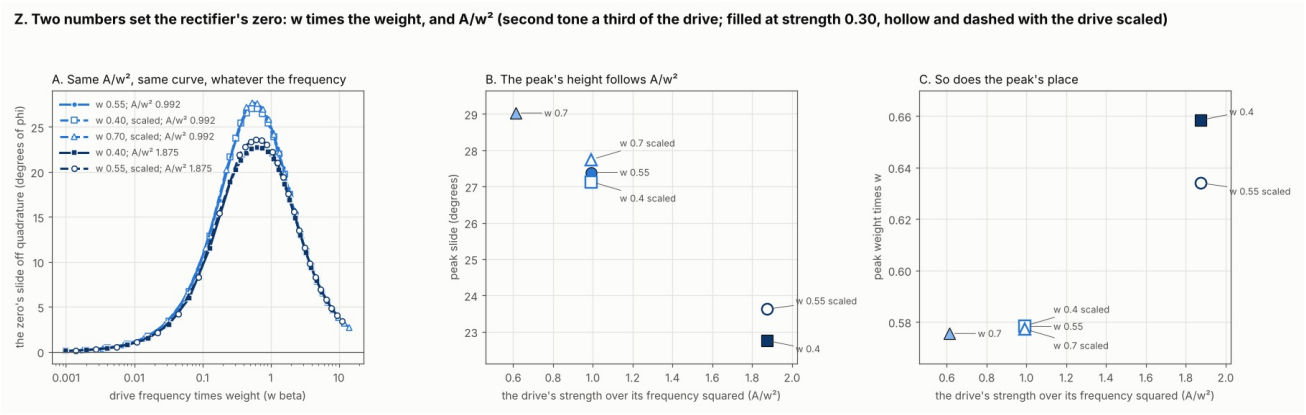


Figure 55. Two numbers set the rectifier's zero: the junction at strength 0.30 with a second tone of 0.10 at frequencies 0.40, 0.55 and 0.70 (filled), and three runs with the drive scaled (hollow, dashed) to borrow another frequency's strength over frequency squared, A/ω². Colour is A/ω², light to dark: 0.61, 0.99, 1.875; the marker names the frequency (square 0.40, circle 0.55, triangle 0.70). A: the zero's slide against the frequency times the weight; runs with the same A/ω² fall on one curve, within half a degree at 0.99 (three frequencies) and within one degree at 1.875. B: each peak's slide against A/ω². C: each peak's weight times the frequency against A/ω². From RATCHET-3 to RATCHET-7.

6.6 Which note: a gear and a half-turn

Knob 6 (§6.7) found that the second tone's note matters. At twice the first tone's frequency it steers; at three times it steers nothing; at four times it steers weakly, and its zero wanders far. This section follows the wanderer. It keeps three rules, and the third is a whole number.

Why four times is weak. To give a steady lopsided push, the second tone has to beat against the first. One way to picture it: the junction bends the first tone into copies of itself at twice, three times and four times its frequency, and the second tone can line up only with the copy at its own note. A copy at four times the frequency takes the first tone four times over, where a copy at twice takes it twice. So with a weak drive the window should grow as the first tone's strength A to the power of the harmonic number: A squared for the second harmonic, A to the fourth for the fourth. That is a picture, not a derivation; the measurement is the law. Halving the drive from 0.15 to 0.075 shrank the two windows by powers of 1.91 and 3.88. Between 0.15 and 0.30, where the drive is no longer weak, the powers were 1.70 and 3.60 (RATCHET-10 and RATCHET-10b). At every strength read, the fourth harmonic carried about two more powers of A than the second, 1.97 and 1.90. That is why it is 118 times weaker at strength 0.30.

first tone's strength A	second harmonic's window, no weight	fourth harmonic's window, no weight
0.075	0.001125	6.49×10^{-7}
0.15	0.004222	9.56×10^{-6}
0.30	0.01367	1.16×10^{-4}

Frequency 0.55, second tone 0.10; windows in units of the critical current.

The gear. Bear drew two pencil lines on figure 57 and wrote beside them that the orange curve looked like a change of rate of the blue. The numbers say something close to that. While the rider is light, the fourth harmonic's zero slides a fixed multiple of the second's, as if the two were geared. At frequency 0.55 the multiple is 5.07, 5.11 and 5.17 at drive ratios A/ω² of 0.50, 0.99 and 1.875, and at every drive it holds to weight 0.64, where Bear drew his second line, and then breaks (figure 56, B). The gear is not the harmonics' alone, though. On one drive ratio, 0.99, it falls steadily as the frequency rises: 5.58 at frequency 0.40, 5.11 at 0.55, 4.73 at 0.70 and 4.44 at 0.85 (RATCHET-11 to 13). The run at 0.70 was registered expecting the gear to stay at 5.1, and it missed; its redesigns at 0.40 and 0.85 were registered expecting 5.3 to 5.7 and 4.2 to 4.6, from the trend, and both held. The middle of the gear tilts with the frequency too: at 0.40 it climbs to 6.38 by weight 0.64 before it breaks, at 0.55 it stays flat, at 0.70 it sags from the lightest weights, and at 0.85 it sags further, to 0.76 of itself by weight 0.32 (registered below 0.90, held). So the fourth harmonic's light end does not follow the two-number rule of §6.5 as the second's does. The frequency has a say of its own here, and a bigger one than it has for the second harmonic.

Coming home. At the heavy end both zeros come home as one over the weight times the frequency. The second harmonic's comes home to quadrature as about 38 degrees divided by ωβ (§6.5). The fourth's comes home to the opposite pole, −180 degrees, as about 181 degrees divided by ωβ. That number was predicted from frequency 0.55 before the runs at 0.40, 0.70 and 0.85, and they gave medians of 183.2, 180.4 and 179.3 (RATCHET-11 to 13; figure 56, C shows 0.40 to 0.70). The drive's strength does not matter at that end: at weight 9.86 and frequency 0.55, the

fourth's zero sat at −146.37, −146.52 and −146.55 degrees at the three drive ratios. Only the middle of the path, its peak, moves with the drive: +137.5, +129.3 and +110.1 degrees, falling as the drive ratio grows, as the second harmonic's peak does.

The half-turn. Picture the timing knob as a clock face, a circle of timings. Every setting of the other knobs has its own clock face, and on each the zero is a hand, with a partner hand half a turn away (the mirror of §6.3). Line the clock faces up along the weight knob, from no weight to very heavy, and the hand traces a path, like a ribbon twisting along a rod. Mathematicians call such a stack of circles over a line of settings a fibre bundle: the circles are the fibres and the line of settings is the base. Bundles are where holonomy lives, the thread that runs through this whole series of books: what matters is how something laid along the base turns on its fibres as it goes, and how far it has turned when it arrives. Here the base is a line whose two ends are pinned, so the hand's whole turn from one end to the other is a count. In §6.12 the base closes into a true loop, the third tone's timing, and the count becomes a winding: how many times the push's direction turns all the way round. Two symmetries pin the hand to the real line at both ends of the rod. With no weight, running time backwards pins it (§6.3); with weight but no honey, running time backwards pins it again (§6.4). So the ribbon's whole twist, from one end of the weight knob to the other, must be a whole number of half-turns. That is a count, not a measurement: it cannot drift a little. The second harmonic's ribbon twists out 27 degrees and back, a count of zero. The fourth's twists out to +129 degrees, comes back through the start near weight 2, and runs on round the other way to −180, a count of minus one half-turn (figure 56, A). The same half-turn appeared at all nine settings run: three drive ratios at frequency 0.55, and the drive ratio 0.99 at frequencies 0.40, 0.70, 0.75, 0.775, 0.80 and 0.85. At 0.785 the weights were run only to 0.64, to place the seam (RATCHET-17), so its count was not read.

A whole number cannot change smoothly. As the drive is turned the ribbon can bend and its peak can rise or fall, but its count can change only where the window closes completely somewhere along the rod, so that the hand has nowhere to point. Such a place would be a dark point: a setting where the fourth harmonic steers nothing at any timing. The count does not change between any two of the settings run, so none lies between them, unless a pair of opposite dark points cancels there, which only a finer scan would rule out. Whether the count grows with the harmonic, minus two half-turns at six times the frequency, say, is open (Part 8).

What it means at the bench. Hold the timing in phase and the easy direction flips each time the zero passes a quarter turn. The second harmonic's zero never does, so its in-phase push keeps one direction at every weight. The fourth harmonic's zero passes a quarter turn three times between no weight and weight 20 at frequency 0.55: between weights 0.32 and 0.64, again between 1.33 and 1.59, and once more between 3.30 and 3.96. A source with a fourth-harmonic second tone, or a stray fourth harmonic, therefore pushes a light rider and a heavy one in opposite directions at the same timing. The pattern itself has a seam. At frequency 0.85 the zero's path no longer reaches a quarter turn on its way out (its largest slide on the weights run is +77.7 degrees), so the in-phase push reverses only once: the first two reversals have met and vanished together. Runs at 0.75, 0.775, 0.785 and 0.80 then located it (RATCHET-14 to 17, figure 58). On fine ladders of weights the path's highest point is +95.5 degrees at 0.75, +91.3 at 0.775, +89.5 at 0.785 and +86.9 at 0.80, falling steadily. So the seam sits between 0.775 and 0.785, near frequency 0.783 (§6.10), where the path just touches the quarter turn. Across it the pattern of reversals changes, and the half-turn count does not. The usual ladder at 0.75 had missed the excursion entirely, its highest rung only +85.7; §6.10 counts that as a wall of the ruler. Since the heavy end follows the weight times the frequency, a fixed junction should meet the same reversals as its drive frequency is swept. How exactly the light end moves with the frequency is the part not yet pinned.

A ratchet's direction reversing is known in the ratchet literature, with the mass or the friction as the knob too: an inertial ratchet's current reverses several times as the drive grows (Jung, Kissner and Hänggi, 1996, checked against its abstract), reverses as the particle's mass changes in a ratchet driven by coloured noise (Lindner, Schimansky-Geier, Reimann, Hänggi and Nagaoka, 1999, its abstract and part of its introduction read), and reverses as the friction per unit of mass changes in a driven one (Carusela, Fendrik and Romanelli, 2009, checked against its abstract); and in a damped, driven ratchet a change in the attractors reverses it (Denisov, Flach and Hänggi, 2014, their §3.5.2). What this section adds, on one real device's standard model, is that here the reversal is a count pinned by two symmetries, the same at every drive run, with a gear of about five to one to the second harmonic while the rider is light. Whether the count has been named before is a question for someone who knows the dissipative-ratchet literature.

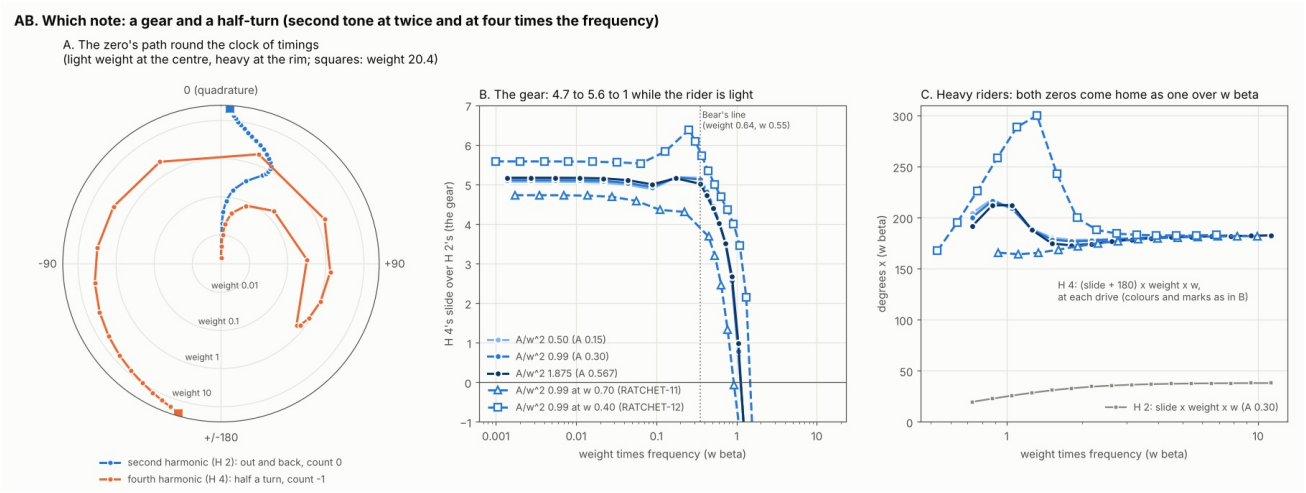


Figure 56. Which note: a gear and a half-turn. The junction's rectifier with the second tone at twice the frequency (H 2) and at four times (H 4). A: the zero's path round the clock face of timings, one ring per weight, light at the centre and heavy at the rim (log scale), at strength 0.30 and frequency 0.55; the second harmonic's zero (blue) goes out and back, a count of zero; the fourth's (orange) swings round and ends half a turn away, a count of minus one half-turn; squares mark weight 20.4. B: the gear, the fourth harmonic's slide over the second's, against the weight times the frequency, at drive ratios A/ω² 0.50, 0.99 and 1.875 at frequency 0.55 (filled circles, light to dark) and at 0.99 at frequencies 0.70 (hollow triangles, dashed) and 0.40 (hollow squares, dashed); the dotted line is Bear's pencil line at weight 0.64, frequency 0.55. C: the heavy end, the fourth harmonic's (slide + 180 degrees) times the weight times the frequency, settling near 181 at every setting; grey, the second harmonic's slide times the weight times the frequency, settling near 38. From RATCHET-3, 5, 6, 7, 8, 10, 11 and 12.

6.7 Every knob, one by one

The move has ten knobs. Here is each one: what it is in a laboratory, what it does, what was measured, how to set it, and what is still missing.

Scientists use knobs too, under other names. In the mathematics of dynamics a knob is a control parameter, and the map of what a system does across its knobs is its bifurcation diagram. A setting where something stops caring about a disturbance, to first order, has a name in each field. To a watchmaker it is a crystal's turnover point, and to a circuit designer a reference's zero-temperature-coefficient point. An early superconducting quantum circuit, the quantronium, was run at a saddle point of its two bias knobs, "an ideal working point" where its frequency is "independent, to first order, of the bias parameters" (Vion and colleagues, 2002; the paper's figure caption read); the field now often calls such points sweet spots. Optical lattice clocks hold their atoms in light of a magic wavelength, where the light shifts both clock levels alike and the shift cancels (Katori, Takamoto, Pal'chikov and Ovsiannikov, 2003). Industrial statisticians call the whole method robust parameter design: set the control factors, the knobs you can turn, so that the output holds steady while the noise factors, the things you cannot hold, wander (Taguchi; as summarized in a reference work). In §6.11 the rider's weight is the noise factor. And a deliberately simple working model, built to show a mechanism plainly, is what physicists call a toy model; this book's toy is one.

Knob 1: the timing (φ). This is the steering wheel: the phase of the second tone against the first. Turn it and the lopsidedness goes round as one clean cosine. It is biggest one way when the tones are in phase, zero a quarter turn later, biggest the other way at half a turn, and zero again at three quarters. Measured: within 0.03% of a cosine with no weight (RATCHET-1b), and one cosine within two parts in ten thousand of its height at every weight from 0 to 20.4 (RATCHET-3). Turning the timing half a turn reverses everything exactly. That held at every weight of every run, to every printed digit, and it is a law of the equation, not a measurement. With weight the whole cosine slides along the circle by the zero's slide of §6.3 to §6.5, the angle by which the zero leaves quadrature, so the marks on the wheel move but the wheel works the same.

Knob 2: the weight (β). The weight is how hard the rider is to hurry, against how fast its honey drains it. In a junction it is the Stewart–McCumber parameter, which grows with the capacitance and with the square of the resistance (knob 8). It does three things.

- It slides the zero. The slide grows in proportion to the weight when the rider is light: the tangent of the slide is about 1.86 times the weight times the frequency when A/ω² is near 1, and about 1.65 times when it is near 1.9. It peaks near the weight times the frequency of 0.58, and comes home as about 38 degrees divided by the weight times the frequency when the rider is heavy.

- It fades the move when the rider is heavy. The move is nearly flat to weight 0.3, then falls nearly as the weight to the minus fourth: by weight 20 it is a sixteen-thousandth of its size with no weight.
- It widens the step a little. At strength 0.30 and frequency 0.55, the step's full width is 1.862 with no weight, 1.865 at weight 0.32, 1.898 at weight 1.106, the rung of the weight ladder nearest the peak, and 1.999 at weight 20.

How to set it: a light rider, below about 0.3, keeps the whole move and needs its timing moved only a few degrees. A rider near the weight times the frequency of 0.6 needs the most re-timing. A rider much heavier than one over the frequency barely moves at all.

Knob 3: the frequency (ω). The frequency works through the two numbers of §6.5: the weight times the frequency, which the manual of §6.8 calls the first number, and the drive's strength over its frequency squared, the second number.

- At a fixed strength, lowering the frequency raises the second number. That makes the move bigger: with no weight, at strength 0.30, its size is 0.0061 at frequency 0.70, 0.0137 at 0.55 and 0.0319 at 0.40.
- It also makes the peak lower and later: 29.0, 27.4 and 22.8 degrees.
- The frequency also keeps a small say of its own. With the ratio held fixed, the peak climbs from 27.1 to 27.4 to 27.7 to 28.2 degrees at frequencies 0.40, 0.55, 0.70 and 0.85.

Knob 4: the first tone's strength (A). It works together with the frequency, as A/ω^2. Along a line of fixed A/ω^2, a stronger drive gives a bigger move, but a little less than in proportion: as the drive scale to the power 0.97 to 0.94, falling as the drive grows. On the first survey, strengths 0.16 to 0.57, the biggest weightless window measured is 0.058 of the critical current (strength 0.567 at frequency 0.55, RATCHET-6), and the smallest is 0.0061 (strength 0.30 at frequency 0.70); weaker drives give smaller windows, 0.0011 at strength 0.075 (§6.6), and the strong drives of RATCHET-83 larger ones, 0.247 at drive ratio 4.375 (R83-H2).

Knob 5: the second tone's size (A_2). This is the volume knob. The equation fixes two facts before any run.

- With no second tone the step is fair: measured, its two edges equal to every printed digit.
- Turning the second tone upside down, changing the sign of A_2, is the same as turning the timing half a turn, so the lopsidedness changes sign with A_2.

So a small second tone should give a window in proportion to its size, and should leave the zero where it is. Both guesses were written down before the run, and both held (RATCHET-9, figure 57, C). The second tone was run at five sizes, from an eighth of its usual size to twice it (A_2 0.0125 to 0.20, the usual being 0.10), at weights 0, 0.32 and 1.106. Doubling it from 0.0125 to 0.025 doubled the window to within two parts in ten thousand at all three weights (1.99968, 1.99982 and 1.99991 times), and moved the zero by no more than 0.0022 degree. At the usual size the new runs gave the earlier runs' edges back to every printed digit. The window per unit of second tone is about 0.137 with no weight, 0.134 at weight 0.32 and 0.076 at weight 1.106.

Where the proportion bends, it bends gently, and in a way the equation predicts. Because the lopsidedness changes sign with A_2, it can contain only odd powers of A_2: A_2 itself, then A_2 cubed, and so on, never A_2 squared. So the window divided by A_2 should fall below its small-tone value by an amount that grows as A_2 squared: four times for each doubling. Read after the data, it does, 3.95 to 4.005 times per doubling at the three weights. At twice the usual size the window per unit of second tone is 1.36% below its small-tone value with no weight, 0.78% below at weight 0.32 and 0.39% at weight 1.106, and the zero has moved back toward quadrature by 0.13 to 0.18 degree.

How to set it: the second tone's size sets the window's size and almost nothing else. Double it and you double the window, with the timing unchanged, up to at least two thirds of the first tone's strength.

Knob 6: which harmonic. This knob picks the note the second tone sings: twice the first tone's frequency, three times, or four times. In the equation of §6.2 the 2 in $2\omega t$ becomes a 3 or a 4. The symmetry argument of Flach and colleagues (§6.1) says what an odd harmonic, such as three times the frequency, must do. A drive made only of odd harmonics has a hidden mirror: shift it half a beat and it is exactly upside down. Flip the steady push's sign and turn the phase over, and that mirror carries the equation into itself, with weight or without, so the step must be fair at every timing. Even harmonics break that mirror and can steer.

To test it, the engine was given the harmonic as a knob. It was checked first against the second harmonic's runs: all 928 edges came back to every printed digit. Then the third and the fourth harmonics were run at the same 29 weights as the second harmonic's runs (their 928 edges are 29 weights, 16 timings and two edges each), each with the second tone at 0.10 (RATCHET-8, figure 57). Five verdicts were written down first, and all five held (R8-a to R8-e); what they show, with one finding nobody wrote down beforehand:

- **The third harmonic steers nothing.** At 28 of the 29 weights the two edges are exactly opposite at all sixteen timings, to all twelve decimals the engine prints. At the 29th, one timing differs by one unit in the last decimal. A symmetry predicted it, and it held as exactly as a computer can show.

- **The fourth harmonic steers, but weakly.** With no weight its window is 118 times smaller than the second harmonic's (1.16×10^{-4} of the critical current, against 0.0137). Its mirror held exactly, and with no weight its zero sat exactly on quadrature, as the second harmonic's does.
- **Its zero wanders far.** Nobody wrote this down beforehand. With light weight the fourth harmonic's zero slides 5.1 times faster than the second harmonic's: 0.75 degree at weight 0.0025, against 0.15, still in proportion to the weight as the symmetry requires. It passes 87 degrees by weight 0.32 and peaks at 129 degrees at weight 0.64, more than a third of the way round. Then it swings back through quadrature near weight 2 and runs on round the other way, reaching −164 degrees by weight 20 and still moving. The second harmonic's zero never strays more than 27.4 degrees. At −180 degrees the fourth harmonic's zero would be back on quadrature with every direction reversed. It closes on −180 as 181 degrees over the weight times the frequency: predicted from one frequency, it came out 183.2, 180.4 and 179.3 at three others (§6.6; RATCHET-11 to 13).
- **The heavy rider forgets it fast.** At weight 20 the fourth harmonic's window is 31,000 times smaller than the second's.

The rules the fourth harmonic's zero keeps, a gear and a half-turn, are in §6.6.

How to set it: use exactly the second harmonic. It gives by far the largest window and the tamest zero. An odd harmonic gives nothing, which makes it a clean blank for a laboratory check: swap the octave for three times the frequency and the window must close at every timing. Borgongino and colleagues report that blank in a device: driven slowly, their junction is reciprocal when the second tone sits at an odd multiple of the first, and its diode weaker at higher even ones (their Figure 3g). A stray fourth harmonic as large as the second tone would add a lopsidedness under a hundredth of the octave's, so it could move the zero by at most about half a degree with no weight (argued from these runs, before the two were run together). They have since been run together (§6.11, and in full in Appendix A). With no weight the zero hardly moves, a tenth of a degree at most, as argued. But a stray off the mirror line puts a slope into the window's weight dependence: at 1% of the octave, a quarter turn off, it alone spends a 0.01% spec at first number 0.066 (RATCHET-48; the spec and the first number are the budget of §6.11).

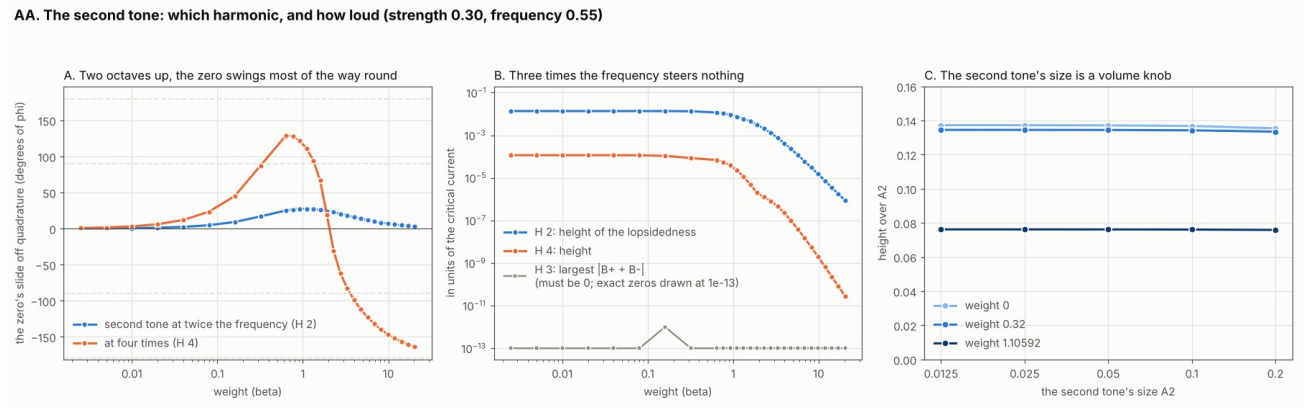


Figure 57. The second tone: which harmonic, and how loud. The junction at strength 0.30, frequency 0.55. A: the zero's slide off quadrature against the weight, with the second tone (0.10) at twice the frequency (blue, RATCHET-3) and at four times (orange, RATCHET-8); the fourth harmonic's zero swings out to +129 degrees, back through quadrature near weight 2, and on to −164 degrees by weight 20. B: the height of the lopsidedness against the weight, both scales logarithmic, for the second and fourth harmonics; grey, for the third harmonic, the largest |B+ + B−| over all sixteen timings, which the equation says must be zero. Its exact zeros, at 28 of the 29 weights, are drawn at 10^{-13}; the one non-zero is a single unit in the twelfth printed decimal. C: the window divided by the second tone's size, against that size, at weights 0, 0.32 and 1.106; flat lines mean the window grows in proportion to the second tone. From RATCHET-8 and RATCHET-9.

Knob 7: the steady push (B), the operating point. The second tone does not push the rider by itself. At zero bias the rider stays locked at every timing (RATCHET-1b's runs, locked at B = 0). What the second tone makes is a lopsided doorway. Set the steady push inside the window, past the easy edge and short of the hard one, and the rider slips the easy way and not the other. The window is the size of the lopsidedness: 0.0137 of the critical current at strength 0.30 and frequency 0.55 with no weight, between a forward edge of 0.9241 and a backward edge of 0.9378 in phase.

You can also use the move without a steady push. Rock the push slowly back and forth, with a swing that reaches past the easy edge but not the hard one, and the rider should slip one way on every swing and never back. With weight, the memory of Part 4C keeps it slipping a little longer on each swing once it starts. This rocking use is argued from the edges and not run. Can we run it: yes, with a slow rocking push on the same engine.

Knob 8: the honey (damping). In the equation the honey is set to one by the choice of units, so it lives inside the weight and the frequency. More honey means a lighter rider and, measured in the rider's own time, a faster drive. In a junction the honey is the leak through its resistance, and less resistance means more honey. In the standard units of Stewart and McCumber (1968), the weight is twice the electron's charge, times the critical current, times the resistance squared, times the capacitance, divided by the reduced Planck constant. The frequency is counted in units of twice the electron's charge, times the critical current, times the resistance, divided by the reduced Planck constant. The equation itself fixes one more fact: with weight and no honey at all, time runs backwards exactly, and the zero stays on quadrature (§6.4).

Knob 9: noise and warmth. Run once, for the drift at zero push (RATCHET-22 and 23, §6.9). With noise of strength D 0.5 (D, the noise's strength, is not a drive like the pin's D* of §4.26) the weightless rider drifts at zero push at about 1.1×10^{-3} in the book's units, its timing curve one clean cosine with its zero on quadrature (-0.8 ± 3.1 degrees), as the symmetry requires; with weight the drift's zero goes out and back, and the drift fades, below what 4,800 riders a timing can read at weight 4.76. The step's corners under noise are still not run. Heat jostles the rider at random and lets it hop over an edge early, rounding the step's corners. The move survives only while the window is wider than that rounding. The windows measured on the first survey are about half a percent to six percent of the critical current (a tenth of a percent at the weakest drive of §6.6, about a quarter at RATCHET-83's strongest), so a cold junction should keep the move, but that is a guess. Can we run it: yes, with noise added to the engine (Part 8).

Knob 10: the drive's shape beyond two tones. A drive of three tones has since been run: the octave with a third tone at four times the frequency, for the dark points, the flat point and the meter of §6.12 (from RATCHET-39 on, the three-tone engine checked first by FIDELITY-3T), and with one at three times, for the count of the beats (BEAT-1 to 4); for the weight laws it has been run at one setting, the 0.55 edge (Appendix A, "A third tone, tried"). For the dancers, an extra tone was heard as one over its harmonic number squared (§4.28), which suggested that higher even harmonics would steer more weakly. Knob 6 has now measured how much more weakly, and the guess was far too kind: the fourth harmonic in place of the second steers 118 times more weakly, where one over n squared would say four times.

knob	what it does	measured over	still to run
1 the timing φ	steers: the move goes as one cosine of the timing; half a turn reverses it exactly	the whole circle, 16 phases, every run	none
2 the weight β	slides the zero out and back; fades the move when heavy; widens the step a little	0 to 20.4, and 24.5 for the heavy end's power (RATCHET-80)	none for the book
3 the frequency ω	through the weight times the frequency and A/ω^2; a small say of its own	mainly 0.40, 0.55, 0.70, 0.85; others from 0.20 to 1.00, and 2 to 24, for single questions	none for the book
4 the strength A	through A/ω^2; size grows a little slower than the drive	0.16 to 0.57; along A/ω^2 from 0.612 to 4.5 at frequency 0.55, strengths to about 1.36 (RATCHET-83, R83-e, R83-F, R83-G, R83-H2)	A/ω^2 past 4.5 (Part 8)
5 the second tone's size A_2	the volume knob: the window grows in proportion to A_2 and the zero stays put; the first bend goes as A_2 squared	0.0125 to 0.20, at three weights (RATCHET-9)	none for the book; A_2 0.40 would test the bend
6 which harmonic	the second steers; the third steers exactly nothing; the fourth steers 118 times more weakly (it grows as A^4 where the second grows as A^2) and its zero makes a half-turn along the weight knob (§6.6)	the second, third and fourth, at 29 weights (RATCHET-8, 10 to 13)	the sixth harmonic's count (Part 8); the fourth harmonic's envelope at other drive ratios (§6.11)
7 the steady push B	the operating point: inside the window, one way only	the step's two edges	a slow rocking push
8 the honey	lives inside the weight and the frequency	through the weight	none for the book
9 noise	at zero push, a drift whose zero goes out and back with weight (§6.9); rounds the step's edges (a guess)	D 0.5, weights 0 to 4.76, the drift only (RATCHET-22, 23)	the step's corners under noise (Part 8); the road to no honey (§6.9)
10 the drive's shape	higher even harmonics steer more weakly: the fourth 118 times (knob 6)	two tones; three at the dark points and in the count of the beats (§6.12)	three tones over the rest of the knob plane

6.8 The manual, step by step

The rectifier's rules are short enough to fit on a card, and the steps below are that card.

To make the rider slip forward and not back:

1. **Know your rider.** Find its weight β and your drive's frequency ω, and multiply them: that is the first number. Below about 0.3 the rider is light. Near 0.6 the timing needs the most care. Well above 1, the move fades fast.
2. **Choose your drive.** Pick the first tone's strength A and divide it by the frequency squared: that is the second number. Near 1 is where most of the runs were made. A bigger number gives a bigger window, but a lower and later peak. If the window's size must not change with the rider's weight, use the weight-proof ratio of §6.11: about 1.325 at frequency 0.40, 1.306 at 0.55 and 1.275 at 0.70. How closely to hold it depends on your spec and your heaviest rider's first number; §6.11's budget gives the band, about 15% either way for a window held to 0.1% with the first number under 0.07. Set the ratio first and trim with the second tone's size, which moves the edge about six times more gently.
3. **Add the second tone** at exactly twice the frequency. Its size sets the window in proportion (knob 5): at strength 0.30 and frequency 0.55 with no weight, the window is about 0.137 times the second tone's size, so a second tone a third the size of the first gives a window of 0.0137. Anything from an eighth of that size to twice it was run. Exactly twice the frequency matters, for two reasons. Three times the frequency gives no window at all, and four times gives one more than a hundred times smaller, with a zero that wanders far (knob 6). And if the two tones are not locked at two to one, their timing drifts, and the move averages away over each drift, as the relay's frozen frame did in §4.6.

4. **Set the timing.** With no weight, put the tones in phase: that is the most lopsided step, with forward the easy way. With weight, turn the second tone ahead by the zero's slide off quadrature (§6.3), read from figure 55's curve for your second number at your first number. The most lopsided step is there. For riders whose weight you do not know, set the timing for the middle of their range: at the 0.55 edge that holds the push within 0.1% from weight 0.04 to 0.12, and then the slide, not the window, is what your budget must cover (§6.11). Better still, use §6.11's held-timing setting (drive ratio 3.25, the second tone about 1.24 times the first, timing in phase), where the push itself is weight-proof.
5. **Set the steady push inside the window.** Push past the easy edge and short of the hard one, or rock the push slowly with a swing between them.
6. **Check the bench before believing it.** Turn the timing half a turn: the window must reverse exactly. Then turn it a quarter turn from your setting: the window must close. For a third check, swap the second tone for one at three times the frequency: the window must close at every timing. If any check fails, the instrument is wrong, not the physics (§7.2, lesson 46).
7. **Find your setting on the envelope** (§6.11, figure 59) before trusting a number from this card. With the second harmonic the card holds everywhere run. With a fourth harmonic in the drive, by design or as a stray, check whether your weight and frequency sit in the island where the in-phase push reverses (the fourth harmonic's island, §6.11), and how close you are to an edge. Then hold its timing in phase or against: off that line even 1% of the octave puts a slope in the window that spends a 0.01% spec by itself (§6.11).

To make it go nowhere, set the timing a quarter turn from step 4's setting: the step is fair, and no steady push inside it moves the rider either way. **To reverse it,** add half a turn.

To use a dark point. A dark point is a setting of the knobs where the step's lopsidedness vanishes at every timing of the second tone, so the timing has nothing to steer, like the still eye of a storm with the wind turning all round it (§6.6, §6.12). It is met in practice with a third tone, at four times the frequency, beside the octave:

1. **Add the third tone** at exactly four times the frequency, locked to the other two, its timing following the second tone's one for one, and set it against the octave, half a turn from in phase.
2. **Find the dark point.** With the first two tones in phase, watch the step's lopsidedness while you lower the octave's size: it changes sign at the dark point. At the edge's drive and frequency 0.55, with a fourth harmonic about the size of the first tone, that is an octave of 1.33% of the first tone. A slower drive needs less (0.82% at frequency 0.20, 1.05% at 0.40) and a faster one more (1.70% at 0.70, 2.74% at 1.00): with the drive's strength over its frequency squared held, the needed octave grows with the frequency as an exponential above 0.40 (§6.12), and at one frequency it grows with the drive's strength roughly as its square, a power of 1.78 at the edge's drive that climbs toward 2 as the drive weakens (RATCHET-53, 54).
3. **Choose the job.** Well below the dark point, the third tone's timing steers the push's best timing all the way round. Above it, the timing only nudges it, within a cone whose half-angle has for its sine the fourth harmonic's arrow over the octave's (each tone's share of the window drawn as an arrow, §6.12). At the flat point, an octave about 0.32 of the first tone at the edge's drive (§6.12 computes it from a small dial), the timing trims the push's size and leaves its direction alone.
4. **To weigh a rider,** set the octave at the dark point found with no weight, then turn the third tone's timing away from against until the push vanishes. For a light rider the turn is in proportion to the weight, about 232 degrees of timing per unit of weight at the edge's drive.
5. **Check the bench.** With no weight the dark point must sit exactly on the against timing, where the push has no sideways part. If it does not, the tones' timing is off, not the physics.

From science to engineering. The rectifier is where this book's equation stops describing and starts steering, and its rules fit on a card. They hold over the settings run: a damped rotor pushed by a tone and an octave from an eighth of the octave's usual size to twice it, one twenty-fourth to two thirds of the tone (RATCHET-9). The octave's size sets the size of the push, in proportion; a tone at three times the frequency gives none, and one at four times under a hundredth. Put the two in phase for the most lopsided push; put them a quarter turn apart for none, if the rotor has no weight. With weight, the timing that gives no drift moves off the quarter turn by an angle read from one curve in two numbers (figure 55). For a light rotor the angle's tangent is about 1.86 times the weight times the frequency when the drive's strength over its frequency squared is near 1, and about 1.65 times when that ratio is near 1.9. For a heavy rotor the angle is about 38 degrees divided by the weight times the frequency. In between it peaks: about 27 degrees where the weight times the frequency is near 0.58, at a ratio near 1, and lower and later as the ratio grows. A rotor much slower than the drive feels almost nothing, the effect fading about as the weight to the minus fourth. And every push goes into whatever holds the floor still: two waves with a phase between them move a rider one way only by pushing on something real, which is the wall of §1.2 seen from its useful side. A junction of known capacitance under a two-tone source could check the card directly in a laboratory.

6.9 Where it is used, and what is ours

Directed motion from two tones with a phase between them is old and well studied, and its name is harmonic mixing. A nonlinear system fed two tones whose frequencies stand in a whole-number ratio answers with a steady part as well; it was seen early with microwaves in germanium (Schneider and Seeger, 1966) and written down for a damped periodic potential by Marchesoni (1986), both as reported in Hänggi and Marchesoni's review (2009; its harmonic-mixing sections read), the 1986 letter itself known to us only by its abstract. For a weightless rider pushed by a tone and its octave, the steady drift goes as the cosine of the timing (their Eq. 17), which is this part's weightless cosine (§6.2). Cold atoms in a lattice of light do it (Schiavoni and colleagues, 2003; Gommers, Bergamini and Renzoni, 2005), rocking ratchets do it, and a fluxon running round a ring-shaped Josephson junction does it under a two-tone microwave drive (Ustinov and colleagues, 2004, who measured the fluxon's mean velocity against the phase between the tones), and an overdamped junction under a quasiperiodic drive has been studied for its symmetries and transport (Neumann and Pikovsky, 2002). The toy's relay did it on 4 October, a kick and its own octave acting as a ratchet with a handedness (§6.1). The ultrasonic motor gets its push from a different pair, two waves of one frequency a quarter turn apart in space and time.

The rules that say when the drift must vanish are written down as well. Flach, Yevtushenko and Zolotaryuk (2000), and the review by Denisov, Flach and Hänggi (2014; sections 1 to 3 and 6 read), name two symmetries. A half-beat shift kills the drift whenever the drive has only odd harmonics; that is knob 6's silent third harmonic (§6.7). Running time backwards kills it at particular timings. The review also reports the effect of §6.3 in its own terms. With weight and honey together, the timing at which the drift vanishes moves: the drift goes as the sine of the timing less a lag, and the lag runs from a quarter turn in the weightless limit to nothing in the limit with no honey, the crossover found by computation (their Eq. 62 and Fig. 6). Cold atoms showed the lag moving as the damping was tuned (Gommers, Bergamini and Renzoni, 2005). That the zero slides with weight is therefore known.

The mathematics of the step is older still. Mawhin's survey of the forced pendulum (1998; read) states the step as a theorem it calls classical (Mawhin and Willem, 1984; Fournier and Mawhin, 1985): for a given repeating part of the forcing, the damped pendulum has a periodic orbit exactly when the steady push lies in a closed interval, the step, and inside it at least two, the locked orbit and its unstable partner (§4.8; §4.28, Scene 6). The survey also records that the interval need not hold zero push at all. Ortega (1987) gave the first forcing with no average for which it does not, and Alonso (1997) showed that at any damping a slow enough one does it, a strong push for a short while and a weak pull for a long one: the lopsided step of this part pushed to its extreme. For forcings with a symmetry, Mawhin points to Schmitt and Sari (1984, 1985) and others, not read here.

The step and the drift, measured side by side. The literature's quantity is a drift, a speed; the book's is the lopsidedness of the step, a threshold of steady push. Running a film of the rider backwards turns a speed round and leaves a push alone, so in the limit with no honey at all the two readings must put their zeros a quarter turn apart. This bench tested whether that difference shows along the book's own weight knob (RATCHET-22 and 23). The junction was given noise (strength D 0.5), and the drift at zero push was measured against the timing at six weights, beside the step's zero from RATCHET-3. It does not show. The drift's zero also goes out and comes home: −0.8 ± 3.1 degrees with no weight, +8.7 ± 3.2 at weight 0.32, +40.4 ± 3.6 at 1.106, +35.3 ± 4.8 at 1.593 and +12.2 ± 4.5 at 2.293, where the step's are 0, 16.9, 27.4, 26.0 and 22.6; at 4.755 the drift is too small to read. The registered expectation, that the drift's zero would pass 45 degrees and stay, failed, and its rival held (R23-b). So along the weight knob, with the drive's frequency held, both readings come home. They differ in the shape of the hump, not by a quarter turn: the drift's swings higher and comes home sooner, and the two zeros sit 8 to 13 degrees apart at the four weights read above zero. The quarter-turn difference belongs to a different road to the same limit: draining the honey with the mass and the drive held, which in this book's units raises the weight while lowering the frequency. That road has not been run (Part 8). What the test gives the engineer: on the weight knob, the timing that stops a noisy rider's drift and the timing that makes a locked rider's step fair differ by up to 13 degrees, and both return toward quadrature as the rider grows heavy.

The same step, reached from the device's side. The lopsided step itself, a threshold and not a drift, is published as well, under another name: a tunable Josephson diode. Scheer, Seoane Souto, Hassler and Danon (2025) drive a junction without weight with a tone and its octave and find the zero-voltage step shifted, so that the supercurrent runs more easily one way. The shift vanishes at one timing and is largest a quarter turn away, and at small drive it goes as the cosine of the timing: this part's weightless law (§6.2), reached from the other side. They treat a slow drive, a faster one as a cascade of two-tone mixing, and larger drives, where the diode turns into a ratchet. Borgongino and colleagues (2025) built it: an indium-arsenide junction at 70 millikelvin, its two tones sent by an antenna or, at slow drives below the antenna's band, through the leads, the timing turning the diode's direction and its strength up to the ideal, with a slow drive throughout and the fast regime left for later. Two roads reached one plateau. Theirs came from the device. Ours came from a toy's relay, which showed the handed kick on 4 October (§6.1), and from one rider mapped in the toy and in the junction's model alike; the meeting is a check that the bench's step is the real junction's step. Neither paper gives the rider weight: the first has none in its equation, and the second models its junction with a small capacitance and does not vary it.

So the step, and its quadrature zero for a rider with no weight, were in print first; this part met them on its own road. What it adds is weight, the map of one real device's standard model across it, and the wiring of that map to the toy and the loops:

- the step's zero along the weight knob, pinned on quadrature at both ends, and beside it on the same bench a noisy rider's drift whose zero also goes out and back, by a taller and narrower hump (RATCHET-22, 23);
- the out-and-back hump, with its clean tails;
- the heavy end's 38 degrees;
- the two numbers, tested blind from both sides and at a third frequency;
- the second tone measured: in proportion to its size, exactly silent at three times the frequency (the known symmetry, confirmed on this medium), and weak at four times, with a zero that wanders more than a third of the way round;
- the card above, the envelope of §6.11 with its three seams and the second harmonic's weight-proof edge, the dark points and the drift-proof weight meter of §6.12, and the arrays of §6.13.

A laboratory could check knobs 1, 2 and 7 at once, with a junction of known weight, two microwave tones locked at two to one, and a reading of the step's two edges as the timing is turned; Borgongino and colleagues have already run its light-junction half. The windows at our settings are tens to hundreds of times wider than the narrowed steps of the pinch that §5.6 asked a laboratory to see. And every slip forward is paid for by the floor: in a junction, by the current source and the circuit that holds the bias; on a skater, by the ice. Nothing in this part moves a closed box. A cheaper bench can come first. Ngatcha and colleagues (2021; read) ran the same junction equation, with one tone, on an Arduino microcontroller, and saw on it, by eye, the chaos their simulation predicted. With the octave added and the step's two edges read as the timing turns, such a board would check the bench's arithmetic and the card's knobs on a desk. It is an analogue computer, not a junction, so it tests the equation, not the material.

6.10 Walls, seams, and what is still open

> *we keep seeing behaviors similar to this… big jumps or transitions or boundary walls. each time, its a seam/wall/knob that needs to be understood.*
>
> — Bear, 15 October 2026

Every result in this part sits on a wall: a place where something stops, starts, flips or vanishes as a knob turns. Laid side by side, the walls the move has shown come in four kinds. Some are set by a symmetry of the equation and hold to the last digit the engine prints. Some belong to the rider itself: they are measured, and some are not yet derived. Some belong to topology: a dark point is the only door through which a whole-number count can change (§6.12). And some belong to the ruler, not to the rider.

wall	what happens there	where it sits	kind, and do we understand it
the step's two edges	the rider starts to slip, forward or backward: two orbits of the passing map meet and vanish (a fold)	at strength 0.30, frequency 0.55, no weight: 0.9331 both ways with no second tone; 0.9241 forward and 0.9378 backward in phase (§6.2)	the rider's; understood: each edge is found exactly as a fold
quadrature, the fair step	the lopsidedness is zero	a quarter turn of the timing, exactly, when the rider has no weight (§6.3)	symmetry (time reversal); exact to every printed digit
the half-turn mirror	every lopsidedness reverses	half a turn of the timing, at every weight (§6.3)	symmetry; exact to every printed digit in every run
the odd-harmonic wall	a second tone at three times the frequency steers nothing at any timing	the whole knob plane (knob 6, §6.7)	symmetry (the half-beat mirror); exact to 12 printed decimals at 28 of 29 weights, one last-digit unit at the 29th
the two quiet ends of the weight knob	the zero comes home to its pin	no weight, and weight with no honey (§6.4); in between, the second harmonic's zero peaks at 27.4 degrees near weight times frequency 0.58 (§6.5)	the ends: symmetry, exact; the peak's place: the rider's, measured on two numbers, not derived
the fourth harmonic's gear break	the fourth's zero stops sliding a fixed multiple of the second's	after weight 0.64 at frequency 0.55; at 0.40 the gear climbs first, at 0.70 and 0.85 it sags from the start (§6.6)	the rider's; measured, not understood

wall	what happens there	where it sits	kind, and do we understand it
the in-phase reversals	with the tones in phase, the easy direction flips	each time the fourth harmonic's zero passes a quarter turn: three times between no weight and weight 20 at frequencies 0.40 to 0.775, once at 0.80 and 0.85 (§6.6)	the rider's; understood once the zero's path is known
the seam	two in-phase reversals are born together, or vanish together	the fourth harmonic on the 0.99 drive ratio, between frequencies 0.775 and 0.785, near 0.783 by a parabola through the fine ladders' three highest rungs (0.782 by a straight line through their highest slides, +91.3 and +89.5 degrees; §6.6, figure 58)	the rider's; located between two runs, not derived
the window's seams	the window's height stops falling steadily with weight: a dip and a recovery appear or vanish together, or a small rise grows out of the light end	the fourth harmonic on the 0.99 drive ratio: a dip pair from frequency 0.40 (0.76 of the weightless height at weight 0.40, back to 0.89 at 0.768) that vanishes between 0.5215 and 0.5235, near 0.5228 by a fold's law on the finest rungs (0.5225 on coarser ones; RATCHET-18, 21, 81); a light rise born where the light coefficient a (§6.11) changes sign, between 0.755 and 0.76, near 0.7578 (0.7576 registered; RATCHET-19), growing to +0.15% at 0.85; the second harmonic rises 0.18 to 0.29% at drive ratio 1.875 at frequencies 0.40, 0.55 and 0.70, and on neither the 0.99 nor the 0.612 ratio (RATCHET-20; §6.11, figure 59)	the rider's; located between runs, not derived
dark points	the window closes at every timing, so the half-turn count may change	none at the nine settings run (§6.6); on the third tone's dial, with no weight, one between octave sizes 0.0025 and 0.0075, where the window's winding round the dial changes from 0 to one full turn (RATCHET-52, §6.12), placed on the against timing at octave 0.00525010, growing with the drive's strength as its square, which is §6.12's law (the fourth harmonic's share falls as one over the drive squared) seen from the octave's side, bent by strong drive: a power of 1.78 at the edge's drive, climbing to 1.95 at a quarter of it (RATCHET-53, 54); with weight it leaves the against line in proportion to the weight and shrinks as its square (RATCHET-57)	topology: the only door through which the count can change; one found and placed, pinned to the mirror line by the reflection at no weight
the gap between rungs	an excursion of the zero can hide between two weights of the ladder	at frequency 0.75 the usual ladder's largest rung was +85.7 degrees (weights 0.32 and 0.64 on either side), so it counted no quarter-turn crossings; a fine ladder found +95.5 at weight 0.45 and two crossings (RATCHET-15). A parabola through three coarse rungs errs both ways: 2.4 degrees high at 0.75, 4.4 low at 0.80. The reverse luck happens too: at 0.775 the zero passes back through quadrature at +0.08 degree right on the rung at weight 1.106 (RATCHET-16); a crossing read between rungs carries the rungs' spacing	the ruler's; closed by a finer ladder
the print floor	the window falls below the last decimal the engine writes	for the fourth harmonic at strength 0.30, frequency 0.55, near weight 35 (an estimate from the last two rungs)	the ruler's, not the rider's; a finer print would move it

The four kinds are treated differently, and that is the use of the table. A symmetry wall is a test of the instrument: if it does not hold to the last digit, the bench is wrong, not the physics (§6.8, step 6). A wall of the rider's own is a result: it is registered before the run, measured, and where it is not derived the book says so. A wall of the ruler is named so that nobody mistakes it for physics: the fourth harmonic does not stop steering at weight 35, the printing does. Bear's seams are the rider's walls, and the one measured most closely so far is the seam in the fourth harmonic's in-phase reversals. It also shows why the kinds must be kept apart: at frequency 0.75 a wall of the ruler, the gap between two rungs, sat right on top of the rider's seam and hid two reversals until a finer ladder was run.

Why the seam moved when the ruler got finer. The seam is where the top of the zero's path touches the quarter turn, and no run ever sees the top itself, only the path at the weights on its ladder. The highest of those can only be at or below the true top, never above it, so a coarse ladder reads every peak low and puts the seam too early. Each finer ladder raised the peaks it read and moved the seam later: near frequency 0.767 from the usual ladder, 0.782 from fine ladders, and 0.783 from a parabola through the fine ladders' three highest rungs, where the estimate stopped moving, to within 0.0005. A parabola through three coarse rungs, by contrast, can err either way, so its error has a size but no direction. Underneath is the resolving power of any ruler: an excursion narrower than the ruler's step is invisible. At

0.75 the path stays above the quarter turn from weight 0.36 to 0.56, at 0.775 only from 0.40 to 0.50, and at the seam not at all, so close to the seam the excursion is always narrower than the usual ladder's step from 0.32 to 0.64. The rule the bench takes from it: refine the ruler until the answer stops moving, and report the step at which it stopped.

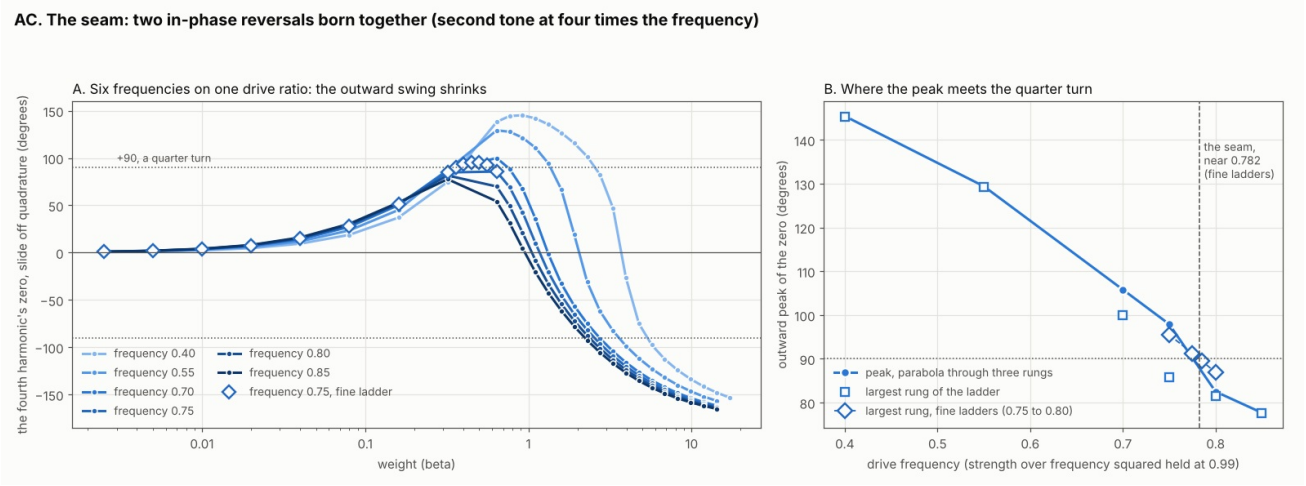


Figure 58. The seam. The junction's rectifier with the second tone at four times the frequency, on one drive ratio (strength over frequency squared 0.99, second tone a third of the first). A: the fourth harmonic's zero, its slide off quadrature against the weight, at frequencies 0.40, 0.55, 0.70, 0.75, 0.80 and 0.85 (light to dark); dotted lines at plus and minus a quarter turn, where the in-phase push flips; hollow diamonds, a fine ladder of weights at 0.75. B: the outward peak of the zero against the frequency: filled circles by a parabola through the three highest rungs, hollow squares the highest rung itself, hollow diamonds the highest rung of fine ladders at 0.75, 0.775, 0.785 and 0.80; the dashed line is the seam, where the fine ladders' peak meets the quarter turn. From RATCHET-8 and 11 to 17.

Still open. The open questions the runs have not settled are listed in Part 8: why the peak falls as A/ω^2 grows, why the move's size grows a little slower than the drive, and why the fourth harmonic's gear depends on the frequency, and whether its half-turn count grows with the harmonic. Knobs 5 and 6 are now run (RATCHET-9, RATCHET-8, and for the fourth harmonic's path RATCHET-10 to 13, §6.6). Knob 7's rocking push is unrun. Knob 9 has been run once, for the drift at zero push (RATCHET-22, 23), and not yet for the step's corners. Knob 10's three-tone drives have been run at the dark points and in the count of the beats (§6.12), and not elsewhere. Each says above whether the rest can be run.

6.11 The envelope: a map for the engineer

An engineer who wants to use the move needs more than the card of §6.8. The card says what the rectifier does at the settings run. The engineer needs to know where on the knobs each of its numbers holds, where it bends, and how sure the book is of each. In engineering that is a performance envelope: the region of operating points inside which a device does what its data sheet says, drawn with its edges marked. This section draws the rectifier's envelope from the seams of §6.10 (figure 59) and reads it region by region. Everything on it is measured on the junction's rectifier with the second tone a third of the first, and every edge was found between two runs, or between two rungs of a ladder of weights. So every edge comes with its ruler: the gap inside which it is known. This section gives the results and the map. Every run behind them is told in full in Appendix A, under the headings this section first had.

The second harmonic: one region, and a soft edge. With the second tone at twice the frequency, at every weight and frequency run on the drive ratios 0.61, 0.99 and 1.875, the zero stays within 29.2 degrees of quadrature, so the in-phase push keeps one direction. Near no weight the window's height changes as $1 + a\beta^2$: it starts flat, and the light coefficient a, a new use of the letter and not the extra tone's size a of §4.28, says which way it bends as the weight squared. Along the drive ratio A/ω^2, a changes sign at an edge: 1.3250 at frequency 0.40, 1.3056 at 0.55, 1.2746 at 0.70, 1.22994 at 0.85 and 1.17228 at 1.00 (RATCHET-24 to 29, EDGE-85, EDGE-100). Below the edge the window only falls as the rider gets heavier; above it, the window first rises by a few tenths of a percent. Each edge was placed by a bowed curve. At 0.70 the first bow registered, carried over from 0.55, missed its bar by 0.0004, and its redesign held; every straight line registered against a bow missed by far more. Across the frequencies run the edge is itself a parabola in the frequency, and it sharpens as the frequency rises: to keep a within 0.001 of zero, the ratio must sit within ±0.009 of the edge at 0.40 and within ±0.0011 at 1.00. At frequency 0.85, from ratio 0.99 out to 1.875, a keeps the edge's sharp bend and gains a gentle third-power twist far out (LIGHT-85, LIGHT-85b).

The weight-proof setting. At the edge the β^2 term is gone, and the window depends on the weight only through the next term, $c\beta^4$, the fourth-power term (this c is new, not the $c = \cos(\pi\, p/q)$ of §3.5): the edge is the rectifier's turnover point for weight, as a quartz crystal has one for temperature. At frequency 0.55 the book predicted from older runs that the window would stay within 0.1% of its weightless height to weight 0.1765; it held to 0.1778, and at weight

0.04 it moved about 60 times less than on the usual ratio, where the edge's own shift sits in the last printed place (RATCHET-26, RATCHET-65). Only the size is weight-proof. The timing still slides with the weight, so step 4 of the card still applies.

What it takes to hold the edge: one number. How far the band reaches is set by the first number, the weight times the frequency, and not by the weight alone. Registered from that number before each run, the band ended at weight 0.2446 at frequency 0.40 (predicted 0.2445), 0.1395 at 0.70 (0.138), 0.11475 at 0.85 (0.11508) and 0.09742 at 1.00 (0.09743 from the first number's drift carried on; its plain mean, 0.09770, missed by 0.00028). Its first number lies between 0.0974 and 0.0979 at all five frequencies, within 0.44% from end to end, and even its faint drift reads as a straight line in the frequency, a reading not checked at finer rungs (RATCHET-30, RATCHET-65, EDGE-85, EDGE-100). So the error budget can be drawn once, against the first number of the heaviest rider:

window held within	heaviest rider's first number	how far the drive ratio may sit from the edge	the 0.99 ratio's own change, for comparison
0.1%	0.05 or less	no edge needed: every ratio fitted, 0.99 to 1.875, passes	−0.071% at 0.05
0.1%	0.07	about 15% below to 17% to 19% above, by frequency	−0.152%
0.1%	0.09	about 3.3% either way	−0.282%
0.01%	0.03	about 10% below to 11% to 12% above, by frequency	−0.024%
0.01%	0.05	about 1.2% either way	−0.071%
0.01%	0.07	nowhere at the 0.55 edge: the fourth-power term alone breaks the spec; reachable at a joint edge at frequencies 0.40 and 0.55, not at 0.70 ("The joint edge" below, and Appendix A)	−0.152%

For a loose spec the drive ratio hardly matters. For a tight spec with a light rider, holding the ratio within about one percent of the edge is enough. For a tight spec with a heavier rider no drive ratio will do on its own, and the joint edge below is the tool. The hard part at the bench is knowing the drive's strength at the device, not at the generator: calibrate it from the Shapiro steps the drive prints on the junction itself, hold the temperature, and check the setting on a test chip whose junctions differ only by an added capacitor (Appendix A, "What it takes to hold the edge").

Hunting a knob for the fourth-power term. Neither the drive ratio nor the second tone's size zeroes c in the range run (RATCHET-31, 32). The second tone's size moves a instead, about six times more gently than the drive ratio, so the ratio is the coarse knob for the edge and the second tone's size its fine trim. A reflection of the equation says why the window holds only even powers of the weight and the slide only odd ones, and why every knob that comes with such a reflection has a fine end: the reflection makes the knob's size enter squared, gentle near its zero (Appendix A).

A weight-proof push at a held timing. For a rider whose weight is not known, the timing cannot follow the slide, and then the slide costs more than the window. At drive ratio 3.25, with the second tone about 1.24 times the first and the timing held in phase, the push itself is weight-proof. With the second tone 1.25 times the first it is 48 times flatter at weight 0.08 than with the second tone a third of the first, and 2.79 times stronger; at the crossing, 1.2353 times, it stays within 0.0773% of its weightless value out to weight 0.32 and leaves the 0.1% band at weight 0.3364 (RATCHET-34 to 37). Just past that it falls off a cliff, and near weight 0.36 the engine stalls on two paths (RATCHET-40).

A third tone at four times the frequency. It mixes with the octave instead of simply adding. The light coefficient moved as adding said, but the fourth-power term stayed where the octave alone has it (RATCHET-41 to 43), because two hands, one that flips with the third tone's sign and one that does not, cancel at the size first chosen (RATCHET-44). In phase near size 0.195 it shrinks c by about 47%: a partial knob, not a zero (RATCHET-46). Off the mirror line its timing becomes a dial, and puts a slope, a term in the weight itself, into the window (RATCHET-45, 47). That is a warning and a knob. A stray fourth harmonic at 1% of the octave, a quarter turn off the line, spends a whole 0.01% spec at first number 0.066 (RATCHET-48), and a stray third harmonic at a tenth of the octave moves the window by 0.57% at the same first number (BEAT-4). Put in on purpose, the same tilt widens the 0.1% band to weight 0.22656, 1.2744 times the untilted edge (RATCHET-70). §6.12 takes up the count the third tone's timing keeps on its dial.

The joint edge. Two knobs together open the budget's empty corner. A stronger drive shrinks c, and the fourth harmonic in phase pulls a back to zero: at ratio 2.75 with the fourth harmonic at 0.143, and at 3.25 with 0.196, c falls to 0.30 and 0.28 of the 0.55 edge's (RATCHET-49, 50). Along that line c is best near ratio 3.2, not at the strongest

drive: by 3.75 the larger fourth harmonic it needs bends c down again (JOINT-LINE). There a 0.01% window reaches first number 0.0737 and 0.0761, where the plain edge reached 0.055 (RATCHET-66). At other frequencies it is not a fixed recipe: the corner opens at frequencies 0.40 and 0.55 and stays shut at 0.70 (RATCHET-51, 56, 82).

The fourth harmonic: four regions and three seams. With the second tone at four times the frequency, on the drive ratio 0.99 (figure 59, A), hold the tones in phase. A light rider is pushed the no-weight way. Above a floor at weight 0.33 to 0.40 lies an island where the push is reversed; its roof falls steeply as the frequency rises, from weight 2.56 at frequency 0.40 to 0.50 at 0.775, and floor and roof meet at the seam near 0.783. Above the roof the push points the no-weight way again until the zero passes −90 degrees, at weight 5.37 at frequency 0.40, falling to 2.22 at 0.85, and past that it is reversed for good. The window has two seams of its own: a dip and its recovery, which vanish together as a fold near frequency 0.5228 (0.5225 on coarser rungs; RATCHET-18, 21, 81), and a light rise, born where a changes sign near 0.7578 (RATCHET-19).

where on the knobs	what the rectifier does	the numbers	known to within	at the bench
second harmonic: every weight, drive ratios 0.61 to 1.875, frequencies 0.40 to 0.85	the in-phase push keeps one direction at every weight	the zero's highest rung 22.7 to 29.2 degrees; the heavy end 38 degrees over ωβ (§6.5)	every turning point the rungs can show, 29 weights a run	the card of §6.8 holds as written
second harmonic, drive ratio above the edge (about 1.33 at frequency 0.40, 1.31 at 0.55, 1.27 at 0.70)	the window rises a little before it falls	at ratio 1.875: +0.19% at weight 0.32 (frequency 0.40), +0.18% at 0.16 (0.55), +0.29% at 0.16 (0.70)	the rise's top between rungs 1.5 to 2 apart	ignore for sizing; a sign of a strong drive
second harmonic, at the edge: drive ratio 1.3250 (frequency 0.40), 1.3056 (0.55), 1.2746 (0.70), 1.22994 (0.85), 1.17228 (1.00)	the window's size hardly depends on the weight	within 0.1% of the weightless height to weight 0.1778 (frequency 0.55), 0.1395 (0.70), 0.2446 (0.40), 0.11475 (0.85) and 0.09742 (1.00), first number 0.098 at the first three, 0.09753 at 0.85 and 0.09742 at 1.00; 0.988952 of it at 0.32 (0.55); about 60 to 4 times steadier than on ratio 0.99 at weights 0.04 to 0.16	the edge between ratios 1.32 and 1.33 (0.40), 1.30 and 1.32 (0.55), 1.27 and 1.28 (0.70), by a parabola through six ratios from 1.21 to 1.26 (0.85) and through eight from 1.14 to 1.21 (1.00); the band at all five	use it where the window's size must not change with the rider's weight; how close to hold the ratio depends on the spec and the heaviest rider's first number (the budget above): about 15% for 0.1% to first number 0.07, about 1% for 0.01% to 0.05; the timing still slides
second harmonic at the edge with a fourth-harmonic tilt a quarter turn ahead (frequency 0.55, ratio 1.3056)	the window rises a little, turns over, then falls, so the 0.1% band covers more riders	third tone 0.014604 (0.1109 of the octave): within 0.1% to weight 0.22656 (first number 0.1246), 1.2744 times the untilted edge, peak +0.000899; a quarter turn behind, to weight 0.09678	the reach 0.9994 of the blind prediction	set the third tone's timing a quarter turn ahead and its size in proportion to the tilt wanted
second harmonic, the second tone's size at the edge (frequency 0.55, ratio 1.3056)	moves the edge: a goes as $-0.02378 + 0.21112\ q^2$, q the second tone's size over the first's (here q is a ratio of sizes, not a dance's denominator); c barely moves	a −0.02280, −0.01819, +0.00001, +0.02971, +0.06998 at q 1/12, 1/6, 1/3, 1/2, 2/3; c/ω^4 at weight 0.16 (c read there as the window's height less $1 + a\beta^2$, over β^4) from −10.505 to −11.857	the law held blind at q 1/2 and 1/12 (RATCHET-32), within 0.0007 in a	a fine trim for the edge, about six times gentler than the drive ratio; not a knob for the fourth-power term
second harmonic, held timing: ratio 3.25, the second tone 1.2353 times the first (frequency 0.55)	the push at a fixed in-phase timing hardly depends on the weight	at weight 0.08: 1.000026 of the weightless push (q 1.25), against 0.998763 (q 1/3); the push 2.79 times larger	the crossing between q 1.20 and 1.25 (RATCHET-35); within 0.0773% to weight 0.32, out of the 0.1% band at 0.3364, first number 0.1850 (read with a curve through the rungs; a straight line had said 0.3356) (RATCHET-36, 37); a cliff and an engine stall near 0.36	build this for riders of unknown weight; timing errors cost only at second order

where on the knobs	what the rectifier does	the numbers	known to within	at the bench
third harmonic, in place of the octave	nothing, at every timing	exactly zero	a symmetry	a blank for checking the bench (§6.8, step 6); beside the octave it is not a blank (the count, §6.12; a stray third harmonic, above)
fourth harmonic, light, below the island's floor	pushes as with no weight; the window falls as $1 + a\beta^2$	a from −1.44 (frequency 0.40) down to −1.61 (0.50), then up to −0.04 (0.755)	rungs 0.0025 apart at the light end	the zero slides about five times as fast as the second harmonic's (§6.6)
fourth harmonic, light, frequency above 0.758	the same push; the window rises slightly first	a from +0.03 (0.76) to +1.47 (0.85); the rise at most +0.15%	the birth between frequencies 0.755 and 0.76	ignore for sizing
fourth harmonic, the island: from its floor (weight 0.33 to 0.40) to its roof (2.56 at frequency 0.40, 0.50 at 0.775), frequencies 0.40 to the seam	the in-phase push is reversed	floor and roof as in the text; the seam near 0.783	each edge between neighbouring rungs, 0.04 apart near the floor and up to 0.46 apart near the highest roof; the seam within 0.01 in frequency	a rider here moves the other way at the same timing: turn the timing half a turn, or keep out
fourth harmonic, the dip: frequencies 0.40 to about 0.5228, weights 0.32 to 0.77	the window dips and partly recovers	0.76 at weight 0.40, back to 0.89 at 0.768 (frequency 0.40)	rungs 0.04 to 0.08 apart	read the window off its curve; never draw a straight line between coarse rungs
fourth harmonic, above the roof and below the −90 line	pushes the no-weight way again	the −90 line at weight 5.37 (frequency 0.40) falling to 2.22 (0.85)	between rungs	the window is falling fast here
fourth harmonic, heavy, past the −90 line	reversed for good; the zero comes home to −180 as 181 degrees over $\omega\beta$	the move fades; the print floor near weight 35 at frequency 0.55	the floor is the ruler's, not the rider's	the move is very weak; a finer instrument is needed to see it
off the map	the fourth harmonic on other drive ratios; the sixth harmonic; the step under noise; three tones beyond the edge mixes of §6.11	not run	(none)	run it before relying on it

Reading a number off the map.

1. **Pick your harmonic.** The second: one region, and the card of §6.8 holds. The third: nothing, anywhere. The fourth, by design or as a stray: go on.
2. **Work out your frequency and your drive ratio A/ω^2.** The fourth harmonic's map is drawn on the ratio 0.99 only. At another ratio you are off the map. The second harmonic's runs say its peak and edges move with the ratio (§6.5), so expect the island to move too.
3. **Find your weight on panel A and read your region:** which way the in-phase push points, and whether the window is falling smoothly.
4. **Read the ruler.** If your setting is within one rung of an edge, or within 0.01 of a seam in frequency, the map has not settled your region. Run a fine ladder there before you rely on it (lessons 155 and 158).
5. **Check the bench before believing any of it** (§6.8, step 6).

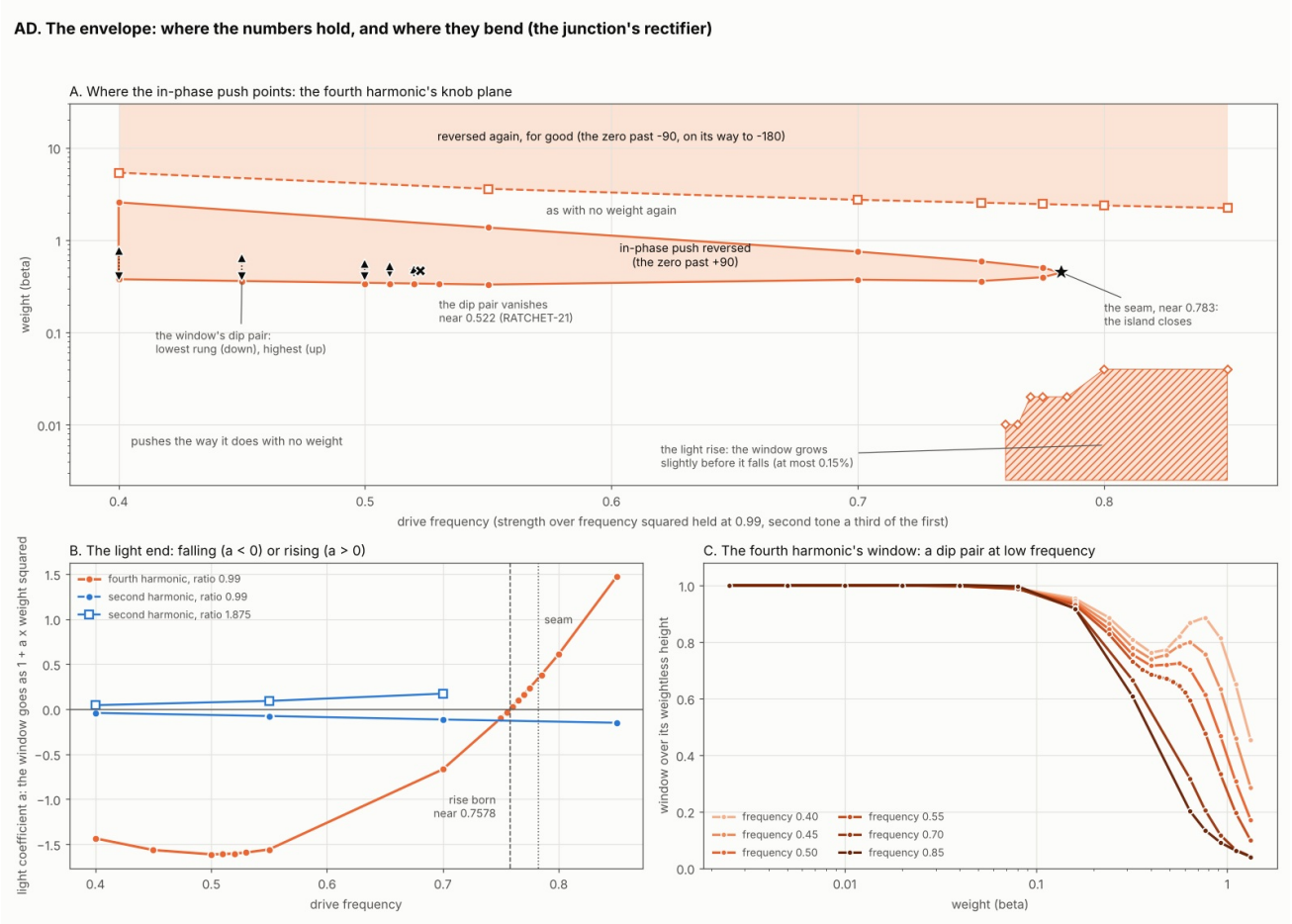


Figure 59. The envelope: where the numbers hold, and where they bend. The junction's rectifier, second tone a third of the first. A: the fourth harmonic's knob plane on the drive ratio 0.99, frequency across and weight up (log scale). Filled circles: the weights where the zero passes +90 degrees, the floor and roof of the island (shaded) where the in-phase push is reversed; the star, the seam near 0.783, where the island closes; hollow squares on a dashed line, where the zero passes −90 degrees, past which (shaded) the push is reversed for good. Triangles: the window's dip pair, its lowest rung (pointing down) and the highest after it (up); the cross, where the pair vanishes, near 0.5225 as read from the coarser rungs (0.5228 on the finest). Hatched: the light rise, from no weight up to the rung of the highest window. B: the light coefficient a, read at weight 0.0025, against the frequency, for the fourth harmonic on the ratio 0.99 (orange) and the second harmonic on 0.99 and 1.875 (blue); the dashed line, where the fourth's a crosses zero, near 0.7578; the dotted line, the seam. C: the fourth harmonic's window over its weightless height against the weight, at frequencies 0.40 to 0.85 (light to dark). From RATCHET-3 to 8 and 11 to 21.

6.12 The dark points: the eye of the storm

At the eye of a storm the air is still, and all around it the wind turns. The rectifier has places like that. A dark point is a setting of the knobs where the push loses its direction, and on any small loop around it the direction turns once all the way round. They earn a section of their own for three reasons. They are walls of a fourth kind, beside the symmetry's, the rider's and the ruler's walls of §6.10: topology's walls, the only doors through which a whole-number count can change. They mark where a knob changes its job, from steering the push all the way round to only nudging it. And, with their sibling the flat point, they are the drive's fine-trim settings: places where a knob's effect is magnified, or switched off, or turned into a null reading. This section gathers what is known: how to find a dark point, what it is, where it moves as each knob turns, and what to build with it. The table after 'Fine trims' is the map so far.

A count on the dial. Carry the window once round the dial, the third tone's timing from in phase through against and back, and watch not its size but the way it points (RATCHET-52); from here on the window has a direction as well as a size. Think of two arrows added together: the octave's, which stays put, and the fourth harmonic's, which turns once with the timing. While the octave's arrow is the longer, their sum only rocks: with no weight and the octave at 0.02, the window's direction swings at most 14 degrees from where it started and comes back, a winding of zero. When the fourth harmonic's arrow is the longer, the sum is dragged all the way round: with the octave at 0.0025, the window turns one full time as the timing goes once round, and set against, the easy direction is reversed. A winding is a whole number, so it cannot change a little. It can change only where the window closes completely somewhere on

the loop, a dark point, and one lies between octave sizes 0.0025 and 0.0075. This is the book's holonomy reading of the rectifier: the thing carried round the loop is the window's direction, and what fails to return to its starting branch is counted, not measured. The geometry is elementary, two arrows, one turning; what the runs add is that the junction's mixing does not spoil the count, and where the dark point sits. For the engineer, the count is a test with only two answers: below the dark point the direction of push is set by the timing of the third tone, above it by the octave.

Where the dark point sits, and what it is. A dark point is a setting of the knobs where the window shrinks to nothing: the second tone's timing no longer moves the push at all, so the window has no direction to point. Opticians know such points as the dark cores of twisted light, oceanographers as the amphidromic points round which the tide wheels while the water there hardly rises, and mathematicians as the zeros of a complex number that depends on two real knobs. Two knobs are needed, one to bring each half of the window to zero, so in a plane of two knobs dark points are single points. Each carries a charge: the number of turns the window's direction makes on a small loop round it. The winding of any larger loop is the sum of the charges inside, so a dark point cannot fade away. It can only move, leave through an edge, or meet one of the opposite charge and vanish with it. The reflection says where to look. With no weight it makes the window at one timing the mirror image of the window at the opposite timing. In phase and against are their own mirror images, so there the window has no sideways part at all, and on the against line it can reach zero only by changing sign. A dark point off that line would have a mirror twin of the same charge, and the pair would change the count by two. The count on the dial changed by one, so the dark point sits on the against line, and finding it takes one knob and a sign. Three old dials put it at an octave size of 0.00525006. Runs at four sizes found it at 0.00525010, seven parts in a million away (RATCHET-53). A rival that left out the mixing, and put it 0.14% lower, failed.

Three arrows, not two. The prediction needed a third arrow. Split round the dial, the fourth harmonic's arrow is its own, the window RATCHET-42 measured with no octave at all (recovered here to every printed digit), plus a part that grows as the octave's size squared and turns round the dial the other way. That part is the two tones mixing: four, less two, less two. At the dark point it is a seventh of a percent. At the full octave, a third of the first tone, it is larger than the fourth harmonic's own arrow, 1.11 times as large at the size RATCHET-45 ran, so there the window's small wobble round the dial goes round backwards. Where the two turning arrows match, the loop goes flat: this is the flat point, where the timing changes only how big the window is, never where it points. Carried on from the small dials, they should match near an octave size of 0.132. Measured on either side, at 0.10 and 0.16, the wobble turns one way below and the other way above, and a straight line between the two put the crossing at 0.1247 (RATCHET-55). Read directly at the crossing, it is 0.1255 (RATCHET-63). The carried-on guess was about 5% high, because the octave does two things at once as it grows. It feeds the backward arrow as its size squared, and it drains the fourth harmonic's own arrow by the same law, to 0.93 of its strength at octave 0.10 and 0.83 at 0.16, so the ratio climbs faster than the small dials said. The rival, that the full-octave dial's ratio carries unchanged to a fourth harmonic about twice as strong, landed within 0.02% of the straight-line reading and 0.6% below the direct one: where the loop goes flat hardly depends on the fourth harmonic's size, as it should, since both turning arrows grow in proportion to it. At the edge's drive the flat point sits at 0.318 of the first tone's size, and the book's usual octave, a third, sits 4.9% past it. For the engineer this is a dial setting of its own: with the octave there, the third tone's timing trims the push's size and leaves its best timing alone. With these three arrows the window's direction at every timing of all three dials comes back within a tenth of a degree, and the largest swing obeys the two-arrow law: its sine is the turning arrows' length over the fixed arrow's, 44.5 degrees at octave 0.0075, where the eight timings caught 44.3.

The flat point in closed form. The flat point does not have to be hunted for. The octave feeds the backward arrow as its size squared and drains the fourth harmonic's own arrow by the same law, so the two meet where one minus the drain times the octave squared equals the feed times the octave squared. That puts the flat point at one over the square root of the drain plus the feed, and both come from a small dial: the fourth harmonic's window with no octave, and one small octave read at three timings of the third tone, four runs in all. At the edge's drive those numbers say 0.12495, 0.4% below the direct reading. At three quarters of that drive they were read first and the flat point then looked for, blind: the small dial said 0.08194 and the flat point came out 0.08207, within 0.16% (RATCHET-63). The small leftover is the next order: the drain and the feed themselves soften a little as the octave grows, and letting them soften, as read at octaves 0.10 and 0.16, lands within 0.002%. The flat point is not a fixed share of the first tone, as the book had guessed: it is 0.318 of it at the edge's drive and 0.277 at three quarters, growing as the drive to the power 1.475. The drain hardly changes with the drive (as its power −0.58), but the feed weakens fast as the drive grows (as its power −3.1), so a stronger drive pushes the flat point to a larger octave (a reading on two drives). For the engineer: to set the decoupler (the octave at the flat point, which trims the push's size and leaves its direction; 'Fine trims' below), run the small dial, compute one over the square root of the drain plus the feed, and set the octave there, with no search.

At fast drives, in swing units. Run at frequencies 2, 4 and 6, with every tone held at the swing it has at the edge's drive (its strength over its own frequency), the dark point and the flat point were each predicted from a small dial before they were looked for, and both landed blind: the dark points within 0.012% and the flat points within 0.14% (RATCHET-64). The flat point's share of the first tone is 0.3151 to 0.3152 at all three frequencies, and 0.3177 at

frequency 0.55: within 0.8% over an elevenfold range of frequency. It is set by the swing, not by the frequency, because the two turning arrows it balances answer the frequency the same way. The dark point's share settles as well, from 1.33% at frequency 0.55 to 0.73% at 4 and 6. For the engineer: in swing units the decoupler is one number, about 0.315 of the first tone at this swing, at any fast drive, and the magnifier, the octave near the dark point ('Fine trims' below), is another, about 0.73%.

The flat point at smaller swings. At fast drive the flat point's share is the same at every frequency, but not at every swing. At frequency 4, with the first tone cut to three quarters of its swing and the fourth harmonic and the probe (the small dial's one small octave) held, the small dial put the flat point at 0.530799, and it was found at 0.53100926 (FLAT-75). Its share of the first tone fell from 0.315176 to 0.246495, to 0.782087 of its full-swing value; at frequency 0.55 the same cut had kept 0.872172, and the lean that the share would fall the same way missed. Cut to half the swing, the small dial said 0.24383 and the flat point was found at 0.24384386, share 0.169789 (FLAT-50). The reason, read after the data, is that at fast drive the drain has almost gone. The feed is 0.9914 of drain plus feed at full swing (against 0.8908 at frequency 0.55), so the flat point is set by the feed alone, and the feed weakens fast as the swing grows: as the swing to the power −3.7226 between full and three quarters, and −3.8430 between three quarters and a half. The flat point grows as the swing to the power 1.8544, then 1.9194, steepening toward the square of the swing as the swing gets smaller. Registered as a single power of the swing carried from the first two swings to the third, the guess missed: the flat point came out 0.97398 of it. The small dial, which assumes no law in the swing at all, landed every time, and closer the smaller the swing: off by −0.1223%, −0.0396% and −0.00575% (FLAT and the flat point both at full precision; a six-decimal print of FLAT had given −0.0057%). Cut to a quarter of the swing, with the probe octave still held, so that it now stands 2.337 times the flat point, the small dial said 0.0622477 and the flat point was found at 0.0622461, share 0.086684 (FLAT-25): off by +0.00258%, the closest yet, its error now on the other side of zero. The swing's power from half to quarter was 1.969902, after 1.8544 and 1.9194, so the flat point is closing on the square of the swing, and its gap to 2 shrinks, 0.1456, 0.0806, 0.0301, the last step by much more than half. That was registered as a window between the last power and 2, and it held; the rival, that the steepening stops, failed. In the gentle limit the feed alone sets the flat point and the feed falls as the swing to the minus fourth power, so the flat point goes as the square of the swing (a reading after the data, matching the feed's measured powers, −3.7226 and −3.8430, closing on −4).

Why the square of the swing. Cut to an eighth of the swing, with the probe now 9.3 times the flat point, the small dial said 0.0156459 and the flat point was found at 0.0156454 (FLAT-12): off by +0.00311%, on the same side of zero as at a quarter and a little further out. The power from quarter to eighth was 1.992246, within a hundredth of 2. How fast that gap should close was registered from a symmetry. Turning the clock by half a beat of the first tone flips the first tone's sign and leaves the octave and the fourth harmonic as they were, so the window cannot tell a swing from its negative. The flat point is then the swing squared times a series in the swing squared, and halving the swings should divide the gap to 2 by four. It shrank to 0.2576 of the last gap, against the quarter predicted. A gap shrinking only in proportion to the swing would have left the power at 1.98495, and that rival missed. The two turning arrows that set the flat point show the same law on their own (a reading). The fourth harmonic's own arrow falls as the swing to the power 3.916, 3.970 and 3.993, closing on four, because it is made of four quanta of the first tone against one of its own. The backward arrow is two octaves against one fourth harmonic, four less two less two, and needs no first tone at all, so it hardly moves: by −0.56% from a quarter of the swing to an eighth, its steps shrinking by 0.2653, near the swing-squared law's 0.25. The small dial's error grew by a fifth while the probe's lead over the flat point grew four times, so the lead does not set it. What does is the gap between two squares. At all five swings the error is one number, 0.00149 to 0.00153, times the probe's square less the flat point's square: it changes sign just where the probe passes the flat point, and a probe sitting on the flat point would be exact (a reading after the data; the reason is that the dial assumes each arrow moves as the octave squared, while their next terms move as its fourth power). Written first from that reading, doubling the probe at an eighth of the swing should raise the error to 1.2533e-4, and halving it should lower it to 7.49e-6. They came to 1.2548e-4 and 7.47e-6, within 0.12% and 0.32%, and the rival, that a floor of the solver sets the error so that it grows as the probe shrinks, failed (DIAL-GAP). For the engineer: at a fixed swing the decoupler is one number at any fast frequency, but it moves with the swing, and faster than at slow drive. Run the small dial at the swing you will use. At gentle swings the rule is simple: halve the first tone and the decoupler's octave falls to a quarter. And the small dial's own error is known: at frequency 4 it is about 0.0015 times the probe's square less the decoupler's, so set the probe near the decoupler you expect, or subtract it.

Counting the beats on your fingers. The square law above came from counting, and the count is a tool of its own. Give each tone its harmonic number: the first tone 1, the octave 2, a third tone at three times 3, the fourth harmonic 4. A steady push can come only from a group of tones whose numbers, each taken with a plus or a minus, add up to zero, and only from a group with an odd number of members. The second half of that rule is the book's mirror. Turn the angle over, from ψ to $\pi - \psi$, while every tone and the steady push flip their signs, and the equation comes back unchanged, so the push must flip when the whole drive flips, and a group with an even number of members cannot make it. A group's push grows as the product of its members' sizes, so counting the members reads a power law off before any run. The octave's push is 1 + 1 − 2: three members, two of them the first tone, so it grows as the first tone

squared times the octave, the form harmonic mixing has long been written in (Denisov, Flach and Hänggi's review gives it as their Eq. 61). The third harmonic alone has only odd numbers to work with, and odd numbers that add to zero always come in an even count, so it is silent: the half-beat symmetry of knob 6, seen as counting.

the group	members	its push grows as	measured
1 + 1 − 2	three	the first tone squared, times the octave	the octave's window (§6.2)
1 + 1 + 1 − 3	four	nothing: silent	the third harmonic alone (knob 6); BEAT-1's control, silent to 9.992e-16
1 + 1 + 1 + 1 − 4	five	the first tone to the fourth, times the fourth harmonic	the fourth harmonic's own arrow: powers 3.916, 3.970, 3.993 as the swing halves (FLAT-12)
2 + 2 − 4	three	the octave squared, times the fourth harmonic, with no first tone	the flat point's backward arrow: moved 0.56% from a quarter of the swing to an eighth (FLAT-12)
1 + 2 − 3	three	the first tone, the octave and the third tone, each once	the third tone heard through the octave: 1.96% of the window at size 0.01, 1.999932 times as much at twice the size (BEAT-1)
1 + 1 − 2 + 3 − 3	five	the octave's push, times the third tone squared	the part of the window that does not turn with the third tone: 3.9994 times as much at twice the size (BEAT-1)
2 + 3 − 1 − 1 − 1 − 1 − 1	seven	the first tone to the fifth, the octave and the third tone once	a second arrow turning the other way, 1.2% of the main one at the book's drive (BEAT-1 to BEAT-3)

The count also predicted something before it was run. With the octave present, the silent third tone has a group of three, 1 + 2 − 3, so it should be heard through the octave, in proportion to its own size, and its timing should turn one new arrow in the window once round for each turn. It was, and it did (BEAT-1): at a third tone of 0.01 the window gained an arrow 1.96% of its size, 1.999932 times as large at twice the size, and with the octave taken away the third tone fell silent again, to the file's last digit. The count missed something too, and then explained the miss. The new arrow came out as two: the main one and a second, turning the other way, 1.2% of its size. The count finds a group for the second only at seven members, 2 + 3 − 1 − 1 − 1 − 1 − 1, five of them the first tone, so its share of the main arrow should fall as the first tone to the fourth. Halving the first tone cut that share 10.15 times, a power of 3.34, and the main arrow, which counting says should halve, fell only to 0.568 of itself, a power of 0.82 (BEAT-2, which missed both its bars). Halved again, both powers closed in, to 3.80 and 0.951 (BEAT-3, which held). Each gap to its counted value shrank to between a quarter and a third of itself, from 0.66 to 0.20 and from 0.184 to 0.049, near the quarter that the reflection's swing-squared law gives. So the count gives each power exactly in the gentle limit, and a strong drive bends it, by an amount that shrinks about as the drive squared. Nonlinear optics uses the same rule to say which crystals can double a laser's frequency: a material with a centre of symmetry has no even-order response, just as this rider gets no push from an even group. For the engineer, counting says before building which stray tones matter. A stray tone that can join the drive in an odd group will steer the push; one that cannot is harmless; and the smallest group it can join says how fast its effect fades as the drive is turned down. A third tone at three times the frequency, which the book had filed as a clean blank, is a blank only when the octave is off.

How the dark point moves with the drive. The octave's push uses the first tone twice, and the fourth harmonic's uses it four times. So where the two balance, the octave needed grows as the drive squared; put the other way, the fourth harmonic's share against the octave's falls off as one over the drive squared. That is the leading order. At the edge's drive the dark point grows as the drive to the power 1.78 just below and 1.74 just above (RATCHET-53): the inverse square, bent by about an eighth by a strong drive. Each arrow alone bends much more, the octave's to about 1.3 powers and the fourth harmonic's to about 3.1, but the two bend together, and their difference, which places the dark point, stays near 2. If that is the leading order bent by a strong drive, the power must climb back to 2 as the drive weakens, and it does (RATCHET-54): 1.95 between a quarter and a half of the edge's drive, 1.85 between a half and the whole, 1.78 just below it. Each arrow climbs toward its own leading power as well, the octave's to 1.85 and the fourth harmonic's to 3.80 at the weakest drive run. So the inverse square is the dark point's law in the drive, exact in the gentle limit and bent by about an eighth at the edge's drive.

How the dark point moves with the frequency. The book's drive unit, strength over frequency squared, carries many of the rectifier's numbers from one frequency to another (§6.5). It does not carry the dark point (RATCHET-58 to 60). With the first tone held in that unit, and the fourth harmonic held in proportion to it, the octave needed for the dark point, as a share of the first tone, is 0.82% at frequency 0.20, 0.92% at 0.30, 1.05% at 0.40, 1.33% at 0.55, 1.70% at 0.70, 2.17% at 0.85 and 2.74% at 1.00. Above 0.40 the share grows by the same factor for every equal step of frequency: its logarithm rises by about 1.6 for each unit of frequency on every stretch run. That is an exponential in the frequency, not a power of it, and it was registered as such before the run at 1.00, which landed within 1% of the exponential and 3% off the power law (RATCHET-60). Below 0.40 the exponential flattens: at 0.20 the share sits 2.7% above where the exponential would put it if carried down at the slope of the stretch just above it (1.371 in the logarithm per unit of frequency, written down before the run, RATCHET-60), so a slower drive keeps helping, but less and less (figure 60, panel A). The two arrows part company. The octave's arrow keeps the unit closely, to within a fifth of a power of the frequency. The fourth harmonic's arrow keeps neither that unit nor strength over frequency: held in the first it grows with the frequency, and held in the second it falls, as the frequency to the power −1.25 to −1.30, steadily from 0.30 to 0.85. Between the two, the fourth harmonic nearly holds still at strength over the frequency to the power 1.32: run that way from 0.30 to 0.85, its window stays within 8% of its size at 0.55, where held in strength over frequency it ranged from 2.2 times to 0.58 times (RATCHET-60). What is left is a gentle bow, lowest in the middle, because the power that would hold it still drifts with the frequency, from about 1.38 on the slowest stretch to 1.28 on the fastest (a reading). Across frequencies 0.30 to 1.00 that drift is even in the logarithm of the frequency, about 0.12 for every factor of e. A straight line through the four stretches, fitted before the next run, put the stretch from 0.85 to 1.00 at 1.257, and it came out 1.262 (RATCHET-61). Past 1.00 the drift slows. Between 1.00 and 1.50 the holding power is 1.24, where the even steps said 1.22, so the test registered for the even steps failed there. Read after the data, the fall for each factor of e had already eased to about 0.09 going into the stretch from 0.85 to 1.00, and eased to about 0.07 after it. The easing begins near frequency 1, where the drive starts to outpace the rider's own pull, whose strength sets the book's unit of time (a reading). Run on to frequencies 2 and 3, the holding power levels off (RATCHET-62). A curve easing toward a floor, fitted through the last three stretches before the run, held within its bar, and both rivals failed: that the fall stops dead, and that it keeps going at its last pace. The data are flatter still than that curve, 1.2315 between 1.50 and 2.00 and 1.2299 between 2.00 and 3.00. So at fast drives the fourth harmonic keeps a unit of its own: strength over the frequency to the power of about 1.23. So the faster the drive, the more octave it takes to balance the fourth harmonic, and the slowest drive run needs the least. For the engineer that is one more number to carry with the frequency, and one more vote for a slow drive.

Why it levels near 1.23. At a fast drive the rider cannot follow each wiggle of the music. It feels each tone through how far that tone could swing it in one wiggle, the tone's strength over its own frequency, and averages the rest away, the way a pendulum shaken fast feels only an averaged pull (Kapitza's pendulum works this way). That averaged pull is the same in both directions, so on its own it cannot rectify. The lopsidedness that makes a push comes one order further on, as one over the frequency. The fourth harmonic's window needs four quanta of the first tone and one of its own, so it grows about as the fifth power of the tones' swings. Put together, the window holds still when the strength grows as the frequency to the power one plus one over that count. This was said before the run, as a guess, and was not graded. Read after the data, both of its parts show. Held at strength over frequency, the window falls as the frequency to the power 1.14, then 1.08, then 1.04, closing on 1. Its power in the tones' strengths is 4.5 to 4.7 rather than the 5 of weak tones, because these tones are not weak. Together they put the floor near 1.23. The leading term of the averaging, a product of two Bessel functions, counts 4.8 to 4.9 and puts the floor 0.010 to 0.014 low, so it carries most of the shape but not the last digit (a reading). Run again at half the drive, the floor came out 1.2231, beside the 1.2242 predicted, but not by the predicted route (RATCHET-63). The window's power in the tones' strengths rose to 4.90, nearly the leading term's 4.96, so the gap that held the floor up shrinks as the tones weaken; and the lopsidedness order rose too, to 1.09, so it is not set by the frequency alone. The two moves partly cancelled. Read on two drives, the order's excess over 1 doubles when the drive halves, which says the fast-shaking picture is reached sooner by a stronger drive. Carried on to frequencies 4 and 6 (RATCHET-64), the order closes on 1 faster than guessed, as one over the frequency squared: its excess fell to 0.264 of its value at both drives. Carried on to frequencies 8 and 12 with that law registered first, it fell again, to 0.254 and 0.252 of its value on 4 to 6 (RATCHET-65, one registration in three parts: this law, the 0.55 band re-read on finer rungs, and the blind 0.40 band of §6.11). One doubling further, on 16 to 24, it fell to 0.2510 and 0.2507 of its value on 8 to 12, so 0.0638 and 0.0635 of its value on 4 to 6 after two doublings, where the law says a sixteenth; those runs kept fifteen decimals, and their reruns at 8 and 12 matched RATCHET-65's files within the files' rounding (K-16). And at weak drive the excess stops growing as the drive weakens: at an eighth of the drive it is within 1.1% of a quarter's. The floor there comes to 1.2076 at a quarter of the drive, close to the one plus one fifth the averaging picture gives for weak tones. At full drive it rises again on 4 to 6, to 1.2455, because the held swings grow large enough that the window's power in them drops to 4.12. So the floor near 1.2 is a weak-tone number, and strong tones lift it. For the engineer: above frequency 1, scale the fourth harmonic with the frequency to the power 1.23 and its window stays put; below 1, the power to use drifts up toward 1.38 at frequency 0.30.

What a dark point is for. On the against line the octave's size is a reversing lever: above the dark point the push goes one way, below it the other, and the dark point is neutral. Off that line the dark point divides two kinds of steering. Below it the third tone's timing swings the push's best timing all the way round; above it, only within a cone whose half-angle has the arrows' ratio for its sine. Near it the direction is touchy, and a small turn of a knob swings it far, as a robot arm's joints race near a singular pose. That is bad for a controller and good for a sensor. A balanced bridge is most sensitive at its null, and the dark point is a null whose place is set by the ratio of two arrows, so its position reads how strong the fourth harmonic is against the octave: a meter for a stray harmonic in the drive, read off as an octave size. With weight, the reflection pairs a rider with its negative-weight twin, not with itself, so the dark point is free to leave the against line, and it does (RATCHET-57). It slides along the third tone's timing in exact proportion to the weight: 0.0129 of a half-turn at weight 0.01, 0.0258 at 0.02 and 0.0513 at 0.04, about 232 degrees of timing per unit of weight. The reason is the two arrows' slides. As the rider gets heavier, each arrow turns, the fourth harmonic's 5.1 times as fast as the octave's, close to the 5.14 it shows alone (RATCHET-39), and the dark point sits wherever the two point exactly opposite ways. Its octave size shrinks as the weight squared, by 1.1% at weight 0.08. Its charge keeps it alive: it moves, and it never vanishes. That makes it a scale. Turn the third tone's timing until the push vanishes, and the distance from the against line reads the rider's weight, in proportion. It is a null reading, the kind a balance makes, where the instrument hunts for nothing rather than measuring something, and null readings are the precise ones.

Fine trims: a magnifier, a decoupler and a scale. Three settings on this map are tools for trimming a drive. Near a dark point the push's direction is touchy: the closer the two arrows are to equal, the further a small turn of the timing swings it. At octave 0.0025 one eighth of a turn of the timing swung the direction by 72 degrees (RATCHET-52). That is a magnifier, a vernier for the direction, paid for with a small push. At the flat point the opposite holds: the timing changes the window's size and leaves its direction alone (RATCHET-55). That is a decoupler, keeping a size trim apart from a steering trim, the way a good control panel keeps gain and phase on separate knobs. And the dark point itself, followed as the rider's weight changes, is a scale read as a null (RATCHET-57). It is a meter as well: the octave at the dark point grows in proportion to the fourth harmonic, 0.3% and 1.3% off proportion at half its size and at twice (RATCHET-61), so the octave needed to darken the push reads how much fourth harmonic the drive carries. The small bend away from proportion grows as the square of the fourth harmonic's size, as the mirror that turns its sign into a half-turn of its timing says it should. That was read on two sizes, then predicted at three times the size, where the octave came out 0.967 of proportion against the 0.964 predicted; a bend in proportion to the size would have given 0.986 (RATCHET-62). At the book's usual octave there is no dark point to use. The dark point's octave is a 25th of it, and to bring a dark point up to it the fourth harmonic would have to be about thirteen times the first tone, so at the usual octave the third tone's timing can only nudge. That was checked as far as eight times the fourth harmonic's size on the full-octave dial (0.176, as RATCHET-45 ran it), 3.57 times the first tone (RATCHET-62). The push at the against timing kept its sign all the way, closing only from −0.0256 to −0.0184. Its turning arrows had begun to fall behind proportion there, by 13%, the shortfall growing about four times with each doubling (the same square law), while the steady part shrank a little, so the push ended slightly nearer dark than proportion alone predicted (−0.0191). An engineer trimming this drive sets the octave by the job: below the dark point to steer all the way round, at the flat point to trim the size without steering, and near the dark point to read small changes with the most gain.

knob turned	what the dark point does	the numbers	run
octave size and third-tone timing, no weight	one dark point, pinned to the against timing by the reflection	octave 0.00525010 at the edge’s drive (A 0.394944, ω 0.55, fourth harmonic 0.3802)	RATCHET-52, 53
the drive’s strength	grows as the square of the drive (the inverse square, read from the octave’s side), bent by a strong drive	its octave grows as the drive to the power 1.95 at a quarter of the edge’s drive, 1.85 at half, 1.78 at the edge	RATCHET-53, 54
the rider’s weight	leaves the against line in proportion to the weight, and shrinks as the weight squared; held where it was, the push comes back a quarter-turn across, in proportion to the weight	1.29 half-turns of the timing per unit weight; its octave 1.1% smaller at weight 0.08; the push brought back 0.00428 of the weight, its size over the weight even in the weight, its direction meeting 90 degrees as the weight goes to zero	RATCHET-57, 67
the frequency, drive held at A/ω² 1.3056	moves: the fourth harmonic grows with the frequency faster than the unit; above 0.40 an exponential in the frequency	its octave 0.82%, 0.92%, 1.05%, 1.33%, 1.70%, 2.17% and 2.74% of the first tone at ω 0.20, 0.30, 0.40, 0.55, 0.70, 0.85, 1.00; above 0.40 its logarithm rises about 1.6 per unit ω; below, it flattens	RATCHET-58 to 60
the flat point, its sibling	the turning part of the loop collapses to a line; the timing trims size only; placed by a small dial	octave 0.1255, 0.318 of the first tone, nearly the same for a fourth harmonic about twice as strong; 0.08207 at three quarters of the drive, where the small dial said 0.08194	RATCHET-55, 63
the fourth harmonic’s size	its octave grows in proportion to the fourth harmonic, bending as the size squared	1.003 and 0.987 of proportion at half and twice the size (fourth harmonic 0.19 and 0.76); 0.967 at three times (1.14), where the square law predicted 0.964	RATCHET-61, 62
the book’s usual octave, a third of the first tone	no dark point within reach	the fourth harmonic would have to be about 12.8 times the first tone (a reading from the full-octave dial); run to 3.57 times the first tone, the push kept its sign	RATCHET-43 to 45, read from their outputs before RATCHET-61 was registered; RATCHET-62
fourth harmonic’s own unit	a power that drifts, then levels	1.376 (0.30 to 0.40), 1.279 (0.70 to 0.85), 1.2435 (1.00 to 1.50), 1.2299 (2.00 to 3.00): the power of the frequency that holds the fourth harmonic alone still	RATCHET-60 to 62
fast drives, every tone held at its swing	both points placed blind by small dials; the flat point keeps its share across frequency, not across swing	flat point 0.3151 to 0.3152 of the first tone at ω 2, 4, 6 (0.3177 at 0.55); at ω 4 with the first tone cut, 0.246495 at three quarters, 0.169789 at half, 0.086684 at a quarter and 0.043576 at an eighth, closing on the swing squared; dark point 0.0077, 0.0073, 0.0073	RATCHET-64, FLAT-75, FLAT-50, FLAT-25, FLAT-12

The scale on the bench. The explorer now carries the rider’s weight as a knob (§3.2), so the scale can be watched. Set the dark point and give the rider a little weight. The push that had vanished comes back, in proportion to the weight: 0.00428 of the weight at weight 0.01, still 0.00416 at 0.08, and 0.00428 again at 0.004, a weight no run had tried. It comes back pointing across its old direction. With no weight, the little that is left of the window at the dark

point (about 3×10^{-10}) points at 180 degrees, along the old direction; at weight 0.01 the push that comes back points at 92 degrees, nearly a quarter-turn round from it. That is the dark point's slide seen from where the dark point used to be. The two arrows that cancelled there now turn by different amounts, the fourth harmonic's about five times as far as the octave's, and what is left over points across them. That much was a reading of RATCHET-57's file and the bench, so it was then put to a blind test (RATCHET-67). The mirror that pins the dark point to the against line with no weight pairs a rider with its negative-weight twin and turns the window into its mirror image. If that is right, the push that weight brings back must start exactly across the old direction, and its size over the weight can have no straight-line part, only a square one. Both were written down from RATCHET-57's rungs before two lighter rungs were run, at weights 0.0025 and 0.005. The size landed 1.59 parts in a million from the square law, where a straight line would have missed by 6.33 parts in ten thousand. The direction met a quarter-turn at 89.999958 degrees as the weight went to zero, where a straight line through the old rungs had pointed at 89.997426. For an engineer this is a phase-sensitive meter. Referenced to the old window's direction, a lock-in reading of the push (a lock-in amplifier reads a signal against a reference wave and splits it into the part in step with the reference and the part a quarter turn off) sees the weight only in the square in its in-phase channel and in proportion in its quadrature channel, so at the dark point the quadrature channel is a linear scale for the rider's weight, and the ratio of the two channels gives it a second time.

The scale as a two-channel meter. A real drive drifts, so a scale that reads the rider's weight must not mistake a drifting drive for weight. The same mirror sorts the two. The drive's strength is not touched by the mirror, so on the against timing a drift with no weight still leaves the window real: it moves the push only along the old direction. Weight's first push is across. So the push at the dark point has two channels, in phase for the drive and in quadrature for the weight, and a drift can never put an offset into the weight's channel; it can only change that channel's gain. That was registered with a number before it ran (RATCHET-71). Near the dark point the window is the octave's slope times its distance from the dark octave, and the dark octave moves with the drive as its 1.7608th power, so a drive 0.1% stronger should move the window by +1.8607e-06, all of it in phase. It moved by +1.862829e-06, and the drive 0.1% weaker by −1.856839e-06, within 0.21% of the prediction, with the quadrature part still exactly zero. With weight, the quadrature channel's reading changed between the two lightest weights only by its own cubic term, the same at all three drives, so there was no offset. The gain did move: the weight reading grows as the drive to the power 3.3766, so a 0.1% drift scales it by 0.3377%. Because the in-phase channel reads that drift directly, it can correct the gain. For the engineer this is a ratiometric scale: read the push with a phase-sensitive detector referenced to the dark point's own direction, take the weight from the quadrature channel, and take the drive's drift from the in-phase channel to correct its gain. A reference one degree off lets a 0.1% drift pass for a weight of 7.580e-06. Phase-sensitive detection and null-balance bridges are standard measurement practice; what is particular here is that the junction's own mirror puts drift and weight at right angles. Because the result looked almost too clean, it was put to a second witness (WITNESS-1): an engine written fresh that shares only the equation with RATCHET_3T, with a different integrator (an adaptive eighth-order one), a different way of finding the edges, and a set of timings no run had used. It agreed on the drift to within 5.07e-07 of the first engine's value, relatively, and on the weight's gain to within 3.70e-08, and it left the weightless window real to below 1e-15. Pushed to ten times the drift (RATCHET-72), the in-phase response kept its straight line to 1.00001 of the prediction, its small bend matched the curvature predicted from two older runs to 0.9999, and the gain followed its power law to within 0.34%.

How many timings the meter needs. The meter's freedom from drift is not the dark point's alone. It is §6.3's zero: on the mirror line, with no weight, the push is exactly zero at the quarter timings, where the octave is turned a quarter turn or three quarters of a turn ahead of the first tone (φ of π/2 or 3π/2, in quadrature), at any drive and any octave. So one timing read at a quarter is already a weight sensor that no drift can offset, and at the weight-proof edge of §6.11 it is 6.083 times stronger than at the dark point (TIMINGS-1). What the dark point adds is the empty background: with no window to tilt, a phase reference that is slightly off has nothing to leak. The timings themselves must respect the mirror. Read at an even number of timings that start at zero or half a step, the meter is clean at every count: eight and four read the weight with the same gain as sixteen, and the two quarter timings alone read the weight at twice the gain with the drift channel exactly blank. Off the mirror, the push's higher harmonics fold into the window turned a quarter-turn, straight into the weight channel. At the edge, where those harmonics are strong, the fold was predicted from sixteen timings and measured by the second witness to within 2.15e-13 at four timings and 7.2414e-14 at two (TIMINGS-2; the first printing of the two-timing figure, 3.41e-12, was only the rounding of a file that kept ten figures). Four timings set a sixteenth of a turn off the mirror then mistake no weight for 3.8101e-04, and two for 5.1128. The engineer's rule: put the timings on the mirror, and if they cannot be, use eight or more.

The ninth note. One leak in that test came out a little off: at eight timings it missed its prediction by −7.21e-03 of itself, while four and two matched every printed figure. Two things were hiding in it. The first is aliasing. On sixteen timings the push's ninth harmonic cannot be told from its seventh, because cos 9φ and cos 7φ agree at every one of them, so a prediction read from sixteen timings carries the ninth inside the seventh, while eight timings set off the mirror see the two with opposite signs. Run at the record's own timings as well, the second witness held thirty-two timings in one engine and resolved the ninth: −1.3072e-13. It sits on a smooth ladder. Climbing the odd harmonics, each is a small fraction of the one before: 5.3097e-04 of it, then 1.6560e-03, 2.2132e-03 and, for the ninth, 2.5353e-03 (NINTH-1). The second thing looked like the engines and turned out to be the record's file. At the same sixteen

timings, the record and the witness differ by at most 7.701e-13 in the push, so the ninth made only 0.699 of the eight-timing miss. Sized from that mixed miss, the ninth came in at 0.6992 of its registered size, and every verdict of that first test failed. The redesign stayed inside one engine and went out of sample: at a third set of timings no run had used, the leaks predicted with the ninth landed within 3.068e-17 at eight timings and 6.512e-16 at four, and without it they missed by 1.850e-13 and 1.843e-13 (NINTH-2). Then the gap itself was taken apart (STEP-1). The record's engine writes each edge to 12 decimals; written to 15, the same run differs from its record by up to 7.521e-13, almost the whole gap. Its step error is small, and when the step is halved the change shrinks 16.527 times, as a fourth-order method's should. With full digits the two engines agree to 2.598e-14, and that last sliver is the witness's own lean: with its tolerances ten times tighter the lean fell to 0.0857 of itself, the engines agreed to 3.997e-15, and the ninth stayed at 0.9985 of its size (LEAN-1). Two lessons for the engineer. A model read from N timings mixes each harmonic n with harmonic N − n, and the pair on either side of N/2, here the seventh and the ninth, is exactly what N/2 timings set off the mirror fold into the weight channel, so check a sampling scheme against a model sampled several times more finely than the scheme itself. And at this level every digit is part of the instrument: a file that keeps twelve decimals cannot judge a difference near 1e-13, and an engine's tolerance shows up as a steady lean that tightening it shrinks.

Odd counts of timings. A rectifier engineer knows why a three-phase bridge draws only the line harmonics 6k ± 1, the harmonics of six pulses: the three equal spacings remove the multiples of three, and the currents' half-wave symmetry removes the even ones. The push has that same half-wave symmetry. §6.7's half-beat shift leaves it only odd harmonics, so half a turn of the octave later the push is the same with its sign flipped, and read through the window's turning arrow it repeats exactly. An even number of equally spaced timings therefore spends half of them repeating their half-turn partners, and an odd number never does, so M odd timings read what 2M read. Set a sixteenth of a turn off the mirror at the edge, three timings read the same window as six to within 3.376e-15, and five the same as ten to within 2.113e-16. Three crooked timings leak only 2.1625e-03 of what four leak, because three fold in the fifth and seventh harmonics where four fold in the third; their leak, −2.147628994e-08, was predicted from the witness's thirty-two timings to 7.525e-08 of itself. Five crooked timings fold in nothing below the ninth and leak −1.3918e-13, where eight leaked −5.145035e-11. Odd grids can sit on the mirror too, if one timing sits on a quarter: three, five and fifteen timings placed that way read the weightless window clean, within 3.788e-15 of the witness's own lean (ODD-1). The engineer's rule gains a line: when the timings must be few, or cannot sit on the mirror, make their number odd.

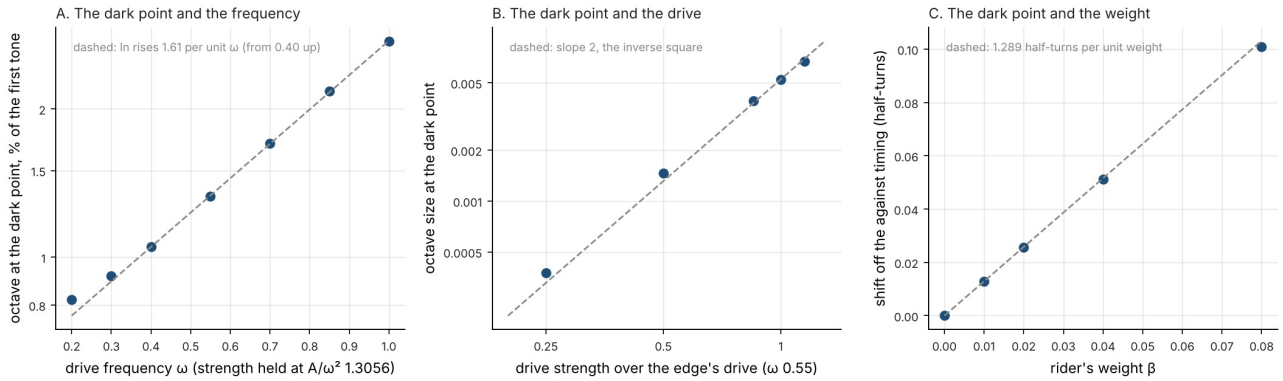


Figure 60. The dark points. The junction's rectifier with a third tone at four times the frequency, set against the octave, no weight unless stated. A: the octave needed for the dark point, as a share of the first tone, against the drive frequency, the first tone held at strength over frequency squared 1.3056 and the fourth harmonic in proportion to it (log scale); the dashed line is a straight line in the logarithm through the points from 0.40 up, an exponential in the frequency. B: the dark point's octave against the drive's strength, both on log scales, at frequency 0.55; the dashed line has slope 2, the inverse square, through the edge's drive. C: the dark point's shift off the against timing, in half-turns of the third tone's timing, against the rider's weight; the dashed line is the light rider's slope through the origin. From RATCHET-53, 54 and 57 to 60.

Left on the map: of the twelve smaller crossings read by a straight line between rungs, five are settled without a run (three re-read here with a curve moved by at most 0.00032, two by a recheck of the record's own numbers), and four needed a run, all four since done (the dip pair's vanishing near 0.5225, where the fold's law reads 0.52250 and a straight line in the depth 0.52098; the 0.70 joint edge's size 0.3795; the 2/5 swap tone's bracket; the hand-off widths), and three are already hedged in the book's own words ("near", "about", "read between the dial's steps"), a reading recorded as TRIAGE-12. The first of the four has since been run, its test written first (RATCHET-81): on weight rungs 0.01 apart the pair is still there at frequency 0.5215, a dip of 0.00014 of the weightless height, and gone at 0.5235. The straight line's place is passed and the fold's holds; read again from the finer rungs, the fold's law puts the vanishing at 0.5228 and the straight line at 0.5222, and a parabola through the rungs moves the fold's place by only 0.00001, so at this stride the answer has stopped moving. The finer rungs also read the dip at 0.52 12% deeper than

the old ones had (0.00044 against 0.00039), lesson 155's one-sided bias once more. The other three have been run too, each test written first. At the 0.70 joint edge's size, 0.3795, the light coefficient a/ω² comes out at −0.00002, where the cubic it was read by said −0.00003 and a straight line between the runs at 0.30 and 0.40 said +0.0038 (RATCHET-82). The 2/5 dance through lag 1/2 lets go at −0.03363 on second tones 0.0001 apart, 0.000004 from the straight line's reading (PINCH-3). And the hand-off widths, read again on tones ten times finer, are 0.00076 at strength 0.10 and 0.00210 at 0.125 (WALK-3); the second had been read as 0.00214 from tone labels printed to four decimals, which put two of its tones 0.00005 off. The lopsidedness order's one-over-frequency-squared law, once on this list, has since been carried to frequencies 8 and 12 with its floor at weak drive (RATCHET-65) and to 16 and 24 (K-16).

6.13 Arrays of riders: the 3-3-9 array

Arrays of riders. Several riders on one drive, each with its own octave timing, add their pushes, provided they do not act on one another (an assumption this book has not tested). Fan their timings evenly round the octave's turn and the half-beat shift decides the sum, now with the roles of odd and even reversed from the meter's timings of §6.12. An even fan holds every rider together with its half-turn partner, whose push is the same with its sign flipped, so the pushes cancel at every shared timing: eighteen even fans read from the witness's runs added to at most 6.994e-15, where one rider's push reaches 2.599775e-02 (ARRAY-1). An odd fan keeps only the harmonics that are multiples of its count, so three riders keep the third harmonic and five the fifth: a small push that the shared timing steers three or five times as fast, a vernier. It is 1.5929e-03 of one rider's swing for three and 4.3965e-06 for five, and every fan matched its prediction from the harmonics, the fans of three to within 9.511e-11 and the fans of five to within 3.043e-08 of themselves. Both sides of that match are read from the same rider's runs, so it checks the bookkeeping of the harmonics; what the array adds rests on the assumption above. Two fans of three make a knob of their own: offset by a twelfth of a turn they keep −1.164422e-05, and offset by a sixth they become one fan of six and cancel to −9.992e-16. For an engineer this is a station-keeping drive: an even array holds still whatever the shared timing does, an odd array holds nearly still with a fine trim left in, and the offset between two sub-arrays sets how fine. Reading and summing use the same arithmetic with opposite verdicts: to read the push, an odd count of timings is worth twice its number; to cancel it, an even count is perfect.

Three groups of three: the 3-3-9 array. Bear calls this arrangement the 3-3-9 array, three groups of three making nine, a name and a picture he has carried since 2012, long before any of these runs. Nine riders split into three fans of three behave like a three-element phased array, with the offset between the groups as the steering. Each fan keeps only its trim, the third harmonic, so the three trims add with the offset's phase, and the swing of their sum, as the shared timing turns, follows a three-element array factor: the sum of three equal arrows turned by the offset, the same sum that aims the beam of a row of three antennas. Aligned, the nine swing 1.266138053e-04; offset by 10 degrees of the octave's timing, 0.910684 of that; by 20, 0.666667; by 30, 0.333333; and at 40 degrees the nine are one evenly spaced fan of nine, an odd fan that keeps only the ninth harmonic, and the swing all but vanishes (NINE-1, figure 61). Every total matched its prediction from the harmonics to within 1.667e-15. Three groups of two hold still at every offset, to at most 2.098e-14, because each half-turn pair cancels on its own. What it costs to hold the array in lock depends on what slips. If a whole group drifts together, the push moves only through its trim: a fan of three's largest slope is 4.7864e-03 of a lone rider's, and the evenly spaced fan of nine's 4.0028e-10 of it. A whole group of three slipping 10 degrees moved the push by 1.642522637e-05, where one rider slipping 10 degrees inside a half-turn pair moved it by −1.297858515e-03. Read after the data from the same harmonics, the aligned nine can drift together 69.642 times further than a lone rider for the same error in the push. If instead each rider's timing wanders on its own, grouping does not help: the spread of the nine's push is 5.8858e-02 per rider per unit of timing spread aligned, and 5.8857e-02 interleaved, adding as the square root of the count. So the upkeep of a phase-locked array is to keep each rider in step with its own group; the group's place against the drive matters far less, and an evenly spaced fan of nine's hardly at all.

The second lobe and the price of the null. Past 40 degrees the trim comes back. The array factor rises to 0.244017 of the aligned swing at 50 and at 70 degrees and to a third of it at 60, then falls to a second null at 80, where the nine are once more one evenly spaced fan of nine; read from the same thirty-six timings, each of these swings matched its prediction within 4.063e-16 (LOBE-1). So a null is a point, not a plateau: 40 degrees (or 80) is the only spacing that holds still. Its price is a V with straight sides. With one group slipped off 40 degrees, the swing returns in proportion to the slip, 4.417636383e-06 at 2 degrees and 2.184670946e-05 at 10, the same on both sides to six figures. These were read at a new set of 180 timings two degrees apart and matched their predictions within 1.057e-15. The swing follows the closed form $6|c_3|\ |\sin(3\pi s/2)|$, a slope of 2.209828e-06 per degree near zero slip, where c_3 is the third harmonic of one rider's push, the trim each group of three keeps, and s is the slip counted in half-turns, so that 10 degrees is s = 1/18: six times the size of c_3, times the sine of one and a half times π times the slip. Holding the nine still therefore asks for the spacing between the groups to be held as tightly as the allowed trim error demands, while the shared timing may wander almost freely.

Twelve riders, and why 80 degrees is 40 again. Add a fourth group of three and the array has four elements. Read from the same thirty-six timings, the twelve aligned swing 1.688184071e-04, 1.333333 times the nine, one third more for one third more riders. Offset by 10 degrees they keep 0.836516 of that, by 20 degrees 0.433013, and at 30 degrees

nothing. The trim comes back to a quarter at 40 degrees, 0.224144 at 50, nothing again at 60, 0.224144 at 70, a quarter at 80, and nothing at 90 (TWELVE-1). Every swing matched the four-element array factor within 1.585e-15. So three groups put their nulls every 40 degrees and four groups every 30, as the array factor says for g groups of three, a null every 120/g degrees (measured here for g = 3, 4 and 5); and the lobes between them are lower the more groups there are, a third of the aligned swing for nine and a quarter at 40 and 80 degrees for twelve. The V at the 80-degree null held its own surprise. Read at the 180 timings with one group slipped off 80 degrees, it matched the 40-degree V in every printed digit, even at its bottom, 1.951845869e-28. The reason is seating, not physics. Folded into one beat, the nine riders at 80 degrees sit in exactly the same nine seats as at 40, and the six that do not slip sit in the same six seats too; only the labels saying which rider belongs to which group have changed (a reading after the data). The twelve do the same: at 30 and at 90 degrees they sit in the same twelve evenly spread seats, and their swings at those nulls agree in every digit, 4.599223222e-31, while at 60 degrees they double up, two riders to a seat on six seats, and the bottom differs, 3.875712080e-31. For the engineer: what an array does is set by where its riders sit in the beat, not by how they are grouped. Two spacings that seat the riders the same way are one setting, so a design table needs only one of them.

Silent sets: a holding pattern with on and off ramps. Bear asked whether the angle could be “a staging ground”, a holding pattern with ramps on and off. The measured swings say it can. Each group of three carries only its trim, and a group’s trim turns with its offset at three times the rate. So two groups 60 degrees apart have trims half a turn apart and cancel, and three groups 40 degrees apart cancel as a set. Take every such silent set out of an arrangement and the groups left over carry the whole push (a reading after the data, from NINE-1, LOBE-1 and TWELVE-1, no new run). Four different arrangements prove it: nine riders at 30 degrees, nine at 60, twelve at 40 and twelve at 80 all swing exactly 4.220460177e-05, one lone group’s swing to every printed digit, because in each the other groups sit in a silent pair or triple. Twelve at 20 degrees swing 1.732051 groups’ worth, the two groups left once the pair at 0 and 60 is taken out; twelve at 30, 60 and 90 degrees leave nothing over and fall silent. For a designer, a silent set is a parking spot. Whole silent sets, six riders as two groups 60 degrees apart or nine as three groups 40 degrees apart, can be added, held or taken away without moving the push at all. Turning one group out of its set is an on ramp, and its trim returns smoothly as it leaves; turning it in is an off ramp. One price stays: a parked rider is silent on average but not quiet, because each rider’s own timing jitter still adds, 5.8858e-02 per rider per unit of its timing spread, whatever the arrangement (NINE-1). Parking costs nothing in push but something in noise.

Fifteen riders: five groups, and a pentagon in the lobes. Add a fifth group of three and the array has five elements, so its nulls come every 24 degrees: at 24, 48, 72 and 96, and again at 144 and 168. Before any swing was read, the seats were folded in exact fractions, and they said something the twelve had not. Five shares no factor with three, so at every one of those nulls the fifteen riders sit on the same fifteen evenly spread seats; no null doubles riders up the way the twelve’s 60-degree null does. All six nulls are therefore one setting, and the read confirmed it (FIFTEEN-1): their swings were equal to the last bit, as were the swings of every other pair of offsets that share seats, such as 12 and 132 degrees. The mirror pairs, 12 and 108 degrees or 36 and 84, are equal by the array factor but sit on different seats, and they agreed only to the run file’s floor, 5.042e-12 and 3.216e-12 of themselves, never to the last bit. Read from LOBE-1’s 180 timings with no new run, the sixteen swings matched the five-element array factor within 1.924e-15, and the forty-four swings with one group slipped off a null within 1.009e-15. Aligned, the fifteen swing five lone groups’ worth. The lobes carry the pentagon’s number: at 12 degrees the fifteen swing 3.236067977 lone groups, twice the golden ratio, because five equal arrows each turned 36 degrees from the last lay out half of a regular ten-sided figure, whose width is exactly that; at 36 degrees, where each arrow turns 108 degrees, they swing 1.236067977, twice its inverse (a reading after the data; this is the textbook five-element array factor). At 60 and 180 degrees the fifteen swing one lone group, 0.200000 of aligned: two silent pairs parked and one group left over, which the staging reading had found after the data and this read held blind. And the V at every five-group null is the nine-rider V of the 3-3-9 array, the same slope to the printed digits, because at a null the other groups add to nothing and only the slipped group speaks. For the engineer: more groups lower the lobes and pack the nulls closer, but the slip tolerance at a null is set by the one group that slips, so a bigger array does not loosen it.

On ramps: turning a group out of a silent pair. Two groups of three set 60 degrees apart are a silent pair: their trims point opposite ways and the pair pushes nothing. Turn one group back toward the other and the pair comes alive (RAMP-1). Its swing follows the two-element array factor all the way, from nothing at 60 degrees to two lone groups’ worth at 0, all forty-seven swings within 1.511e-15 of their predictions, and the ramp’s first few degrees are the 3-3-9 array’s V, the same slope to the printed digits, because leaving a silent pair and slipping off a null are the same move. The ramp also steers. With one group turned, the trim’s direction swings 1.5 degrees for every degree of offset, 87 degrees at 58 and 45 at 30, within 5.000e-09 degrees of that rule. Turn both groups instead, half each way, and the swing is the same while the direction holds still. For the engineer, one pair gives two ramps: the one-sided ramp is a size knob and a steering knob at once, and the split ramp is a pure size knob, a gain with no phase, the decoupler of §6.12 again, now in an array. The same read settled one more number. The fifteen’s null bottomed out at about 1.0189e-15 in FIFTEEN-1, and that was the run file’s last digit, not the riders: read with sixty shared timings 2 degrees apart, so that the 24-degree seats repeat every twelve timings, the bottom fell to 1.235112566025243e-31,

sixteen orders lower (FLOOR-1). A sampling grid that steps evenly into the seats lets the file's rounding cancel; one that does not leaves it in. For anyone reading an array this way: choose the number of samples so that the riders' spacing is a whole number of steps.

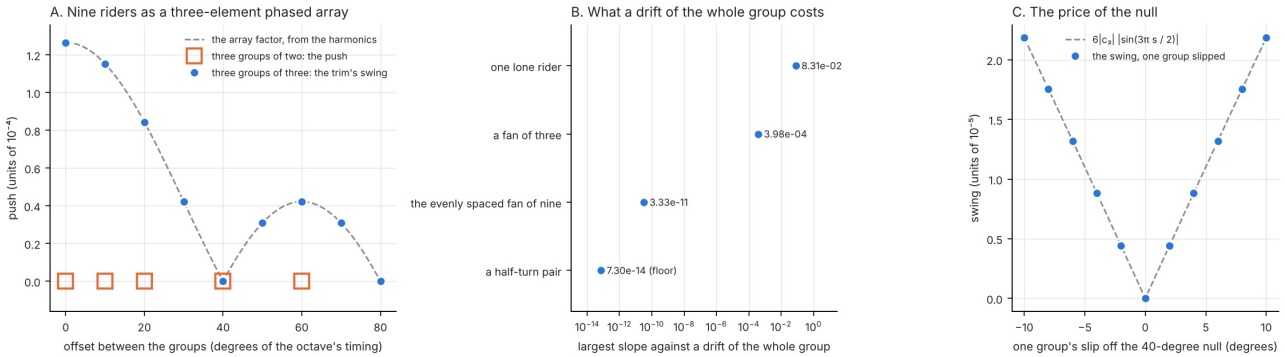


Figure 61. Nine riders as a phased array. Riders on one drive at the weight-proof edge, no weight, their pushes added (the riders assumed not to act on one another). A: three groups of three; dots, the swing of the nine's push as the shared timing turns, read from the witness at thirty-six timings, against the offset between the groups out to 80 degrees; dashed, the three-element array factor from the push's third harmonic; squares, three groups of two (half-turn pairs), whose push is zero at every offset. B: the push's largest slope against a common drift of the timing, for one lone rider, a fan of three, the evenly spaced fan of nine and a half-turn pair (the last at the witness's floor), read from the harmonics. C: the swing brought back when one group slips off the 40-degree null, read from the witness at 180 timings; dashed, $6|c_3|\,|\sin(3\pi s/2)|$. From NINE-1 and LOBE-1.

Part 7 — What failed, and what it taught

> *the rider's footsteps are seen through a picket fence. When a footstep lands near a picket, which gap you see it through depends on where the fence starts. The walker's stride is the real thing.*
>
> — no-physics read 3

Bear's frame for this part: "we share where it fits and where it fails…where it shines and where its light is hidden." Every lesson below was paid for by a miss, a slip or a test that held for the wrong reason, and every one names the run or the slip it rests on, so it can be checked in the record, whose public extract is published with the book. This part is where the method's lessons live in full. The Field Guide's chapter "Two Readers and a Ruler" and its rows carry them to other work and cite this part; §7.10 says which row rests on which lesson.

Each lesson keeps, in brackets at its end, the label it was first written under: B6.n for the lessons of sittings up to B6 (5 and 6 October), B7 for the instrument lessons of sitting B7, B8.n and B9.n for the series that runs from 1 to 100 across sittings B8 and B9, B10.n for sitting B10's (from 101), and PR8, PR9.n and PR11.n for the pattern-recognition lessons of sittings B8, B9 and B11; P2 marks a lesson of the second publishing sitting. A label that says "plan" points to the book's plan: the working outline kept beside the record while the book was written, whose numbered rows listed the runs still needed and the questions they answer (its file is named in the Declaration). Where one lesson was learned more than once, it is written once with every time it happened. The lessons are numbered through this part, 1 to 168, ten of them now kept in the Field Guide instead (§7.6); a number in brackets with a prefix (B9.81) is the record's, a bare number (lesson 81) is this part's. The limbs are the Field Guide's: L1 what you read, L2 where it lives, L3 what you write, L4 what you owe.

7.1 The ruler: a verdict is only as fine as what measured it (L1)

1. Print what the ruler's step allows, and say what it does decide. THIRD-FACE-2's first ratio read 7.33 against a registered band of 3 to 7 and was written down as failed; but the dial's step allowed anything from 5.25 to 11.5, which crosses the band's edge, so the failure stands as registered and decides nothing. The same coarse ratio was decisively above Feigenbaum's 4.669, at 1.12 to 2.46 times it: a ruler too coarse for one question still answers another. Under every judged number, print what the step allows and whether that range crosses an edge of the bar. The off-scale flag that does this caught an overstatement on its first use and, on its second (AD-2c, 33 of 36), showed a held verdict resting on one case. A step's range covers the dial, not the settle: read again with a settle three times as long, the first doubling moved and the ratio came to 6.18, still above 4.669 (THIRD-FACE-3, TF3-1b). (B8.1, B8.2)

2. A failed prediction often failed on its ruler, not its idea. BRIDGE-1's far-bank prediction failed because one weight's bank had been read on a grid four times coarser than the others; on one ruler (BRIDGE-1b) all seven banks fell in order. WAYSTATION-1's "two beats away per miss" failed because a sorted list of lags loses the order in time; counting trips in the time-ordered strobes (BALANCE-1) held 208 of 208. The redesign check found the ruler both times. (B8.3)

3. A coarse ruler reads the grid, not the world. On the coarse catchment map the second saddle looked surrounded by one gear; a fine map showed a ribbon of the other gear about a third of a coarse cell wide, and the same coarseness had sunk three seam tests (SV-3, SV2-2, SZ-2; GEAR-BASIN, SEAM-ZOOM). Later, BAL-H's law predicted lifts of 0.0003 to 0.0034 on a pull grid 0.01 apart, so "exact" became a question of which cell the edge fell in; the same edge was then read on a grid 0.001 apart (FINE-H). Zoom before judging a seam, and match the grid to the size of the effect. (B6.7, B9.54)

4. A missing step may be hiding between two ticks. On a pull dial of 0.00005 the steps 1/20 and 1/23 were missing from a staircase of 1/n locks, and the slips on either side of each gap bracketed them. The measured rule (n^2 times height constant) predicted both, blind, and a dial of 0.000001 found each exactly where predicted, each narrower than the old step (TRICKLE-1, TRICKLE-1F). When a sequence has a hole, predict the missing member and look there before calling it absent. (B8.29)

5. A crossover read on a grid inherits the grid's rounding. The map reads an edge as the lowest pull that is not resting, which rounds every lift up by as much as one step. Near a sign change that bias moves the apparent crossover by a column: the map read 0.6775 where the fine dial found 0.672, and the fine law rounded the map's way reproduced the map exactly (DIAL-1, DIAL-2). Print what each reading allows, and look for the rounding rule in every crossover. (B8.36)

6. The edge of a map, or of a chart's window, is not physics. The memory floor at weights 5 and 10 sat on the map's own floor (MEM-5d); a lower band of pulls fixed it, and the same trap came back at weight 3.00 (SEED-3e) until the low band fixed it again. Memory on the pad atlas seemed to saturate near 0.31, but heavy riders remember in about 93% of the cells under pull 1.0 and in none above it, and the near band's rows under 1.0 are 40 of 121: the ceiling was the chart's range. Before explaining a limit, check whether the map's own edge sets it. (B8.23; plan, B9)

7. Set a search's range from the theory, and print where the answer fell in it. FBP-1's first run at ripple strength 0.25 crashed: its root scan, 1.40 to 1.50, had been set by habit, and the root was at 1.509. At weight 0.30, PIN3-2's best drive sat on the window's edge, so its 0.0015 is only a bound. The off-scale flag applies to searches as well as to maps. (B9.50)

8. A reading window reports its middle. A ramp's leave line is a window's middle. A 200-passing window spans about 0.0024 of the drive, so a departure can be read up to about 0.0012 before it happens; the tolerance's sign had been set the wrong way (SV-2, GEAR-SOLVER). (B6.8)

9. A window that listens for a beat must span whole beats. Over 20 passings, a fifth of a beat, the hum's span depended on where the window fell in the beat, and that sank a prediction (TS2-4, TRAIN-SLIP-2). Judge a state by a window that holds a whole number of its cycles, or by its fraction and period, never by a slip read over a record that cuts a cycle (LONG-HOLD's check, sitting B7). (B6.5, B7)

10. Near an edge, count slips; do not average speed over a short window. Near its edge a light pendulum slips about once every 126 time units, so a 20-unit window may or may not hold a slip, and lines read that way are quantized (MCCUMBER-RIG). (B6.6)

11. A measuring window can make fake steps. One turn in 2,000 looked like a step. A staircase read through a window inherits the window's own steps: in sitting B8 the slips between two treads were the window's, not the rider's. (sitting B8's notes, as the book's plan, docs/LOOP_ATLAS_RF_BOOK_PLAN_2026-10-12, carries them)

12. A reader's threshold must lie beyond what the ruled-out state can do on its own. A climb-back reader that called a slip at 0.02 turn per passing over 20 passings was tripped by a band that wanders inside the gear without slipping. Its readings of memory were withdrawn in a correction (16:12 UTC, 5 October), with one paragraph of a letter (SCISSORS-MASS-4). Read slipping on windows long enough that only slipping can pass. (B6.2)

13. Compare two riders at the same phase of the pattern, never at the same clock time. Two walkers that reached the same corner by different routes disagreed by up to 2.56 radians at one clock time. Compared at the next passing of the pattern they agreed within 0.009 radians, and their drift became a whole number (389.998 and 184.999 turns; LOOP-WALK-2, LOOP-WALK-2B). A locked rider's lag swings within each passing, so "the same place" includes the pattern's phase. (B6.1)

14. How many starts a map uses is a knob of the instrument, not of the world. Bear's coin is a band of drives where a rider of weight 0.40, the ripple turned to run with it, lands on one of two faces depending only on where it starts: heads, the conveyor (§4.19), or tails, the 1/2 groove; at drive 1.444, just above it, some starts land on a third face, a 2/3 dance. REV-WEIGHT-2's four starts per drive placed the coin's band too narrowly; 100 starts found both faces outside it (COIN-1), and 1,000 found islands of each face at the edges between them (COIN-2). A chart drawn from one start cannot show two states at all: on the textbook tongue map with 64 starts per drive, all starts agreed at all 241 drives below the critical line (strength 0.9), and at strength 1.2 they disagreed at 42 drives while the single start stepped backwards at 6 (SINE-CIRCLE-MAP). When two outcomes coexist, raise the starts before reading their boundary. A second specimen: a fin on the coin that seemed to end on the 100-start map runs on at 1,000 starts, a stripe two to four thousandths of a turn wide that had slid past the hundredths the coarser map sampled (COIN-3). (B8.4, B6.9)

15. A short average of a tumbling rider is not a state. Four riders at one drive disagreed on their slip by 0.006 to 0.067 turn per passing. Most of that was tumbling, whose short-time averages differ by start; only one drive held two landed states at once, and the test had read the disagreement as memory (PAD-4, LANDING-PADS, STACKS). Separate landed riders, whose snapshot repeats, from riders in flight before counting states. (B6.10)

16. Say which side a switch was read from. The bare ball's memory switch was bracketed at weights 0.70 to 0.71 by stepping the pull up toward the edge. The Loop Atlas then showed that a switch approached from below the edge reads high, moving as the square root of the distance left, and found 0.70256366 by shooting at the edge itself (MEM-5f, ours; MEMORY_SWITCH and MEMORY_SWITCH_2, the Loop Atlas's). A bracket is honest; a number inside it also carries the direction it was approached from. (B8.32)

17. Medians are not columns. "The floor never sits below the launch" was drawn from medians. Column by column, the edge dips below the launch in 10 of 41 columns at weights 3.00 and 5.00, exactly where the balance law's lift turns negative (LOW-2, BAL-H). A median over a smooth ramp only picks the middle column. A median supports "typically"; "never" needs every case. (B9.47)

18. A measure of the tool is not a measure of the curve. WRAP-1c's whole-string fall read where the tool started and stopped a curve, not the curve itself. Before reading a number off a string, check that the string's ends are the curve's ends and not the linker's. (plan, B9)

19. A quantity that looks pinned at one setting may be the zero of a drift; move the knob you did not move. At ripple strength 0.30 the pin was sharp to 0.00032 on the dial across weights 0.10 to 0.40, and to 0.00020 read between the dial's steps; at 0.25 and 0.35, read between the steps the same way, it was 7.5 and 5 times blunter

(SHARP-PIN-1a), because at 0.30 the next order nearly cancels. The toy's other side looked tidy at 0.30 too, the pair 0.027 to 0.028 inside the frozen points at every root; at 0.20 that band was ten times wider and at 0.40 it ran the other way, because 0.30 sits near where the drift with drive changes sign (TOY-SIDE-3d, 4f). Before calling a coincidence a law, try one setting either side. (B9.57, PR9.11)

20. An offset has two ends: before giving a measured motion to the dancers, check whether the ruler moved. TWO-TONE-2 found the pair's offset from its frozen point odd and straight, and a read gave the motion to the couples. Measured end by end, the frozen point did the moving, 0.21 to 0.22 of a turn per unit of the tone, and the couples followed a tenth to a sixth as far. The same evening it happened in weight: held at the weightless root, the dances seemed to slide sideways (WEIGHT-WALK-1), but the root was drifting off the middle of a widening tongue; with the bias following the tongue most of the slide went away (WEIGHT-WALK-2). Any reading against a reference that can move needs both ends measured. (B9.75)

21. Two routes to every key number. The grip M′ by finite difference and by its integral agreed within 0.006%; the roots from a weightless scan and from direct bisection within 3.6×10^{-7}; the leap's start from the half passing and from the full passing's matrix within 3.9×10^{-7}. When two routes agree that closely, the arithmetic is right; when they part, one of them is wrong. Routes that share an orbit, an integrator and a step cannot check the integration; a finer step or another integrator does (§3.3). (B9.60)

22. When two machines disagree by the same amount at every point, compare their constants before their integrators. The gap of 5.3×10^{-9} between the Loop Atlas's machine and ours was the rounding of the golden ratio in the passing speed; the step count carried only -1.3×10^{-10} of it (KNOBS-2). The Loop Atlas measured the same rounding from its side at about 5.4×10^{-9} (its FOLD_OR_FLIP runs). (B7)

23. A fidelity bar must allow for the rounding of its own inputs, and a comparison for each route's printing. SELECT-1 asked weightless orbits to close within 10^{-8}, but the root drives were carried to 7 decimals and the map's drive sensitivity reaches 4,667, so rounding alone allows about 2×10^{-4}; the bar failed twice for that reason, as predicted before the re-run. Later, FAREY-2b's and FAREY-3b's checks compared an integral route printed to twelve figures with a finite difference printed to six places, and forgot that six places carry up to 5 parts in ten million of rounding: three parents differing by 2.3 to 4.9 parts in ten million failed as coded, though the runs were sound. A comparison's allowance is the sum of everything that separates the two numbers, the printing included. (B9.65, B9.89)

24. Rounding a root can move you off it where the window is narrow. An audit read a root from a table kept to eight decimals and found a different pair and a different M′. At that setting the 1/2 window is only 3.5×10^{-7} wide in bias, so eight decimals left $\Phi 11$ at -0.095 instead of zero. Measuring how fast $\Phi 11$ changes across the window explained it, and that became a result of its own (WINDOW-1, the window narrowing as $\exp(-c1/\omega)$, e to the power of minus a constant c1 divided by the frequency; §5.2). (B9.71)

25. When a check fails only at the edge, audit the check before trusting or dropping the rows. JJ-CORNER-2's per-setting check failed at the slow settings because the finite-difference route to M′ disagreed with the integral route by up to 16%. Doubling the resolution changed nothing, so the time step was not the cause; with the root re-bisected in place the finite difference closed on the integral route as its step shrank (9.3% to 0.8%), then drowned in rounding. The rows stood; the check's fixed step was the ruler at fault. (B9.70)

26. A small difference of two numbers carries their error, not its own. The excess δ is minus the stable grip plus its partner's, divided by the partner's (the grip ratio being $-(1 + \delta)$): a small difference of two nearly equal numbers. At strength 0.05 the finite-difference route's disagreement reached 5 parts in a hundred thousand at 5/8, where the excess itself is 2 parts in ten thousand; FAREY-2's single graded row was half noise, and its reader scaled the warning by the excess instead of taking it as it is, so the precision check passed and the verdict is void. At 0.03 three children read negative. Carry the uncertainty of the parts in the difference's own units, and when it is too large change the route: the integral route (JJ_11) takes no difference. A floor read from a slope between two nearly equal points is a floor read from noise, too (KNOBS-2's note). (B9.88, B7)

27. If halving the step moves the answer by the same amount in no steady direction, the ruler is out of digits, not out of steps. LADDER-1b halved the step twice to test whether an offset near 10^{-11} in the deepest rungs was truncation, which falls sixteenfold with each halving and keeps its sign. It moved by 1e-11 to 5e-11 each time, either way, with successive changes in ratios from 0.56 to 3.11: rounding, from hundreds of thousands of additions to a phase many turns long. The same engine in long double (JJ_16) moved by 1e-14 to 5e-14 when the step was halved, and three of four rungs landed in their registered bands. (B9.95)

28. Before believing a small, steady effect, replay the reading on a known answer. GROOVE-1 found the dancers giving each groove's width 3 to 6% short, the same at every weight, and the first story was the hump's shape. Recomputed on starts even in the straightened lag, the humps are sines within 0.14%; replaying GROOVE-1's exact steps on the selector, with no table, no weight and no gear, reproduced its shortfall within 0.4% (SHAPE-1b, 1c). The tables' starts were even in the raw lag, which is coarse in the straightened lag just where the dancers stand. An effect that is small, steady and the same everywhere is the fingerprint of the ruler. (B9.98)

29. Before comparing two estimates, remove the bias they share. GROOVE-1 registered that the mean of the steady and wobbly dancers would beat either alone at 2/5. It failed: both estimates carried the same grid bias, about 3% short, and the steady dancers' own lean, about 2% high, happened to cancel it. Correct for the shared part first, then compare what is left. (B9.97)

30. A tracker that picks by distance from a mirror cannot tell twins apart; follow a branch by continuation. WALK-1 took, at each second tone, the stable stop nearest lag 0. At A_2 −0.056 two stable stops sat near lag 0, mirror partners at −0.016 and +0.014, and the rule jumped between them; the one-way walk failed on the tracker, not on the dance. Followed from its last position the walk is one way, −0.0068 to −0.0251. The moment both twins exist is the interesting one. (B9.80)

31. An engine that assumes how many things exist will average a newcomer into the old ones; count before you average. JJ_5 expects two self-mirrored dances and hands every zero of the selector to the nearest. Inside the hand-off window the selector has 4q zeros, because the walking pair exists there even without weight; its stops were averaged into the old dances' grips and every grade of WINDOW-0 failed. The engine had printed the count, 20 zeros instead of 10, on every polluted row, and that count was also the finding. (B9.93)

32. Near a period doubling, read with the doubling in mind. Beside a doubling the strobe's period test reads the doubled dance early, because the rider settles slowly there (RS-6). A follower that watches one orbit reads the birth of that orbit in a doubling as a fold, because its multiplier there is +1, the square of its parent's −1 (KN2-2). At dances of 112 passings and longer the multipliers scatter by about 10^{-4} around their line, a scatter not traced to its source (no finer step or tighter tolerance was run to tell the integrator's share from the dance's), so a doubling is read from a fitted line with the scatter stated, and every Newton call on a doubled dance carries the parent guard (FLIP-7). (B7)

33. Carry the lag modulo a turn. The two-path stretch ruler, with the lag carried as a running total, gave wrong rates on strongly held riders (up to 0.16 per passing) and, at the extreme, not-a-number; carrying the lag modulo a turn with the tangent equations fixed it (NAN-CHECK, RULER-CHECK). (B7)

34. A stand-in at a cheaper setting can cross a stability line the real run never meets. The rectifier's engine integrates in phase with a fixed step. The dry run's stand-in, at a coarse step (1,000 a passing) and the lightest weight, 0.0025, put the step over the weight times the frequency at 4.57, and the integration went unstable; the real runs never meet that line. The engine now stops itself, saying STOPPED, whenever that ratio passes 2.5 (RATCHET_2). Before trusting a stand-in, check that it lives inside the engine's stable range. (B10.101)

35. Bracket by a property you can test before handing a guess to Newton. A fixed starting guess for the weightless edge slid onto whole-turn orbits at strength 0.20, the stand-in's setting. A bisection on whether the zero-slip orbit exists, followed by Newton, fixed it, and the reader now sets aside any row whose residual is not below 10^{-9} (RATCHET-2). (B10.102)

36. A formula exact on paper can lose its digits just where the answer sits. The fold's lag was first taken from the passing matrix's double fixed point, $z^* = (\Phi 11 - \Phi 22)/(2\Phi 21)$, the difference of the matrix's two diagonal entries over twice its lower-left one. Near lag 1/2 that form loses its digits: residuals of 35 and 4.7 at phases 0 and π, found by the fidelity check. The same fixed direction taken from the better-conditioned row of the matrix reproduced every row of record (RATCHET_2B). (B10.107)

7.2 The registration: guesses, bars, rivals and readers (L3, L4)

37. Write the reader before the returns arrive, and keep a list of what it has seen. Every pad atlas reader judges only chunks absent from a seen list kept by name and checksum. The returns are cumulative, so a chunk read twice would vote twice; with the list, 897 chunks across two folders were read once. The LOW readers were written after the returns arrived, though before any value was read, and that was declared as a lapse: write the reader at registration, in the same exchange. (B8.5, B9.49)

38. Every miss gets its redesign check, said aloud, and the misses are listed at the end of every exchange. The question is asked even when other parts held: where did it break (the data, the bar or the method), and can it be redesigned? At least a dozen failed predictions in sitting B6 became follow-on runs this way, and several found that sitting's best results (SEAM-ZOOM's threads, LANDING-PADS's staircase, LOOP-WALK-2's whole-number drift). In sitting B9 five misses in four tests went without a check until an independent read of the record found them (LN-8c, WRAP-1x, JOIN-1's two and GRID-1c). At the end of every exchange, list the FAILED lines and match each to its redesign. (B6 review, B9.44)

39. Keep failed bars visible; the misses made the best results. Three laws came out of registered failures, each found by asking where the miss broke: the window count (red stops inside the frozen points = $q - p - 1$) from READ-E2 and PIN7-1g; the slope ratio's steps with $\min(p, q - p)$ from PIN7-1h; the limit of the lever reading, CATCH-1's lever (lesson 71), from an independent check. Each time the bar was wrong and the data held a better rule. (B9.58)

40. The ruler can sit in your own registration; check its definitions before the data. PINCH-1L-b (a line of the pad atlas, no relation to §4.28's PINCH-1) was graded with the passing speed taken at the plain's edge, a choice written into the registration itself. Every square of the pad atlas has its own passing speed, so lines high above the edge read too slow, and the lean failed as worded. With the speed taken at the line the drift went the lean's way, and two blind columns (PINCH-1B) held. Three registrations in sitting B8 carried a ruler built in the same way: MEM-5a's bar too strict for its grid, MEM-5d's map bottom, TRICKLE-4's yes-or-no bar. (B8.28; the notes handed on from sitting B8)

41. Grade by the registered words, and print the script's verdict beside them. CHECK-E's script added a duplicate-count condition the registration never had, and it printed FAILED from a merge tolerance of 10^{-6} against six-decimal inputs; SELECT-1's first run printed failures from a 0/0 in its root finder. Both verdicts were kept as printed and explained. In sitting B10, R3-0 printed FAILED (320 of 352) because its reader also counted 32 rows at a weight outside the registered ladder; graded by its words, by a saved script, it held, 320 of 320, and the printed verdict stands beside it (R3_0_WORDS_CHECK). (B9.64; B10)

42. A verdict starts false and turns true only when every graded row is found and passes. FAREY-2's reader started each verdict true and turned it false row by row, so a file without the graded rows left it standing: a file holding none of them printed two verdicts HELD. In sitting B10 the dry runs caught the same fault twice: RATCHET-4's reader passed two verdicts with a weight missing, and RATCHET-5's crashed on a missing weight instead of failing. Both now require all 29 registered weights. A dry test should include a file that lacks the graded rows. (B9.86; B10)

43. Every verdict needs one stand-in on which it should pass and one on which it should fail. R3-0's dry run used a stand-in on which it was bound to fail anyway, so it could not show the scope fault of lesson 41. From then on each dry run was built so that every verdict had a passing case and a failing case (RATCHET-4, RATCHET-5), and each found a fault before its run. (B10.105)

44. Make the stand-ins with the engine, and check that the smallest prediction is bigger than the ruler. LADDER-1's reader was dry-tested on stand-ins written by hand, so every row, 1/2 included, carried one grip of each sign; the engine cannot do that at 1/2, where an even number of beats puts both starts on one orbit. One real row of each kind would have shown it. And each predicted excess times its grip came to about 8×10^{-12} at 7/15, under the 10^{-11} error assumed for each grip, which could have been worked out before the runs. Both checks cost a minute; the miss cost an exchange's main test. (B9.94)

45. Rerun the dry test after every edit to a reader. RATCHET-5's dry run found two faults: a missing weight crashed the reader instead of failing it, and an edit had commented out a working line. RATCHET-4's found a tuple read in the wrong order. Each was fixed and the dry test rerun before the registration was written. (B10.111)

46. Register the exact symmetries first; they are the cheapest bug detector. RATCHET-1 wrote two exact symmetries down from the equation before any run: the step at $\varphi + \pi$ mirrors the step at φ, and without weight the step is symmetric at $\varphi = \pi/2$. The first reading broke the mirror at every phase but 0 and π. Nothing in the physics can do that, so the instrument was wrong: it read the lock from $\Phi 11$, which equals half the trace only when the drive is symmetric in time about the strobe. In sitting B10 the same mirror held to every printed digit at every weight of every run, and time reversal told in advance which part must grow with the weight and which with its square (RATCHET-2). (B9.99; B10)

47. Register the rival you half believe as well as the guess. The power law $\delta_2 = \delta_1^2$ was seen after TOY-SIDE-2's data. Before the next rows were opened it was registered with two rivals, a smaller claim and a joint curve through both media. The guess held blind (8 of 8, then 4 of 4 in the junction); both rivals failed. The joint curve as first drawn, straight lines between the junction's points, missed by 0.047 at δ_1 0.446 (0.397 measured against 0.350); a smooth curve fitted later on all 28 two-step rows then held blind in both media (GAP-1, §4.28). A rival registered after its idea came from the data is still a test while its rows are unopened. (B9.72)

48. A rival that cannot lose is not a test: work out both predictions before registering. FAREY-1 registered the Farey product against the old law δ_1^3, each with a band of 15%, expecting them to part at 4/7 and 5/8. They differed by 0.6 to 5.2% at the four rows; the product held within 6% and the rival within 11%, and the run could not say which law the junction follows. A roots-only pass of a couple of minutes would have shown it. Register only where the two differ by more than twice the band, or choose cases where they do. (B9.83)

49. When a bar and its rival both fail, say where in the gap the answer fell. T4c's bar (the twin within 0.01 of lags 0 and 1/2 at weight 0.01) and its rival (farther than 0.05) both failed. The twin was 0.017 away and closes in at about 1.7 turn per unit weight. The gap between a lean and its rival is where the measurement lives. (B9.52)

50. Two stories that fit every row need a row where they part. "The grip is set by strength over passing speed squared" and "each fraction's grip is smooth in strength" fitted all 21 rows at three strengths, because those strengths never overlapped on the first story's scale. Two in-between strengths chosen to overlap their neighbours on that scale made the stories disagree by 7 to 17%, and one failed at every overlapping row (TOY-SIDE-4). (PR9.12)

51. Before betting on an old law, ask what the newest held reading says about it. DELTA-WEAK-1 registered the old short-way count for the grip's excess as the lean and the count of turns as the rival. FAR-1 had held an hour earlier, and its reading, counting the drive's quanta, predicts the rival in two lines. The rival held. The turn count had been on file since 13 October, in JJ-SMALL-1's rows (at strength 0.10 and frequency 0.60, 2/3's excess was 0.45 of 1/2's and 3/4's 0.25), hidden by the old law's straight line in bias. (B9.82)

52. Do not put a yes-or-no bar on a change that fades. TRICKLE-3 and TRICKLE-4 asked where the plain's edge switches from a crawl to a stumble. The data showed a fade and then a collapse, with the stretch rate rising at every one of nine weights, so every bar failed one way or the other. Register a measure that moves and lean on its trend; keep thresholds for things that really jump. (B8.33)

53. Size a bar against the part of a quantity that can change, not against its total. HOLD-1's drift range was mostly the two bands' fixed alternation, so its 1.5-times bar could not have caught a slow drift (HO-2). (B7)

54. Sign and size are separate claims: grade the sign and the zeros first, and the size only in the coordinate where the motion is even. SLOPE-1 registered that each start's multiplier minus 1 is one gear times minus the wrinkle's slope, read in the raw lag. The sign held at 99.7% of 10,584 starts and the zeros matched within one start in all 53 cells, but one gear fitted the sizes poorly (the fit explained 26 to 85% of their spread), because the raw lag is a crooked clock face. Read in the straightened lag (SLOPE-2), one gear fitted every cell (explaining 99.6 to 99.97% of the spread), within 1 to 2% of the value the passing matrix gives with nothing fitted. The Field Guide's row 712 adds the other side: blind reads of the three-body pictures, made from their pixels, were borne out on the three-body work's own data in sign and place, while their sizes were off in opposite directions. An eye proposes sign and place; the size belongs to the data. (B9.91)

55. A prediction can hold for the wrong reason: test the mechanism with the simplest separate run. MEM-4 put the weight 2 memory floor at the retrapping pull and held (floor 0.86, band 0.85 to 0.95). A one-ball run with no ripple showed a rolling ball keeps rolling down to 0.78 to 0.80, as MCCUMBER-RIG had measured on 5 October; the floor is instead where the half-turn start gets rolling at all (0.86 to 0.87). The band was wide enough to catch the wrong mechanism. The floor's dip mid-map, below that launch pull, was then found the same way, with the ripple put back: there the ripple shakes the ball held in a ledge's dip near the frequency at which it would rock in that dip, and read ten times finer the floor is lowest, 0.8455 at weight 2, where the ripple passes at 0.81 of that frequency, a little below it, as written down first for a dip that softens as the swing grows (MEM-7). The escape of a shaken particle from a potential well is known to go the same way: escape comes easiest a little below the well's own frequency, and it is the well's softening that moves it below (Gendelman and Karmi, 2019, checked against its abstract). Two guesses at a single frequency had missed first (MEM-6). (B8.26)

56. Run the control that could kill the explanation before writing the chapter that rests on it. A literature search run with Claude inferred that only a moving wave makes fractional locks, which would have made only travelling-wave media our twins. One column with the moving ripple swapped for an added drive, five times finer than the map, showed the added drive locking more fractions, not fewer (ADDED-1); on the map's own grid the two tie at one each and could not have decided it. The theorem behind the result was checked against a paper before it was written as fact. (B8.34)

57. A mechanism named from recall for a measured law is a claim like any other; the cheap test is the knob it says cannot matter. SHAPIRO-1's "ponderomotive" was named for a measured law; the wave's direction, which that mechanism says cannot matter, was the knob to turn, and the word was corrected (KNOBS-3). (B7)

58. Search the record before registering, by the registration's numbers and not its label, and for the same object facing the other way. Two predictions modelled stage two's floor as its steady-drive fold, though the record had said since 3 October that the bottom edge holds only when the filter is slower than one over the room (TRAIN-SLIP). A registration of 9 October (LATTICE-TEST-1) took the conveyor's birth from a coarse dial's 1.4585, not citing WAYSTATION-1's 1.4582 on a dial five times finer. FOLD-1 borrowed the reversed rider's fold weight (0.25) for the forward rider, whose own fold line (0.33 to 0.40, MAP-K-F) had been on file since 7 October. On 12 October the plan asked whether the pin sits at the mirror at ripple strengths 0.25 and 0.35, and FBP-1's T5 printed the answer three hours later, though the row was never marked done. On 14 October the question was opened again as PIVOT-2, and its lean was registered without that answer, which was already on file. At pull 1.0 the memory edge at clock ratio 2/3 looked like a coincidence until the record's PA-4a, from 7 October, explained it. Search by strengths, drives and fractions, and mark a row done in the same step that answers it. (B6.13, B8.16, B8.35, B9.77, PR8)

59. Name a reused idea's history. The reading "the groove holds because its rider is nearly still at the strobe" came back for a new question. It had already failed once, as BAND-1b, and PIVOT-1 showed the pin is not the drive of rest. An idea that returns is cited with its failure and given a sharper test, not offered as new. (B9.53)

60. A law fitted on one kind of input is a fact about that kind of input. One curve carried the gearing of both media blind, across every strength and frequency run, and every one of those runs was driven by a single tone. A second tone at twice the frequency kept every rule the dancers follow and pulled the gears 0.05 to 0.075 below the curve at both signs (TWO-TONE-1). Before calling a law general, list what every run had in common and vary one of

those things. The junction taught the same twice: the toy's "outside the frozen points" was written into the junction's prediction, and the rule travelled while the side did not (JJ-1b); and strength over frequency, guessed from three settings, failed at a fourth, while strength over frequency squared then held blind (JJ-3e, JJ-4a). A band set from one setting's numbers failed the same way (JJ-4b's speed). (B9.73; plan, 13 October)

61. When a law holds at one new setting and misses at another, ask which other knob moved between them. RATCHET-4 registered that the rectifier's peak weight times the frequency stays fixed. It held at frequency 0.70 (0.9952 of the reference) and missed at 0.40 (1.1386). At 0.40 the drive's strength over its frequency squared was also nearly twice as large. Rerun at 0.40 with that ratio held (RATCHET-5), the peak landed at 1.0003 of the reference and the whole curve within half a degree. The miss belonged to the knob that had moved with the frequency. (B10.109)

62. A line the reader prints but does not grade can be the next registration. DANCE-3's reader graded one combining rule and printed a second beside it. The graded rule failed at two rows; the printed one came within 15% at all six, and across all eighteen rows on file within 11% in the per-step factor. Printing rivals costs nothing and turns a miss into a ready redesign, but a printed rule seen after the data is tested next, not counted. (B9.74)

63. Carry the question to where the ruler reaches, not the number you fitted. The question whether the two arms of children of 1/2 meet could not be read at strength 0.05, where its deepest rung would carry about 10% uncertainty. It was moved to strength 0.10, and there it held: the gap closed to under a quarter of its period-9 value, and the rival in which it levels off failed (LADDER-2). The number that rode along, the gap's power of the period read at 0.05, missed by 16 to 31%. A law read after the data at one setting describes that setting. (B9.96)

64. A straight line is a promise only over the stretch it was drawn on. LAB-MAP-2's starting guesses at heavy weight were straight lines in the second tone fitted between -0.12 and -0.07 and used out to -0.22; the tongue's centre bent away and seven far cells failed to converge. Its lean that weight 0.3 would swap near -0.182 came from a straight line through two tones of a multiplier that turns over instead. Step by continuation, and register a lean from a line only within the range that drew it. (B9.87)

65. When a lean fails cleanly, look for the sum before running more cases. The ripple's crossover did not move with the weight at all. Rather than more weights, a two-line average of the ripple's push explained why (the weight dulls the drag and the shaking alike) and predicted the edge in 51 columns at four weights not yet looked at; the blind test held (CROSS-1, THEORY-1). A lean that fails exactly is often a law showing itself. (B8.37)

66. Answer "how is this possible" with a sum the run can check. Asked how one number could place every step, the crawl over a vanished resting place was given in words and checked with one sum: the weightless crawl time at the 1/30 step, 189.4, against 30 ticks, 184.9. The plain account then named what is still open: the measured constant is 0.9425 of the weightless value. A plain explanation that carries a number the run can contradict is a test, not a story. (B8.31)

67. Predict the next member of a family before computing it. The quarters, fifths, sevenths, sixths and eighths were each predicted from rules found on the members before them. A new member is the cheapest out-of-sample test a family offers, and it is honest only if its bars are written first. (B9.61)

68. Test a rule where it predicts a negative as well as a positive. STACKS-2's window B was chosen where the rule says no stack should land, and its answer there was as informative as the stacks it found. (B7)

69. When a theorem guarantees the answer, say that the run tests the instrument. Wells count $+1$ and saddles -1, and on the rider's tube-shaped space they must sum to zero. The run tested whether the solver finds every fixed point; it does, at eight drives and two windings (INDEX-COUNT), and that became a free completeness check on every later solver run. (B6.12)

70. Rules every rotation obeys will match on every substrate; carry what is specific. These hold for any rotation in disguise, so a match on another medium proves nothing: q stops per dance, the numerator as the number of stops skipped, red and purple alternating, the window count once the pair is given. What is specific, and can be carried to a medium: the pair just outside the frozen points (carried to the junction, it did not hold: there the pair sits inside them, §5.2), the 0.013 gap, weight choosing the dance that holds the leap, the ratio's steps. Sort a family's rules into universal and specific before taking them to a substrate. (B9.63)

71. A mechanism for "moves little" is not a mechanism for "moves with". CATCH-1's lever explained why the leap's start moves 44 to 128 times less than the other mirror step. It did not explain why it moves at nearly the frozen point's own rate (0.095 against 0.104 turn per unit drive), which keeps the gap steady. An independent check caught the overclaim. Insensitivity and tracking are two claims. (B9.62)

72. Revise a tool only with a fidelity check. The old version is retired by its checksum, and the new one must reproduce every number the old one wrote. Version 2 of LOOP-WALK-2's tool reproduced every column of version 1 at both paces. In sitting B10 each new rectifier engine reprinted the old one's rows before its first new run: 928 of 928 identical for RATCHET_2C against RATCHET-3. (B6.30; B10)

73. A short trial run before the real one (a smoke run) only checks that the tool finishes, and its time limit is written into the registration. Once a trial run's screen output was seen, and once a revised tool ran with no trial run first; both were recorded where they happened and neither changed a registration. When a trial run prints error messages, read the error messages, and only those. (B6.29, B7)

7.3 Reading pictures: two readers and a ruler (L1, L2)

A detour before the lessons: how the pictures were read, and how to read yours with an AI. Most of this book's results did not start as equations. They started as pictures, and as guesses about pictures written down before anyone measured anything. Bear would look at a figure and say what he saw: a weave, a pinch, a slow swell inside a busy band, "half a dipole". Each guess became a run, registered in his own words, and then the numbers had their say. Some of his reads held and some missed, and the misses were often as useful as the hits.

The habit is worth borrowing, and it has one trap that took months to notice. For four months Bear assumed that when Claude showed him a graph, Claude had already seen what he saw in it, so he kept the patterns to himself, thinking they were already collected. They were not. An AI reading a figure sees what a sentence would name, the big shapes and the labelled curves. It misses faint lines, crowded corners, empty lanes and the slow drift under a busy pattern, and it does not notice unprompted (lesson 74). A person and a machine see different things in the same picture, and neither knows what the other saw until it is said out loud. Once Bear began to say what he saw, the work sped up. On the triple-star bench a swell he traced inside a band became the beat of two notes, a one-line remark, "the ratio stays fixed, the rates don't", became a law tested across eight slices, and a squeezed, tilted look at a strobe picture showed where a star's island really survives (from the triple-star work's own notes and runs, not ours; that work is published as *Held, Rocking, Locked*). On this bench two sentences that came to him on waking became the count on the dial and the dark point's inverse square (§6.12).

It runs the other way too. When the triple-star bench held its data back and sent only pictures, Claude read three patterns from the pixels: links between dots five passes apart, a lopsidedness that flipped sign across a run, and a slow scallop about 680 passes long. Each was judged on the numbers behind the picture, and each was borne out, two of them at a different size than the pixels suggested (as the triple-star work's notes record). Two readers who each say what they see, before either looks at the numbers, catch more than either alone.

So, to a reader with an AI beside them: show it the data, and then tell it what you see. Say it in your own words, however rough, and draw on the picture if you can. Ask it what it sees, and have it write that down before it looks at your marks. Then turn each claim into a test with a bar it could fail, and let the numbers decide. The eye proposes and the ruler decides. The lessons below are what this partnership cost to learn.

Bear's way of reading is set out as steps in §3.4. These are the lessons that made it a method. The method is very quick at rediscovering known physics (Shapiro steps, saddle-node bottlenecks, hysteresis, tori); when a shape matches a named result, the first stop is the literature, and a match is not a discovery.

74. Say plainly what you cannot see. Asked whether Claude sees every curve Bear sees, the honest answer is no: in a picture Claude sees the shapes a sentence would name, misses faint or crowded ones, cannot tell close colours apart, and does not notice unprompted. Each of Bear's findings in one exchange of 9 October (two stops of one dance moving differently, three footholds shared by every fraction, ghosts in the stripes) was measured in seconds once he pointed. Eyes point; numbers measure. (B8.9)

75. Read blind, write it down, then compare. For every picture Bear sent, Claude wrote its read into the record by the clock before opening his marks. What both found independently held up best (the pictures are the three-body work's; what was read from them is kept beside the record, in PATTERN_RECOGNITION_LESSONS_RF_SPOKE_B8): the hooks paired across the V by height, the beat at the centre, the pinch near psi 0.42, the flank's nodes. What only one saw became a test (his woven strings; his three phase-shifted waves against Claude's shrinking wave, both partly right). Two blind readers who agree are the guard against the Texas sharpshooter. Two steps complete the protocol: declare what you have already seen before reading blind, which is the Loop Atlas's; and after the shuffle, run the ruler over a made-up case whose answer is known, which both books reached separately on the same day. In sitting B9 the order became: check where each picture came from against the project's own files, read blind, check the pattern, read his marks, score. (B8.17, PR8, PR9)

76. Shuffle before believing a pattern, and run the ruler over a made-up case. The weave test passed as registered (72% of links on two gaps), but the bar was too easy: nearest neighbours on a page are always close in pass. Against the same dots with their heights shuffled 200 times, the real links preferred a gap of 5 passes at +12.4 sd and shunned gaps of 1 and 3. On made-up cases the weave linker found planted gaps of 5 and 3 at +54.0 and +57.0 sd and nothing above +2.6 sd where nothing was planted, so the real weave's gap 5 stands and its gaps 13 and 15 (+2.7, +3.3) are about chance (the guide's row 711). (B8.18, PR8)

77. Name the picture before naming the pattern: each picture has its own neighbours. The same three-body readings link to dots 5 passes apart on the time page and 47 passes apart on the opened torus (two of the three-body work's pictures). A picture's axes decide which dots sit side by side. (B8.19)

78. Zoom out before naming a shape; demagnify to see what a feature belongs to, magnify to measure it. At 600 passes the opened torus showed strands and a weave; at 6,000 they filled a solid hourglass with sharp walls, and a slow clock of about 680 passes appeared. Bear: “most of the time we want to demagnify.” Demagnified, every weight’s S curve lay on one master curve, and that one sheet led to the band, the pin, the mirror and the weightless root (BAND-1, PIVOT-1). His “zoom and double” was not a request to magnify: it names the period-doubling renormalization, which composes a map with itself and rescales it (M. C. Cross’s notes). (B8.20, B9.56, PR9.3)

79. When the data is held back, measure the picture, and know the plotter’s traps. Calibrated on the picture’s own ticks and checked on known marks, pixel counts (PICTURE_PIXEL_TOOLS) turned eye claims into numbers: the lopsidedness flip, the grey drift, the dense top (3.7 to 6 times), the rider lingering mid-swing, and they reproduce when rerun. Two traps: matplotlib draws the last points on top, so colour counts favour late passes; and dots merge where crowded (43 of 505 blobs on one panel), so pixel and blob counts bracket the truth. A pixel count is only as good as its colour rule: the rule that found all 39 columns of one panel found 16 of the paler other panel, so count what the rule caught. (B8.21, PR9.18)

80. The eye keeps what a summary throws away. In one of the three-body pictures the blue core’s centre held within 0.0004 across a run, so the average said nothing happened; Bear’s “short spiky top left, long spiky top right” was a flip in the ripple’s lopsidedness from −0.5 to +0.5, measured afterwards. He reads negative space too: “I unfocus my eyes and I allow the white patches to fill my vision. They are each bordered by little 4+ dot line clusters.” Look at the shape of a signal, not only its mean and edges. (B8.22)

81. Hold back the data while the eye is learning. Bear: “let’s finish the examples and then I will give you all the data. I want you to see it otherwise the learning is shortcut.” With the numbers in hand Claude measures instead of looking; without them it has to look first, and the looking is the skill being taught. (B8.24)

82. A colour scale that lumps classes fools the eye. On the weight 2 map, colouring by denominator paints the rest (0/1), the conveyor (1/1) and the 2/1 and 3/1 grooves the same orange. Claude’s blind read named a chain above pull 1.0 as the rest; the data said conveyor, 37 cells at lag 0. Check the class before naming what a colour shows. (B8.25)

83. Explaining it simply is a test. Drawing the weight 2 map for Bear “like to a smart 12 year old on a chalkboard” (Figure 2) found three mistakes while it was being drawn: a chain lumped with the rest was the conveyor, the whole-number steps sit on Shapiro lines (the conveyor within 0.003 in clock ratio from pull 0.90 to 1.20), and a memory test had held for the wrong reason. Every label had to name what the data says. Bear’s plain questions (who sets the clock; does the honey only steal speed) separated the outside tick from the rider’s own pace. (B8.27)

84. Follow one object from frame to frame before calling it growth. Bear read the single fractions on the plain’s edge as seeds that grow into ramps as the weight drops. Tracking each fraction across six weights showed that each fraction’s line slides toward the slower tick (34 of 38) rather than growing where it sits (SEED-1, SEED-2). “The same place, bigger” and “the same thing, moved” look alike across separate frames. (B8.30)

85. The size of the holes decides what a string is; keep a relation only if it survives a change of scale. Linking holes into strings is where the lattice method is fragile. At 100 rows the 1/2 groove’s upper stop had two parallel relatives (LN-6); at 50 rows its string turned off after the groove and had none. At both scales the conveyor’s string lost pads to strings born in the staircase, because a hole joins one string and the first claim wins. The numbers read from a curve’s own points (the S’s first and last thirds, 16.4 and 3.7 times as steep as its middle, and the staircase’s order) did not move. On made-up crossings the strings tool returned planted lines exactly, but at a crossing it swapped partners and made two bent strings, and its “parallel” rule called a curve and a line relatives because their mean slopes agreed; the second version lets many strings share a hole and passed only on its third design (§3.4). (B8.14, PR8)

86. Read a birth from both sides. Below the conveyor’s birth at weight 0.26 the staircase’s 1 − slip falls as the distance to the birth to the power 0.507; above it, the conveyor’s lag leaves the birth as a square root (c 1.023). The two fits named the same birth, 1.458137, to about a millionth, finer than the dial that measured them. The coarse dial had made the slope at birth look gentle (26.8 against 160 on the fine dial). (B8.15)

87. A split is a tree with three roads. Where the two-step’s top foot splits, Bear drew three roads, not two: up, down, and one straight on between them, the road the walker would take if it could still stand there. Where the walkway’s eye opens he drew six offshoots, three from each lid, and the dance found further down the dial has twelve stops in six close pairs, three by each lid. He read the third face’s histogram as “splitting probabilities 1/5,1/5,3/5”; the map measured shares of 0.219, 0.263 and 0.500. The estimate’s shape, two equal small bars and one large, told the reader what to count. The eye defaults to a fork. (record, 9 October)

88. Long reach beats many short chains. On his test picture Bear drew a few long strokes, “about 80%” sure of all but one “maybe”. Where a curve stands alone (the upper stop, the S, the conveyor) his strokes and the tool’s strings agree point for point; through the staircase, where he drew one curve, the tool drew three to six pieces. A string earns

its place by how far it carries one smooth shape, through clouds and across gaps (BEAR_STROKES_MATCH). The Field Guide's row 707 adds that the strings the eye does not follow have to be looked for on purpose. (record, 9 October; PR8)

89. The person who sees patterns should draw in the instrument's own numbers. Bear's screenshots with lines drawn on them had to be read by eye; the explorer's pen keeps each line in drive and lag, numbered, with his word on it, saved where Claude reads it directly. A drawn line becomes a registered prediction before it is checked. His marks are also answer keys for the tools: his "many strings can overlay a lattice hole" built the linker's shared mode, which lifted coverage on his curves from 10 to 25 of 35 drives and from 16 to 23 of 27. (B8.8, PR9.7)

90. Where dots crowd, follow a curve by its bend into the nearest attractor. In Bear's words of 12 October: "where dots crowd, follow a curve by its bend into the nearest attractor (a ring, a birth), rather than ending it at the crowd; a family of converging curves is read by where they meet before it is counted." Claude stopped where the dots crowd; he followed both curves into the ring. (PR9.1)

91. A count by eye is a guess; count by machine before saying a count, and ask what each count is in the problem's own integers. Claude's by-eye counts went wrong three times in one sitting: a "period-3 dance", the crest counts (23 against 37, 38 against 43), and the rope check's "about four" (about every 80 passes, by pixels). Bear's dot counts in figure 31 became q, p and $q - p - 1$. (PR9.2, PR9)

92. Symmetry is a stalking point: find the transformation that makes it, and test that. Bear's "bilateral symmetry is another stalking point" led to the mirror in the 1/2 dance's stops, then to the time mirror $(u, t) \to (-u, -t)$, exact to 10^{-13} for a weightless rider (T1), which placed the pin with nothing fitted and carried the whole dancers' thread. (B9.40, PR9.4)

93. A fan of curves through one point means something holds that point fixed. His staircase curves all end at the conveyor's birth ring; figure 34's stop curves all pass through the weightless root. When many curves cross at one point whatever the knob, ask what leaves that point unmoved. (PR9.5)

94. Crowds mark slow places; gaps mark fast ones. The middle of the turn is empty because a circulating rider crosses it fastest. The mirror dance crosses the strobe slowly (0.14 of a turn per passing, against its average of 0.5) and is flung over the top. A picture's density is a picture of one over speed. (PR9.6)

95. Pattern sense works on text too; file side notes where they can be found. The frozen balance point was a "reported, not graded" line in BAND-1's registration. Bear picked it out of a long reply ("this is MASSIVE"), and it opened the night's work. Keep side notes in the record with their numbers. (B9.41, PR9.8)

96. Watch what moves and what stays. The wrap was pinned (1.4415 to 1.4435) while the ring moved, so the descent steepened until it missed; the groove holds its place while its unstable twin slides more than a thousand times further per unit weight. The pinned thing is where a symmetry or a law is hiding. (PR9.9)

97. A picture points; numbers measure. "Data visualization as a best practice not a crystal ball" (Bear). Every pattern he pointed to in sitting B9 was measured within the hour: the mirror, the fan, the second S, the ring, the doors. (PR9.10)

98. A remark that fails as worded can still be right as a shape, and his physical words point at theorems. "The biggest jumps happen when nobody is dancing in the centre" failed at 3/5 and became the exact window count $q - p - 1$ once the stop the mirror forces at lag 0 was understood. "Repulsion" was interleaving; "does the dot coming around the circle change its frame?" was the multiplier as a holonomy. Name the theorem in his words, then check it. (PR9)

99. Squeeze the axis that repeats to see the outline; find with a grazing look, never measure with it. A picture of many columns is a comb, and the eye follows teeth; squeezed sideways it becomes one silhouette whose waist and flares can be read. A grazing look compresses the far side more than the near and bends lines, so anything compared across a picture is checked on a uniform squeeze or on the original. Mark the layout before the data: Bear's pink data bands and black spacer rungs kept a plotting choice from being read as structure. Draw the axis first, then both edges, then check the outline by numbers: every part of his eyeball outline on the three-body work's strobe picture of its island 7 (a dot each pass, a column of dots for each run) survived a pixel count (STROBE_PIXEL_CHECK; the waist's spread 0.155 turn against 0.306 elsewhere, the narrowest column at 7.525 where he drew a small ellipse). (PR9.13 to 16)

100. A picture word needs its frame, and the same word can name opposite things. Asked whether the second tone moves the tongues "up", a reply said yes "with bias running up the page", while the book's map runs bias across; in that map the tone slides the tongues sideways. The next evening a reply said weight "doesn't move the chalk line at all", true at a fixed bias but not at a tongue's middle. And "a pinch" was a waist in a phase's scatter in one picture (holding tightest) and a tongue's width collapsing in the junction (holding most weakly). Before "up", say which axis runs up; before "does not move", say what is held still; before carrying a word across, name what narrows. (B9.76, PR9.17)

101. Count the holes; a flat picture cannot show linking; a whole number that cannot change smoothly is a latch. A ring-shaped surface seen at an angle shows one hole; two hollows with a filled middle mean two loops overlapping, or one band crossing itself, his "twisted piece of conduit" (EYE_PIXEL_CHECK: middle-strip densities 0.21 and 0.38 of the rims on one side, 0.32 and 0.51 on the other, the middle about as full as the rims). In a three-dimensional flow two orbits cannot pass through each other, so whether they are linked cannot change while both exist: his "can't cut that math" (Birman and Williams, 1983, checked). A winding or linking number holds until an event the smooth motion cannot make releases it; a dance's p/q holds across its tongue and changes only at the tongue's edges. (PR9.19 to 21)

102. A label moved to make room can name the wrong dot. Figure 44's first draft printed 3/7 and 4/9 on top of each other; pushing labels apart made them legible and wrong, with 3/7's label beside 5/9's dot. A leader line from each label to its own dot fixed it. Check each label against the number it names. (PR9.22)

103. Crest or slope: name the curve before placing anything at its crest. His saddles-and-wobblers diagram put the steady dancers on the crests of a wavy line. On the wrinkle the steady stand on its climbing sides and the wobbly on its falling sides; on the landscape whose slope is the wrinkle less the bias, the steady sit in wells and the wobbly on ridges (SLOPE-1, SLOPE-2). His notebook's +1 well and −1 saddle are exact. (PR9.23)

104. Keep every word of the picture, the small ones too. Asked about the tilted floor, he answered "two U-shaped curves offset side by side (intersecting)". The registration took the first half and left out the word in brackets. The runs showed the arms offset at every period but the gap closing (0.0587 at period 5, 0.0056 at 13), as if the curves cross at 1/2: the bracketed word, then held at strength 0.10 (LADDER-2). (PR9.24)

105. When he counts the parts of a picture, count them in the runs. His four-legged skater has four blades on the two lips of a groove. In the runs each tongue has two edges, and at each edge a steady and a wobbly stop meet and vanish: four. The count matched before the meaning was settled, and the meaning followed: the four blades are two dances, a skating couple. (PR9.25)

106. He reads a limit the way a mathematician does. Of figure 45's gap closing period by period he wrote: "it will never touch but it defines the gap." No rung ever touches; the shrinking gaps pin the meeting point. When he says something never touches but defines, test it as a limit. (PR9.26)

107. When he says the phase between two parts does the work, ask which two parts. His vehicle on two pads, with "the phase offset or quadrature between the blades doing work", is right in kind: directed motion from two oscillations with a phase between them is real. Which phase works depends on the pair. In the ultrasonic motor two standing waves of one frequency 90 degrees apart carry the rotor; in the junction a tone and its second harmonic rectify most in phase and not at all in quadrature when the rider has no weight (RATCHET-1b), and with weight the zero slides off quadrature and back (RATCHET-2 to 5). The instinct names the mechanism; the run names the phase. (PR9.27; B10)

108. Turn each of Bear's pictures into a run. The ferry and rope (his drawing of a cable ferry, a waisted ring riding a twisted rope, cocked by the current until it sticks) became GEAR-BASIN; the orchards' 50-50 stripe the uncertainty-width test; the landed pad against a vector in flight LANDING-PADS; the pentagon and heptagon INDEX-COUNT; the two walkers LOOP-WALK; switchbacks and trucks SWITCHBACK; brake heat and rest stops SWITCHBACK-2. Read the picture, find the part of the rider it maps to, and register the part that can fail. Correct a picture's physics in conversation when it needs it. (B6.26)

7.4 Theory before the scan (L2)

109. When a theory is a limit, test the limit first. FBP-1's T5 put first-order bars on weights 0.10 to 0.40 and failed at two of three ripple strengths. T6 and CONV-1 tested the limit itself, at weights 0.02 and 0.05, and held at all three: each mirror drive equals the weightless root to five decimals. Register the limit, then measure how fast the world leaves it. (B9.38)

110. Compute the structure before you scan. The weightless rider is a Riccati equation, so one passing is a 2×2 matrix; the half-turn root and the mirror dance came as closed forms with nothing fitted. The same root found the 1/2 groove at strength 0.25 after it had moved 0.073 in drive: 586 of 648 riders landed in a window set from the root alone. A grid search would have needed the answer to set its own window. (B9.39)

111. Solve the exact limit, then the first order, then check the full problem at a small parameter. Weight entered as a first-order selection function (SELECT-1), and Newton's method at weight 0.01 checked it at thirteen roots. Where first order and Newton parted (4/5, off by 0.029), the next order is the next question. (B9.59)

112. Before linearizing about a point, check that the point stands alone. KERNEL-1's first draft treated a dance as a fixed point of the weightless map and divided by one minus its multiplier. At a root every orbit closes and the multiplier is exactly 1; the dance is picked out by the passing matrix's shape, not by a fixed point. Rewritten on the matrix, the same idea reproduced 56 measured moves with nothing fitted. (B9.78)

113. An engine copied for one regime carries that regime's quiet assumptions; walk it to the new regime's edge on a stand-in first. JJ_9 was JJ_3 with one line added, and JJ_3 had only run where the least current stays below 1, so the strobe always has frozen points. At strength 0.05 the 3/5 root sits at B − A = 1.0000085, and JJ_9 stopped on the arccos of a number above 1; JJ_5 already handled it. A dry run for a new regime uses that regime's hardest row, and an engine's cousins are searched for fixes they already carry. (B9.84)

114. A strongly hyperbolic map defeats a Newton start; when an old route held, take it. At frequency 0.40 the rectifier's weightless map is strongly hyperbolic (half-trace 1,090 at zero bias, against 136 at 0.55), and the fold-Newton start failed. Sitting B9's own route, the half-trace of the passing matrix crossing 1, which had held at every setting, gave the edges, finished by a short Newton polish (RATCHET_2B). (B10.106)

115. When continuation fails, shorten the stride before blaming the object. RATCHET-4's weighted continuation failed at the doubling from weight 0.16 to 0.32 at frequency 0.40. The engine was taught to redo a failed stride in 4, 16 and then 64 smaller steps and to print how many it used (RATCHET_2C). Every weight then converged, 21 in one stride and 7 in four: the failure was the stride, not the edge. With the weaker drive of RATCHET-5 every weight converged in one stride. A continuation that follows a dance must also keep its steps inside the dance's window and guard against falling onto the parent (FLIP-2 to FLIP-6). (B10.108, B7)

116. Guard iterations against the converged state. A secant step divides by the difference of its last two values; once both are exactly zero that is 0/0, and the nan travels into every later number (SELECT-1). Take no step where the difference is zero. (B9.66)

117. Change the pace to separate dynamics from geometry. Walking the same loop at three speeds gave drifts of 797, 390 and 185 turns; fitted as a constant plus a slope times the walking time, the constant was −18 turns, under 5% of the largest drift (LOOP-WALK-2). Geometry decides whether; dynamics decides how much. (B6.15)

118. A rest is useful only when something builds up; put in the variable your picture depends on. With no build-up, rest stops changed nothing. With a heat that drains by Newton's cooling and fades the grip, rests gave back what the heat took, at one overshoot exactly to the cold value, and still could not save a leg that was simply too long (SWITCHBACK, SWITCHBACK-2). (B6.16)

119. To tell memory from lag, hold the rider; do not ramp it; and start a held ensemble on the branch the ramp carries. At weight 0.75 the Loop Atlas's ramps read an opening that halved as the ramp rate halved, the mark of a lag; our ramp, sixteen times slower than their slower one, read none, and their held test confirmed it (SWAP_STATE and CLIMB_HOLD, theirs; SWAP-RIDER and GEAR-HELD, ours). Twice a natural-looking start sat nowhere near the state under test: HELD-SCISSORS started at the drive's lab speed and fell into the weak gear; GEAR-ONSET started from lag 0 at gear 2's speed, off gear 2's branch, so its counts were only lower bounds. Prepare the state with the ramp itself, then step to the test drive. (B6.3, B6.4)

120. Check the book plan when the cases start confirming instead of teaching. Bear: "how far in the weeds are we?" The thread had answered two rows the plan had left open for what is now Part 4, and the next two runs only confirmed rules that already had reasons. That is the moment to turn to the plan's thinnest chapter, then the substrates, now Part 5. (B9.68)

7.5 What the map itself taught (L2)

121. Chaos can keep a count. The riders shaking on the coin's third face all keep two turns in every three passings (slip within 0.0004 of 2/3), and the dances inside that chaos are 2/3 too; the shaking just below the coin keeps no count. Whether a chaotic orbit's winding is locked is a read worth making before calling it disorder. (B8.10)

122. A window can open and close by the same road. The third face's 2/3 doubles at both ends: chaos, 32/48, 16/24, 8/12, 4/6, 2/3, 4/6, 8/12, 16/24, 32/48, chaos, its opening read on a finer dial (THIRD-FACE-3); the coin's tails opens the same way, 2/4 below 1/2. Look for a cascade on both sides of a window, not only the side the textbook draws. (B8.11)

123. Two maps can share a line. The pad atlas's plane of pull against clock ratio carries the golden-ray column the RF maps run up. Before treating two instruments as separate, find the parameter line they share and check that the two agree on it. (B8.12)

124. A ghost is visible before the thing is born. Riders in flight crowd toward the lag where the conveyor will appear, more and more as the drive nears its birth (WAYSTATION-1 at weight 0.26; the stripes at 0.10). Where a lock is about to form, look for the crowd first. (B8.13)

125. A staircase can run backwards. The prediction assumed the slip only climbs with the drive; it stepped back at 18 of 61 drives, and the 2/5 pad sits below the 1/3 pad in drive (PAD-2, LANDING-PADS). When the map folds, monotone is not a law. (B6.11)

126. A weightless rider cannot fold; weight is what lets memory and tumbling in, and memory lives between two locks. A first-order rider's once-per-passing map can never fold, because two starts cannot cross, so it has one state per drive (ONE-GEAR). With weight, stage two has two gears at one drive; their overlap opens near

weight 0.61 and is 0.123 of the drive wide at weight 1.0, with gear 1's end at a fold at 1.413316 and gear 2's at 1.290373 (GEAR-OVERLAP, GEAR-SOLVER). No edge between a lock and free running kept a memory the readers could see. The Loop Atlas, working the same rider with the strengths swapped, so that the strong wave stands and the weak one passes, wrote down before our run where it would leave first gear and come back (SWAP-RIDER, written first; three predictions theirs, one of which missed in part and was withdrawn, and one ours). Ramped up and down at nine weights from 0.25 to 3.0, sixteen times slower than their slower ramp, it left first gear just above 1.6180 (one over 0.618034) at every weight, between 1.6225 and 1.6410, below their folds, as they said. From weight 1.0 up it came down through a band of gear 2 before first gear returned, and below 1.0 it did not, as their caution said. Every return lay within 0.0045 of their table, each a little higher, as ours said. Their lower bound on the opening missed at weight 0.75, where our ramp read none; their own held test then found no memory there, and they withdrew the bound (lesson 119). From weight 1.0 up the opening a ramp reads is gear 2's bottom, as they warned, so it cannot test the bound. (B6.17, B6.18)

127. Above the fold, a rapid with pools. Past gear 1's fold the swapped rider mostly tumbles and lands on pads at simple fractions (1/7, 1/4, 2/5, 1/3, 2/3, 3/5, 1/2), and the 1/2 and 1/3 pads double as the drive comes down (LANDING-PADS). Sitting B7's finer grid changed the picture: the tumbling's mean slip climbs from 0.19 to 0.76 turns per passing across 1.640 to 1.700, the pads are narrow windows that open in it, neighbouring fractions need not be neighbouring in drive (2/5 and 1/3 lie 0.0026 apart with tumbling between), and some pads carry a fraction on an orbit longer than its own: 4 in 6 and 8 in 12 side by side at 1.6756, a memory a reader keyed by fraction alone cannot see (STACKS-2, STACKS-3). (B6.19, B7)

128. Wells and saddles balance, and they are born and die in pairs. Bear read the saddle as a heptagon (curvature −1) and the well as a pentagon (+1). The count sums to zero at every drive, and a well and its saddle walk into each other toward a fold before both vanish. A pentagon need not be a well: from drive 1.35 up, gear 2's +1 point is a flip saddle (INDEX-COUNT). (B6.20)

129. Which door you come home through depends on the direction you walk. A loop of drive and weight round the tip of the overlap brings the rider home in gear 1 one way and gear 2 the other; the two walkers agree at every far corner, where only one gear exists. A loop that crosses the overlap without going round its tip brings both home in the same gear (LOOP-WALK, LOOP-WALK-2). The split-frame measurement does the same on purpose: send one walker each way and compare them where they meet, at the same phase; the difference is a whole number of turns. (B6.21, B6.24)

130. The edge's ghost survives being chopped into teeth, and a near miss is a loan. A zigzag drive past a lock's edge keeps its grip only if its average stays below the edge and each leg is short enough; the longest leg scales as the ghost law predicts, within a factor of 1.5 (SWITCHBACK). Near a simple fraction, hold time × detuning × tooth rate climbs to the good band's half-width (1.53 rad), not to a full turn; near a lock's edge the time between slips goes as one over the square root of the offset (FREE RIDE, CLIFF-HUM-4). (B6.22, B6.23)

131. Whole numbers are the surveyor's marks. On a blindfolded hike the readings to trust are the integers: the whole-turn drift, 64 of 64 riders landing on one pad, index sums of zero, a pad's period counted in whole passings (LOOP-WALK-2B, STACKS, INDEX-COUNT). When a reading lands on a whole number, the map is telling something real. (B6.25)

7.6 The record and the ledger (L4)

Ten lessons first written in this section, about running the work rather than about the physics or the way of reading it (132, 139 to 143 and 148 to 151: the clock in the notes, watching and stopping a running job, folders and grouped commands, a cloud workspace that restarts, working copies, the kit that rebuilds the baseline, the laptop's limit, letters between sessions and keeping the project's files in order), now live in the Loop Walking Field Guide, which keeps such lessons from every project. Their numbers are left unused here, so that every other citation of this part's lessons still points true.

133. Line counts and small counts are claims; count with the tool in the same step. A histogram's 42 lines were written as 44; five FBP-1 outputs got counts from memory (25 as 31, 61 as 64, 36 as 42, 35 as 33, 40 as 39); BAND-1's "4 of 16" was 3 and its "14 of 16" was 15; BAL-H's "5 of 10" was 3. Even after the lesson was written a read-table section went in with six counts from memory, and a machine line count caught them before it shipped. (B8.7, B9.46)

134. A range written from the head is a guess, even inside a results note. On the night of 13 October three summary numbers went into the record uncomputed: a size of matrix entries ("1e4 to 1e5"), a span of ratios ("−1.019 to −1.069", by code −1.0778 to −1.0227), and in the book a miss of the old line from a row the fit had left out (63% where the right row gives 46%). The record's two were corrected by dated lines, never by editing the note. In sitting B10 two plan values were typed in the same command that printed them; the printed values agree, and the slip is recorded. (B9.69; B10)

135. Write the line that cites a check only after the check has printed. A correction to DW-b's redesign check was appended in the same command as the sympy check meant to support it. Sympy was not installed; the import failed, and the line, already written, said "(sympy now)". The formula was right, as a numpy check showed a minute later, but the attribution was false when written. Run the check, read its output, then write the line in a separate step, naming the tool that printed the number. (B9.81)

136. Provenance is a claim: check its source line. Notes first gave the frozen balance point a provenance the record did not support; the record's lines where its calculation was first written said otherwise, an independent read caught it, and the record carries the correction (09:56, 12 October). (B9.42)

137. Have the work read by someone who did not write it. Two separate AI readers that had written none of it read sitting B9's work. Between them they found a provenance error; 0.000014 written as "within 0.00001"; a verdict reported more kindly than registered (T5a); four tests whose failed predictions had no redesign check (five misses in all, lesson 38); five wrong line counts; a wrong date; and several sentences stronger than their outputs. The author found none of these. A signposts page written from recall that called a cousin a twin was caught the same way, by two readers independently. (B9.43; plan)

138. A run file is a snapshot until it says it has finished; read the done line of the file you grade. Version 5 of the pad atlas was recorded as stopped from a file taken mid-run, and the next answer pointed Bear at the wrong batch. Two scans ran side by side and a tail over both showed the shorter one's done line, so PINCH-2's reader ran while the longer was at its last weight and its convergence check failed for that reason alone. In sitting B10 a fidelity check was first run on an unfinished file (864 of 928 rows missing) after a wait loop timed out; it printed FAILED, was kept under its own name, and was rerun after the done line (928 of 928). (B9.48, B9.79; B10)

144. A one-off reading is saved as a script before it is shipped, and rerun to prove it. WALK-2c's read from its table and SLIDE-1's root-free slide ratio sat in the record as numbers with no file that could make them again, and the run queues they leaned on lived outside the project's files. Each was written as a script, rerun and checked against the record (the table byte for byte, the ratios to every printed digit), and the queues were copied in. A number nothing in the project's files can reproduce is a citation without a source. (B9.90)

145. A check that does not gate the build is only a report. On 15 October, in sitting B9, the count of every table row's cells ran in the same command as the PDF builds, so the builds went ahead before its result was read. It found one bad row, which was fixed and recounted before anything shipped, but only because the output was read before sending. Chain the build behind the check; the same holds for a reader's verdict before its figure, and a fidelity check before a new engine's first run. (B9.100)

146. A table cell cannot hold the table's own divider; count every row's cells before every build. HYPER-1's row in the paper and one picture row in the book wrote an absolute value with bars, which a markdown table reads as cell walls; the PDF printed the rows torn. A count of the bars in every row against its header found both. Inside tables, write "Φ11 beyond ±1". (B9.92)

147. Count the images in a built PDF. A PDF built from the wrong folder dropped two figures, and counting the images against the markdown's image tags caught it before shipping. The book's count carries a known offset of one, from one image split in two. (plan, 13 October)

152. Material from another project stays out of this project's files; the three-body work is pictures and lessons here, never tests. Bear's control-room toy was read for a comparison and not filed; only its update rule was used, written out in the record. The three-body pictures taught Part 3's reading, and every check of them counted the pictures' own pixels, never the three-body work's data. (B6.33; Bear's standing ruling)

153. Names are claims, and so are citations. 0.236068 is 2R2 − 1, R2 being the second stage's speed ratio 0.618034, and that equals R2 cubed, not R2 squared: a reader judged against the wrong name, and the error was corrected at the read (GEAR-OVERLAP). Citations named in conversation are not yet checked: a paper checked against an authors' hosted copy, or read through a tool's summary, is marked so, and every outside name is checked against its publisher's record before a deposit. (B6.31, B6.34)

154. One book running two signs of the ripple must name the sign everywhere. The atlas, the chalkboard and the record run A cos(u − W t); the S, the wrap, the pin and the dancers run A cos(u + W t). The Field Guide's chapter quoted only one, and asked. Turning the ripple around is the same as turning the pull around (KNOBS-3, 296 of 296), so both are one rider; the book now names the side in every figure (§2.4). The junction runs, including the rectifier, have no travelling ripple at all. (plan, 13 October; answered again 15 October)

7.7 The ruler again, and the quantity: what the seam and the literature taught (L1, L4)

155. Refine the ruler until the answer stops moving, and report the stride (L1). The seam in the fourth harmonic's in-phase reversals (§6.6, §6.10) is the frequency where the top of the zero's path touches the quarter turn. Read from the usual ladder of weights it sat near 0.767; from fine ladders, 0.782; from a parabola through the fine ladders' three highest rungs, 0.783, where it stopped moving (RATCHET-14 to 17). The seam never moved; the reading did, and always the same way, because the highest sampled point can only read a peak low, never high. Near the seam the excursion above the quarter turn was also narrower than the usual ladder's stride (0.32 to 0.64), and anything narrower than the stride is invisible. This is the rule of Bear's note *A Coincidence Has No Single Price* (2026), its check zero, run on a second instrument: there, "widen the window until the answer stops moving, and report how wide you had to go"; here, refine the ruler until the answer stops moving, and report the stride. The note's spectroscopy sentence, that two neighbouring states are distinguishable only when their spacing exceeds the width of the line, is the same rule turned inside out: a feature is seen only when it is wider than the ruler's step. The same idea showing up in two unrelated instruments is worth a lesson of its own. (The method is cited here, not the note's running specimen.) (B10.112)

156. When a fitted peak and the highest measured point fall on opposite sides of a wall, run a fine ladder before reading the wall (L1). A parabola through three coarse rungs read the fourth harmonic's outward peak 2.4 degrees high at frequency 0.75 and 4.4 low at 0.80 (RATCHET-15, 16), so its error has a size but no direction, and it produced a surprise, "the peak falls fastest at the seam", that the fine ladders withdrew. At 0.75 the highest coarse rung said no excursion and the parabola said one; only a fine ladder settled it (+95.5 degrees, two crossings). (B10.113)

157. A list after a paragraph needs a blank line, and a page that holds a new list is a page to look at (L4). Four bulleted lists in Part 6 shipped as one run-on line of dashes, because pandoc's markdown starts a list only after a blank line; the word-by-word readback cannot see shape, and the pages looked at that exchange were not the ones holding the lists. A list-gap check now gates every build beside the table-cell gate, and its dry run found the four gaps in the shipped book. (B10.114)

158. A threshold is a ruler too: beside every "none", say how small a thing it could have missed (L1). The scan for more seams (§6.10, the window's seams row) counted a turning point only where the curve moved by more than a ten-thousandth of its own range, and it reported no light-weight rise in the fourth harmonic's window at frequency 0.80. Printing the heights themselves showed one, 1.00027 of the weightless height at weight 0.04, smaller than the threshold. Rerun with no threshold, the scan found the rise at 0.775 and 0.785 as well, so it is born between 0.75 and 0.775, at 0.7578 by RATCHET-19, 0.025 below the zero's seam rather than at it (Appendix A); a draft of the walls table had said "not at 0.80" and was corrected before it shipped. A threshold is a stride of its own, and a "none" means none larger than the threshold. The same rule as lesson 155, met a third time in one afternoon. (B10.115)

159. When a guess with no law behind it misses, let the miss choose the law: register the candidate laws as rivals (L1). RATCHET-18 guessed, with no law, that the fourth harmonic's window would lose its dip pair by frequency 0.50. It was still there, its depth shrinking 0.125, 0.060, 0.0085 at 0.40, 0.45 and 0.50. The redesign set two laws against each other before the run: a straight line in the depth, which put the vanishing near 0.508, and a fold's law, the depth to the power 2/3 on a straight line, near 0.519. The fold held (a pair at 0.51 and 0.52, none at 0.53), and its forecast from the coarser rungs still fell early by about 0.004, lesson 155's one-sided bias once more. (B10.116)

160. A prior-art pass matches the quantity, not only the effect (L1). The harmonic-mixing literature reports the effect of §6.3, a zero that slides as weight and honey change, and a pass that matched on the effect alone would have filed the whole of Part 6 as known. Read closely, its quantity is a drift, which running time backwards turns round, and the book's is a threshold, which it leaves alone, so at the limit with no honey the two zeros must sit a quarter turn apart. The same reading kept the rest honest: the slide itself is known, and the book says so. The difference was then tested, and along the weight knob it did not show (lesson 161). (B10.117)

161. An argument about a limit says nothing about the road to it: test it along the knob the reader will turn (L1). The symmetry argument of §6.9 is exact at the limit with no honey. The book wrote that the drift's zero therefore moves over a quarter turn as the weight grows, and registered it (R23-b). Measured with noise along the weight knob, the drift's zero went out and came home instead (+40.4 degrees at weight 1.106, +12.2 at 2.293), and the rival, that it follows the step's, held. Raising the weight at a held frequency is a different road to the limit from draining the honey at a held mass and drive, and the argument spoke only of the end of the road. (B10.118)

162. A verdict on an angle needs a gate on the amplitude it is the angle of (L1). RATCHET-22's reader graded the drift's zero at weight 4.755, where the drift's amplitude was smaller than its own error, and two verdicts held by the letter on noise. Before the run, a blind check had given an amplitude only 1.6 times its error, which says "not yet readable" and not "this size"; it was read as a size. RATCHET-23's reader gates every angle on an amplitude at least four times its error. (B10.119)

There was a crooked man, and he walked a crooked mile,
He found a crooked sixpence against a crooked stile;
He bought a crooked cat, which caught a crooked mouse,
And they all lived together in a little crooked house.

— a traditional nursery rhyme

163. Straight lines lost all day: between two known ends a straight line is a guess, not a law; let the data's own shape choose (L1). On 15 October three registered guesses that were a straight line, or had no law behind them, missed: the dip pair's vanishing guessed with no law (R18-c), the drift's zero carried along the weight knob by an argument about a limit (R23-b), and the second harmonic's edge on a straight line between drive ratios 0.99 and 1.875, 0.088 too high (R24-b). Each redesign that let the data's own shape choose the law held: a fold's law against a straight line (RATCHET-21), the drift following the step (RATCHET-23, by its rival), and a bowed curve through five ratios that placed the edge within 0.001 (RATCHET-25). Between two known ends a straight line is the least-informed guess there is; when a third point bends it, the bend is the law to test next, with the straight line kept on the record as its rival. Bear asked for this one as a Field Guide note. (B10.120)

164. Read a fit's numbers by evaluating the fit at a point you know, never by their place in a printed list (L1). A least-squares fit of a parabola prints its three numbers highest power first. Writing up RATCHET-27's redesign, the record took the last number of the fit at frequency 0.70, its constant term, for its curvature, and called the bow there seven times as deep as at 0.55. A check, first written to correct it, went wrong the other way and read the fit at 0.55 backwards. Evaluating the two fits settled it at once: read highest power first, the fit at 0.55 is zero at its own edge and the fit at 0.70 passes through its own data. Read the other way, the fit at 0.55 misses its own edge by 0.24. The bow at 0.70 is 1.45 times as deep, not seven, and the R27-c verdict, graded before any of this, stands. (B10.121)

165. To trust a new path through old code, give it an old job and swap the roles (L1). The three-tone engine reproduced two old runs exactly, but only with the third tone silent, so its own path had never carried a number. Bear asked whether a surprising sign was a bug. Running the same mix with the octave in the third tone's place and the fourth harmonic in the second's must give the same edges, since the drive is the same sum; it gave all 288 to every printed digit (RATCHET-43). A swap needs no new reference, only a symmetry of the job, and it tests the new path at the very setting in doubt. (B10.122)

166. A symmetry says where a zero must sit; find it there with one knob and a sign (L2). RATCHET-52's count round the dial changed by one between two octave sizes, so a dark point lay between them, somewhere on a plane of two knobs. Searching a plane for a point is slow. The reflection made it a line search: with no weight, the window at the mirror timings has no sideways part, a dark point off the mirror line would have a twin of the same charge, and a single change of the count rules the pair out. So the dark point had to sit on the against line, where the window is a plain signed number, and four runs at one timing placed it within seven parts in a million of the prediction (RATCHET-53). Before hunting for a zero in two knobs, ask what symmetry pins it to a line, and whether its count lets it leave. (PR11.39)

167. When every rate bends, look for the ratio that holds (L2). Bear put a long afternoon of the triple-star thread into one line, "the ratio stays fixed, the rates don't"; that thread reports it held at three of eight slices over 600 passes and as registered over 2,400, and that its misses found where the star's ellipse stops and turns backward (from the triple-star work's own notes and runs, not ours; that work is published as *Held, Rocking, Locked*). On this bench the same shape turned up in the dark point's law. With the drive moved, the octave's arrow grew as the drive to about the power 1.3 and the fourth harmonic's to about 3.1, both far from the simplest picture's 2 and 4, while the ratio of the two, which places the dark point, grew as about the power 1.78 and climbed to 1.95 as the drive weakened (RATCHET-53, 54). When every rate is bent, the quantity that decides an event may still be a ratio that holds. Name it first, as he did, then measure where it holds and where it cannot. (PR11.40)

168. Read a run's dial as it was set, not as it was printed (L1). WALK-2 stepped the second tone by 0.00025, and the engine prints each tone to four decimals, so the run at −0.03825 was printed as −0.0382 and the run at −0.04025 as −0.0403. Its reader took the tones from the printout and placed the hand-off window's edges between tones that were not the ones run: 0.002143 wide, where the run's own file, read with the tones as run, gives 0.002107, and a scan ten times finer 0.002097 (WALK-3). The verdict it graded held inside its 25% bar either way, so nothing flagged it; it surfaced only when a new reader was written against the file. Read the numbers from the file the run wrote, at the precision it worked in; a printout is a picture of the run, rounded for the eye. (P2)

7.8 Where each tool fits, where it fails, where it shines, and where its light is hidden

Bear’s four questions, asked of each tool. The first seven rows are sitting B8’s first pass; the last two were added in sitting B10.

tool or habit	where it fits	where it fails	where it shines	where its light is hidden
lattice strings	a curve standing alone: Bear’s strokes and the strings agree point for point	crowded places: through the staircase the greedy linking split his one curve into three to six strings, and relatives change with lattice size (lesson 85)	the S measured: its slope 190 at birth, 2.8 in the middle, 53 at the end	relations that only show at another lattice size
two blind readers	geometry: hooks paired by height, the beat at the centre, the pinch near psi 0.42, the flank’s nodes	classes and positions by eye: the conveyor named as the rest; dashes placed at clock ratio 0.72 to 0.74 that sit at 0.680 to 0.695	agreement reached without seeing each other’s reads	a blind spot both readers share, such as a picture’s draw order
the shuffle	any pattern of links on a page	only as good as what it scrambles: without it the registered weave test passed on a widened window that smeared the real gap of 5	gap 5 at +12.4 sd; gaps 1 and 3 shunned (−6.2, −5.6 sd)	in the same output nobody looked at gaps 13 and 15 (+2.7, +3.3 sd); made-up cases later put them at about chance
measuring the picture	lines, envelopes and clean colours, calibrated on the axis ticks	crowded dots merge (43 of 505 blobs on one panel); colour counts favour the last-drawn passes	the lopsidedness flip, −0.49 to +0.47, reproduced exactly on a rerun	whatever lies under the top layer of the drawing
the off-scale flag	any verdict with a band	it cannot see a wrong mechanism: MEM-4 passed with its ruler fine and held for the wrong reason	coarse dials caught before a verdict was trusted (PA-5d, AD-2d, LN-1)	—
the simplest separate run	checking a mechanism: one ball, no ripple	it cannot speak to what needs the missing ingredient, such as the tongue	found the Shapiro lines and MEM-4’s real reason	mid-map the memory floor dips below the no-ripple 0.86 to 0.87, because there the ripple shakes the ball in its dip near the dip’s own rocking frequency; the floor is lowest, 0.8455, at 0.81 of it (MEM-7)
explaining it simply	finding where an account is wrong	a plain picture leaves things out: the chalkboard’s staircase is one pull and a frozen ripple	three mistakes found while drawing one map	Bear’s plain questions surfaced the outside tick and the rider’s own pace; the light came from the questions
the exact symmetry, registered first	any equation with a mirror: the rectifier’s φ → φ + π, the weightless step’s quadrature, time reversal	it says nothing about size: the mirror held at every weight while the zero slid 27 degrees	it caught a wrong instrument at once (RATCHET-1) and told in advance which part must grow with the weight (RATCHET-2)	where weight and honey are both present the symmetry that pins the zero breaks, and only the runs say by how much
the dry run on engine-made stand-ins, every verdict passing once and failing once	any registered reader	stand-ins written by hand hide what the engine cannot do (LADDER-1); a stand-in at a cheaper setting can leave the engine’s stable range (lesson 34)	it found a fault before every rectifier run of sitting B10	a fault that only the real setting’s hardest row shows

7.9 The techniques that travel

- **Predict the missing member of a family from its rule, and look there** (lessons 4, 67).
- **Judge a transition by smooth measures, not yes or no** (lesson 52).
- **Test a new tool on made-up cases with planted answers until it passes,** then turn it unchanged on the real map (lessons 76, 85).
- **Read blind, write it down, then score against the marks** (lesson 75).
- **When counts hold everywhere, look for the coordinate in which they are obvious** (the straightened lag; lessons 28, 54).
- **Hunt the mirror.** Symmetry found the pin, and the pin led to the dancers; registered first, an exact symmetry is also the cheapest bug detector (lessons 46, 92).
- **Carry the question to where the ruler reaches** (lesson 63), and **when a law misses at one new setting, ask which other knob moved** (lesson 61).

7.10 Where the Field Guide’s rows rest

The Field Guide’s rows below cite this part. The guide’s wording of each row is its own, and some rows also draw on other sessions’ cases; the lessons here are R2W’s part of each, stated in full above.

guide row	its short form	this part’s lessons	the runs
686	a verdict is only as fine as the ruler under it	1, 2, 5, 16	THIRD-FACE-2, AD-2c, BRIDGE-1, WAYSTATION-1, BALANCE-1, DIAL-1, DIAL-2, MEM-5f
696	chaos can keep a count	121	the third face (sitting B8)
697	a window can open and close by the same road	122	the third face, the coin’s tails
698	two maps can share a line	123	the pad atlas and the golden ray
703	the size of the holes decides what a string is	85	LN-6 at 100 and 50 rows; the made-up crossings
705	a split is a tree with three roads	87	Bear’s marked pictures, sitting B8
707	long reach beats many short chains	88	BEAR_STROKES_MATCH, sitting B8
711	shuffle before believing a pattern	76	the weave test against 200 shuffles; the made-up cases
712	sign and size are separate claims	54	SLOPE-1, SLOPE-2; the three-body blind reads
713	name the picture before naming the pattern	77	the three-body weave, 5 and 47 passes
714	when the data is held back, measure the picture	79	PICTURE_PIXEL_TOOLS
715	the eye keeps what a summary throws away	80	the lopsidedness flip; the white patches
716	hold back the data while the eye is learning	81	Bear’s ruling, sitting B8
717	a colour scale that lumps classes fools the eye	82	the weight 2 map, 37 cells at lag 0
722	a lock depends on its reader: name the reader with the count	15, 127	PAD-4, STACKS, STACKS-2, STACKS-3
726	when the whole toy turns out to be a textbook object, its map becomes a bench	57; §7.3’s opening	SHAPIRO-1, KNOBS-3, MEM-5f
727	a map transfers in two senses: number for number where the equation is the same, and as a calibrated ruler elsewhere	56, 70	ADDED-1; the swing equation (§5.3)
728	judge a transition by a measure that changes smoothly	52	TRICKLE-3, TRICKLE-4

Part 8 — What is open

> *two quiet ends of one knob, each holding the needle on north, and a loud middle where they argue.*
>
> — no-physics read 191, its last line

Each question says why it is open and whether it can be run.

question	why it is open	can we run it
Why does the grip multiply where it is small ($\delta_2 = \delta_1^2$, $\delta_3 = \delta_1^3$), and why does its step follow the one curve it does?	the power law and the curve were each found after the data, then held blind in both media under a single tone, on both sides of the toy's plane, from δ_1 0.12 to 1.89 (GAP-1, TOY-WALK-1; §4.28, figure 37). At the junction's weakest drives the excess counts turns, not short-way steps, as counting quanta predicts (DELTA-WEAK-1, the rival that held), and a middle dance's excess is close to the product of its Farey parents', which held blind one level further down (FAREY-1, within 6%), though at that strength the old law held too; at weaker drives, where the laws part, the product carried every child nearer than the old law at strength 0.03, all seven, and all but 4/7 at 0.05 (FAREY-2b, 3b, by the integral route), with one extra factor κ wherever 1/2 is a parent, read from the first generation and held on the second, blind at five of its seven rows (X-d); κ grows as the drive weakens, on a straight line in the strength, from 0.98 at 0.10 to 1.16 at 0.02, where the line predicted it within 0.0011 (KAPPA-1, figure 44)	partly; why 1/2 carries κ (the place reading failed: 5/12, between two children of 1/2 but not one itself, sits with the others, PLACE-1; neither the parent's excess size nor an even period explains it, on numbers on file; the half-step test, 3/2's children one interval up, needs a stronger drive, since the dances there grip hundreds of times more weakly); how the two arms of children of 1/2 meet (at strength 0.10 they close toward one top, LADDER-2, but by no single power of the period, and the next rungs there sit too near the ruler's floor to tell a settling closing from a creeping one; a ladder at a stronger drive, with its own check that the ruler reaches, can be run); the curve is not derived
Where is the plain's edge when the turned ripple is strong?	the exact fold method lost the resting motion's branch at seven settings, all with the ripple turned around at strength 0.45 or 0.60 (FOLD-DIR); walking the strength up found three and lost four, weights of 1 or less at 0.60, and lost even the one whose fold settling had found (weight 0.75), so the branch itself may change shape there (FOLD-DIR-b); mapping the whole branch lost it as well (FOLD-DIR-c). Read directly, ball by ball from rest, the edge at strength 0.60 is one drive, 0.88 to 0.89, for every weight from 0.05 to 1.00, and at strength 0.45 a ball at rest slips off 0.05 to 0.11 before the fold (FOLD-DIR-d): the finer reading found the cause: the resting dance doubles at the drive where the ball slips off (weights 0.75 and 1, strength 0.45), and under a strong turned ripple heavier balls slip off sooner, weight 5 at 0.713 at strength 0.60 (FOLD-DIR-e); the heavy balls double too at strength 0.45, but at 0.60 they leave by the fold of a second resting dance, at 0.860 and 0.713, while the dance followed up from a gentler ripple goes on to 1.21 and 1.53 (FX-2b); open are the second dance's exact fold, whether a ball started in the upper dance stays on the plain past 0.86, and why the light balls' edges at strength 0.60 forget the weight	partly answered: the edge from rest is the fold of the dance a ball at rest falls into, within 0.0001 (FX-2c); the walked dance holds a ball from its fold down 0.00042 of drive at weight 2 and 0.063 at weight 5, then doubles, and the doubled dance holds a ball too at both weights (FX-2d, FX-2e); both sit far above where a ball set down at rest leaves, and the walked dance widens as the ripple weakens (FX-2e); how fast its multipliers fall at the fold foretells its width at weight 2, not at weight 5 (FX-2f). Can we run it: yes, a ball set down at rest at weight 5 between drives 1.47 and 1.52; minutes of computing
Does every extra step pass through the same gear as the second?	read after the data across 18 rows (0.97 to 1.11 times); the rule registered before it failed at the junction's four-step dances	answered: written first and held blind at eight rows in the junction's far bend, 1.01 to 1.04 times (DANCE-4); why each extra

		step is a few percent gentler than the second is a derivation, not a run
How does the shape of the music regear the steps?	partly answered: the gearing is a hump in the second tone with its top just below zero, read at five settings and confirmed blind at four more (HUMP-1), a shelf and a drop on one side and a steady fall on the other; the frozen point moves with the tone and the pair drifts after it a tenth to a sixth as far, both in straight lines (§4.28, figure 38); why the top sits below zero, and why the drop, are open; near −A/4 the drop steepens and the plus side's fall keeps its pace, both written first (HUMP-2)	no further run on this engine, which holds only inside a second tone of A/4; why the top sits below zero and why the drop steepens toward −A/4 are derivations
What sets the side of the chalk line the pair lands on?	partly answered: in the junction A/ω^2 does, flipping between about 0.33 and 0.42 (§5.2); in the toy the ripple's direction does: inside on the atlas's side at every strength from 0.20 to 0.70 (§4.27; 0.60 and 0.70 printed by TOY-BEND-1, not graded), outside on the turned walkway at 0.10, 0.20 and 0.30, and at every drive read at strengths 0.40 to 1.00, written first at 0.80 and 1.00 (CATCH-2, CATCH-3, §4.24); why near 1/3 is open	the walkway at stronger ripple, answered by run: it never flips; why near 1/3, a derivation
Does the toy's first gear keep falling as the ripple grows stronger?	answered (§4.28): yes; written first and run at strength 0.80, its size over the strength at the 1/2 dance came to 0.90, below 0.93, each step down larger than the last (TOY-BEND-2)	done
Where do the dancers' rules stop as weight grows?	mostly answered: read at the middle of each dance's tongue on a ladder a sixth of a decade apart, with the pair outside, 1/2, 1/3 and 2/3 all change their count on one rung, weight 0.32, while still stable, the rung for 1/2 and 1/3 called before the run from the weightless margins; with the pair inside, no count changes while its dance is stable (1/2 holds to weight 1; 1/3 and 2/5 change only after turning unstable); at the tongue's edges it is always one different (§5.2); the narrow tongues are followed further but not all the way, and the engine's edges there sit at its own probe points (WEIGHT-WALK-3); an edge finder that halves its weight step where a dance is lost follows both 2/3s to weight 1, the two computers agreeing, and the stable-dance rule held inside at 3/5 and 3/7, while outside the margin rule missed again at 3/7 and the guess for the middle missed at stable dances too (WEIGHT-WALK-4)	yes: the guess for the middle from three weights, in the weight squared and its fourth power; the margin rule with each stop followed in the weight squared from two rungs, and tried at 3/5 from the dance now found at weight 0.01 (LOST-3/5); the widths' growth read only where the dance is stable
Does a tongue's centre slide as the weight squared at other strengths?	answered (§4.28): yes; written first at strength 0.25, the ratio of the centre's changes came to 4.0024, 4.0006 and 4.0028 at 2/5, 3/7 and 4/7, the widths doubling within 0.0005 (SLIDE-2)	done
Why does the grip ratio follow A/ω^2?	partly answered: to first order in strength the excess is $-0.85/\omega^2 + 0.16/\omega$ times A, measured and held blind (§5.2); the two numbers are not derived	yes, by perturbation theory
Why does the pair sit where it does: 0.013 outside on the toy's turned walkway, inside on its atlas side and in the junction's strong zone?	the side is the setting's, not the medium's (§4.27, §5.2), and on the walkway not the strength's either: outside at every drive read from strength 0.25 to 1.00, the gap	the side on the walkway, answered by run; the size, by derivation from the passing matrix

	growing with drive and strength and its growth slowing, written first at 0.80 and 1.00 (CATCH-3, §4.24); the size is not explained; in the junction's strong, slow corner it closes as about ω^2 and never crosses	
Why does the 1/2 window narrow as $\exp(-c1/\omega)$, and what sets the constant c1?	measured over five orders of magnitude (§5.2); half the slow cycle's below-threshold action does not give it	yes, a slow-drive expansion
Why does the second harmonic fall so far below the closed form?	the first-order theory reproduces it at every dance (§4.28, KERNEL-1), but its ratio to the closed form, 0.29 to 0.45, sits far below its neighbours' (the first harmonic 0.75 to 1.00, the third 0.88 to 0.94). A candidate: the response over a beat carries the drive's own harmonics, and the second harmonic meets the strongest of them	yes: split the response into its harmonics
How does the second tone's push on the tongues depend on the radio wave's strength?	no single power fits; the power falls from about 3 to 1.2 as the strength rises from 0.10 to 0.30, highest at the slow dances (§4.28); weaker still it keeps rising, to 4.7 at the 1/4 dance between 0.05 and 0.07 (PUSH-3); answered: the push is a straight part growing as the strength squared, as frequency mixing says, less a bent part, -0.21 to -0.23 times the second tone squared at every strength, both written first (PUSH-4)	answered by run; the bent part's size is a derivation
Why, under a single tone, is the far-side leaper always the stable dance, and where else do the dances pinch?	it held at every root of every single-tone run in both media, and the second tone has shown it is not forced by the geometry (§4.28, PINCH-1); partly answered: one tone builds the grip one way, as A^p, so it cannot flip it; a second tone builds it a second way, and the pinch moves as A^2 (§4.28, FAR-1, held blind); the swap passes through a window held by a walking mirror pair (WALK-1). The sign of the single way is not derived, and the map of pinches is not drawn	yes: the leading term's sign, the first-order grip's, by perturbation theory; the same engine over strength, second tone and fraction
Why does the rectifier's largest slide fall as the drive's strength over its frequency squared grows?	partly answered: at a held ratio the peak's place and height travel with it, tested from both sides (RATCHET-5, RATCHET-6; §6.5, figures 54 and 55): weight times frequency 0.578 at the peak at ratio 0.99 and 0.634 to 0.658 at 1.875, the peak 27.1 to 27.4 and 22.8 to 23.6 degrees; why the peak falls as the ratio grows is not derived, and the frequency keeps a small say of its own: at ratio 0.99 the peak climbs 27.1, 27.4 and 27.7 degrees at frequencies 0.40, 0.55 and 0.70 (RATCHET-7, figure 55), and 28.19 at 0.85, 0.45 degree above 0.70's, inside the band registered before the run (RATCHET-13); along the ratio at frequency 0.55 the peak falls at every step from 0.612 to 3.25, steepest near 1.3 to 1.5 and slowing past it, which the curve written first did not foresee (RATCHET-83); at 3.25 it is 20.4 degrees, within 0.04 of the forecast a curve through 1.50 to 2.75 made before the run (R83-e); past it, at 3.75, 4.0 and 4.25, it falls on, slowing, to 19.7 degrees, and the curve that had it level off near 4.0 missed	yes: an edge follower whose step guard grows with the window, to read 5.0; why the peak falls and then turns is not yet derived

	(R83-F); at 4.5 it turns back up, to 19.73 (R83-G), and at 4.375 it is 19.707, which places the bottom near ratio 4.32 (R83-H2); at 5.0 the window is open and widening where the engine stops, and the peak is not read (R83-H)	
How does the rectifier's size depend on the drive along a ratio line?	not in proportion: scaled by 0.53, 1.62 and 1.89, its weightless height went to 0.54, 1.58 and 1.82 of itself; RATCHET-6's guess of proportion failed, and the slower growth, registered after that miss, held at the third frequency (RATCHET-7); along the 0.99 line the local power falls from 0.97 to 0.95 as the drive grows, and to 0.917 between frequencies 0.70 and 0.85, inside the band registered before the run (RATCHET-13); why is not derived	yes, by derivation; and a longer stretch of the line
Why does the heavy rider's rectifier fade nearly as the weight to the minus fourth?	read after the data, the local powers running −3.86 to −3.97; a whole number suggests a count of factors, not yet found; written first, the next rung steepened to −3.9771 against −3.977, and at heavy weight the height is in proportion to the second tone to 1 part in 10,000 (RATCHET-80)	answered by run; which factors make the fourth power is a derivation, not a run
What sets the fourth harmonic's gear, and does its half-turn count grow with the harmonic?	partly answered (§6.6, figure 56): the fourth harmonic's weakness is its weak-drive law, a window growing as A^4 where the second harmonic's grows as A^2 (powers 3.88 and 1.91 between strengths 0.075 and 0.15, RATCHET-10b); its zero makes the same half-turn along the weight knob at all nine settings run, and comes home to −180 degrees as about 181 degrees over the weight times the frequency (medians 180.4 and 183.2 at frequencies 0.70 and 0.40, RATCHET-11 and 12). Open: why the light-end gear falls as the frequency rises (5.58, 5.11, 4.73 and 4.44 at frequencies 0.40, 0.55, 0.70 and 0.85 on the drive ratio 0.99, in steps that shrink; RATCHET-11's registered gear failed, RATCHET-12's and 13's redesigns held) while barely moving with the drive ratio (5.07 to 5.17 at 0.55), and why its middle climbs at 0.40 and sags at 0.70; and whether the count is minus one half-turn per step of two in the harmonic	yes: by derivation from the weak-drive picture of §6.6; the seam located between frequencies 0.775 and 0.785 (RATCHET-14 to 17), not yet derived; the window's two seams located (RATCHET-18 to 21, §6.11); the sixth harmonic at a strong drive, after a check that its window is readable
What makes the window's seams?	the fourth harmonic's dip pair vanishes near frequency 0.5228 as a fold's law says (0.5225 on coarser rungs), and its light rise is born smoothly where a, the window's bend in the weight squared, changes sign near 0.758 (§6.11, RATCHET-18 to 21; bracketed by runs at 0.5215 and 0.5235, RATCHET-81); the second harmonic's a is positive on the drive ratio 1.875 and negative on 0.99 and 0.612. The second harmonic's a changes sign near drive ratio 1.306 at frequency 0.55, on a bowed curve (RATCHET-24, 25), and near 1.275 at 0.70, on a bow 1.45 times as deep (RATCHET-27, 28), and near 1.325 at 0.40 (RATCHET-29), the bow deepening in a straight line with the frequency. At the edge the window is weight-proof, within 0.1% of its weightless height to first number about 0.098 at	yes: a knob that cancels the fourth-power term, which neither the drive ratio, the second tone's size, nor a third tone at four times the frequency does (RATCHET-31, 32, 41, 42): a silent third tone at three times the frequency has since been run (BEAT-4, Appendix A: it moves the light coefficient a and tilts the window); next, the ratio past 3.25; which of the step's edges give way near weight 0.36 at the held-timing setting (two paths stall there, RATCHET-40), and the setting at other ratios (RATCHET-34 to 37 found it at 3.25 and its band to weight 0.3364); the fourth harmonic's envelope on the ratio 1.875; a by perturbation theory in the weight

	frequencies 0.40, 0.55 and 0.70 (RATCHET-65, 26, 30); the edge has since been placed at frequencies 0.85 and 1.00, 1.22994 and 1.17228, its fall steepening (EDGE-85, EDGE-100, §6.11). None of it is derived	
The junction at frequency 0.10	its window is too narrow for double precision	yes, with extended precision
What law does the pair’s speed follow?	no simple law fits the plane	yes, by derivation from the passing matrix
The tongues’ widths against q and p/q together	answered (§4.23): per unit weight the width goes as q to the −2.39, times the sine of π p/q to the 0.71, times e to the 0.49 p/q; all nineteen rows within 10%, and 4/9 and 5/9 placed within 3% before their run (TONGUE-1)	answered by run; why these powers is a derivation
The second order in weight at 4/5 and 5/8	first order is large there	yes
The 3/7 ladder’s last doublings	answered (§4.16): span ratios 5 and 6, 4.6187 and 4.6570, closing on 4.669 as the 1/2 groove’s did (LADDER-B-UP, eight doublings, the last where the 896-passing dance itself doubles)	done
The door past the room’s closing	read from weight 0.55 to 1.40 (DOOR-1 to DOOR-4, §4.14): straight to 0.90, bending past 0.905 where gears 1 and 2 overlap; past the closing a parabola in the weight, written first and landed to the drive step at 1.30 and 1.40 (DOOR-4)	answered by run; why a parabola, and where it turns (its slope would reach zero near weight 2.75), is a derivation and two more weights
Noise; the waves’ shapes	the drift under noise is measured along the weight knob (§6.9); the step’s corners under noise, and the waves’ shapes, have no data	yes, with noise added to the engine (§6.7, knob 9)
The re-aim at a real medium	not built	yes, after this deposit
The literature pass	done for this deposit, with what it could not reach: every outside work the text leans on is listed under Sources checked with how it was checked, the four debts this row named before included (the step against the drift, Ustinov and colleagues, Neumann and Pikovsky, Ortega and Alonso). The step as a threshold apart from the drift is drawn by Scheer and colleagues (2025); the measured gap between the two zeros along the weight was not found in print. Not reached from here: Ortega’s own record and Kautz and Monaco beyond their abstract; the two 2025 diode papers were read in full. Names given without a mark, as on the Signposts page, are from recall	not from here; a reader with library access can
Do riders sharing one drive act on one another?	§6.13’s arrays add the riders’ pushes as if they did not; every array number rests on that, and nothing in this book tests it	yes, in the model, once a coupling is chosen (a shared load or bias line): riders run together, against the sum of their runs alone
Does a slow rocking push slip the rider one way on every swing?	argued from the step’s edges (§6.7, knob 7) and not run	yes: a slow rocking push on the same engine
At weight 0.3, is the 3/5 swap gone or only far out?	it moved outward with the weight, from −0.092 at weight 0.03 to −0.114 at 0.2, and at 0.3 it was not found down to −0.22 (LAB-MAP-1b, LAB-MAP-2, §5.6)	answered: gone, not far out. Carried to −0.30 with every start settling, the lag-0 multiplier peaks at 0.9625 near −0.17 and falls to 0.742; lag 1/2 stays above 1.20 (LAB-MAP-3, §5.6)
Why does the memory floor dip mid-map?	answered (lesson 55, §7.8): the ripple shakes the ball held in a ledge’s dip, and at	answered by run; the same law at other weights needs the atlas’s floor read finer

	weight 2 the floor is lowest, 0.8455, where the ripple passes at 0.81 of the dip's own rocking frequency, as written first for a dip that softens as it swings (MEM-7, after MEM-6's two single-frequency guesses missed)	than its 0.005 grid
Do four smaller crossings, read by a straight line between rungs, sit where they were read?	answered, all four by runs written first (§6.12): the dip pair is still there at 0.5215 and gone at 0.5235, as the fold's law (0.52250) said and a straight line in the depth (0.52098) did not (RATCHET-81); at the 0.70 joint edge's size, 0.3795, a/ω² is −0.00002, as the cubic said (RATCHET-82); the 2/5 swap tone sits at −0.03363 on tones 0.0001 apart (PINCH-3); the hand-off widths are 0.00076 and 0.00210, the second corrected from 0.00214 (WALK-3)	answered by run
Does the clutch keep the bottleneck's square-root law?	near second gear's lower edge the catch time falls as the distance to about the power −0.32, not −0.5 (CLUTCH-2, §4.1), and a five-degree catch rule cannot see the squeeze	yes: the time between slips just below the edge, times the square root of the distance, in windows long enough to hold many slips
Does a deeper rocking open a longer run of the dance's steps?	rocked at its own rhythm, the six-passing dance at 1.6756 loses its riders from a run of neighbouring steps, two at depth 0.00002 and three at 0.00005 and 0.0001, read after the data (SONG-3, §4.13); SONG-4's shifts follow from a symmetry of the setup and cannot tell	yes: six dancers, one on each step of the dance, rocked at six passings at depths stepped from 0.00001 to 0.0005
How soft must the board's legs be, and how light its damper?	partly answered (BOARD-2, BOARD-3, §4.6): both knobs decide; a tenth more drag alone keeps none of the band, a damper of 0.1 costs it on legs at 0.05 and 0.1, legs at 0.2 cost it even with a damper of 0.01, and soft, lightly damped legs keep 369 to 407 of the 418 phases; the soft board's floor lies below 0.796 and goes with its damper, and a lightly damped soft board slips at every steady drive from 0.796 to 0.812	yes: the band and the floor on a map of leg rate (0.01 to 0.3) and damper (0.005 to 0.1); the rescue below the soft board's own floor
Can a toothed second stage compound gears?	moved closer to its shaft, the second stage lost its own gear, its pattern running near and above what its grip of 0.6 carries alone, and showed none in between (TRAIN-2, §4.3)	yes: the same stage with a grip of 1.2, deep enough for its pattern at every pace
Does a needle keep a net turn equal to the solid angle its field's polarization walks round? (The nearest run is RF-10's needle in the field of three emitters, §4.10.)	worked out, not run: emitters of fixed phase bring the field back to itself after any loop on the sphere of polarizations, so a needle locked to it keeps no net turn; Pancharatnam's phase shows only against a reference carried round by parallel transport (Pancharatnam 1956; Berry 1984), so a run here would show only that a locked needle follows its field	yes, with a needle that has a memory of its own, such as a spinning top whose axis follows the field's ellipse
How fast does a bead drain a standing wave it rides, when the wave is a finite store, and does a dim one drain slower for the same travel? (The nearest runs are the two stations' standing pattern and its envelope bead, §4.10.)	not run: every field in this book is imposed and feels nothing of what it carries. For an overdamped bead the work is the drag times its speed times the distance, so a dimmer wave, which can only carry it slower, costs less for the same travel; that is arithmetic, not a test	yes, with a store that feels its rider: the standing wave's strength drained by the bead's work
Do the two-station riders keep their split on a lattice rail, and does an array of emitters on a dodecahedron's faces leave a residual when a bead is walked round one vertex?	not run: both need a field in two or three dimensions, and the book's tools make fields on a line and a ring	yes, with a lattice tool: a crossed standing wave for the rail, and twelve emitters for the pad array

(The two-station riders are in §4.10; arrays of riders on one drive in §6.13.)		
Does any lock jam, gripping harder the further it is pushed off its line, as Bear’s cocked ferry binds on its rope? (His drawing of 6 October: a cable ferry, a waisted ring riding a twisted rope, cocked by the current until it sticks; lesson 108.)	partly answered: the ferry’s sliding ring and its trapped ring have the rider’s counterparts, gear 2’s window sliding down with weight and a gear held as memory at one drive (GEAR-ONSET, GEAR-HELD); but every lock in this book lets go past a quarter turn of lag, the opposite of a jam (FERRY, Bear’s drawing of 6 October)	yes: a coupling whose grip grows with the lag, like a rope round a capstan, whose hold grows with each wrap
How much does the relay steer for what it spends, and how many bits a passing can its comb carry? (The relay and its comb are §4.6’s.)	not run, a definition first: the kick never powers the second stage, it only chooses its gear, a valve rather than a pump, as Bear’s four-wire dimmer carries its signal on two thin wires beside two heavy ones (BUS, 5 October); a steering gain, the flow a gear choice moves against what the relay spends to make it, is not yet defined	yes, once defined: the power each gear carries against the relay’s own work, and the comb’s exact fractions read as a channel
Inside one band held at a fraction, do the inner pads come in the fraction tree’s order, or in the order of windows inside a one-humped map’s chaos?	not run: Hofstadter’s butterfly rhymes with the tongue map only in that the picture repeats inside itself at a fraction (BUTTERFLY, Bear, 6 October); around the 2/3 band the coastline’s scan found pads of 18, 24, 30 and 36 passings, and the zoom found 2/3 on a dance of 30 well below it (§4.13)	yes: the pads inside one band read in drive order, against both orders written first
Does a grip fed by a pad’s own slip hold a pad the plain rider cannot, and does a mixture fed by the swing tell the two dances at 1.6756 apart?	not run: MIXTURE was run on the plain rider with one wave (§4.10, ‘Bear’s cornstarch’); this question’s pads and dances live on the relay’s second stage	yes, with the relay engine: the second stage’s grip made to stiffen with its own slip, at the pads and at 1.6756
Is the rider chaotic on its partner’s tread, does a chain of such riders show its first doubling as neighbours swinging opposite, and does a rider with inertia find the paper’s step at twice the partner’s frequency?	not run: RESONATOR-TREAD gave the plain rider a partner (§4.10, ‘A partner that answers back’), with no stretch-rate reader, no chain and no inertia	yes: the stretch rate on the tread, a chain of riders with partners, and the rider given a mass
Where does the true window close at the watershed, read where the sideband’s trap cannot pass for the gear?	at 1.01 times the tooth rate the second stage’s apparent burst of rescue was the shake’s first sideband, caught inside the relay’s ruler; read four times longer the window was still open, at pushes 0.36 to 0.42 (WEDGE-2, FAUX-W2, §4.6)	yes: 1.001 to 1.01 times the tooth rate, each read long enough that the floor lies under the sideband’s offset
Does a stack of three or more relays need a ladder of time constants, each layer slower than the room of the layer below?	answered (§4.11): no ladder; each filter need only be slow against the room of the stage it feeds (STACK-1)	done
Where does the third stage’s pocket close, and does the pocket’s top edge keep one over the room?	answered (§4.11): it closes between rooms 0.10 and 0.11; its top edge is ragged, near one over the room in size only, its bottom edge stands still at small rooms, and it is striped with bands of holds (STACK-3, STACK-4)	done
What is the coin’s third face’s first Feigenbaum ratio, and does its window open by a cascade run backwards?	answered (§4.17): the first ratio is 6.18, the 3 to 6 doubling between drives 1.4439900 and 1.4439925 (TF3-1b), and the window opens by a cascade run backwards, 48, 24, 12, 6, 3 going up (THIRD-FACE-3); why the first ratio sits so far above 4.669 while the next is near 4 is not derived	answered by run; the why is a derivation
Is the coin’s sudden end, where the last tails starts all turn heads at drive 1.44284, the 1/2 groove itself ending?	answered (§4.17): yes; a carried tails rider leaves the groove between 1.44282 and 1.44284, as at a fold, and a finer dial reads the fold between 1.442834 and 1.442836	done

	(COIN-END, COIN-FOLD)	
How does the coin's flickering edge scale with the dial's step?	at the top of the band, on a dial 0.00002 of drive apart, starts turn both ways between neighbouring drives (1,089 from tails to heads and 722 back over 50 steps), and 0.88 of the new heads are born within two starts of a start already heads, at every edge rather than between the two heads arcs (COIN-4, §4.17)	yes: the same band on dials of several steps, where an edge with uncertainty exponent 0.592 says the flips per step fall as a power of the step
Does something chaotic live on the coin's seam, holding riders a while before they choose a face?	the seam is fractal (exponent 0.592) and its slowest starts sit on its crinkles, but every one of 4,000 starts settles within 18 passings, too few lingering to read how fast they leave (SETTLE-1, §4.17)	yes: a hundred times as many starts, each read for about 40 passings; a chaotic set on the seam predicts its escape rate over its stretch rate equal to 0.592, a plain seam about 1
Do phase-locked loops and injection-locked oscillators pinch their sub-harmonic locks under a second harmonic of the reference, as the dancers' rules predict?	answered for the loop (§5.6): the 3/5 lock pinches and swaps in a behavioural circuit model with the carrier's ripple, and a carrier fifty times the tone keeps the swap within 0.0005 (PLL-2)	done
Does an injection-locked oscillator, whose amplitude may play the weight, pinch the same way?	the loop does (PLL-2); an oscillator locked by injection has no loop filter, and whether its amplitude's slow return acts as the weight is untested	yes, in a behavioural model
Past weight 0.025, where both self-mirrored 3/5 dances are stable over a stretch of the second tone, does a rider keep whichever dance it had, and what becomes of the walking pair?	WINDOW-WEIGHT read the two dances' multipliers only; they sit at biases about a tongue's width apart, and at these weights the wrinkle's fine shape is close to the engine's rounding. The slope rule predicts the walking pair is still there, now as the unstable partner, since the wrinkle must climb at both self-mirrored dances and fall between them	yes: the wrinkle in extended precision, then a slow sweep of the tone each way
Does the quarter-turn gap between the step's zero and the drift's zero appear on the other road to the light limit, draining the honey with the mass and the drive held?	along the weight knob the two zeros sit 8 to 13 degrees apart at the four weights read above zero, not a quarter turn (§6.9); the other road raises the weight while lowering the frequency, and has not been run	yes, in the model
What sets the small dial's error at fast drives?	answered (§6.12): the gap between the probe's square and the flat point's; the error is 0.00149 to 0.00153 times it at all five swings run at frequency 4, read after the data, and it held blind with the probe doubled and halved (DIAL-GAP)	answered by run; why that number drifts by 3% across the swings, and what it is at other frequencies, are not read
Does the slope a stray tone puts in the window, a quarter turn off the mirror line, grow in proportion to the tone's size?	answered (Appendix A): for a fourth harmonic, 2.13 times for the size's 2.16 and 1.003 times proportion at 1% of the octave (RATCHET-47, 48); for a third harmonic, within 0.16% at half and double BEAT-4's size (SLOPE-SIZE)	done
Where along the line a = 0 is c best, and can the octave's own size serve as a third knob?	partly answered (Appendix A): c over ω^4 at the joint point is −4.33, −3.18, −2.95 and −3.44 at ratios 2.25 to 3.75, best inside that range, near 3.2 (JOINT-LINE, after the lean that it keeps rising missed); the octave's own size as a third knob is not run	yes: ratios 3.0 to 3.5 to place the best, and the octave's size, on the same engine

Appendix A. The envelope in full: every run behind §6.11

This appendix keeps §6.11's runs as they were told, with nothing cut; §6.11 carries their results, the budget, the table and the map.

The second harmonic: one region. With the second tone at twice the frequency the envelope is a single region. At every weight and frequency run, on drive ratios A/ω² of 0.61, 0.99 and 1.875, the zero stays within 29.2 degrees of quadrature, so with the tones in phase the push keeps one direction at every weight. The window's height has one soft edge. Near no weight it changes as 1 + aβ². In words: it starts flat, with no term in the weight itself, and bends up or down as the weight squared, and the number a, the light coefficient, says which way and how fast. On the 0.61 and 0.99 ratios a is negative, −0.04 to −0.30, and the window only falls as the rider gets heavier. On the 1.875 ratio a is positive, +0.044, +0.091 and +0.171 at frequencies 0.40, 0.55 and 0.70 (the last registered between +0.10 and +0.18, RATCHET-20), and the window first rises, by 0.19, 0.18 and 0.29 percent, before it falls. Between them a changes sign, at frequency 0.55 near the ratio 1.306 (RATCHET-24 and 25). A straight line through 0.99 and 1.875 put that edge at 1.396 and missed: a was already positive at 1.35 (+0.0092). Read against the ratio, a follows a bowed curve, not a straight line. A parabola through five ratios put the edge at 1.3065, registered before two more runs, which found a at −0.0012 at 1.30 and +0.0030 at 1.32 and the edge at 1.3056. So the second harmonic's light rise has an edge in the drive ratio near 1.31: below it the window only falls as the rider gets heavier, above it the window first rises. To an engineer a rise of a few tenths of a percent changes nothing. The rise matters only as a sign that the drive is strong. The edge itself is useful, as the next paragraph shows.

The weight-proof setting. At the edge the β² term is gone, and the window's height depends on the weight only through the next term, cβ⁴. Engineers hunt for points like this. A quartz watch crystal is cut so that its turnover, the temperature at which its frequency stops rising and starts to fall, sits near room temperature. Near the turnover a few degrees either way change the frequency only as the square of the distance. A bandgap voltage reference adds a voltage that falls with temperature to one that rises, so that their first-order changes cancel and only a small bow is left. The edge in the drive ratio is the rectifier's turnover for weight. Before running it, the book read c from older runs, about −1.03. With it the book predicted that at the edge the window would stay within 0.1 percent of its weightless height out to weight 0.1765, and be 0.98919 of it at 0.32. RATCHET-26 ran the edge at frequency 0.55, ratio 1.3056, on fourteen weights. The window held within 0.1 percent out to about 0.1778, and at 0.32 it was 0.988952 of its weightless height. (A straight line between the rungs at 0.16 and 0.20 first read the edge at 0.1745. The departure bends upward between them, as the weight to the power 4.07, so the line reads early; a curve through the four rungs from 0.12 to 0.24 read 0.1778. Run again on rungs 0.005 apart, the edge is 0.1778 (RATCHET-65), within 0.0001 of the curve.) All four verdicts held. Compare the usual ratio of 0.99 at the same frequency (RATCHET-3). At the edge the window moves about 60 times less at weight 0.04 (where the edge's shift sits in the last printed place), 13 times less at 0.08, 4 times less at 0.16 and 1.7 times less at 0.32. The gain is largest where the rider is lightest, as a β⁴ against a β² says it should be. Only the size is weight-proof. The timing still slides with the weight as before, +8.8 degrees at weight 0.16 and +16.3 at 0.32, so step 4 of the card still applies.

The edge moves with the frequency. At frequency 0.70 the edge is lower. The book first carried the bow measured at 0.55 over to 0.70, pinned through the two ratios already run there. It registered an edge at 1.2953, within 0.02, with the straight line's 1.3463 as the rival (RATCHET-27). Runs at ratios 1.25, 1.30 and 1.35 put the edge at 1.2749. The carried bow missed its bar by 0.0004, and the straight line by 0.05. The bow at 0.70 is deeper: its curvature is 1.45 times the one at 0.55, and that alone moves the edge by 0.020. The redesign let the data's own shape at 0.70 choose the curve, as lesson 163 says. A parabola through its five ratios put the edge at 1.2750 and predicted a at ratios 1.27 and 1.28. Two more runs, registered first, found −0.00167 and +0.00196 against the predicted −0.00183 and +0.00184, and the edge at 1.2746 (RATCHET-28). So the weight-proof ratio is a line across the knob plane, not a point: 1.3056 at frequency 0.55, 1.2746 at 0.70. Two points on a line are not yet its law (lesson 163 again). An engineer working at another frequency should run a short ladder of ratios before trusting a number between them. The band in the weight is measured at 0.55 and 0.70 (below).

A third point came next, at frequency 0.40, registered as a contest of three guesses (RATCHET-29). The bow, pinned through the two ratios already run there with its curvature carried in a straight line in the frequency from 0.55 and 0.70, said 1.3284. A straight line in the frequency through the two measured edges said 1.3366, and a straight line in the ratio said 1.4263. Seven runs put the edge at 1.3250. The carried bow held, 0.0034 away inside a bar of 0.004, and both straight lines missed. So the edge falls faster as the frequency rises: by 0.019 from 0.40 to 0.55, and by 0.031 from 0.55 to 0.70. The bow itself deepens in a straight line with the frequency, its curvatures −0.053, −0.093 and −0.135 (the first fitted after the run). Followed down past 0.40, where nothing has been run, that line would reach zero near frequency 0.2, and there a straight line in the ratio might at last be right. The straight line is not wrong everywhere: it is wrong where the thing it describes bends. The edge also sharpens as the frequency rises, about as its square. To keep a within 0.001 of zero, the ratio must sit within ±0.009 of the edge at frequency 0.40, ±0.005 at 0.55 and ±0.003 at 0.70. A slow drive is the forgiving one to build. A fourth point, at frequency 0.85, was registered as a contest between the parabola through the three edges, 1.2320, and the straight line through 0.55 and 0.70, 1.2436. Six runs at ratios 1.21 to 1.26 put the edge at 1.22994, 0.00206 below the parabola and inside its bar of 0.004; the straight line missed by 0.01366 (EDGE-85). The edge's fall keeps steepening, 0.0194 from 0.40 to 0.55, 0.0310 to 0.70 and 0.0447 to 0.85, and at 0.85 the ratio must sit within ±0.0017 of the edge to keep a within 0.001. A fifth point, at frequency 1.00, was registered the same way, between a parabola through all four edges, 1.17316, and the straight line through the last two, 1.18528. Eight runs at ratios 1.14 to 1.21 put the edge at 1.17228, 0.00089 below the parabola, and the straight line missed by 0.01300 (EDGE-100). The fall from 0.85 is 0.05766, but the amount by which

each fall outgrows the last has stopped growing: 0.0116, 0.0137, then 0.0130. Across the frequencies run, the edge is a parabola in the frequency. A cubic through the four earlier edges, which read the last step's growth as a trend, landed furthest away, 0.00272 above. At frequency 1.00 the ratio must sit within ±0.0011 of the edge, eight times tighter than at 0.40.

Far out on the ratio at 0.85. Does the bow keep deepening with the frequency? At frequency 0.85, far from the edge at ratio 1.875, the light coefficient a was registered as a contest of three guesses (LIGHT-85). The parabola read beside the edge, carried out, said +0.297. A straight line in the frequency through 0.55 and 0.70 said +0.252. A straight line in the ratio, with no bow at all, said +0.379. The run gave +0.321, and all three missed their bars of 0.015. The bow is there, but over the long stretch it is gentler than beside the edge. Its curvature from the edge out to 1.875 is −0.139, beside 0.70's −0.135, where the straight line in the frequency would have put it at −0.176. So either the bow stops deepening past 0.70, or at 0.85 a curls with a third-power term. Runs at ratios 0.99, 1.55 and 1.70, registered first, told the two apart (LIGHT-85b). The third-power term won at all three: it said −0.1534, +0.1708 and +0.2419, and the runs gave −0.1527, +0.1704 and +0.2414, each within the bar of 0.001, while the bow that stops deepening missed all three, by 0.003 to 0.004. So at 0.85 the curve keeps its sharp bend beside the edge, and a gentle third-power twist straightens it far out.

What it takes to hold the edge. An engineer does not need a to be exactly zero. What they need is a window that stays within some tolerance for every rider they will meet, from the lightest to the heaviest. That turns the edge into an error budget, and two terms spend it. The first is the light coefficient's share, a times the weight squared, which grows the further the drive ratio sits from the edge. The second is the next term's share, c times the weight to the fourth, which no drive ratio removes. Both turn out to be set by the first number, the weight times the frequency, and not by the weight alone. On the 0.99 ratio, c divided by the frequency to the fourth is −11.746, −11.688, −11.644 and −11.577 at frequencies 0.40, 0.55, 0.70 and 0.85 (read at weight 0.08); at the 0.55 edge it is −10.9 to −11.5, read from RATCHET-26's heights at weights 0.16 to 0.32 (at weight 0.08 it is −10.614, RATCHET-31). If the first number sets the band, the window at the 0.70 edge should stay within 0.1% out to weight 0.138; if the weight sets it, out to 0.1745, as at 0.55. Registered before the run, the band ended at 0.1395 (RATCHET-30). The band ends where the first number reaches 0.098 at both frequencies: 0.0977 at 0.70 and 0.0978 at 0.55, once the 0.55 edge is read with its curve. The straight-line reading had put the 0.55 edge at 0.096, and the two frequencies 1.8% apart; read with the curve they agree within a fifth of a percent, and the guess registered from the first number would have landed almost on the measured edge. The rule was then tested blind at a third frequency. From the first number alone, the band at the 0.40 edge was predicted to end at weight 0.2445; it ended at 0.2446 (RATCHET-65). Its first number is 0.0979, beside 0.0978 at 0.55 and 0.0977 at 0.70: one number, read three times, spread over 0.21%. A fourth reading was registered from the first number before its run, at frequency 0.85: the band should end at weight 0.11508. It ended at 0.11475, first number 0.09753, and the rival, the 0.70 band carried over by weight, missed by 0.02475 (EDGE-85). Read together after the data, the four first numbers fall a little at each step up in frequency, 0.09785, 0.09778, 0.09765 and 0.09753, by 0.33% from end to end. The fifth band, at frequency 1.00, was then registered two ways: from the mean first number, which said the band would end at weight 0.09770, and from the drift carried on in a straight line, which said 0.09743. It ended at 0.09742, 0.000009 from the drift's prediction, and the mean missed by 0.00028 (EDGE-100). So the first number sets the band to under half a percent from frequency 0.40 to 1.00 (0.44% from end to end), and its faint drift reads across that range as a straight line in the frequency, falling 0.00074 for each unit of frequency, which fits all five readings within 0.00003 and predicted the fifth; no finer rungs or tighter tolerance were run on it, so it stands as a reading, not a law. The rule an engineer carries is the first number; the drift is its correction, small enough to ignore for a 0.1% window and worth keeping for a tighter one. So the budget can be drawn once, against the first number of the heaviest rider, and it comes out nearly the same at every frequency run (§6.11's table).

For a loose spec the drive ratio hardly matters. For a tight spec with a light rider, holding the ratio within about one percent of the edge is enough. For a tight spec with a heavier rider no ratio will do, because the fourth-power term needs a knob of its own, which this book had not found when this paragraph was written; the joint edge, below, later opened that corner. Holding a drive's strength to one percent is everyday work for a signal generator. The hard part is knowing the strength at the rider, not at the generator. In a Josephson junction, written in the lab's quantities, the drive ratio is the microwave current times the critical current times the shunt resistance squared, over the drive frequency squared, up to constants. The frequency comes from a synthesizer and is exact, and a cold resistor barely drifts. What moves is the microwave current that actually reaches the junction after the cables have lost part of it, and the critical current, which drifts with temperature. So the build has three steps. Calibrate the drive at the device, from the steps the drive itself prints on the junction's current–voltage curve (Shapiro steps), which grow and shrink with the drive's strength in a known way. Hold the temperature, so the critical current stays put. Then check the setting on a test chip whose junctions differ only by an added parallel capacitor: their windows should match. In the same quantities the first number is the drive frequency times the junction's RC time, so a slower drive or a smaller capacitance makes a lighter rider and a wider band. That is the honest form of "a slow drive is the forgiving one to build", three paragraphs back: the forgiveness lives in the first number. The values of c at the 0.40 and 0.70 edges are taken from the 0.55 edge, and the budget's ratio bands from the fitted curves of a, each read after the data.

Hunting a knob for the fourth-power term. The budget's empty corner, a tight window on a heavier rider, waits on a knob that moves c in a different proportion to a than the drive ratio does. That is how a bandgap reference gets its curvature correction: a second adjustment, with a different mix of effects, cancels what the first one leaves behind. Two knobs in this engine were tried at frequency 0.55 (RATCHET-31). The drive ratio does shrink c past the edge. Divided by the frequency to the fourth, c goes from −10.614 at the edge to −7.926, −6.322, −4.680 and −3.512 at ratios 1.875, 2.25, 2.75 and 3.25 (read at weight 0.08). But it slows as it goes and has not crossed zero by 3.25, while a grows the whole way, to +0.19 there. A bowed curve, registered to cross zero near 2.98, missed, and so did the straight line. The second tone's size was expected to move neither, because the weightless window grows in proportion to it, within 2.6% from a twelfth of the first tone's size to two thirds. It moved a strongly instead, from −0.018 at a sixth to +0.070 at two thirds, and c only a little, by 2.6% and 9.1%. So the size is a knob, but for a: it moves the edge, not the fourth-power term. Turning the second tone upside down is the same as turning its timing half a turn, so a and c can only depend on the square of its size. Registered against a straight line at two new sizes, that law held: a +0.02971 and −0.02280 against +0.02900 and −0.02232 predicted (RATCHET-32), and both straight lines missed. The hunt did not find what it was after, but it handed the engineer something useful. The second tone's size moves the edge about six times more gently than the drive ratio does, so the ratio is the coarse adjustment and the second tone's size the fine trim, like the coarse and fine knobs of a laboratory supply. Within these two knobs no setting zeroes both a and c in the range run. The next places to look need the engine extended: a small third tone at four times the frequency, whose own window bends with weight either way (a from −1.61 to +1.47, §6.11's table), and the ratio past 3.25, toward where the zero-push step itself closes.

Why the window keeps only even powers: the reflection. Run the rider's film backwards and its honey pushes the wrong way. Divide the equation through, and that is the same as giving the rider a negative weight. Shift the clock by half a beat of the first tone and turn the angle half a turn, and the equation comes back to itself, with the weight's sign flipped, the steady push reversed and the timing mirrored. So the window's size cannot tell a weight from its negative, and it holds only even powers of the weight: β^2, β^4, β^6 and on. The slide must change sign with the weight, so it holds only odd powers: β, β^3, β^5. No rider has a negative weight; the argument is about the series. It is why the window starts flat, with no term in the weight itself, and why the slide starts as a straight line. The data agree. At weight 0.08, an odd term in the window is at most 1.7% of the even term beside it, and an even term in the slide at most 3%, at all four settings tested, inside the fit's own noise. So no fifth power hides in the window behind the fourth; the odd powers live in the timing. The same reflection explains the fine knob. Turning the second tone upside down is a half-turn of its timing, so its size enters squared, and a squared knob is gentle near its zero and steep far from it. Every knob that comes with such a reflection has a fine end.

Holding the timing still. The budget of §6.11's table is for the window's size, with the timing turned to follow the slide, as step 4 of the card says. A rider whose weight is not known, or varies, cannot have its timing followed, and then the slide costs more than the window. At the 0.55 edge, with the timing set for no weight, the push at that timing falls 0.311% by weight 0.08 and 1.248% by 0.16, while the window alone falls 0.065% by 0.16. Set the timing for weight 0.08 instead, and the push stays within 0.1% from weight 0.04 to 0.12; it is 0.303% low with no weight and 0.347% low at 0.16. On the ordinary ratio 0.99 the same setting gives 0.277% low with no weight. So at a held timing the edge buys almost nothing: the slide's loss swamps the window's. The reflection says where a held-timing weight-proof point would be. At the in-phase timing the push is also even in the weight, and its β^2 term cancels where a/ω^2 equals $k^2/2$, with k the slide's light-end number. Along the drive ratio that gap closes slowly, from −1.6046 at the edge to −0.5311 at ratio 3.25, as a grows and the slide shrinks (k from 1.8613 at 0.99 to 1.5227 at 3.25). That is by a model that treats the window as one cosine in the timing; read straight from the push at ratio 3.25, the gap is −0.6381, because the window's higher timing harmonics slide too. The second tone's size, a fine trim near its zero, grows steep far from it. At the edge, from a third of the first tone's size to the same size, it adds +0.6204 to a/ω^2. At ratio 3.25 that law did not carry, but the knob did (the next paragraph). A third tone has its own pattern to offer. The fourth harmonic weighs the fourth power 2.84 times more heavily, relative to the square, than the second harmonic does (the simplest picture, a rider filtering each harmonic, says 4), with the same sign. So a fourth-harmonic window set against the second's, at about 1.7% of its size, would cancel c, if the two windows simply added and neither slid (call this the still picture). They slide and they interact; see "A third tone, tried" below. All of this paragraph is read after the data.

A held timing made weight-proof. At ratio 3.25 the gap, read straight from the push at the in-phase timing, is −0.6381 with the second tone a third of the first. Registered from the edge's squared law, it should have crossed zero near q 1.01. Instead it climbed slowly at first and then faster: −0.5826, −0.4933, −0.3477 and −0.1424 at q 0.70, 0.85, 1.00 and 1.15, a miss (RATCHET-34). Redesigned on the ratio's own last three points, a parabola put the crossing at 1.2348 and a straight line at 1.2540. Runs at q 1.20, 1.25 and 1.30 put it at 1.2353, and the line missed (RATCHET-35). At q 1.25 the push at a held timing is 1.000026 of its weightless value at weight 0.08, against 0.998763 with the second tone a third of the first. That is 48 times flatter, and the push is 2.79 times larger. Its next term is about half the window's at the edge (c over ω^4 about −5.5), which said a 0.1% band would reach weight 0.211. Run out to weight 0.32, it did better (RATCHET-36). The push dips to 0.999227 at weight 0.24 and then climbs back, so it stays within 0.0773% of its weightless value all the way to 0.32. Just past that it falls off a cliff: 0.999906 at weight 0.33, 0.998297 at 0.34 and 0.995890 at 0.35, and at 0.36 the engine stalled and was stopped (RATCHET-37). Both registered guesses for the heavy end had the push rising out of the band; it fell out instead. The band ends at

weight 0.3364, first number 0.1850, 1.89 times as far as the weight-proof edge reaches (read with a curve through the rungs 0.33 to 0.345; a straight line between 0.33 and 0.34 had said 0.3356 and 1.92 times). What stops it near 0.36 was then looked at (RATCHET-40). A finer ladder retraced every value to the last printed place and added rungs between: 0.999234 at weight 0.335, 0.997168 at 0.345 and 0.994490 at 0.355. The push falls ever faster, by 0.134 to 0.280 per unit of weight from rung to rung, but its steepening slows, with no sign of the square-root collapse a vanishing fold would show at this timing. The ladder stalled again at 0.36. A second path, one stride straight from no weight to 0.36, also failed to finish in its time. Two paths failing at the same weight point to the rider, not the instrument; which of the step's edges give way there is not yet read. For a rider whose weight is not known, this is the setting to build: drive ratio 3.25, the second tone about 1.24 times the first, the timing held in phase. The held timing is itself a fine knob. With no weight the push is exactly even in the timing, to the last printed place at both settings checked, so a small timing error costs only at second order. This is the second tone's size, the edge's fine trim, run far from its zero, where a squared knob is steep: the fine knob became the strong one.

A third tone, tried. The engine was extended to carry a third tone beside the octave. Before any new number was trusted, it reproduced two old runs exactly, every printed digit of 192 rows and of 128 (RATCHET_3T). Alone at the edge's drive, the fourth harmonic's window is 71.7 times smaller than the octave's, its slide 5.14 times faster, and its fourth-power term 114 times its square term, close to the 131 it shows at ratio 0.99, as registered (RATCHET-39). If the two tones simply added timing by timing, each window sliding as it does alone (call this the sliding picture), the fourth harmonic set against the octave would make c worse, because its fast slide swings the two windows apart as the rider gets heavier. Set in phase, at 2.89 times that size, it would cancel c. The run said otherwise (RATCHET-41). The light coefficient moved as adding said, to −1.2912 against −1.2166, but c stayed at −10.5313, where the octave alone has it. Measured alone at the same size, the fourth harmonic still adds up to c at −3.8337 (RATCHET-42), so the two tones interact. The mix's window is 2.94% larger than the two alone summed, and its c sits 98.8% of the way back to the octave's. In a rider that bends its drive, a tone at four times the frequency beats against the octave, and four less two is two: part of the third tone folds back into the octave itself, and the mix follows the octave's weight law. Was the backwards direction real, or a bug? Two checks answered it (RATCHET-43). The same mix, run with the two tones' places in the engine swapped, gave the same edges to every printed digit, 288 rows of 288, so the third tone's path through the code is the second tone's. And the fourth harmonic set against the octave, at the size the still picture said would cancel c (1.34 times its calibration size), made c worse: −22.1855, where the octave alone has −10.614. Adding the tones had said −17.324, so the harm came out 1.72 times what adding said, while in phase the help adding promised (+6.7803) came out at +0.0827. Both ways, the mixing pulls c down. A parabola through those three sizes peaked at about 0.19 in phase, with c near −6.33, but three points cannot test a shape. Bear read the backwards sign another way: as half a dipole, or a reflected image across a bilateral symmetry, and he asked that it not be fixed out. His picture became two runs, each measured size run again with its sign flipped (RATCHET-44). Any effect of a knob with a sign splits into a part that flips with the sign and a part that does not; what the runs could test is how big each part is and how it grows. Both are big. At RATCHET-41's size the part that flips, the dipole hand, is +15.06, and the part that does not, the mirror hand, is −14.98. They cancel, and that is why the mix seemed to keep the octave's c: the size chosen sat almost exactly where the two hands balance (0.3817, read through all five sizes). The mirror hand grows as the size squared, within 3.4%. The dipole hand grows more slowly than the size, 15% short between the two sizes, so the parabola missed its larger flip: against the octave at 0.38, c came out −40.66 where it said −44.67. In phase at 0.176 the fourth-power term came out at −5.7121, 46% smaller than the octave's alone. Read through all five sizes, the in-phase best sits near 0.196, about 47% smaller, and c reaches zero nowhere: a partial knob for the fourth-power term, not a zero. A ladder of eight more sizes then tested that reading before it could be tuned (RATCHET-46). The curve drawn through the first five sizes landed on all eight within 0.02, the two beyond its range included, and the in-phase best came out at size 0.195, with c at −5.66. The dipole hand tires steadily, its share per unit of size falling from 48 to 38 between sizes 0.10 and 0.42, while the mirror hand barely bends. In the light coefficient a both hands are nearly pure: the fourth harmonic moves a almost in proportion to its signed size, down in phase and up against. That makes it a second knob beside the drive ratio, moving a and c in a different mix, which is what a reference designer needs to cancel two terms at once; the joint edge, below, puts it to work. So a fourth-harmonic third tone is not the knob that zeros the fourth-power term; like the second tone's size, it moves a (to +0.9895 against the octave, where adding said +0.6284). The next candidate is a third tone the octave cannot hear by itself, one at three times the frequency, whose own window is exactly zero by symmetry (§6.7, knob 6). Anything it does to the weight laws would come through mixing alone; that was run later ("A stray third harmonic at the edge", below).

The third tone's timing: a dial. So far the third tone sat on the mirror line, in phase or against. The reflection that keeps the window's powers even says what happens off that line. Running the film backwards mirrors every timing as well as flipping the weight, so a third tone at a quarter turn is the mirror image of one at three quarters, and between the two the weight's odd powers may now enter the window. They do (RATCHET-45). With the fourth harmonic at 0.176 turned a quarter turn, the window no longer starts flat: it changes in proportion to the weight from the very lightest rider, by 0.22 per unit of the first number, and the mirror timing gives the same slope with the opposite sign. Average the two and the slopes cancel, leaving a window that starts flat, as the reflection says it must, within 0.1% in all three pairs. The fourth-power term follows the timing round as one cosine and a small second one: −5.71 in phase, −12.88 at the quarter turn, −22.19 against. The cosine is the dipole hand, an arrow that turns with the timing and

points along the mirror line; the second, about a fifteenth of its size and turning twice as fast, says the mirror hand is not quite blind to the timing. With no weight the window hardly notices the dial: its zero moves off quadrature by at most a tenth of a degree, so the bench check of §6.8 still holds. For the engineer the dial is a warning and a knob. The warning: the weight-proof edge removes the β^2 term, but a fourth harmonic off the mirror line puts back a β term, which no drive ratio removes. If that slope grows in proportion to the third tone's size, a stray fourth harmonic under one percent of the octave, at a quarter turn, would use up a 0.01% spec at first number 0.07 by itself (a reading then; tested two paragraphs below, RATCHET-47 and 48). So at the edge a stray fourth harmonic must be held in phase or against, or kept far smaller. The knob: the slope is a third shape of weight dependence, beside the square and the fourth power, and a reference designer uses just such a tilt to flatten a curve over a range rather than at a point. Whether a small tilt widens the band over a range of riders was run later ("The tilt, built", below). Later the mirror of §6.3, which already ties the window's direction to the weight for two tones, was checked with a third tone in full (RATCHET-68). Running the film backwards turns the whole window into its mirror image, so on the mirror line weight's first push lands exactly across the window: it turns the window and cannot change its size. At an octave of 0.16 with the fourth harmonic at 0.38, the opening started 90.000000 degrees round at timing 0 and 89.999999 at timing 1, and 75 earlier weighted runs of record, from frequency 0.40 to 0.85, all started within 0.001 degrees of square. A quarter turn off the line the opening started 14.433956 degrees short of square, so there the size moves in proportion, and the two mirror timings' openings are each other's mirror images. RATCHET-67 saw the same full mirror at the dark point, where there is no size to keep and the push comes back exactly across.

The count on the dial is the subject of §6.12. Carried once round, the window's direction keeps a whole-number count, which can change only at a dark point, a setting where the push vanishes and its direction is lost.

The dial at twice the size, and a stray measured. Run again at twice the size, 0.38, the dial kept its shape (RATCHET-47). The cosine grew less than twice, as the tiring dipole hand does, while the second cosine, the part turning twice as fast, grew 4.19 times, close to the 4.65 a size squared gives: it belongs to the mirror hand. The slope off the mirror line grew 2.13 times, against the size's 2.16, so it does not tire. That made the warning above testable at the size that matters, and it held (RATCHET-48). A fourth harmonic at 1% of the octave's size, a quarter turn off the mirror line, put in a slope 1.003 times what proportion to its size said, left a and c where the octave alone has them, and moved the window by 9.9 parts in a hundred thousand at first number 0.066: the whole of a 0.01% spec, spent by a stray that changes nothing else.

A stray third harmonic at the edge. A tone at three times the frequency is silent on its own, but beside the octave it is heard (the count, §6.12), and at the weight-proof edge it works the same dial as the fourth harmonic, harder. Run at the 0.55 edge with a third tone a tenth of the octave's size, its timing held (BEAT-4), it moved the window's weight law as one cosine round its timing, with nothing left over on average, as the count says it must. On the mirror line it puts no slope into the window and moves the light coefficient by -0.068 in phase and $+0.074$ against: it shifts the edge itself. A quarter turn off the line it leaves the light coefficient near zero and tilts the window instead, by 0.087 per unit of the first number, so that at first number 0.066 the slope alone moves the window by 0.57%, where a fourth harmonic at a hundredth of the octave moved it by 0.0099% (a reading after the data). Whether that slope grows in proportion to the third tone's size was then registered and run at half and double the size: it came within 0.16% of proportion at both, 0.0434 and 0.1732 per unit of the first number, and the mirror timing gave minus each (SLOPE-SIZE). So a stray third harmonic a quarter turn off the mirror line spends a 0.01% spec at first number 0.066 by itself once it passes about 0.17% of the octave's size, where a fourth harmonic takes 1%. For the engineer this matters more than the fourth harmonic did, because a third harmonic is the distortion a sine source most often makes: symmetric clipping makes odd harmonics. With the octave on, the drive's third harmonic must be held far down, or held on the mirror line and used, like the fourth, as a trim.

The tilt, built. The warning has a second face. A slope that a stray can put in by accident, a designer can put in on purpose, the way a reference designer adds a small tilt to flatten a curve over a range rather than at one point. At the edge the window's size falls only as the fourth power of the weight, so a small rise in proportion to the weight, added first, keeps more riders inside the band: the size climbs a little, turns over, and only then falls through the limit. The first try (RATCHET-69) took the slope from RATCHET-45's lightest rung and assumed it stayed a straight line. The band did widen, from weight 0.1778 to 0.22198, but it fell 4.7% short of the prediction, because the tilt tires as the rider gets heavier: at weight 0.2 it gives 0.78755 of a straight line. RATCHET-45's own ladder, with its much larger third tone, had shown that same shape to within 0.15%; the first try had used only its first rung. Predicted again from the measured shapes, with a tenth of the spec kept in hand at the peak (RATCHET-70), a third tone of 0.014604, about a ninth of the octave (0.1109), set a quarter turn ahead, held the window within 0.1% to weight 0.22656. That is 0.9994 of the prediction and 1.2744 times the untilted edge; in first numbers the band now runs to 0.1246 instead of 0.0978. The same third tone a quarter turn behind shortens the band to weight 0.09678, again as predicted, so which side of the mirror line the third tone sits on sets the sign of the tilt. The tilt's own tiring is the limit: with that margin no tilt carries the band further. For the engineer this is a trim knob with a dial: the third tone's timing picks the tilt's sign, its size the tilt's strength, and the band to design against is a quarter wider.

The joint edge. The fourth harmonic in phase pulls a down, and a stronger drive pushes a up while it shrinks c. So the two can be set against each other: run a stronger drive, where the fourth-power term is smaller, and use the fourth harmonic to take the light coefficient back to zero. Engineers do this with two knobs whenever one knob alone cannot hold two terms. A probe first measured how hard the fourth harmonic pulls at stronger drives, because carrying its strength over from the edge was a guess (RATCHET-49). The guess was wrong in an instructive way: its pull on a hardly changed with the drive, but its effect on c shrank, roughly as the octave's own c does. Aimed with those slopes, the joint edge was then found at two drive ratios (RATCHET-50). At ratio 2.75 a returns to zero with the fourth harmonic at 0.143, and there c over ω^4 is −3.18; at ratio 3.25 it returns at 0.196, with −2.95. That is 0.30 and 0.28 of the 0.55 edge's −10.61. The registered guess for 3.25 missed by 0.45: along the size, c peaks near 0.16 and turns down before a reaches zero, so the joint point sits just past c's best. Its rival, that the curvature would eat the whole gain, failed too. For the error budget this is the first tool for the empty corner. Through the fourth-power term alone, a 0.01% window now reaches first number 0.075 at ratio 2.75 and 0.076 at 3.25, where the 0.55 edge reached 0.055, so the budget's row "0.01% at 0.07: nowhere" becomes reachable at a joint edge, with a held at zero. Read from the window itself, which also carries the next term, the margin is a little smaller. Run at the joint points themselves on fine rungs (RATCHET-66), the window leaves a 0.01% band at first number 0.0737 at ratio 2.75 and 0.0761 at 3.25. An earlier straight line between weights 0.12 and 0.16 had read 0.0715 and 0.0734, early, as such a line reads on a curve that bends upward. The corner opens, with room. The physics is ordinary harmonic mixing; the map of where the knobs cancel is this book's. Where along the a = 0 line c is best was then run at two more ratios, written first that c would keep rising with the drive (JOINT-LINE). It does not. At the joint points c over ω^4 is −4.33, −3.18, −2.95 and −3.44 at ratios 2.25, 2.75, 3.25 and 3.75, with the fourth harmonic there at 0.100, 0.143, 0.196 and 0.276. The octave alone keeps improving with the drive, to −2.56 at 3.75, but the larger fourth harmonic it then needs bends c down, as it began to past size 0.16 at 3.25. Read after the data, a parabola through the last three puts the best near ratio 3.16, at about −2.94. Whether the octave's own size can serve as a third knob is not yet run (Part 8).

The joint edge at other frequencies. Does the joint edge carry to other drive frequencies at the same drive ratio? Not as a fixed recipe (RATCHET-51). At ratio 2.75 the octave alone already differs from one frequency to the next: its a/ω^2 is +0.44, +0.53 and +0.71 at frequencies 0.40, 0.55 and 0.70, and its c/ω^4 is −3.43, −4.68 and −5.71. So at a drive this strong the weight laws do not follow the two numbers of §6.5 alone. The fourth harmonic needed to bring a back to zero, as a fraction of the first tone's strength, climbs steeply with the frequency: 0.124 at 0.40, 0.172 at 0.55 (the 0.143 above, in absolute strength) and 0.380 at 0.70. The first was read through four sizes and then run at exactly that size (RATCHET-56): a/ω^2 came out at −0.00006, and c/ω^4 at −2.025 against the predicted −2.027. The last was run at six sizes from 0 to 0.40 and read between the runs at 0.30 and 0.40 by a cubic, then run at that size, 0.3795, its test written first (RATCHET-82): a/ω^2 came out at −0.00002, where a straight line between 0.30 and 0.40 said +0.0038, and c/ω^4 at −6.384 against the cubic's −6.383. At 0.70 the crossing came out where an extrapolation had put it, but the price is high. Past a size near 0.19 the fourth harmonic stops helping the fourth-power term and starts hurting it, and at the crossing c/ω^4 is −6.38, worse than the octave alone. Through c alone, then, a 0.01% window reaches first number 0.084 at frequency 0.40, 0.075 at 0.55 and 0.063 at 0.70. The budget's empty corner, 0.01% at first number 0.07, opens at the two slower drives and stays shut at 0.70. Once again a slow drive is the forgiving one.

The fourth harmonic: four regions. Panel A of figure 59 is the fourth harmonic's knob plane on the 0.99 ratio, frequency across and weight up. Hold the tones in phase. A light rider is pushed the way it is with no weight. As the weight grows the zero slides out, and where it passes +90 degrees the in-phase push reverses. On this ratio that first reversal sits at nearly the same weight at every frequency, 0.33 to 0.40: a floor that hardly moves. Above it lies an island where the push is reversed. Its roof, where the zero swings back under +90, falls steeply as the frequency rises: weight 2.56 at frequency 0.40, 1.37 at 0.55, 0.75 at 0.70, 0.59 at 0.75 and 0.50 at 0.775. Floor and roof meet at the seam near 0.783, and from 0.785 up the island is gone. Above the roof the push points the no-weight way again, until the zero passes −90 degrees on its way home to −180 (§6.6). That last line sits at weight 5.37 at frequency 0.40 and falls to 2.22 at 0.85. Past it the push is reversed for good, and the move fades as the weight grows.

The window's two seams. The fourth harmonic's window has two seams of its own, of two different kinds. At low frequency the window dips and partly recovers (panel C). At 0.40 it falls to 0.76 of its weightless height at weight 0.40 and climbs back to 0.89 at 0.768 before falling for good. The pair of turning points shrinks as the frequency rises: a depth of 0.125 at 0.40, 0.060 at 0.45, 0.0085 at 0.50, 0.0044 at 0.51 and 0.0004 at 0.52, and none at 0.53 (RATCHET-18 and 21). A guess that it would be gone by 0.50 missed. Its redesign set two laws against each other: a straight line in the depth, which said the pair would vanish near 0.508, and the law of a fold, where two turning points meet and vanish and the depth shrinks as the distance to the meeting to the power 3/2, which said near 0.519. The fold held, and on RATCHET-21's rungs it puts the vanishing near 0.5225; runs registered first later found the pair still there at 0.5215, 0.00014 deep, and gone at 0.5235, where a straight line in the depth would have had it gone by 0.521, and on those finer rungs the fold's law re-placed it at 0.5228 (RATCHET-81). At high frequency the window instead rises a little before it falls. That rise is born where a changes sign (panel B), and a passes through zero smoothly: near 0.7576 by a straight line through older runs, registered before the runs at 0.755 to 0.77, which put it at 0.7578 (RATCHET-19). It is not born at the zero's seam, which sits 0.025 higher. So on the 0.99 ratio the plane has three seams, each

with its own kind of birth: two reversals born together where the zero's path touches the quarter turn (near 0.783), a dip and a recovery dying together as a fold (near 0.5225), and a rise growing out of the light end where a crosses zero (near 0.758).

Glossary: the words used in this book

Big words are explained where they are first needed, and gathered here. This list is for looking one up again, with the sections where it matters most. Several words carry more than one sense here; each entry lists them, and where the companion Field Guide uses a word in another sense, the entry says which sense this book means.

The toy's own words

- *Weight, honey, pull, ripple, ledges*: the toy's mass, its drag, the slope it sits on, the moving wave and the standing wave (§2.1). *Pull* has other senses, each said where it stands: the locking pull of mode locking in general (§4.1); a coupling's strength, such as K in Adler's equation or a follower's answer to the gap it sees (§4.10); the junction's own pull, the restoring push of its supercurrent, which plays the ledges (§4.28, §6.12); and an averaged force, as in Kapitza's pendulum (§6.12).
- *The plain*: rest, where the ball sits on a ledge. *The plain's edge*: the least drive at which the ball leaves it, read along the drive's ray (§2.4, §4.18).
- *Fins*: whole-number locks, so many ledges per tick (Shapiro-like steps). *Ramps*: the level lines of one smooth field, ledges per tick, widened where the ball locks. Fins are the ramps at whole numbers (§4.18).
- *Treads*: the flat steps of a staircase, one for each lock (§4.7, §4.8). *Seams*: in the atlas, the whole-number lines of the ball's own pace (§4.18); in §4.17, the boundary between two basins; in Part 6, Bear's word for any wall where a knob's behaviour changes (§6.10); in §4.6, the window of push in which a relay's kick rescues a gear (figure 14). *Seeds*: the fractions that hug the edge. *The trickle*: the 1/n locks near the edge (§4.18).
- *The outside tick*: each passing of the ripple (§2.1). *The rider's own pace*: how fast it slips with no ripple (§4.18). *Pace* in Part 4A: the gearbox's drive knob, the speed at which the strong pattern sweeps past, the weak one sweeping at 0.618034 of it (§4.1). *Event*: each passing of the ripple, or each cycle of the radio wave; the book reads the rider once per event ("Read in event time", in "The book in one page").
- *Memory*: four senses, each right in its own place; three of them are the Field Guide's. Route-dependence, or hysteresis: two states side by side, and which one the rider ends in depends on the road it came by (§4.14, §4.18). A rider that does not forget where it started: a slipping rider, its stretch rate at zero, remembers, where a gripped rider forgets (§2.6). The memory floor: the least pull at which the memory test's two starts part, which is where a start half a turn round gets rolling at all (0.86 to 0.87 at weight 2), not the retrapping pull at which a rolling ball stops (0.78 to 0.80) (§4.18, lesson 55). And a needle's memory, its inertia, in the Loop Atlas's needles (§4.7). Other uses fall under these: 'a tread with memory' and the weight that 'opens a memory' at a tread's edge (figure 15) are route-dependence (§4.8, §4.10), and an actuator that is 'a slip with memory' keeps every error in its count, the second sense (§4.10). 'The grip remembers' (§4.10) is a use outside the four: a stiffness that fades with a time constant. The Field Guide's fourth sense, a medium whose per-cycle reading moves with the clock, is not used in this book.
- *Lag*: the ball's place on its ledges at a flash of the strobe, counted in turns; a dance's stops are lags. *Clock face*: the lag drawn round a circle; *straightened*, stretched by one number so that a weightless dance ticks evenly round it (§2.1, §3.5, §4.25). Two other senses are times, not places: a fixed offset in a drift law, the drift going as the sine of the timing less a lag (§6.9), and a delay, as in a loop filter's lag (§5.6), the lag a coil would give the shuvit's push (§4.10) or a ramp's (lesson 119); the Field Guide's lags are of this second kind.
- *Dance*: a motion that repeats. *Stops*: where a dance sits at each flash of the strobe (§3.3, §4.22). *Groove*: a locked fraction, a dance on the record (§4.12).
- *The S, the foothold, the second S, the ring, the wrap*: landmarks of the 1/2 groove (§4.19).
- *The conveyor*: the 1/1 lock, the ball carried one ledge for every passing at the ripple's own speed; at the fair, surfing (§4.19). RING CONVEYOR, a run named in §4.10, is another use, a name only.
- *Frozen balance points*: the two lags where the push at the strobe is zero (§4.21).
- *Chalk line, couples*: the dance hall's words for a frozen balance point and a dance's pair of stops; the chalk line answers the drive at the strobe beat, the couples the whole bar (§4.25, §4.28).
- *The pin*: the drive, 1.4360, at which every weight's lower stop of the 1/2 groove crosses; it is the mirror point of a dance that weight barely moves (§4.20, §4.26).
- *Red and purple*: the two mirror dances at a root; with a touch of weight red is the one that stays (stable) and purple the one that runs (unstable) (§3.5, §4.23).
- *The far-side leaper*: the dance whose big step crosses the far side of the circle (§4.23).

- *Grip, excess*: how firmly a dance holds (§3.5, §4.28); also a coupling's strength, as between two stations, twice the product of their strengths (§4.10), or a stage's hold on its pattern (Part 8); *the grip ratio*, $-(1 + \delta)$, sets the stable dance's grip against its partner's, and *the excess* δ is how much the stable dance outgrips it (§4.28). This grip ratio is not a friction ratio, the sense some fields give the phrase. In "the grip forgets and the slip remembers" (§2.6), grip means being locked in a gear. *Slip*: the ledges slipped per tick, the count, the number this book trusts most (§2.2); in Part 6, the rider slips when it leaves the step.
- *Comma*: the remainder that does not close (§1.3). *Gear, gearbox*: Part 4A's words (§4.1). In Part 4B, *gear 1* and *gear 2* are the record's two wide locks, the needle running at 0.618034 and 0.236068 of the drive (§4.14). In §4.28, *first gear* is δ_1, how hard a medium's simplest dance grips, and SLOPE-2's *gear* is q times how fast the rotation per beat grows with the bias; in §6.6, *the gear* is the ratio, about 5 to 1, between the fourth harmonic's zero slide and the second's. *Relay*: two gearboxes in a row, coupled one way through a filter (§1.3, §4.3). *Room*: the distance left to the downstream gear's edge (§4.4); in §4.14, the stretch of drive between gear 1's edge and gear 2's door. *Toll*: the nearly fixed cost each time the strong pattern passes (§4.9).
- *Watershed*: a value that divides two kinds of behaviour, as a ridge sends rain to one river or the other; the relay's kick rate over the next stage's tooth rate has one at 1 (§4.5).
- *Needle*: the rider in Part 4B, drawn as the needle of a record player (§4.12). Also the Loop Atlas's needles, angles driven round a circle as the rider's lag is (§4.7), and RF-10's three-armed needle, turning in the field of three emitters (§4.10).
- *The turned walkway*: the ripple running with the ball instead of against it (§2.4).
- *First number, second number*: the weight times the frequency, and the drive's strength over its frequency squared (named in §6.7, knob 3, and set out in §6.8, steps 1 and 2; the law they carry is §6.5).
- *Drive*: the one knob that turns the pull and the passing speed together along a ray: the pull is the clock ratio times the drive and the passing speed is the rest, so the two add up to the drive; on the golden ray the pull is 0.618034 of the drive and the passing speed 0.381966 of it (§2.3). For the Josephson junction the drive is the radio wave that pushes it, with its strength A and its frequency ω (the fair's stunt show and finale, §4.28, Part 6). In Part 4A's relay the drive is the second stage's steady push, against which the relay's kick rescues a gear (§4.6).
- *Clock ratio*: the pull over the drive, which picks the ray the drive turns along; the golden ray's is 0.618034, the fractional part of the golden ratio (§2.3). *Pad atlas*: the map of the whole plane of clock ratio and pull, at weights 0.05 to 10 (§2.3, §3.2).
- *Floor*: several senses, each said where it stands. *The floor* a rider pushes on: the medium that holds the toy or the junction still, held in turn by whatever holds the laboratory down (§1.2, §1.4, §6.1, §6.8, §6.9). *The stage floor*: the least drive at which the second stage holds its gear on its own (§4.6). *The memory floor*: see Memory. *The island's floor*: the lightest weight of Part 6's island, where the push is reversed (§6.11). *An error floor*: in Part 4A, the largest error a true lock can show on a run of a given length, which falls as the run grows; a pace counts as in gear only under it (§4.1, §4.2). *The rounding floor*: the computer's own, near 10^{-13} a passing (§3.6). *A map's own floor*: the lowest setting a map was run at, where a reading can sit without meaning anything (§7.1, lesson 6). *The floor near 1.2*: the level the holding power of §6.12 eases toward as the frequency rises. *The print floor*, or the ruler's: the smallest effect a reading can show ("the floor is the ruler's, not the rider's", §6.11); only this one is the Field Guide's floor, how much a measurement moves when nothing is happening. At the fair, the floor is the dance floor.
- *Window*: a band of settings where something holds. The window of push, in which the relay's kick rescues a gear (§4.6); a periodic window inside chaos (§4.8); the window count (§4.25); the hand-off window (§4.28); and Part 6's window, the band of steady push in which the rider slips one way and not the other, whose size is the lopsidedness B+ + B− (§6.7). From §6.12 on, that window has a direction as well as a size: the best timing of its push, drawn as an arrow; and in §6.6's table 'window' is that band's size, the rectifier's height. Also a reading window, the stretch of passings a reading averages over, which reports its middle (lessons 8 to 11); the census's window, 0.02 wide in pace, the size of its error (§4.1); the band of drive in which the launch-and-catch gap holds (§4.24); and WINDOW-1's window, the band of bias in which a tongue's dance lives (§5.2).
- *Edge*: where a behaviour starts or stops. The plain's edge (above); a gear's top and bottom edges (§4.4); a tread's edge, which is a fold (§4.8); in Part 6, the step's two edges, B+ and B−, the pushes at which the rider starts to slip; and the weight-proof edge, the drive ratio A/ω^2 at which the light coefficient a changes sign, also called the 0.55 edge (§6.11, §6.12).
- *Door*: a way out, or a way through. Gear 2's door, the straight line in weight where its own edge lies (§4.14); Bear's doors, the two gears a start can choose between (§4.17); the three doors out of the plain, the fold, the doubling and the fold of another dance (§4.18); and the dark point, the only door through which a whole-number count can change (§6.10, §6.12).
- *Weightless root*: a drive at which, with no weight, every start is a p/q dance (§3.5, §4.22). *Selector M*: the first-order function whose zeros are the dances that survive a little weight; a dance is stable where M slopes down (§3.5).

- *Wrinkle*: a tongue's wrinkle: for each starting lag, the bias at which an orbit starting there closes after q beats, drawn round the circle; its spread is the tongue's width, and its lean decides which dance holds (§4.28). *Walking pair*: two mirror-image dances that hold at once inside a pinch and carry the hold from the dance through lag 0 to the dance through lag 1/2 (§4.28, figure 42). *Hand-off window*: the stretch of second tone across which that hand-off happens (§4.28).
- *Flat point*: the octave size at which the third tone's timing changes only the window's size, never its direction (§6.12). *Small dial*: the short way of placing the dark and flat points before looking for them: the fourth harmonic's window with no octave, and one small octave read at three timings, four runs in all (§6.12).
- *Arrows*: in §6.12, each tone's share of the window drawn as an arrow, the arrows added head to tail; the octave's stays put and the fourth harmonic's turns with the timing. They are not the carried arrow of a holonomy.
- *Magnifier, decoupler, probe*: in §6.12, the octave set near a dark point, where a small turn of the timing swings the push's direction far (the magnifier); the octave set at the flat point, where the timing trims the push's size and leaves its direction alone (the decoupler); and the small dial's one small octave, held while other settings change (the probe, or probe octave).
- *Swing*: the vine swing of the fair's stunt show; the swing equation of a power grid (§2.5, §5.3); a tone's swing, its strength over its own frequency, how far it could swing the rider in one wiggle (§6.12); the largest swing of the window's direction (§6.12); and one rider's swing, the size of its output, in the arrays (§6.13).
- *Card*: the rectifier's rules, short enough to fit on one (§6.8).
- *Bead*: the rider drawn as a bead on a wire, pushed along by the patterns (Part 4A); an envelope bead feels only where the sound is loud and soft (§4.10).
- *Direct drive, first gear, second gear*: the bead locked one for one to the strong pattern; or locked making one, or two, slips for every beat of the two patterns (§4.1). Gear's other senses are under Comma, above.
- *Census, faux lock*: the census counts the paces at which the rider holds each fraction (§4.1); a faux lock is a stretch that a count reads as held but a longer read shows drifting off, often a smooth curve passing near a fraction (§4.2, §4.6).
- *Launch and catch*: the red dance's one passing that leaves from just outside the knife-edge frozen point and is caught on the far side, a trapeze act (§4.24).
- *Even fan, odd fan*: riders sharing one drive with their timings spread evenly round the octave's turn; an even count cancels every push, an odd count keeps only the harmonics that are multiples of its count (§6.13).
- *The step, the zero*: in Part 6, the band of steady push over which the rider stays locked and does not slip, and the timing between the two tones at which that step is fair (the opening of Part 6).

Words from the trade

- *Arnold tongue*: on a map of two knobs, the wedge of settings in which a rider stays locked to one fraction (§2.5, §4.2).
- *Bias, bias current*: the steady push on a Josephson junction, which plays the toy's pull (§2.5, §5.2).
- *Bifurcation*: a setting where the kind of motion changes, as when a lock is born or a dance doubles (§4.16).
- *Chaos*, which this book also calls *shaking*: motion that follows exact rules yet never repeats, so that two riders started a hair apart soon disagree (§2.6).
- *Critical current*: the largest steady current a junction carries with no voltage across it; the book measures the junction's pushes as fractions of it (§6.2).
- *Damped pendulum, tilted washboard*: the rider's family, a ball on a tilted, corrugated slope with drag (§2.5).
- *Dark point*: a setting where the push vanishes at every timing, so the timing has nothing to steer; the only place a whole-number count can change (§6.6, §6.12).
- *Devil's staircase*: a rider's slip against the drive, flat on every lock, with steps at every scale (§2.5).
- *Farey tree, Stern–Brocot tree*: the family tree of fractions, each born between two parents by adding the tops and adding the bottoms (§4.2, §4.12).
- *Feigenbaum's ratio, 4.669*: how fast the gaps between successive doublings shrink, the same in many systems (§4.16).
- *Fibre bundle*: a circle, or another shape, standing at every point of a line or a loop of settings; it is where holonomy lives (§6.6).
- *Fold, saddle-node*: where a lock ends, the stable motion and its unstable partner meeting and vanishing together. *Ghost*: the slow spot a fold leaves behind (§4.8).
- *Harmonic, octave*: a tone at a whole-number multiple of the first tone's frequency; the octave is the second harmonic, at twice it (Part 6).
- *Harmonic mixing*: a steady push made by two tones whose frequencies stand in a whole-number ratio, where the symmetry allows it: a tone and its octave push; a tone and its third harmonic alone do not (§6.7, §6.9).

- *Holonomy*: what has changed when something is carried once round a loop and comes home; in this book it is counted, not measured (§6.6, §6.12).
- *Hysteresis*: memory in a machine: two states side by side, and the road taken decides which one it is in (§4.14, §4.18).
- *Josephson junction*: two superconductors with a thin barrier between them, whose phase obeys an equation of the rider's family, with its capacitance as the weight (§1.4, Part 5).
- *Light cone*: in relativity, the border between the places a signal can reach and the places it cannot; in the weightless junction, the line $|\Phi 11| = 1$ between passings that rotate and passings that lock to a whole slip; with weight, each step's own line $|\tau| = 1$ between settling straight and ringing (§4.28).
- *Lyapunov exponent*, the book's *stretch rate*: how fast two neighbouring motions spread apart; above zero is chaos; below zero a gripped rider forgets where it started, and at zero a slipping rider remembers it (§2.6, §3.3).
- *Mediant*: the fraction born between two parents by adding the tops and adding the bottoms, 2/5 from 1/3 and 1/2 (§4.12, §4.28).
- *Mode locking, phase locking*: keeping a fixed whole-number rhythm with a drive (§1.3, §4.1).
- *Möbius map*: a simple map of the circle that keeps points in order, set by a small table of four numbers; one passing of a weightless rider is one (§3.5).
- *Multiplier*: the factor by which a small nudge grows or shrinks over one lap of a dance (§4.16).
- *Performance envelope*: the region of settings inside which a device does what its data sheet says (§6.11).
- *Period doubling, the flip*: a dance that starts taking two laps to repeat, then four, then eight (§4.16).
- *Poincaré section, strobe*: looking at a motion only once per beat (§2.2).
- *Quadrature*: a quarter turn between two tones (Part 6).
- *Rapidity*: the size of a boost, as an angle is the size of a rotation; inside a weightless step it is the step's grip, the step's multiplier being e to the minus twice it (§4.28).
- *Rectifier*: something that turns a back-and-forth drive into a push one way (Part 6).
- *Rotation number*: the average slip per beat, a whole number or a fraction for a locked rider (§2.5, §4.15).
- *Shapiro steps*: the flat steps of locking that a Josephson junction shows under a radio wave (§1.3, §2.5).
- *Stable, unstable*: a stable motion pulls nearby starts in; an unstable one pushes them away (§4.21).
- *Stewart–McCumber parameter*: a junction's weight, which grows with its capacitance, its critical current and the square of its resistance (§6.7, knobs 2 and 8).
- *Symmetry, mirror, time mirror*: a change, such as a flip, a shift or running the film backwards, that leaves the equation as it was, and so forces exact facts (§3.5, §6.3).
- *Uncertainty exponent*: how much aiming a start more finely shrinks the doubt about where it ends; 1 for a smooth border, below 1 for a fractal one (§4.17).

Words from the record

- *Registered, written first*: a prediction put into the dated record, with a bar it could fail, before its run (§3.1).
- *Lean, rival*: the registered prediction most expected to hold, the record's word for it, and the named alternative written down beside it (the rival, §3.1, step 2; the lean, first in §4.2). Elsewhere a lean is a tilt: a wrinkle's lean decides which dance holds (§4.28); an engine's lean is a steady bias that tightening its tolerance shrinks (LEAN-1, §6.12), as the Field Guide's ruler has a lean; and a number "leaning a shade above" another sits just over it (§4.28).
- *Held or landed, failed or missed*: a prediction that met its written bar, or did not (§3.1).
- *Blind*: tested on data not yet seen (§3.1).
- *Stand-in, dry run*: made-up files on which a reader is tried before the real data; the trial is the dry run (§3.1, step 3). *Fidelity run*: a rerun that must give the old numbers back after a tool changes (§3.1, step 4).
- *Sitting, exchange*: the record's units of work: a sitting is one working session (B1 from 3 October 2026, on to P4), an exchange one message from Bear and its answer; a lesson's mark says where it was first written, so (B9.44) is lesson 44 of the series kept across sittings B8 and B9, written in B9, and (PR9.22) the twenty-second of its pattern-recognition notes (§3.2).
- *The explorer, the witness*: the Depth Axis Explorer, Bear's interactive maps of the toy (§3.2); the witness, WITNESS_FOLD, a second rectifier engine written from scratch to check the first (§3.2, WITNESS-1); WITNESS_COIN, written the same way for the coin (§4.17, WITNESS-COIN-1); and WITNESS_EAR, for the couples' ear (§4.28, WITNESS-EAR-1).
- *Names in capitals* (KNOBS-3, RATCHET-26, ROTATION-CHECK): the names of runs and checks in the record, where each can be looked up (§3.5).
- *No-physics read*: a reading of the newest data with its labels and laws taken away, told as what the numbers are doing; one opens seven of the eight parts of this book (§3.8).

Signposts

The rider’s family, and the mathematics this book leans on, belong to others: Shapiro, Stewart and McCumber, Adler (for the locking of oscillators, the phase locking of §1.3), Chirikov, Escande and Doveil, Arnold, Bohr, Bak and Jensen, Buchstaber, Karpov and Tertychnyi, Ilyashenko and Glutsyuk, Poincaré, Morse and Hedlund, Christoffel, Steinhaus and Sós (for counting the steps of a turning circle, the counts of §4.25), Melnikov and Malkin, Guckenheimer and Holmes, Devaney, Lamb and Roberts, and, for the forced pendulum’s own mathematics, Hamel, Mawhin, Ortega and Alonso, with Schneider and Seeger, Marchesoni, Hänggi, Flach, Denisov, Yevtushenko, Zolotaryuk and Renzoni for harmonic mixing and its symmetries. They are named so that a knowledgeable reader can place this work at once. Except for the few glossed here, each is met in the text where its house stands, most in §2.5 and §6.9. What this book claims is narrow: an instrument that reads the locking integer off the orbit, a time-mirror reading that places the pin and the dancers with nothing fitted, the counts read in a straightened lag, the junction’s one-knob grip and its three zones, the grip’s multiplication where it is small; in Part 6, a two-tone rectifier mapped knob by knob, with its weight-proof setting, its dark points and flat point placed by symmetry and by a small dial, a weight meter that a drifting drive cannot fool, and a design rule for riders in arrays; and an honest record of how it was found.

One line belongs here because it separates two families that look alike. Systems without drag can show a mixed phase space, regular islands in a chaotic sea, as in the planar two-electron quantum dot of Schröter’s thesis (2013), which characterises that system’s dynamics as mixed regular-chaotic. Those islands are the KAM kind, kept by the absence of drag. Our rider has drag, so its islands are attractors: every orbit nearby settles onto them. Results about one family do not carry to the other without that check.

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- Kin in the program: *Loop Walking: A Field Guide to Hacking the Universe with Holonomy* (Schulz, 2026), concept DOI 10.5281/zenodo.21811588, checked on 5 November against the version list of its Zenodo record: five versions, the latest of 22 September (the guide’s own version 6); the rows this book cites belong to its next version, not yet deposited
- Kin in the program: *The Latch Atlas: Holonomy and Locks that Can’t Be Picked* (Schulz, 2026), concept DOI 10.5281/zenodo.23068382, checked on 5 November against the version list of its Zenodo record: one version, of 30 September
- Kin in the program: *The Loop Atlas* (Schulz, in preparation), the companion book of which this one is the radio spin-off (§1.1); not yet deposited
- Kin in the program: *Does Claude Dream of Holonomic Sheep?* (Schulz, in preparation), the collected no-physics reads (§3.8); not yet deposited
- Kin in the program: *Taking a Loop Walk Through the Integer Forest* (Schulz, 2026), concept DOI 10.5281/zenodo.21449569, its title checked against its Zenodo version pages
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- Kin in the program: *Held, Rocking, Locked: The Islands of a Light Star Around an Eccentric Binary* (Schulz, 2026), concept DOI 10.5281/zenodo.23755473, checked against its Zenodo record on 31 October (version 1, published 28 October)
- Kin in the program: *Loop Walking Cascadia: a Triangle Lattice, its Rogues and Witnesses, and One Opening of the Vault* (Schulz, 2026), concept DOI 10.5281/zenodo.22076451, checked against its Zenodo record on 24 August, when it was published, and again on 31 October
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Declaration

Status. This is a preprint from a part-time volunteer research team. It has had no peer review. Its literature pass is limited: the text says of each outside work whether it was read, read in part, or checked only against an abstract or an index record, and names given without such a mark are from recall.

Provenance.

what	where
every registration, verdict and redesign check	relay/PREREG_RELAY_STATION_2026-10-03, the working record, kept privately; its public extract, below, is published with the book
the book's plan, from which its numbers were first carried	docs/LOOP_ATLAS_RF_BOOK_PLAN_2026-10-12
the dancers' paper, with its figures and files	docs/LOOP_ATLAS_RF_FROZEN_BALANCE_POINT_2026-10-12
the engines and their outputs, published with the book as A_Rider_in_Two_Waves_engines_scripts.zip and A_Rider_in_Two_Waves_engines_outputs.zip	engines_B/, engines_B7/ to engines_B12/, engines_P2/ to engines_P4/, explorer/, relay/ and lissajous/ (the figure-8 file), with README_ENGINES.md and two manifests that give each file's checksum in the record and as published
why many files are named LOOP_ATLAS_ or RF_SPOKE_	the first radio engines were built in the Loop Atlas, and the project's later files carry its working name; the names were kept so that every reference still finds its file
the method's lessons as first written, sitting by sitting	docs/FIELD_GUIDE_LESSONS_RF_SPOKE_B6, B8, B9 and B10; docs/PATTERN_RECOGNITION_LESSONS_RF_SPOKE_B8 and B9
how to rebuild the baseline	RF_SPOKE_DANCERS_SEED_CRATE_2026-10-13
how the count of "The book in one page" was made: the record read by script, then by eye, and the ledger of every result the book names	tools/RECORD_VERDICTS_P3_2026-11-01, the reading script's fifth version; tools/RECORD_COUNT_BY_EYE_P4_2026-11-03, which runs tools/RECORD_COUNT_BY_EYE_P1_2026-10-24 (the count to 22 October, carrying the earlier corrections word for word) unchanged and adds the readings by eye from then on (docs/receipts_P3/ and docs/receipts_P4/); the ledger, docs/RF_SPOKE_HEADLINES_B12_2026-10-18. The scripts and the readings by eye are published with the book as its counting kit, which gives the count from the public extract alone; the ledger is kept beside the book
the record's public extract, published with the book	made from the record after its last entry by tools/RECORD_PUBLIC_EXTRACT_P4_v8, which keeps every registration, verdict and result
the Depth Axis Explorer, published with the book as depth_axis_explorer.html	explorer/depth_axis_explorer_P4_2026-11-05: the sitting B11 build, its headings renamed for this book on 5 November

Dual use. The work is a classical dynamical toy and its published cousins (a pendulum, a Josephson junction, a power grid's swing). No risk category beyond ordinary engineering use has been identified; the propulsion question is answered by the wall in §1.2.

Tool and computational resource disclosure

Tools used. The runs, the engines, the figures and much of the prose were made with Claude (Anthropic), working with Matthew “Bear” Schulz, who set the questions, read the pictures and made the rulings. Claude was also used for literature search and to write the readers that graded every registered prediction. The engines are Python programs built on NumPy, with SciPy in some and extended-precision arithmetic in one (JJ_16); the figures were drawn with matplotlib; the live bench of §3.2 runs in a web browser, and the toy of “A day at the fair” (play_the_toy.py) is plain Python, checked against 2,911 riders of record (KNOBS-3, BRIDGE-1 and COIN-1). Typesetting used pandoc and wkhtmltopdf.

Responsibility. Responsibility for the correctness and adequacy of the arguments and results presented here, and for the completeness and accuracy of the citations, rests entirely with the named human author. No automated system is an author of this work.

Verification performed. Every number traces to a named run in the record. Each prediction was written into that dated record, with a bar it could fail and usually a named rival, before its run, and graded by a reader written before the run and tried first on made-up files (§3.1). An engine was changed only behind a fidelity run that gave back the numbers of record, and the weight meter of §6.12 was reproduced by a second engine written independently (WITNESS-1), Bear’s coin of §4.17, rider for rider, by another (WITNESS-COIN-1), and the couples’ ear of §4.28, root for root and couple for couple, by a third (WITNESS-EAR-1). The count of predictions landed and missed was made by a script that read the whole record to 5 November, and then every landing the book did not count, every test a later version of the script changed or added, and every test from 23 October on, was read by eye, one by one (“The count”, in “The book in one page”). Outside works are marked in the text as read, read in part, or checked against an abstract or index record, and are listed under “Sources checked”. This PDF was checked with lint_deposit.py, version 1.9.

Attribution limits. Three questions of priority are open, and the book claims none as new: whether the measured gap between the zero of the step’s lopsidedness and the zero of the drift along the weight (§6.9) is written down elsewhere (the step as a threshold apart from the drift is drawn for a junction without weight by Scheer and colleagues, 2025), and whether the counting rule of §6.12 has been stated in general form (Denisov, Flach and Hänggi give its first case). Whether the half-turn count of §6.6 has been named before in the literature on dissipative ratchets is not known. Outside names given without a mark of what was read are from recall and have not been checked against their publishers’ records.

Computational resources. Every run was made in a single cloud workspace or, for the longest surveys (the pad atlas, the door line, the 3/7 ladder and the survey of 32,080 riders of record at twenty weights), on the author’s laptop with at most eight worker processes.